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# United States Patent
Kigoshi et al.

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(45) Date of Patent: Sep. 9, 2025

(54) **SEMICONDUCTOR LIGHT EMITTING DEVICE**


(71) Applicant: **ROHM CO., LTD.**, Kyoto (JP)

(72) Inventors: **Satohiro Kigoshi**, Kyoto (JP); **Yuki Tanuma**, Kyoto (JP); **Gen Muto**, Kyoto (JP); **Minoru Murayama**, Kyoto (JP); **Okimoto Kondo**, Kyoto (JP); **Chikoto Ikeda**, Kyoto (JP); **Yusuke Nakakohara**, Kyoto (JP)

(73) Assignee: **ROHM CO., LTD.**, Kyoto (JP)



( * ) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 758 days.

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§ 371 (c)(1),
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PCT Pub. Date: **Oct. 29, 2020**

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Nov. 14, 2019 (JP) ................................ 2019-206439

(51) **Int. Cl.**
***H01S 5/183*** (2006.01)
***H01S 5/0239*** (2021.01)
***H01S 5/042*** (2006.01)

(52) **U.S. Cl.**
CPC ........ ***H01S 5/04256*** (2019.08); ***H01S 5/0239*** (2021.01); *H01S 5/183* (2013.01)

(58) **Field of Classification Search**
CPC .................................................... H01S 5/0239
See application file for complete search history.


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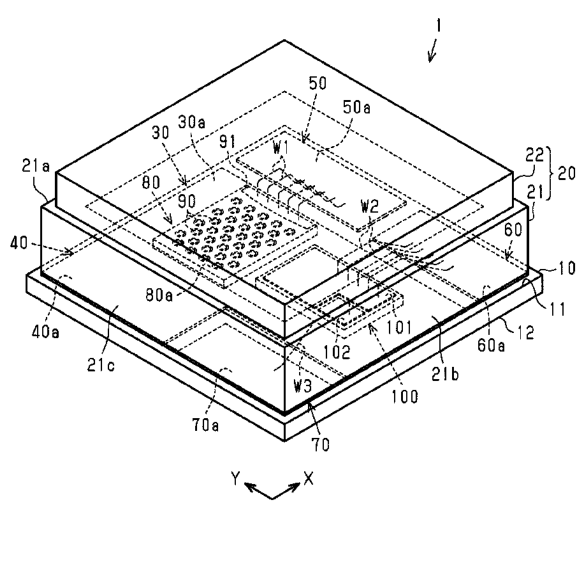
*Primary Examiner* — Minsun O Harvey
*Assistant Examiner* — Stephen Sutton Kotter
(74) *Attorney, Agent, or Firm* — HSML P.C.


(57) **ABSTRACT**


A semiconductor light emitting device includes a substrate, a common conductive portion formed on the substrate, a semiconductor light emitting element mounted on the common conductive portion, and an electronic component mounted on the common conductive portion and electrically connected to the semiconductor light emitting element by the common conductive portion. This structure shortens the conductive path between the semiconductor light emitting element and the electronic component, thereby reducing capacitance caused by the conductive path between the semiconductor light emitting element and the electronic component. Thus, while reducing parasitic capacitance, the semiconductor light emitting element and the electronic component are electrically connected.


**16 Claims, 81 Drawing Sheets**



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| JP | 2013041866 | 2/2013 |
| JP | 2015023229 | 2/2015 |
| JP | 2015115389 | 6/2015 |

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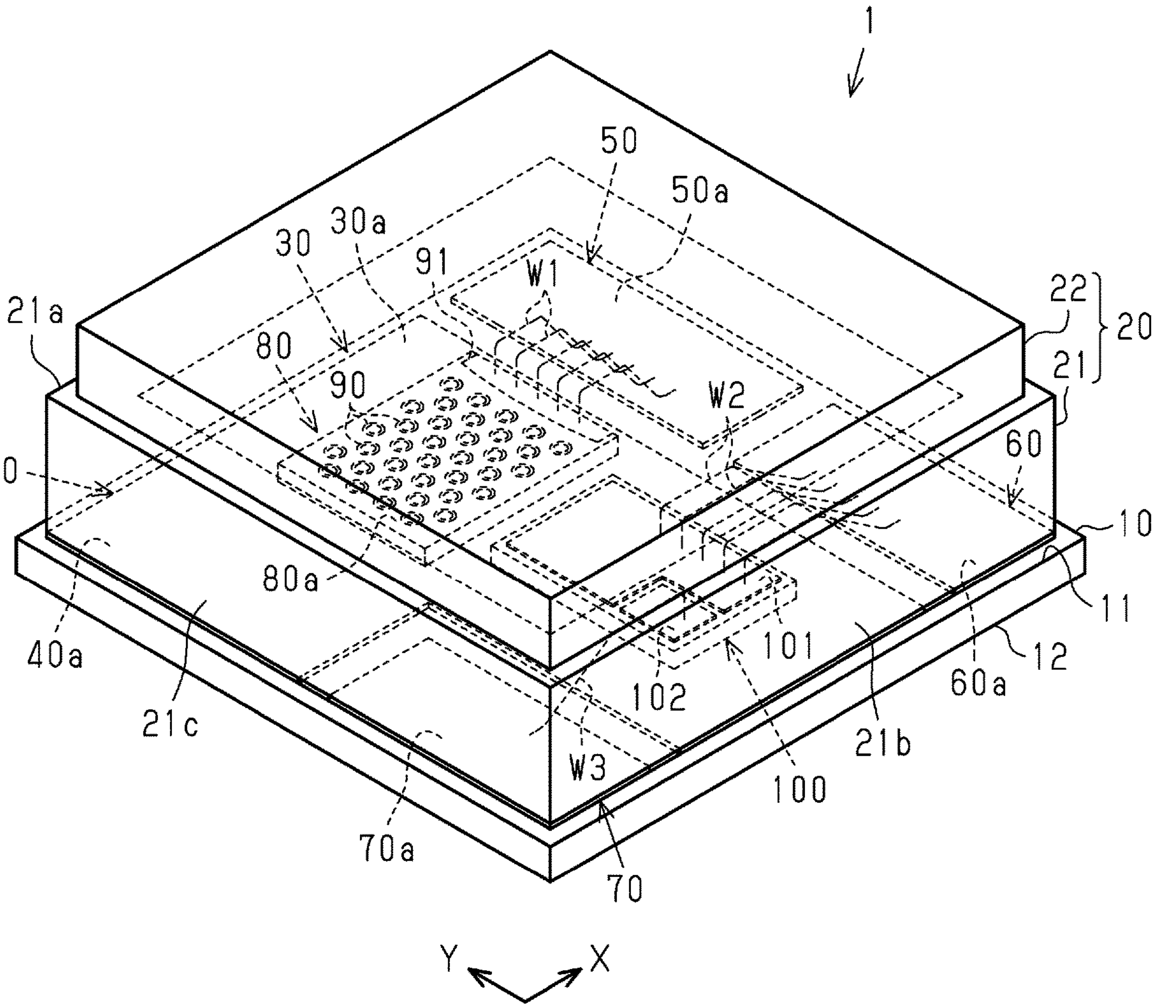
International Preliminary Report on Patentability issued for International Patent Application No. PCT/JP2020/016810, Date of mailing: Nov. 4, 2021, 11 pages including English translation.
Notice of Reasons for Refusal issued for Japanese Patent Application No. 2023-211301, Dispatch date: Feb. 25, 2025, 7 pages including English machine translation.

* cited by examiner

Fig.1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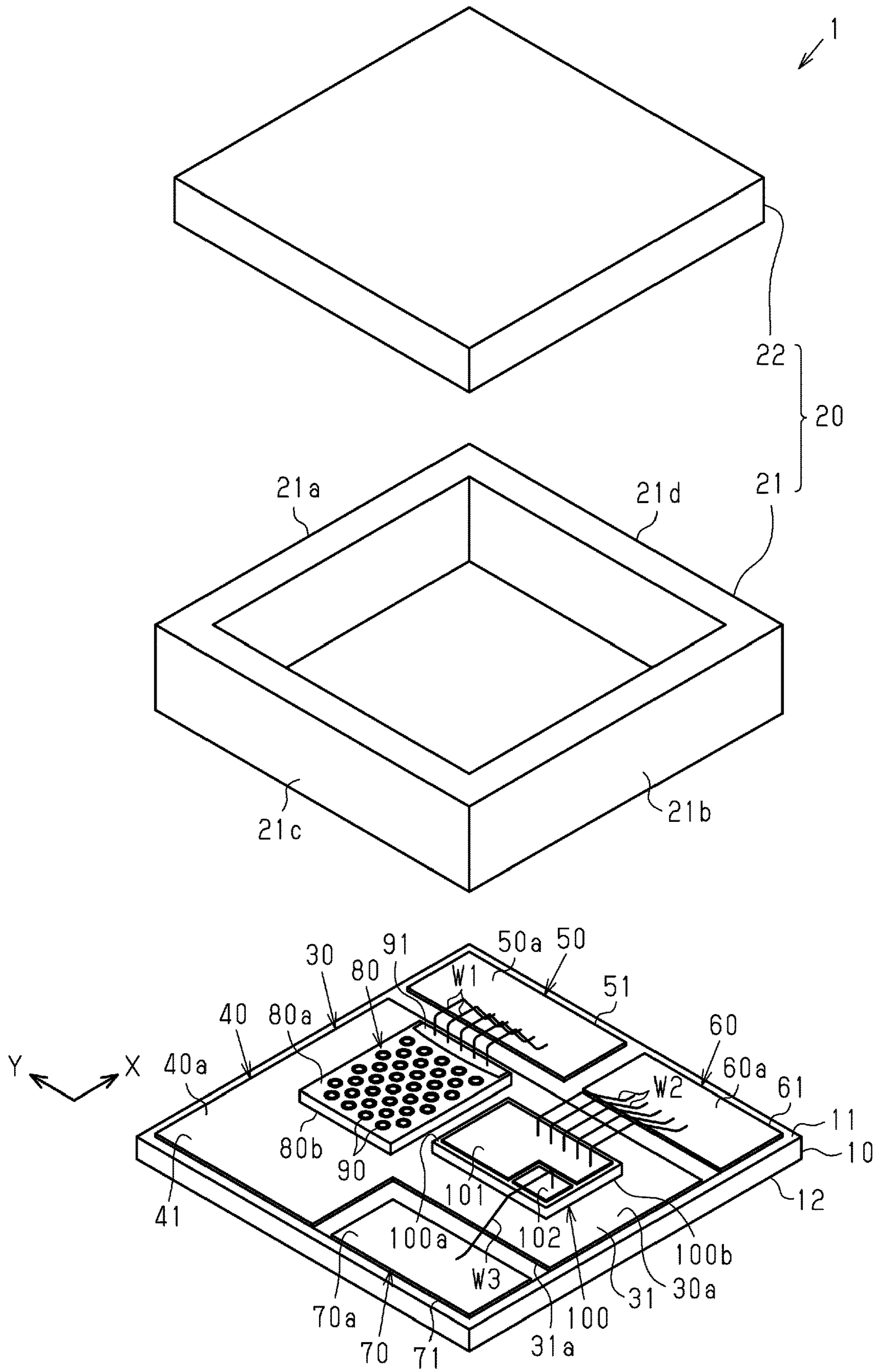
50
50a
30
30a
91
W1
21a
22
20
21
80
90
W2
60
40
10
11
80a
12
40a
101
60a
102
21c
21b
W3
100
70a
70
Y
X

Fig.2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21a
21d
21c
21b
91
50a
50
30
80
W1
51
40
80a
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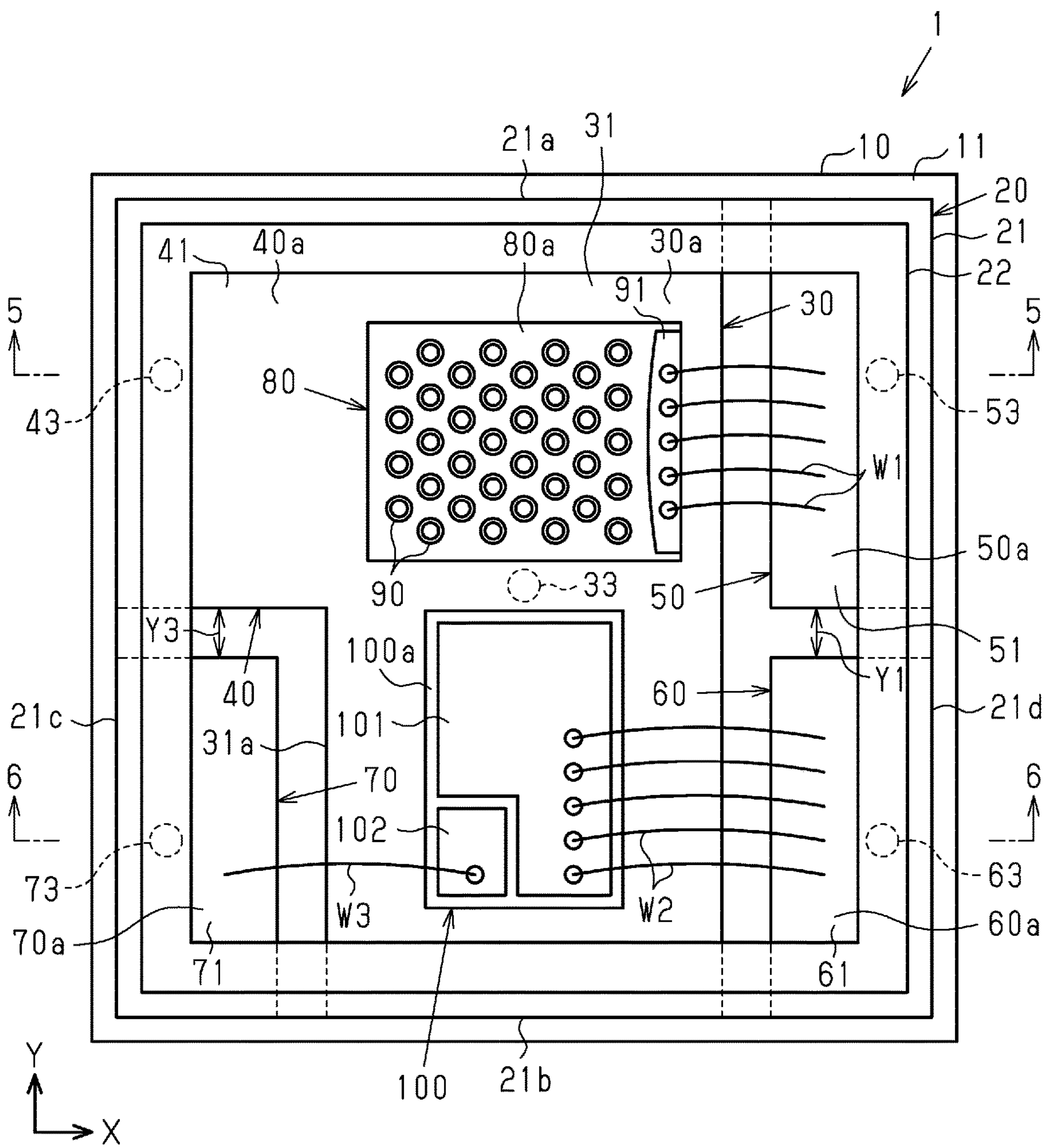
X
40a
60a
W2
61
11
10
80b
90
101
12
41
100a
102
100b
W3
31
30a
70a
100
70
71
31a

Fig.3



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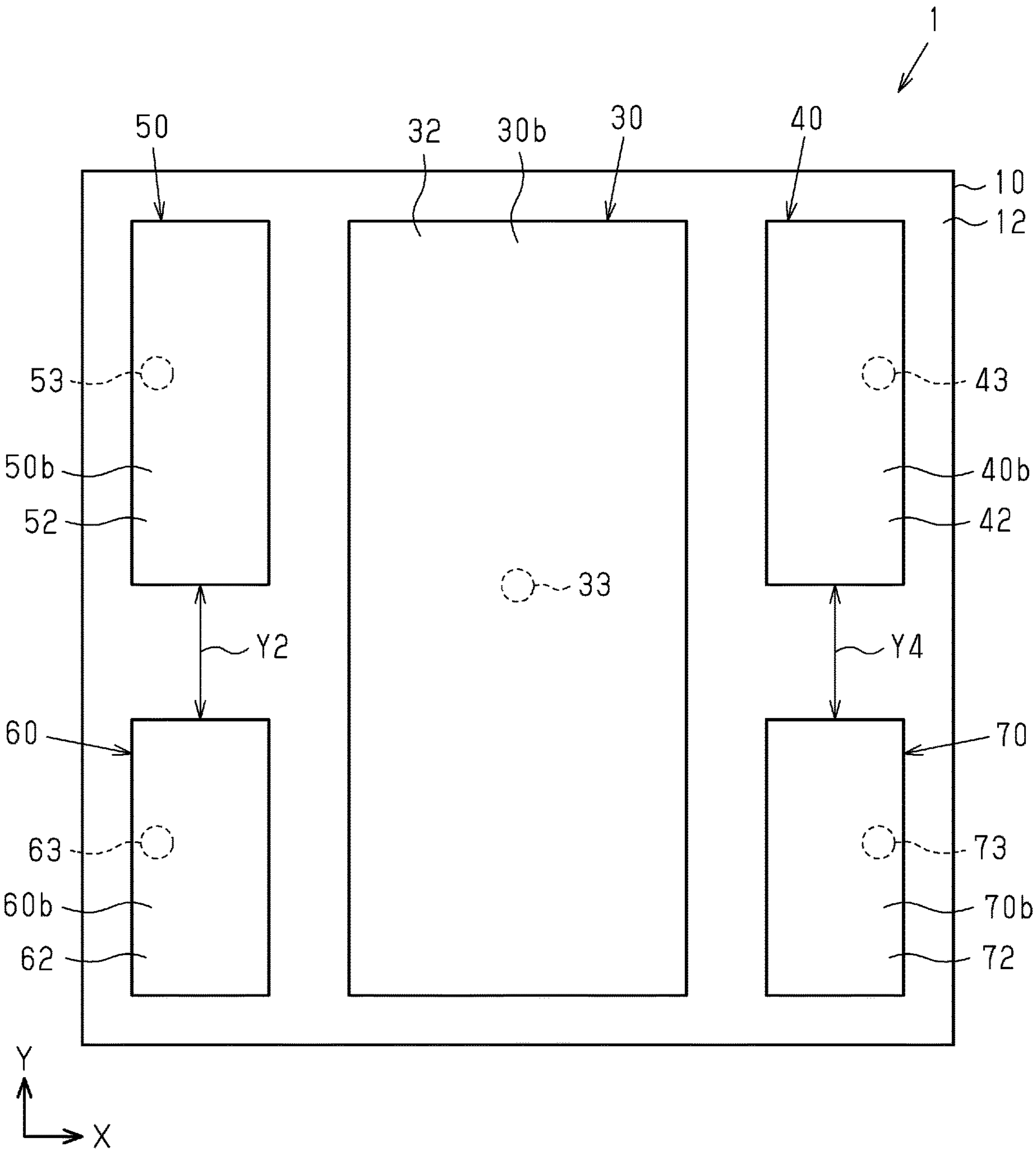
21
22
21a
31
41
40a
80a
30a
91
30
5
5
80
43
53
W1
50a
90
33
50
51
Y3
Y1
40
100a
60
21c
101
21d
31a
70
6
6
102
73
63
W3
W2
70a
60a
71
61
100
21b
Y
X

Fig.4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12
53
43
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52
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33
Y2
Y4
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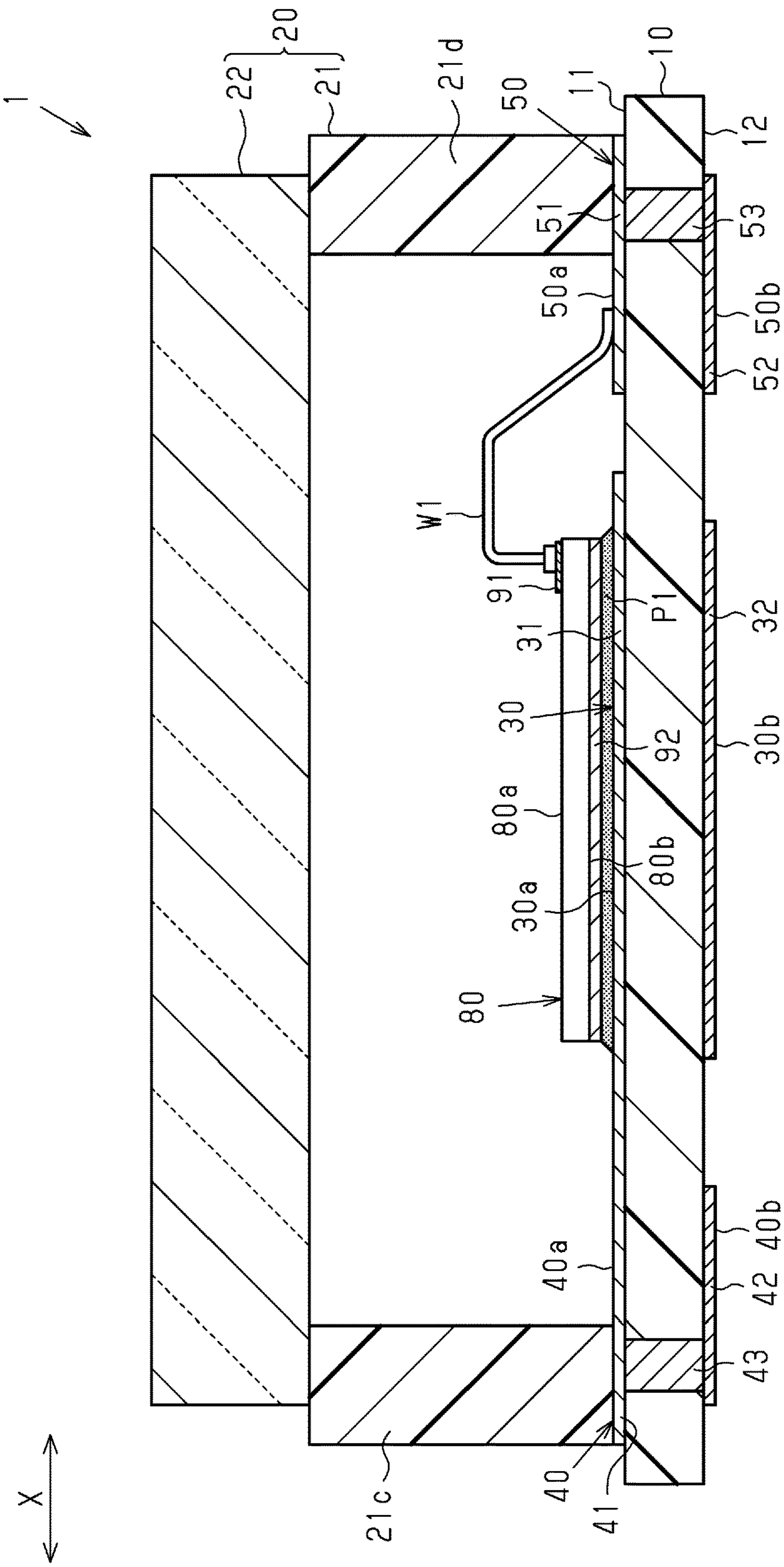
70
63
73
60b
70b
62
72
Y
X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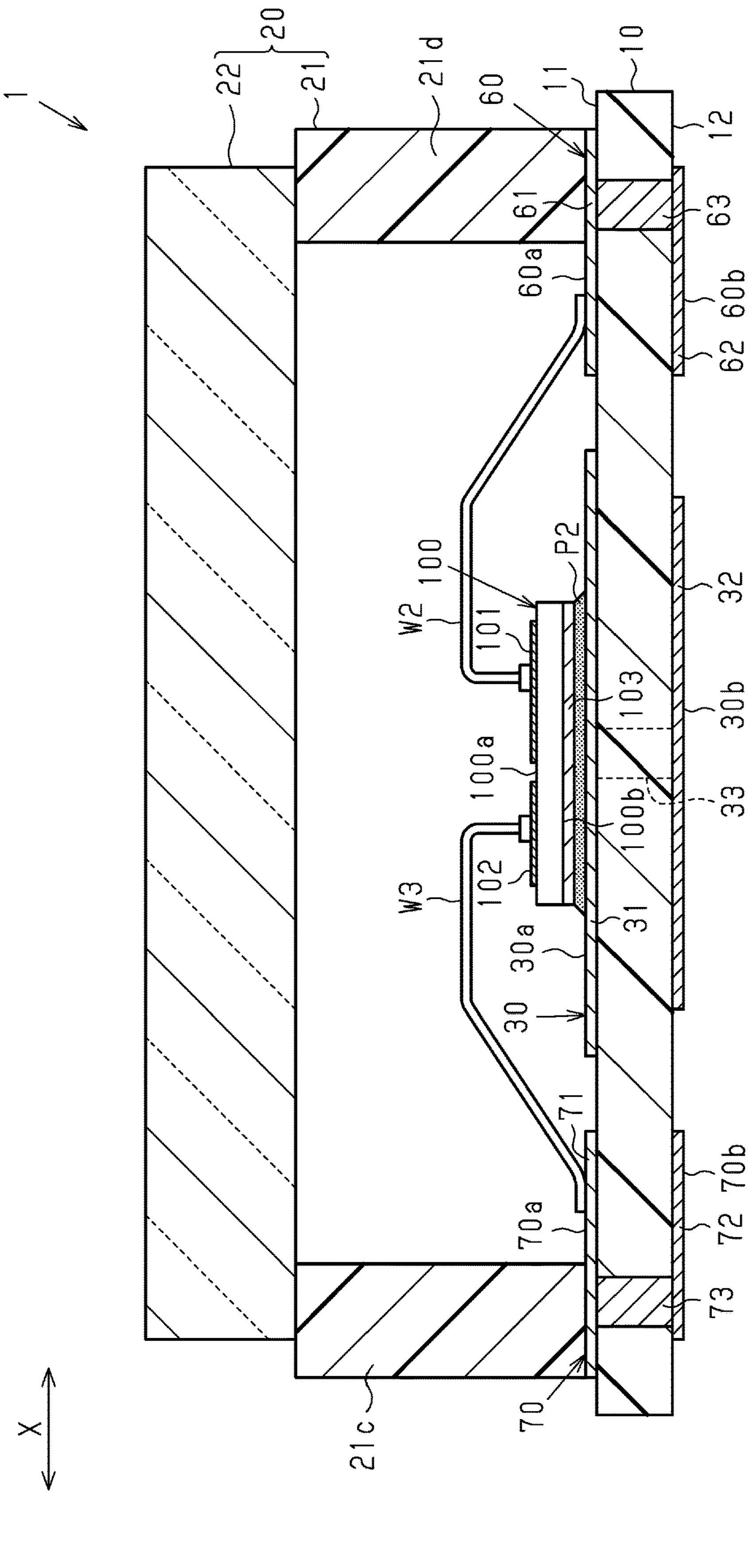
20
22
21
21d
50
11
10
12
51
53
50a
50b
52
W1
91
P1
31
32
30
92
30b
80a
80b
30a
80
40a
40b
42
43
40
41
21c
X

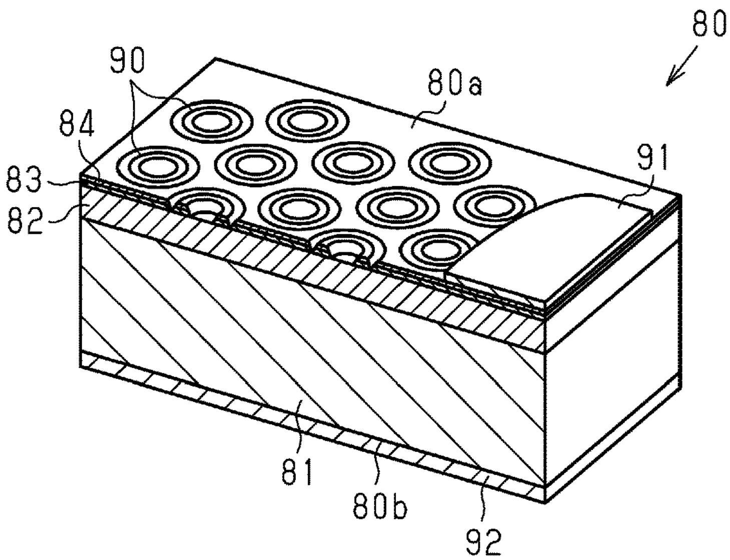
Fig.5



1
20
22
21
21d
60
11
10
12
61
63
60a
60b
62
100
P2
W2
101
103
30b
100a
33
100b
102
32
W3
31
30a
30
71
70b
70a
72
73
70
21c
X

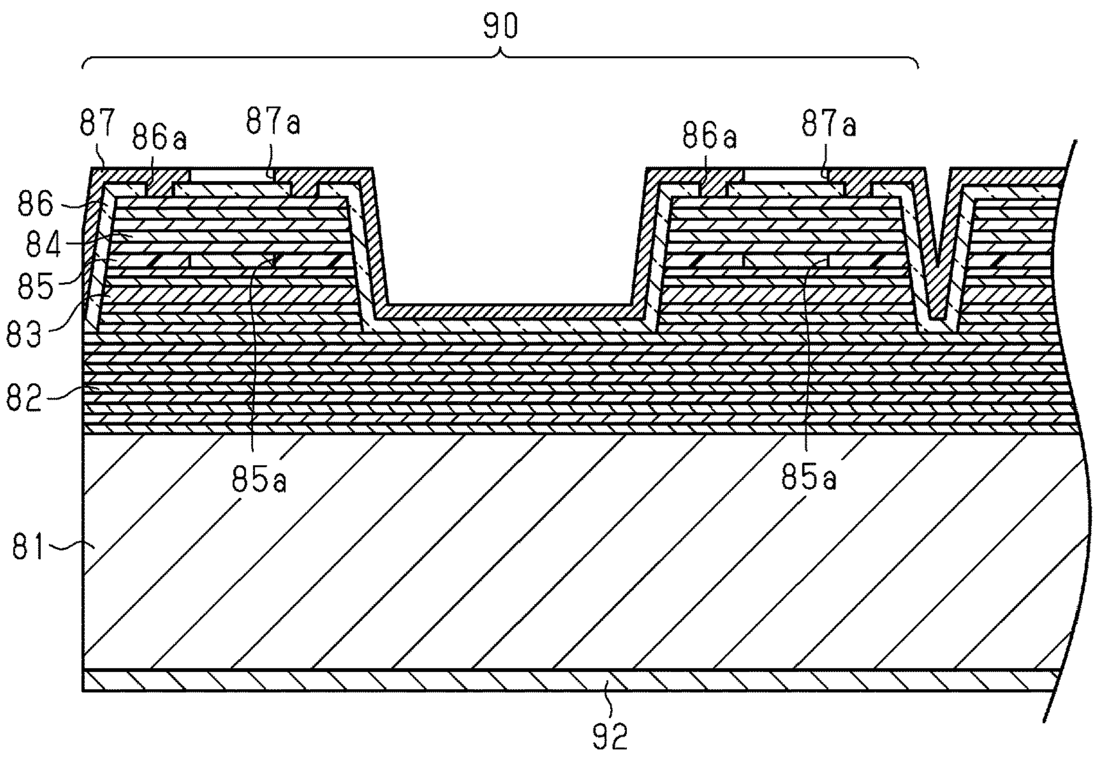

Fig.6

Fig.7



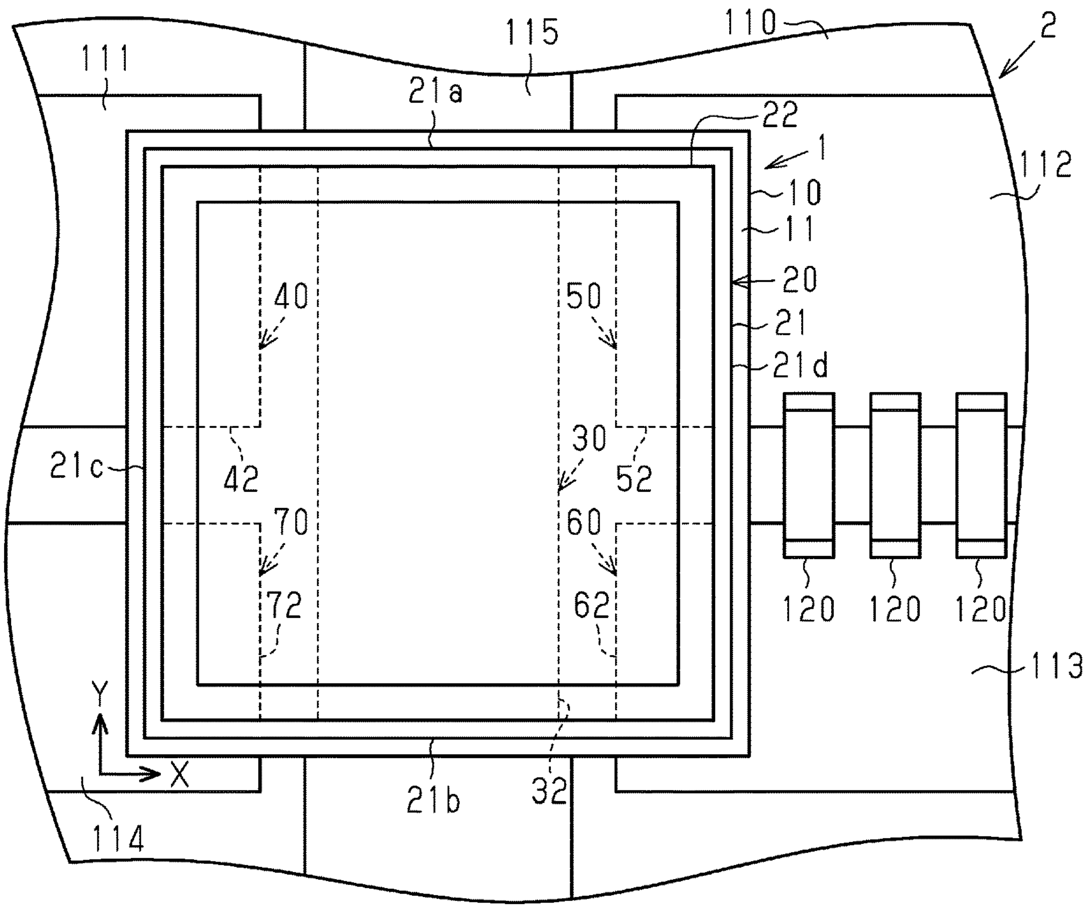
80
90
80a
84
91
83
82
81
80b
92


Fig.8



90
87
86a
87a
86a
87a
86
84
85
83
82
85a
85a
81
92

Fig.9



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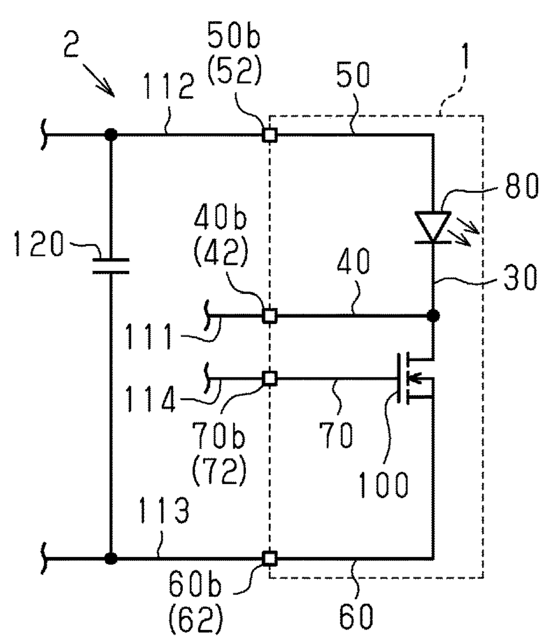
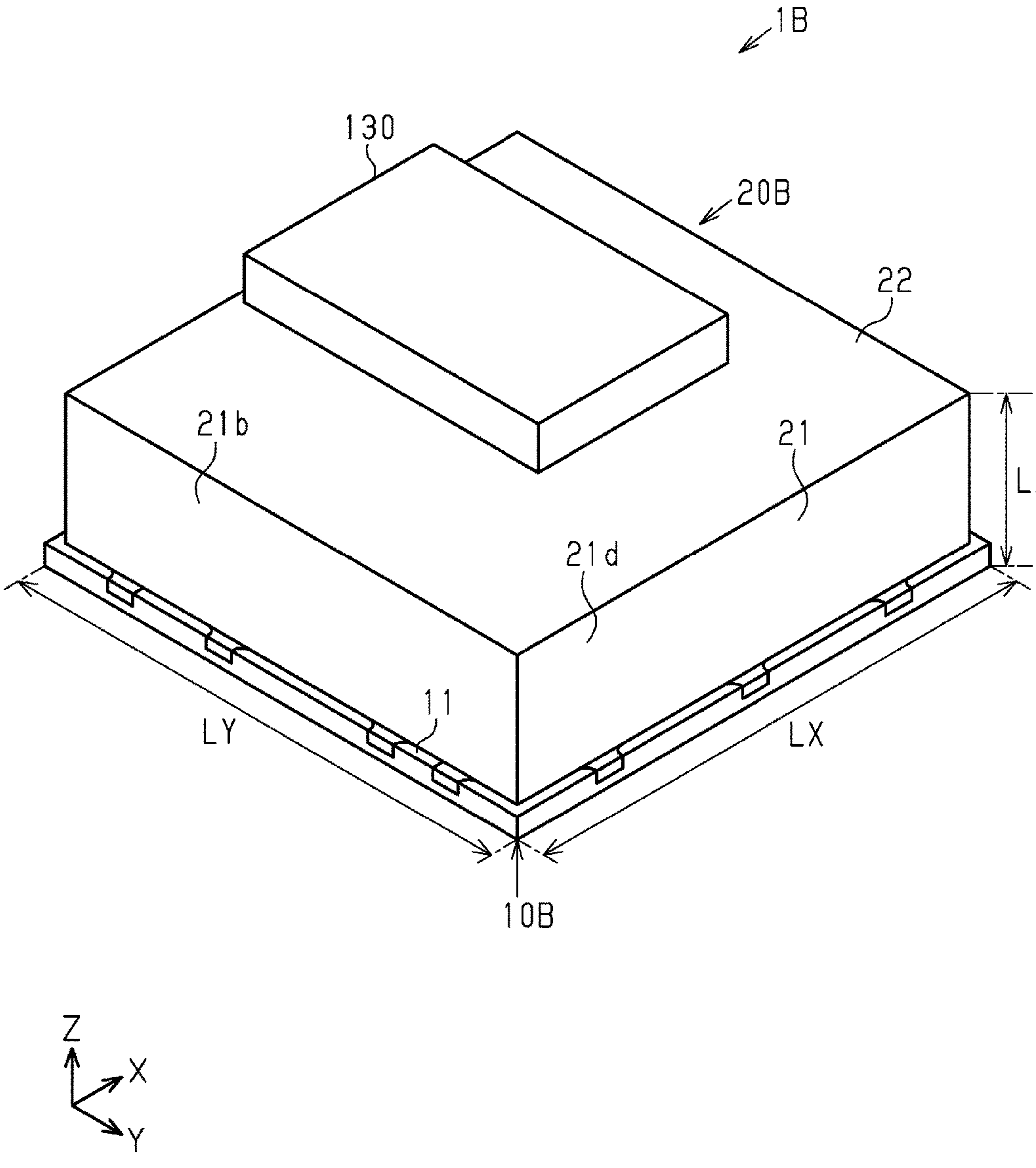
2
111
115
21a
22
1
10
11
112
20
21
21d
40
50
42
30
52
21c
70
60
72
62
120
120
120
113
Y
X
114
21b
32


Fig.10



2
112
50b
(52)
50
1
80
40b
(42)
120
40
30
111
114
70b
(72)
70
100
113
60b
(62)
60

Fig.11



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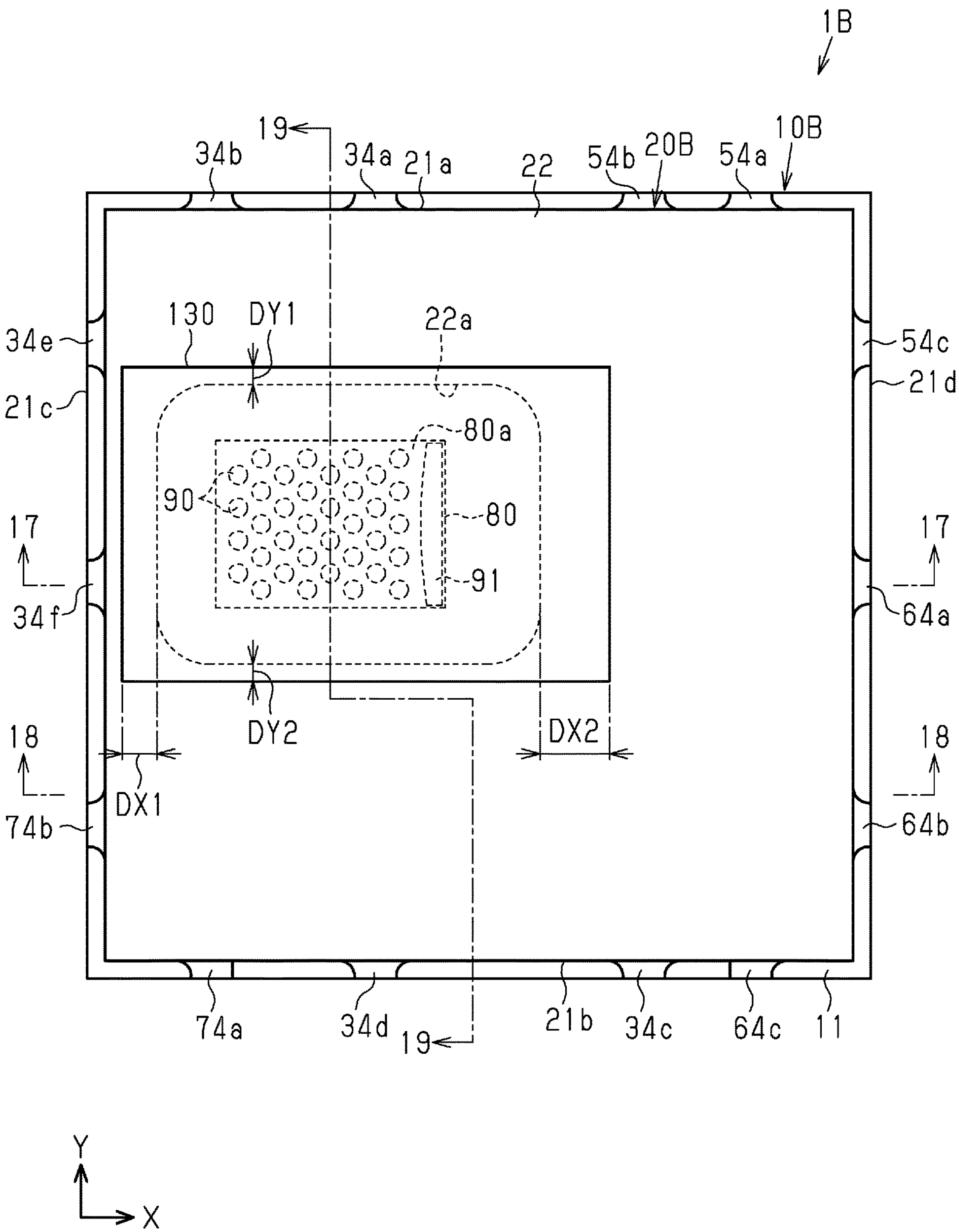
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21d
80a
90
80
17
17
91
34f
64a
DY2
DX2
18
18
DX1
74b
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74a
34d
21b
34c
64c
11
19
Y
X

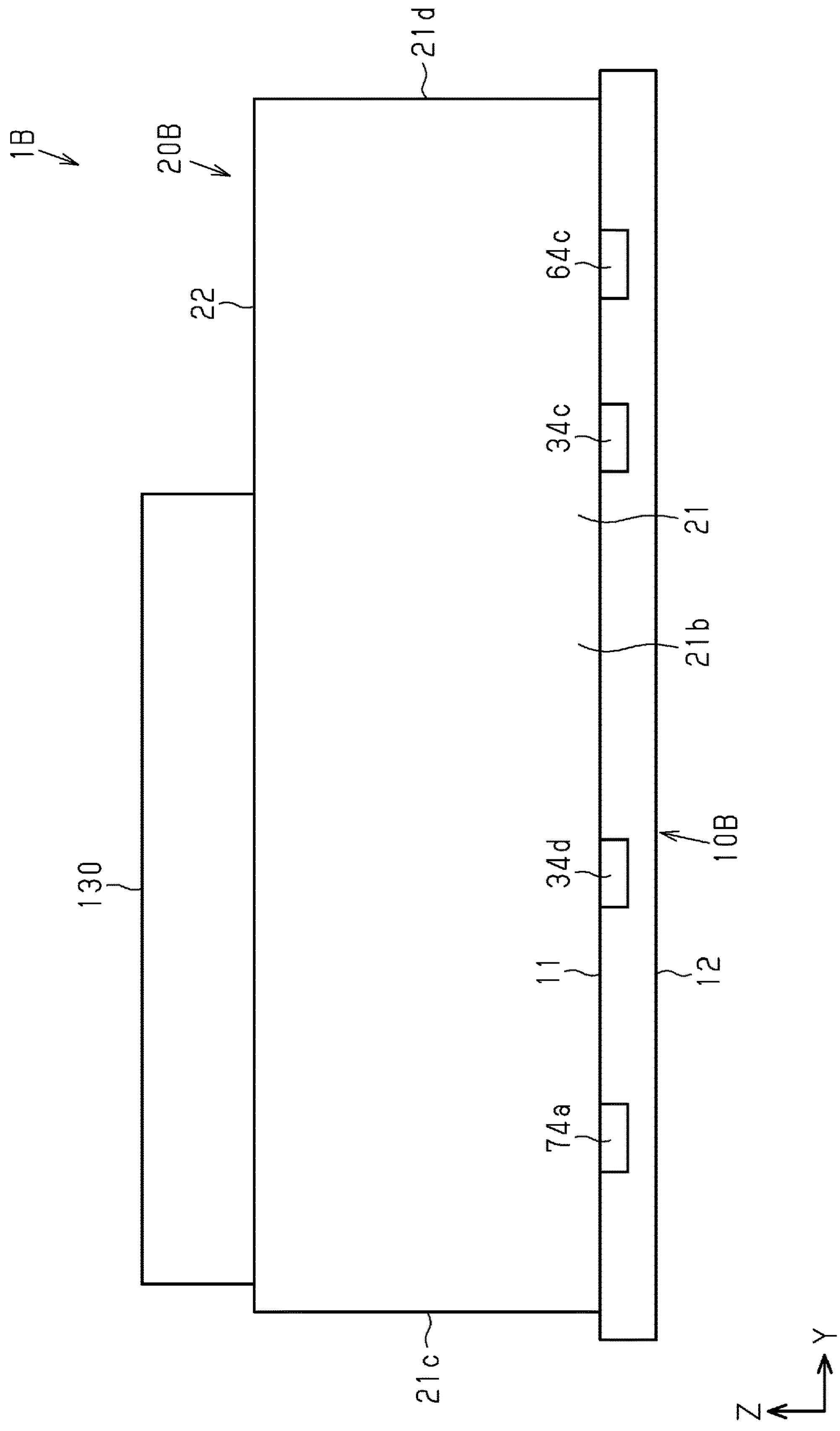
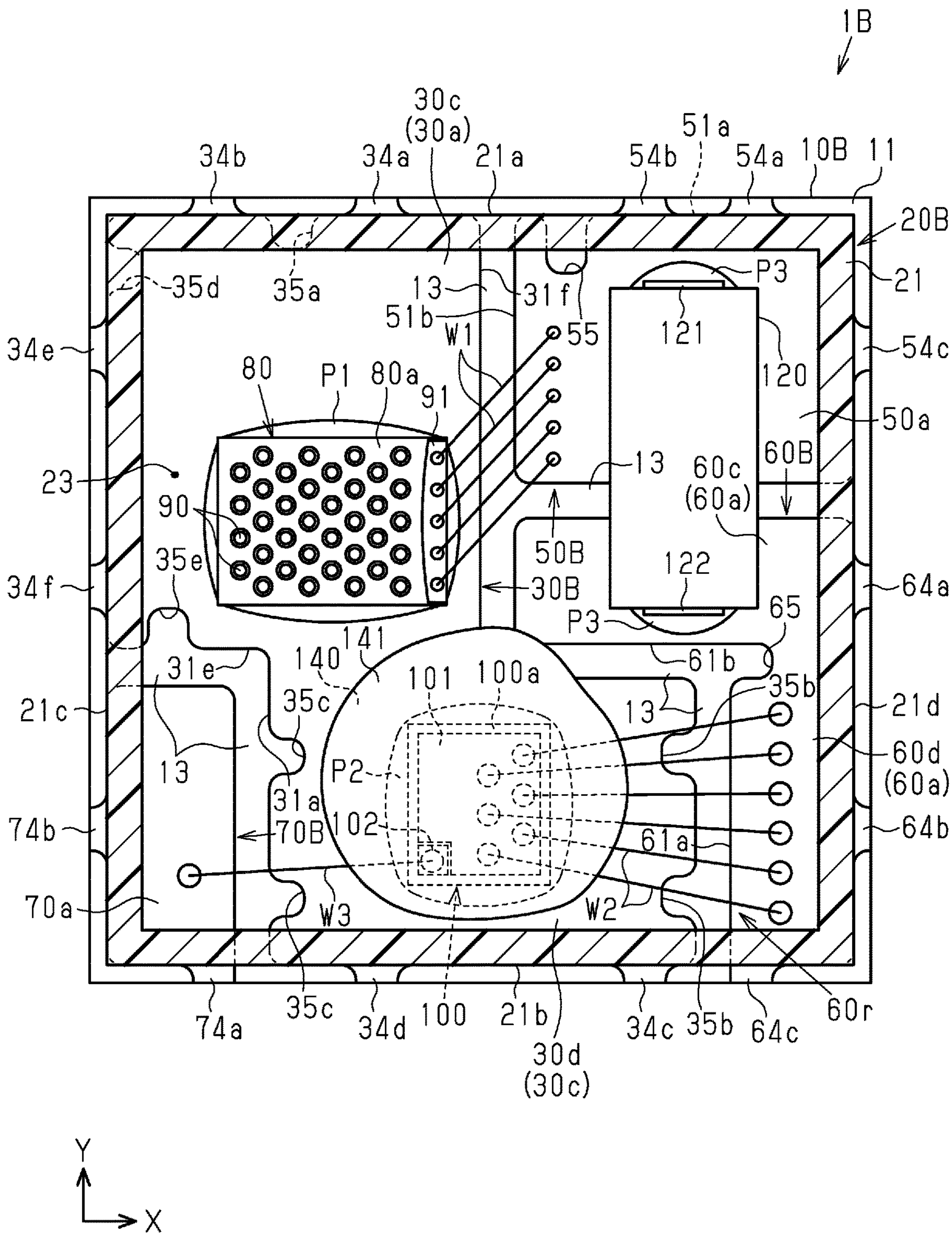



Fig.13

Fig.14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P3
21
51b
55
121
34e
W1
54c
80
P1
80a
91
120
50a
13
60c
(60a)
60B
23
90
50B
122
35e
30B
34f
64a
P3
65
140
141
31e
101
100a
61b
35b
35c
13
21c
21d
13
P2
60d
(60a)
31a
102
70B
74b
61a
64b
W3
W2
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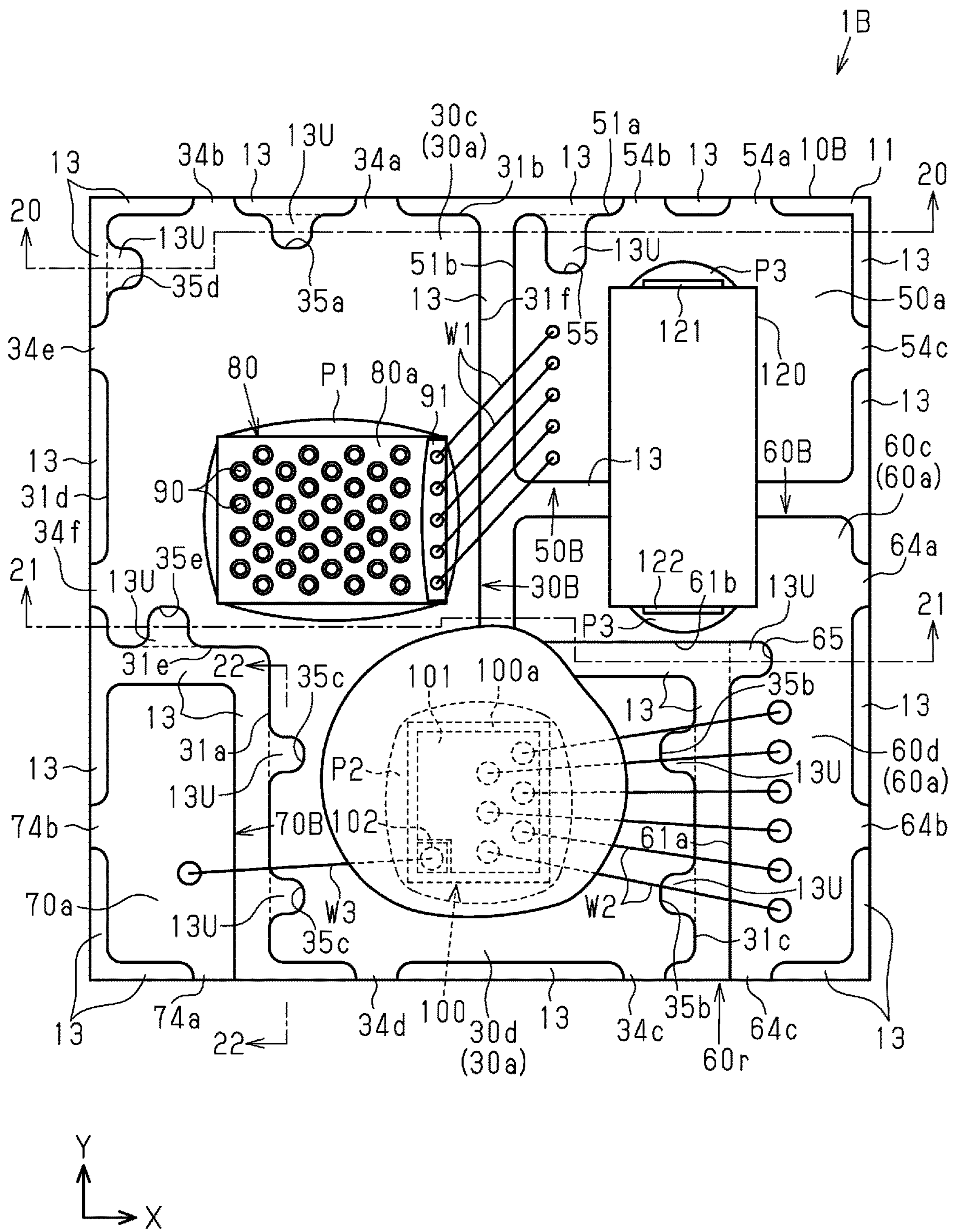
35b
64c
60r
74a
30d
(30c)
Y
X

Fig.15



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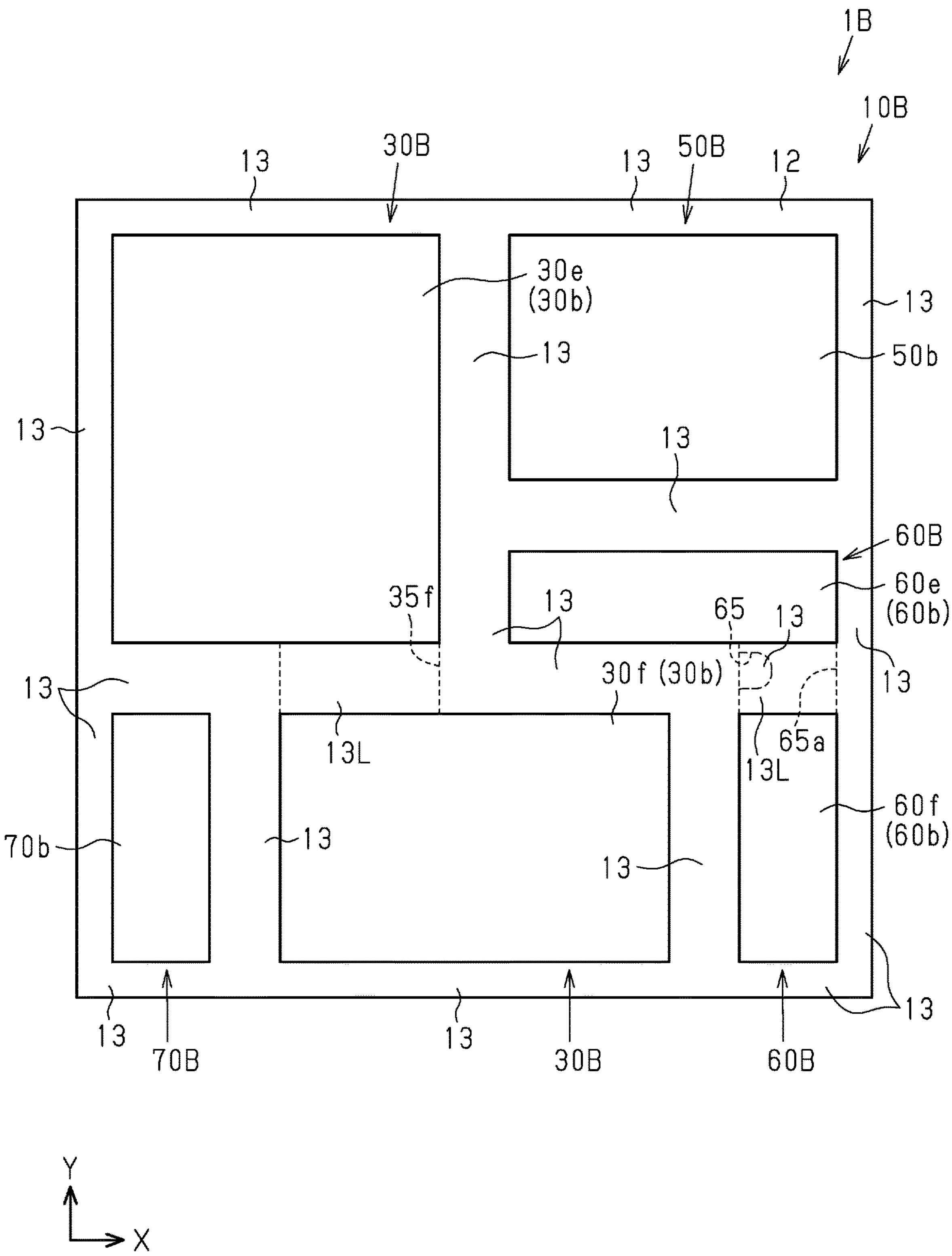
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50B
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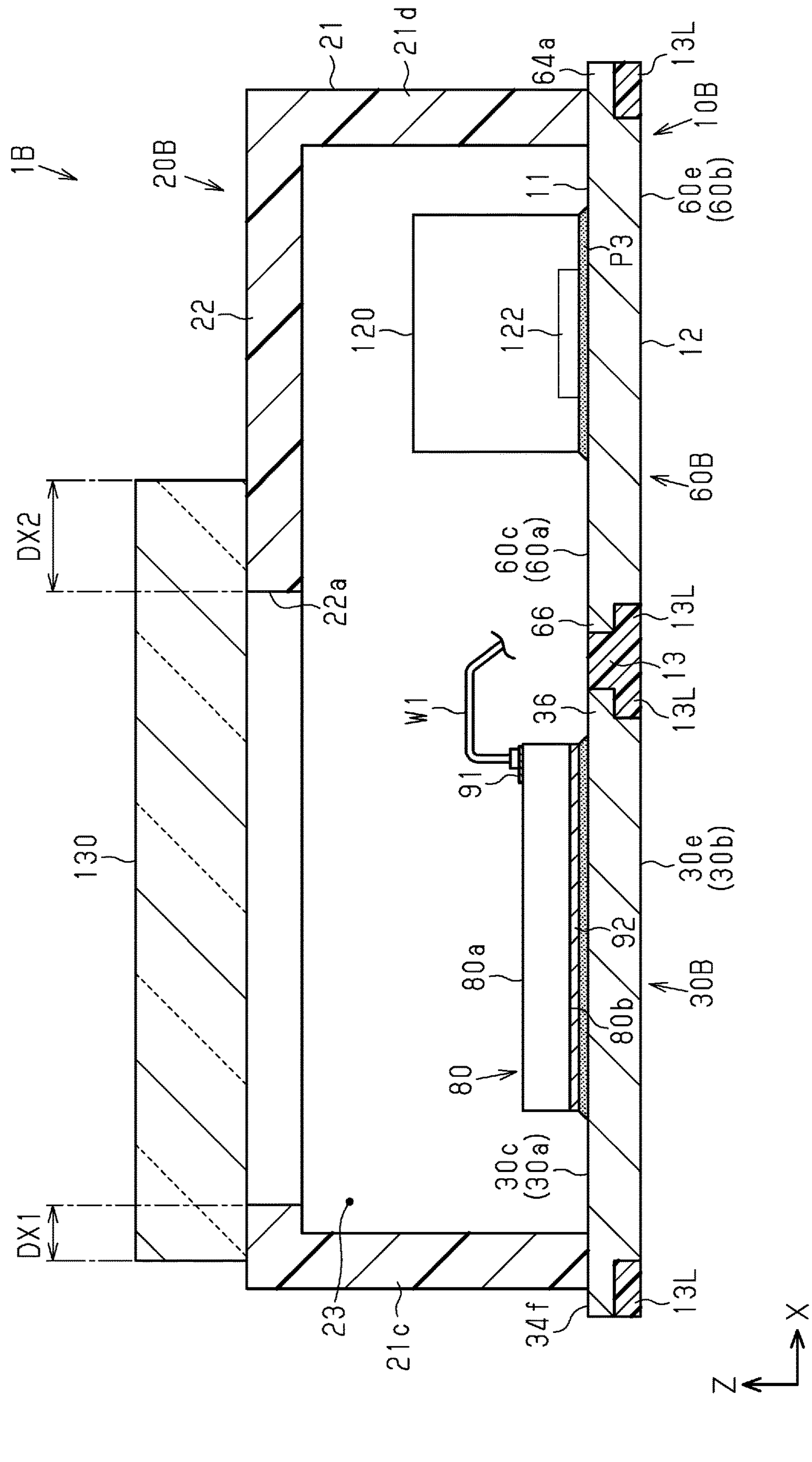
30e
(30b)
13
13
50b
13
13
60B
60e
(60b)
35f
13
65
13
13
30f (30b)
13
13L
65a
13L
60f
(60b)
70b
13
13
13
13
70B
13
30B
60B
Y
X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21
21d
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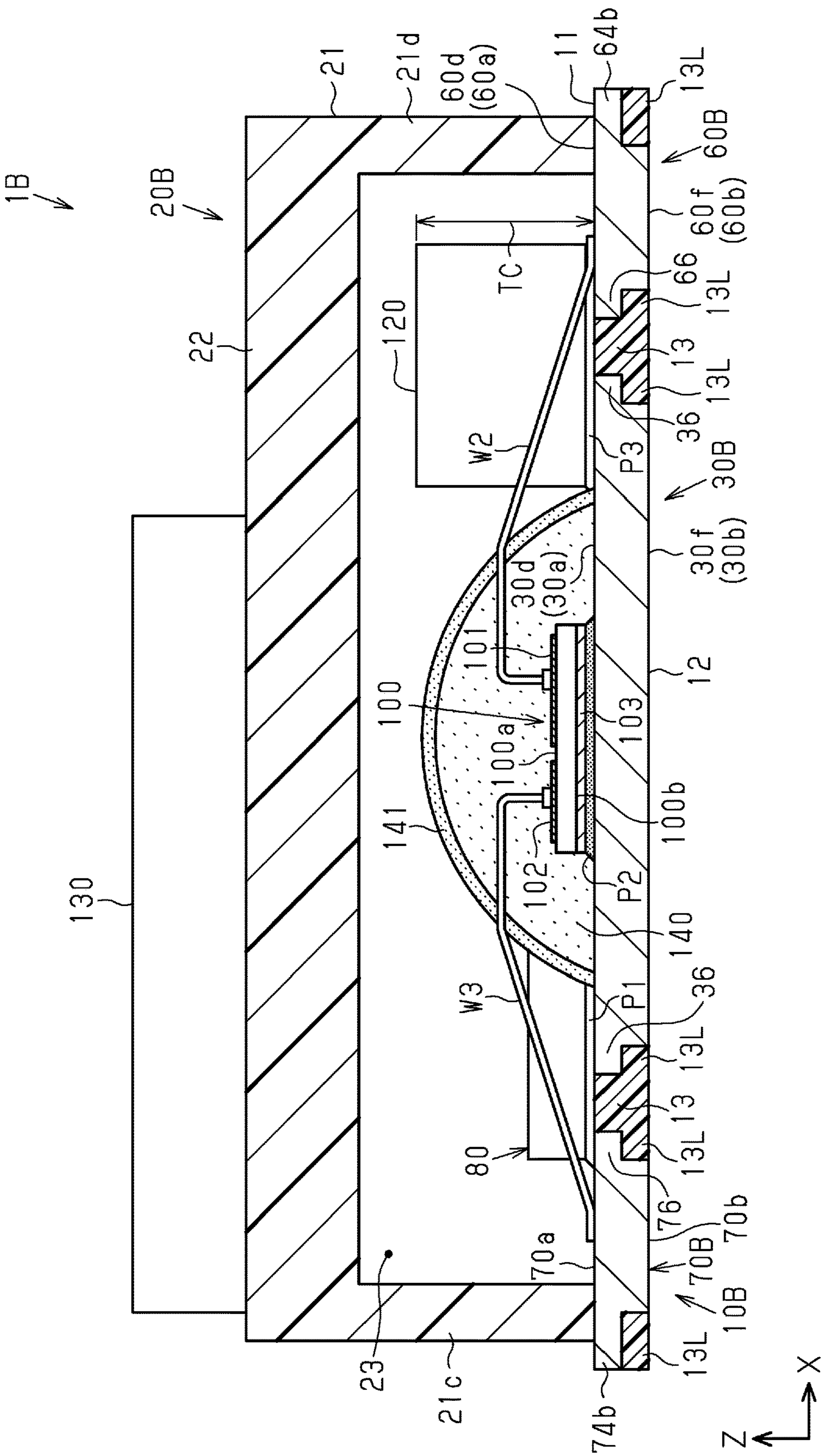
13L
10B
11
60e
(60b)
P3
22
120
122
12
60B
DX2
60c
(60a)
22a
66
13
W1
36
91
130
30e
(30b)
92
80a
30B
80b
80
30c
(30a)
DX1
23
21c
34f
X
Z

Fig.17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21d
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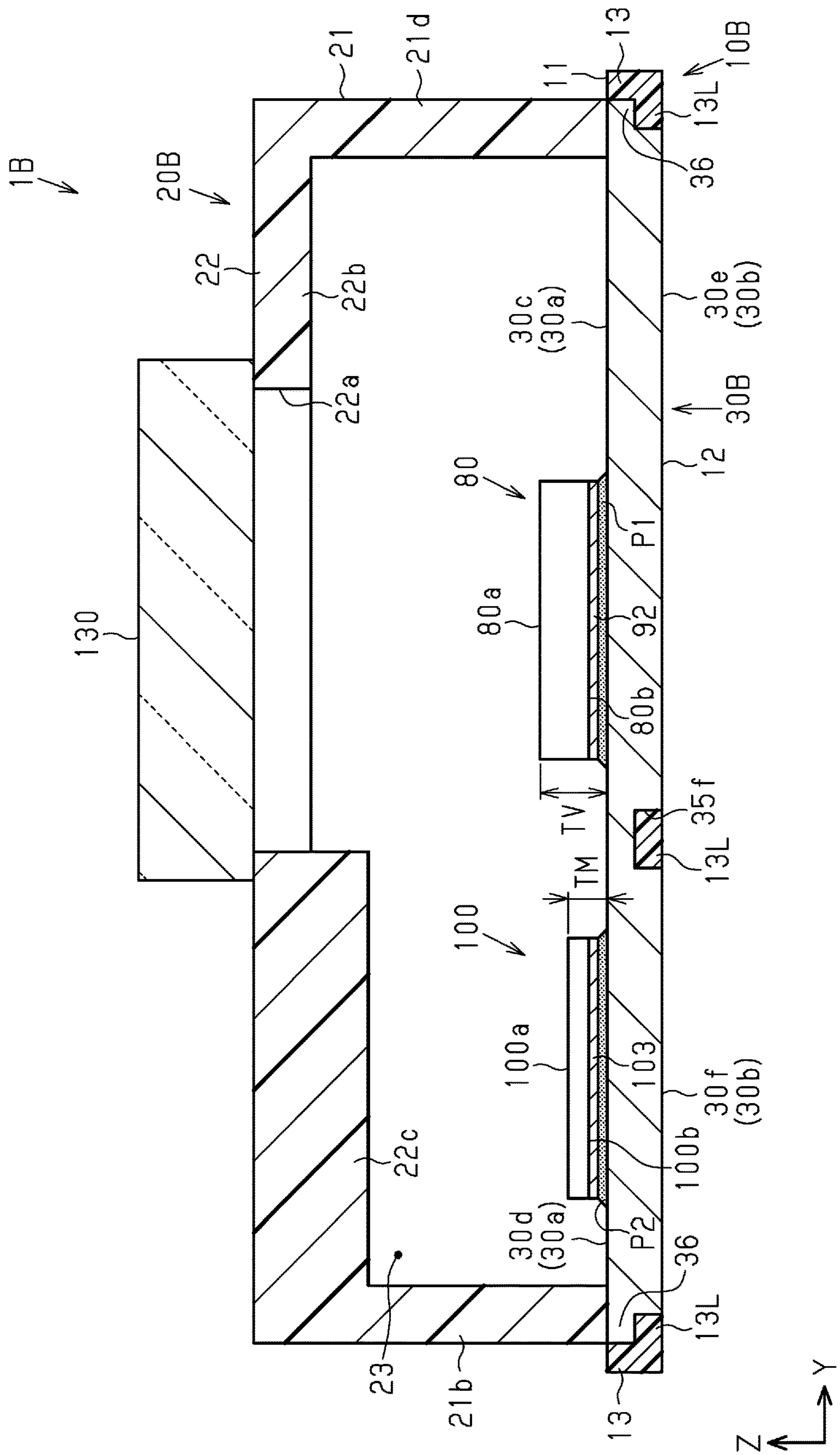
(60a)
11
64b
13L
60B
60f
(60b)
TC
66
13L
120
22
13
13L
36
W2
P3
30B
30f
(30b)
30d
(30a)
101
12
100
103
100a
100b
141
102
P2
140
130
36
W3
P1
13L
13
80
13L
76
70b
70a
70B
10B
23
13L
21c
74b
X
Z

Fig.18

Fig.19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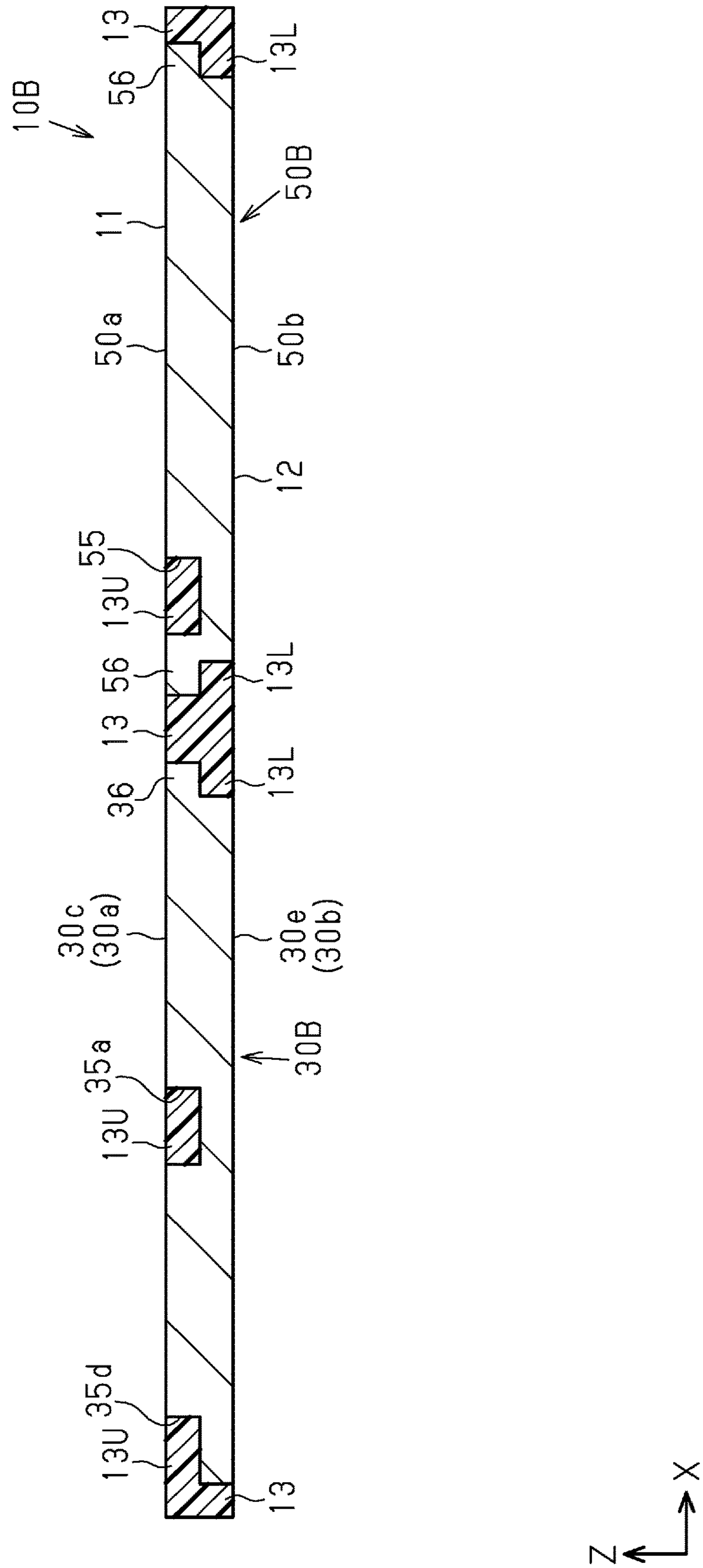
22
22b
22a
30c
(30a)
30e
(30b)
30B
12
80
P1
80a
92
80b
130
TV
35f
13L
TM
100
100a
103
30f
(30b)
22c
100b
30d
(30a)
P2
36
13L
23
21b
13
Y
Z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50b
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56
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13L
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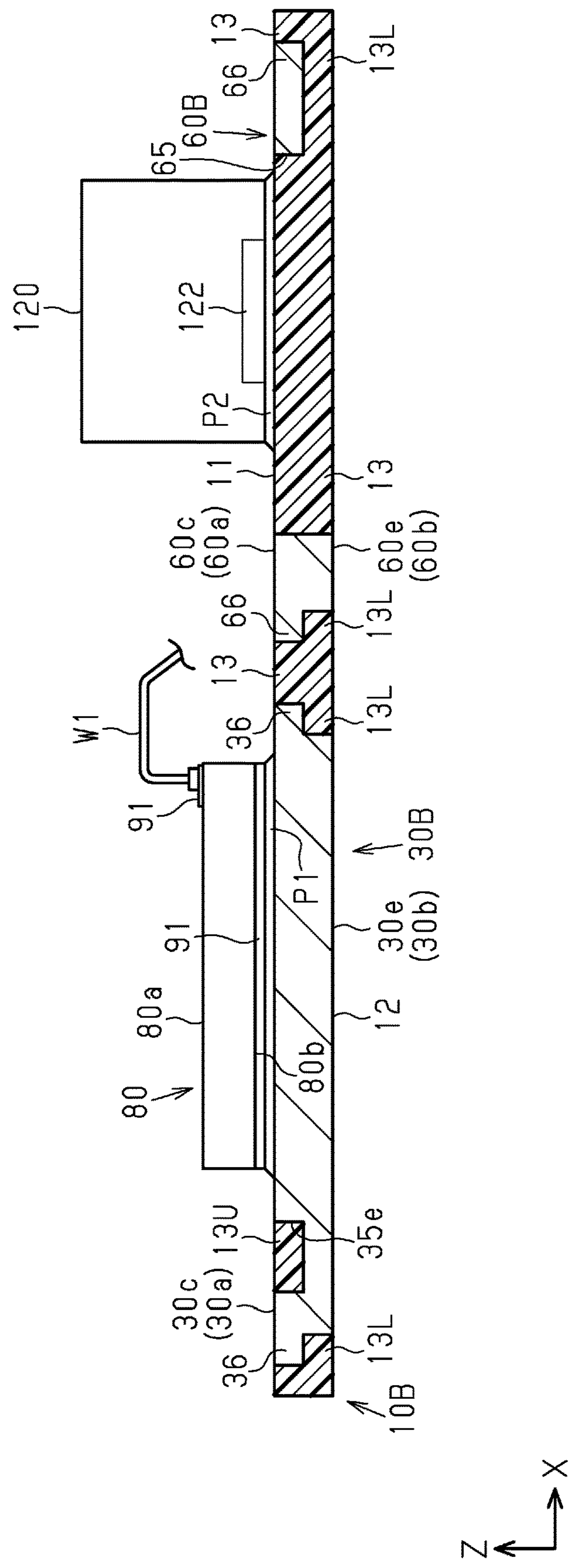
30e
(30b)
30B
35a
13U
35d
13U
13
X
Z

Fig.20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P2
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60c
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60e
(60b)
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13
13L
36
W1
91
P1
30B
91
30e
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10B
X
Z

Fig.21

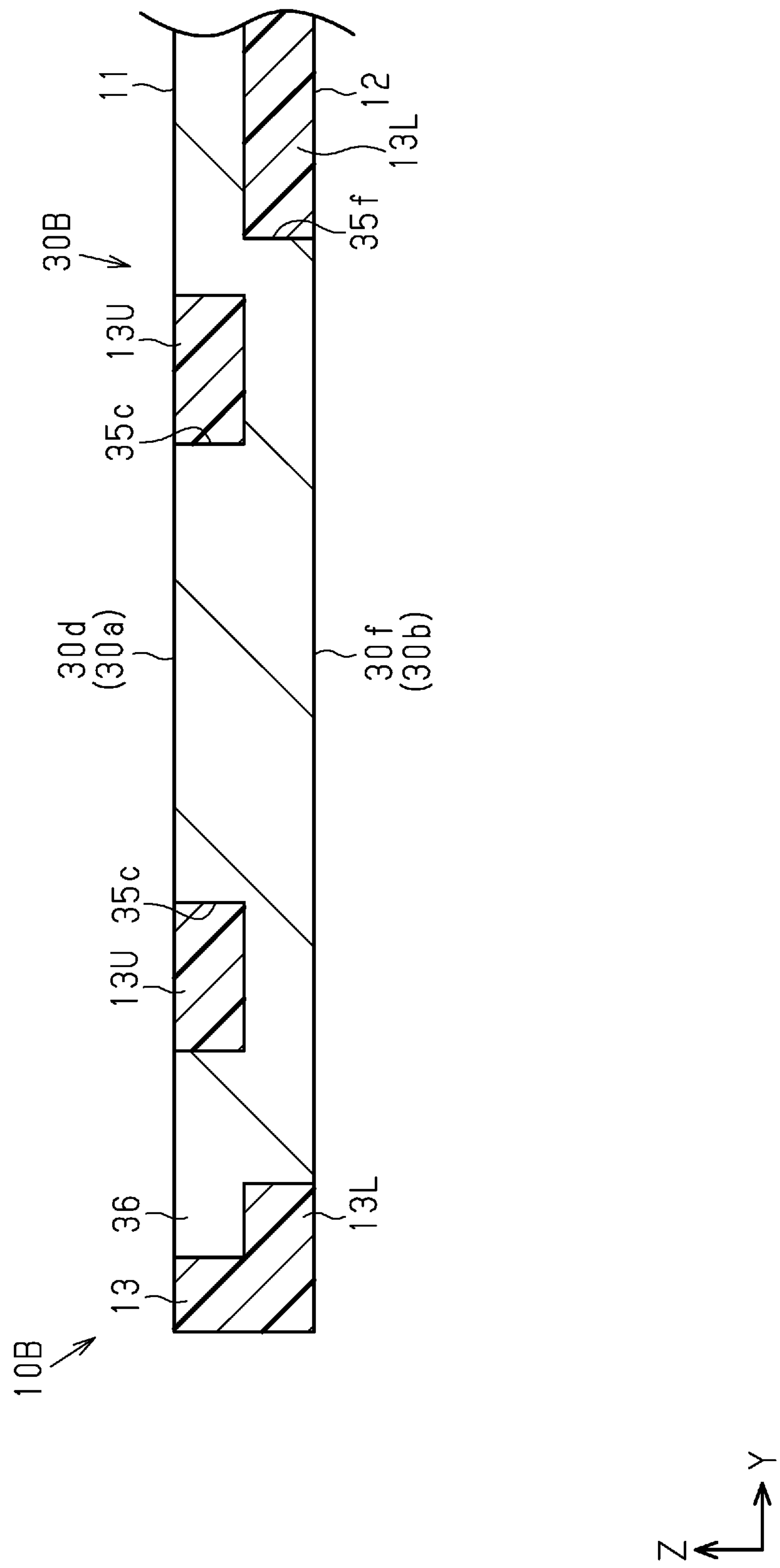
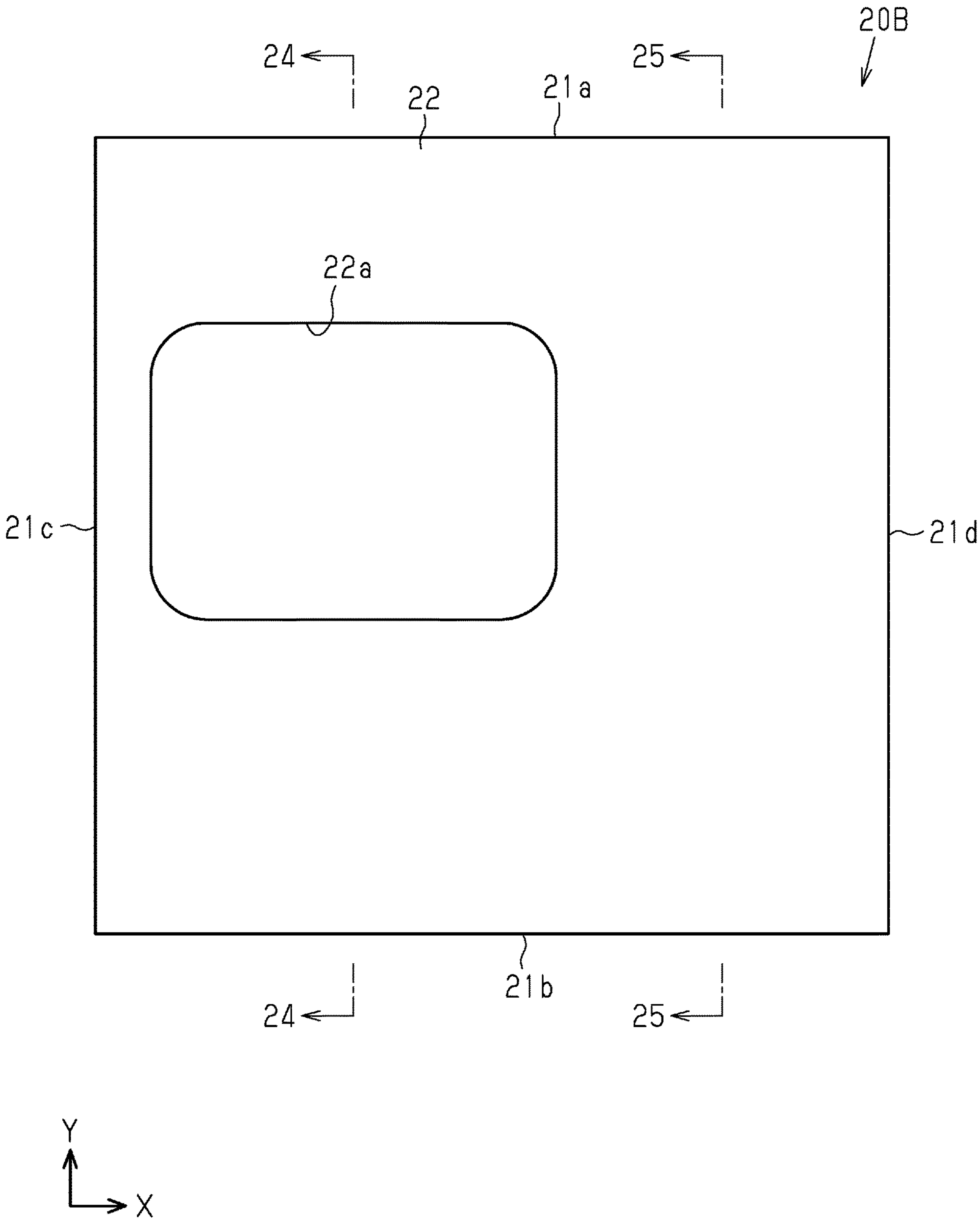



Fig.22

Fig.23



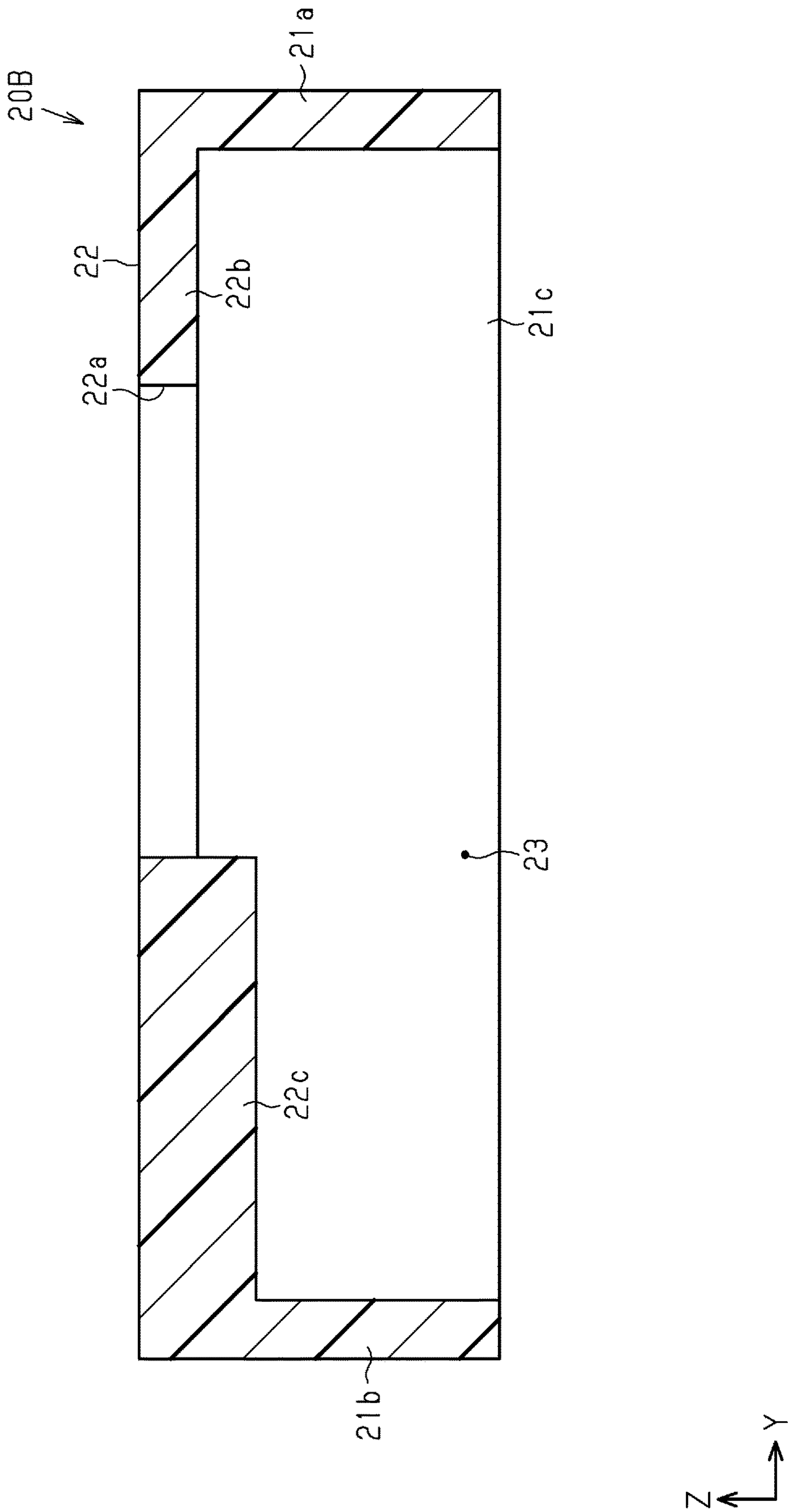



Fig.24

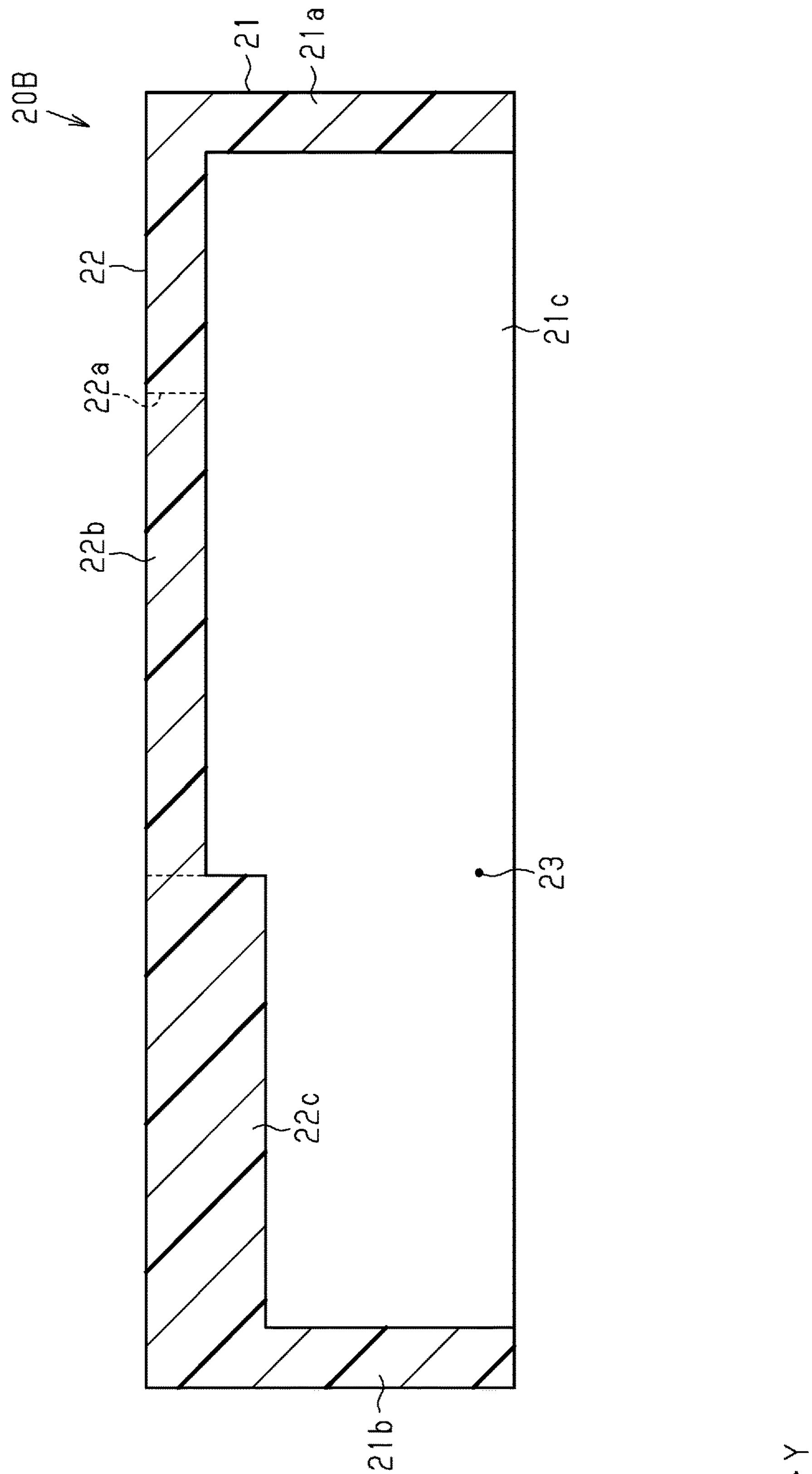
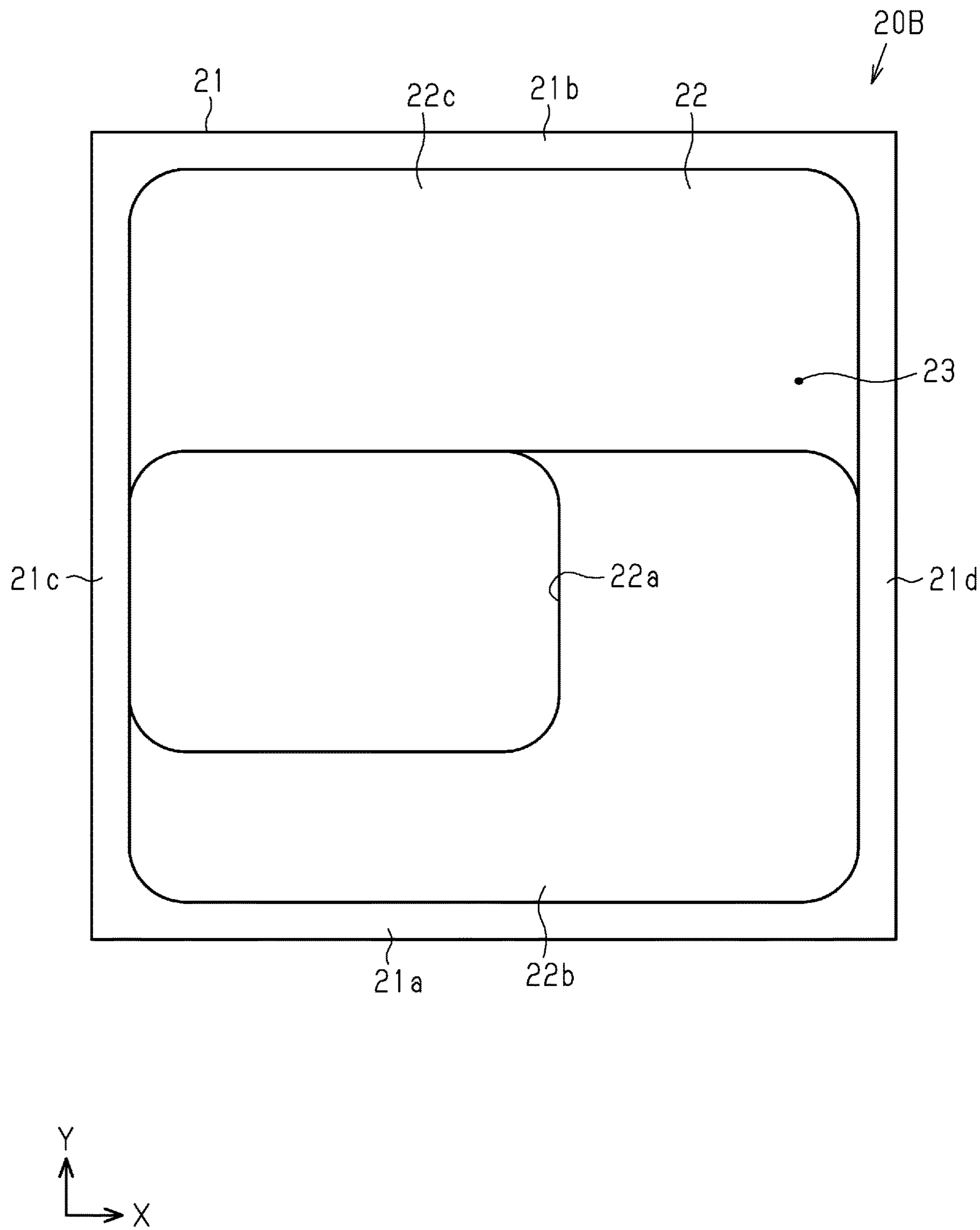



Fig.25

Fig.26



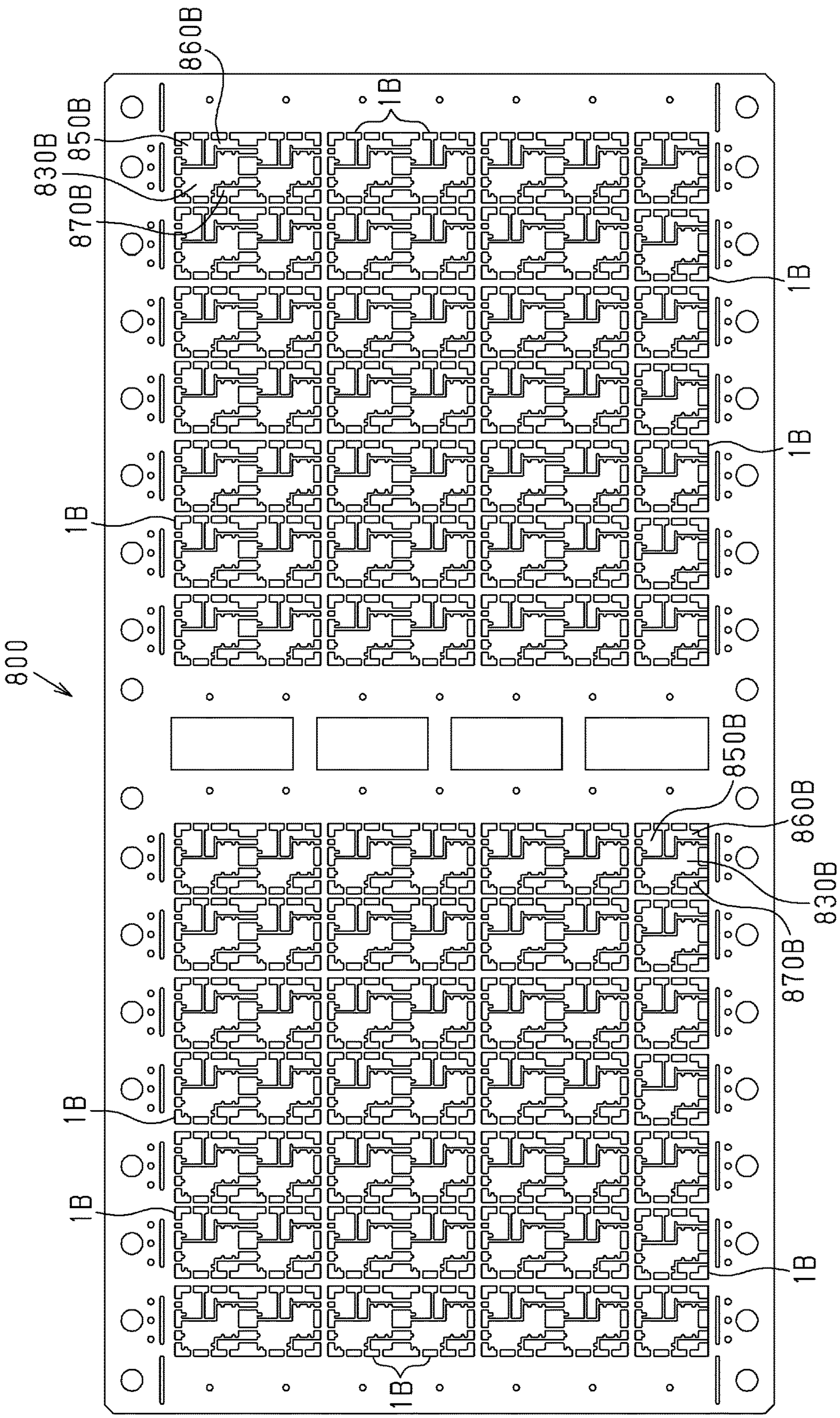
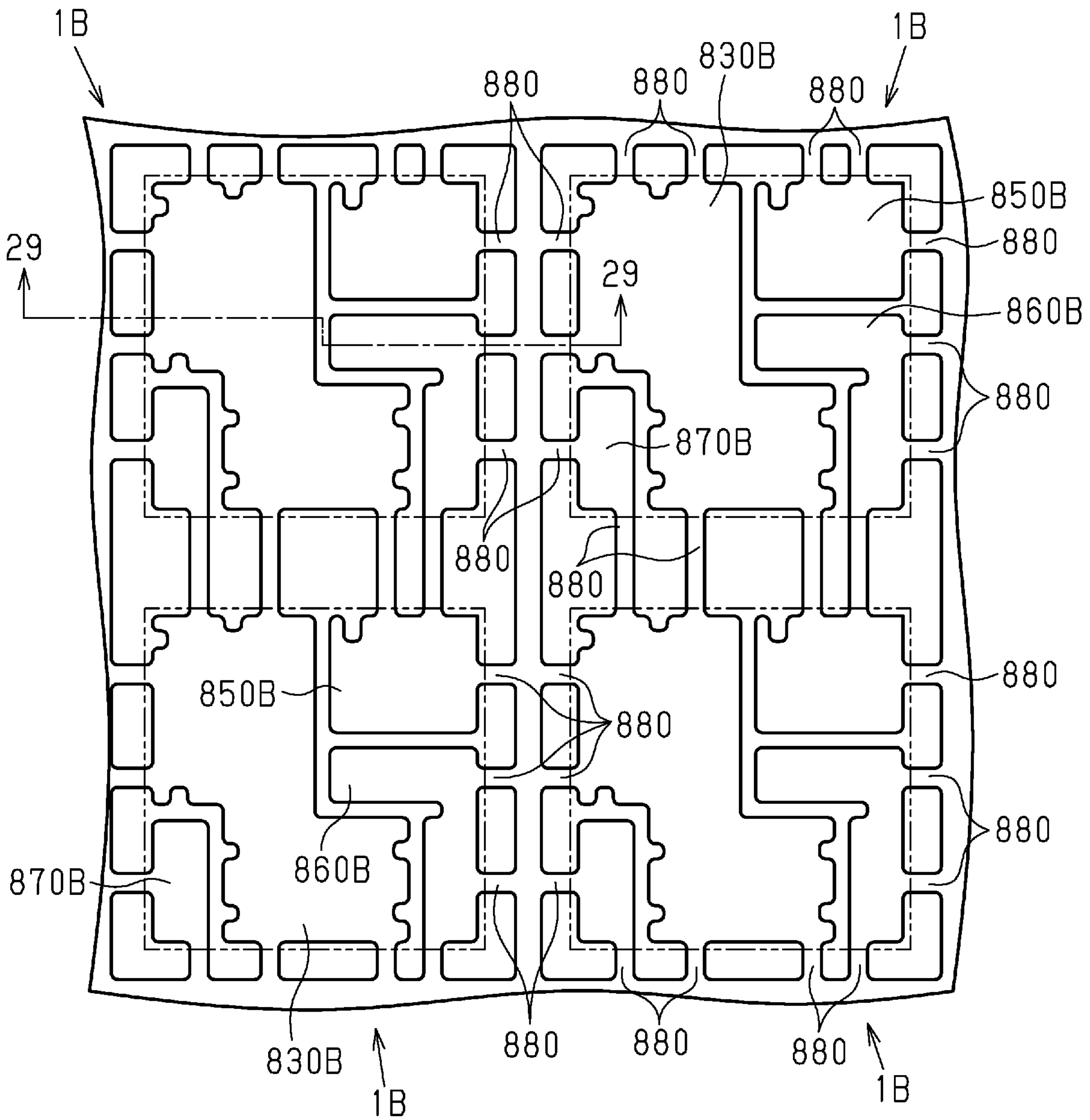



Fig.27

Fig.28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880
880
1B
1B

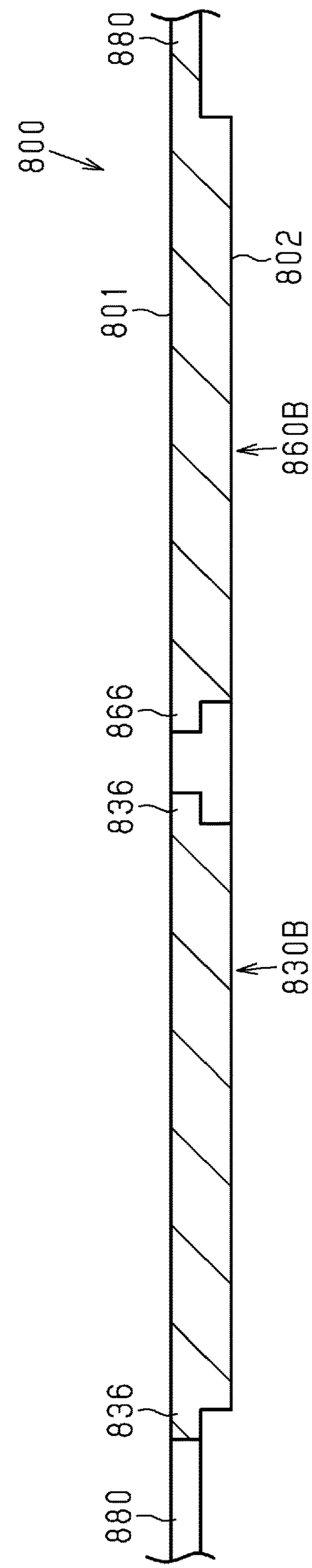
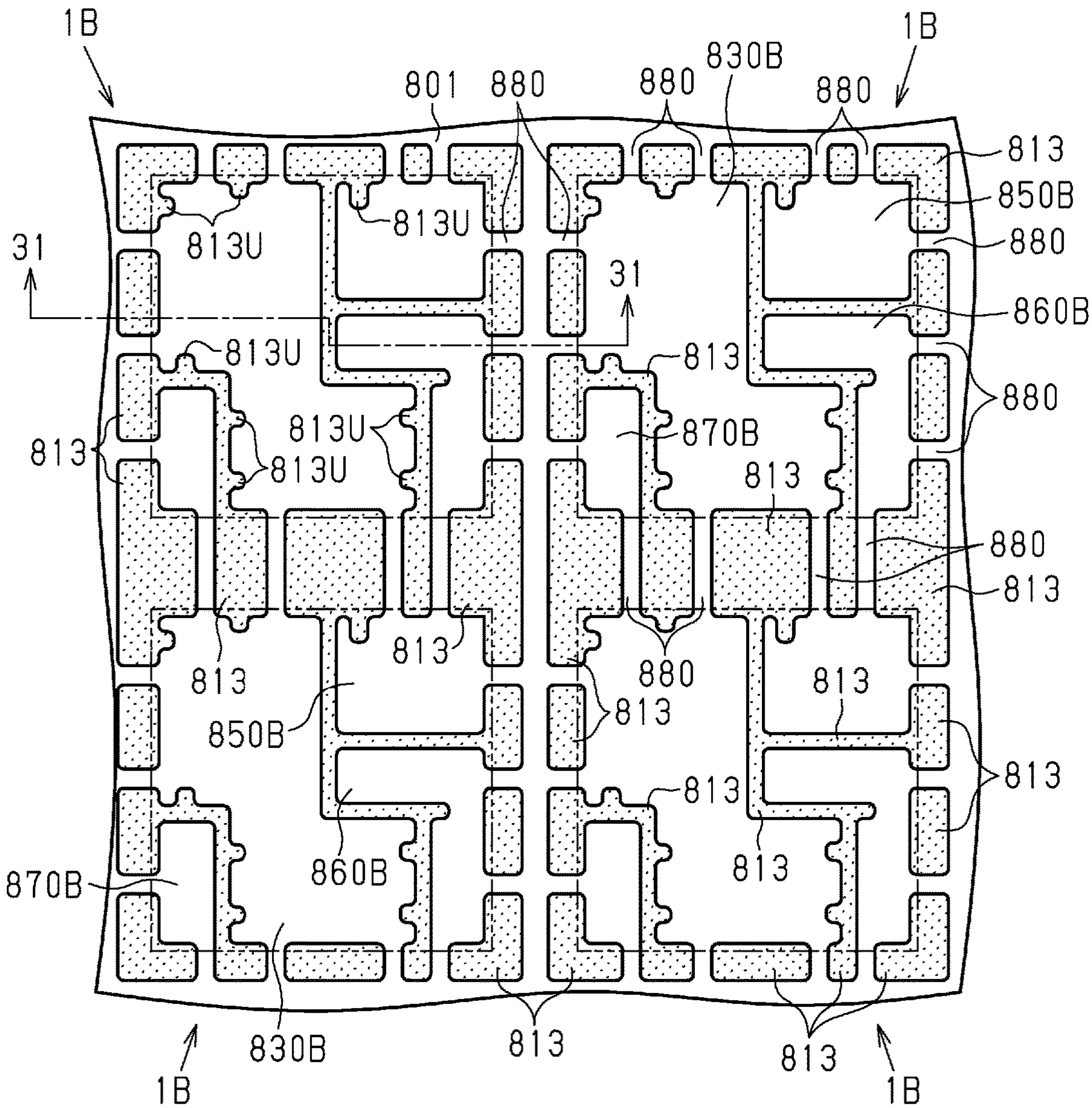



Fig.29

Fig.30



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830B
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850B
880
31
813U
813U
31
860B
813U
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870B
813
813U
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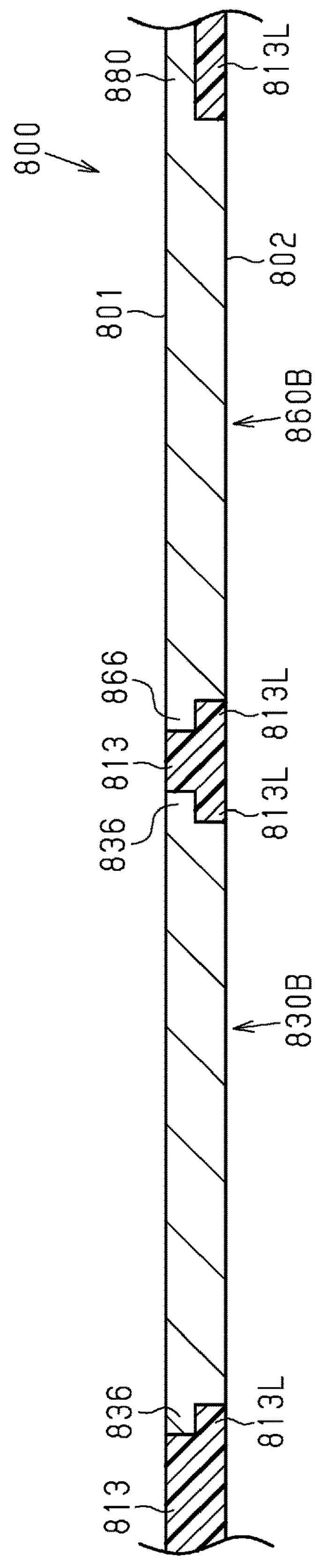
813
813
1B
1B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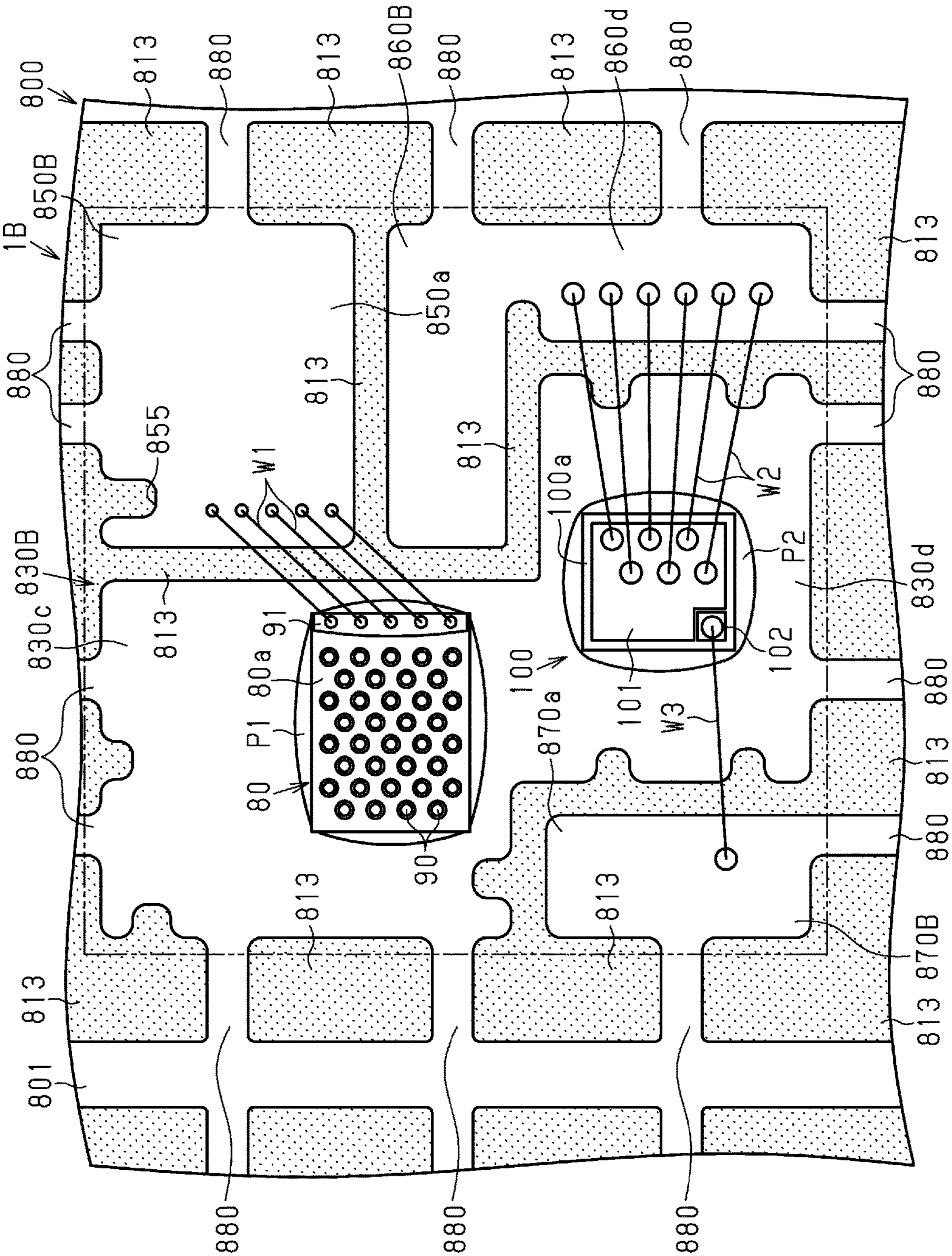
813L
813

Fig.31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830c
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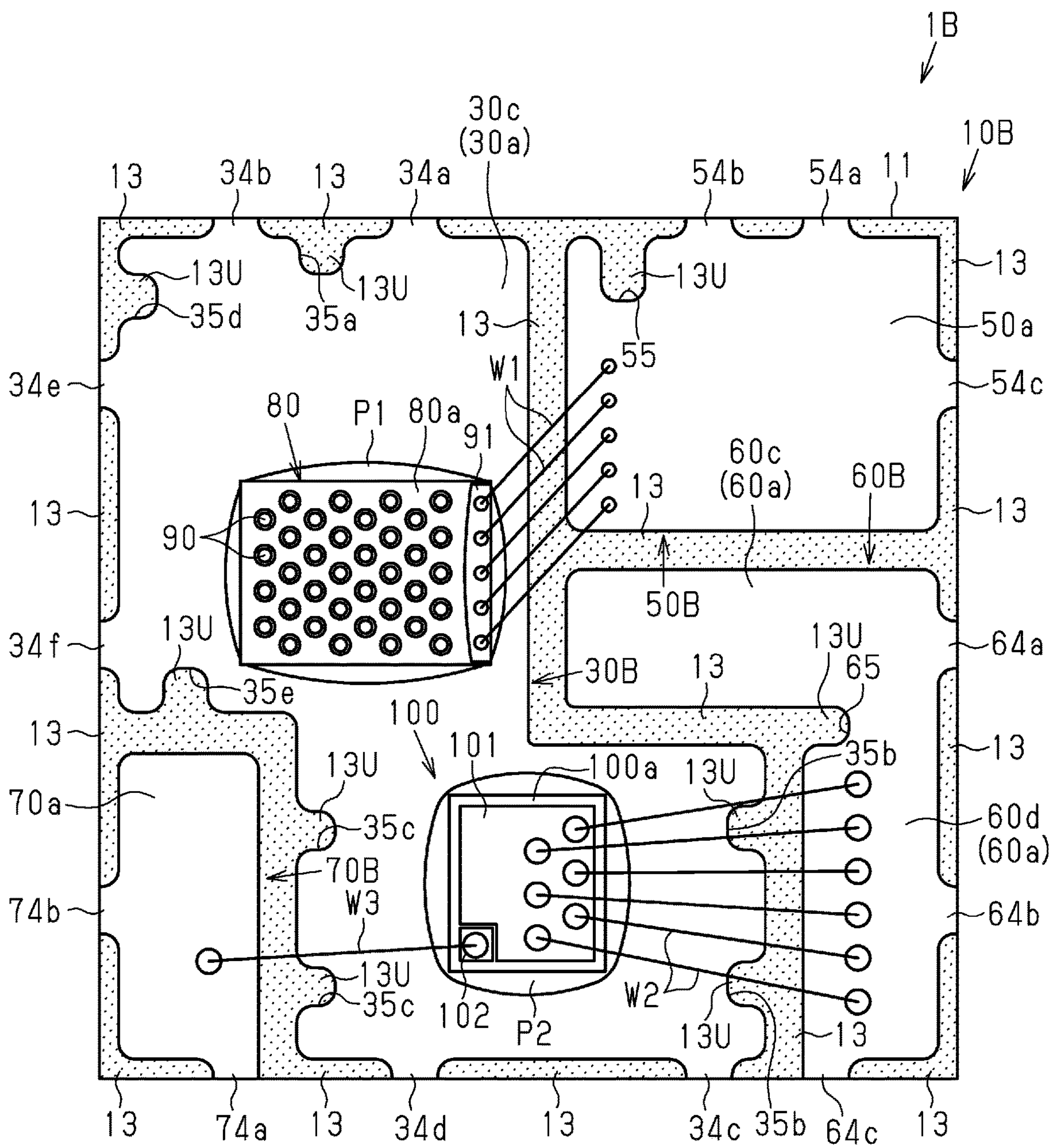
100a
100
101
102
W2
P2
830d
W3
870a
870B
801

Fig.32

Fig.33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10B
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(30a)
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34b
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34a
54b
54a
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13U
13U
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55
50a
13
W1
34e
54c
80
P1
80a
91
60c
(60a)
60B
13
13
13
90
50B
34f
13U
13U
65
64a
30B
13
35e
13
100
13
101
100a
13U
35b
13U
70a
35c
60d
(60a)
70B
W3
74b
64b
13U
W2
35c
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P2
13U
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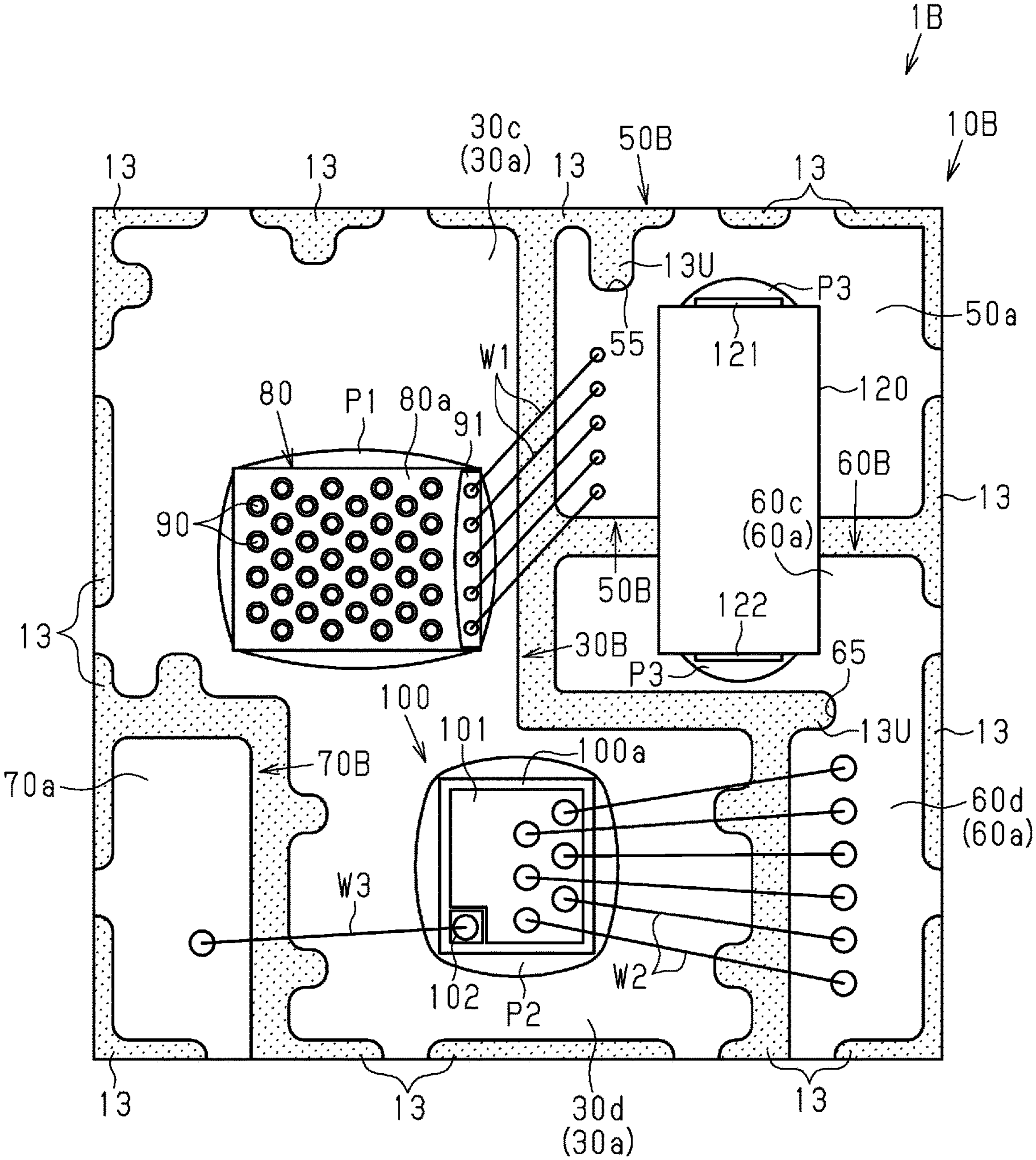
74a
13
34d
13
34c
35b
64c
13

Fig.34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30c
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50B
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P3
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W1
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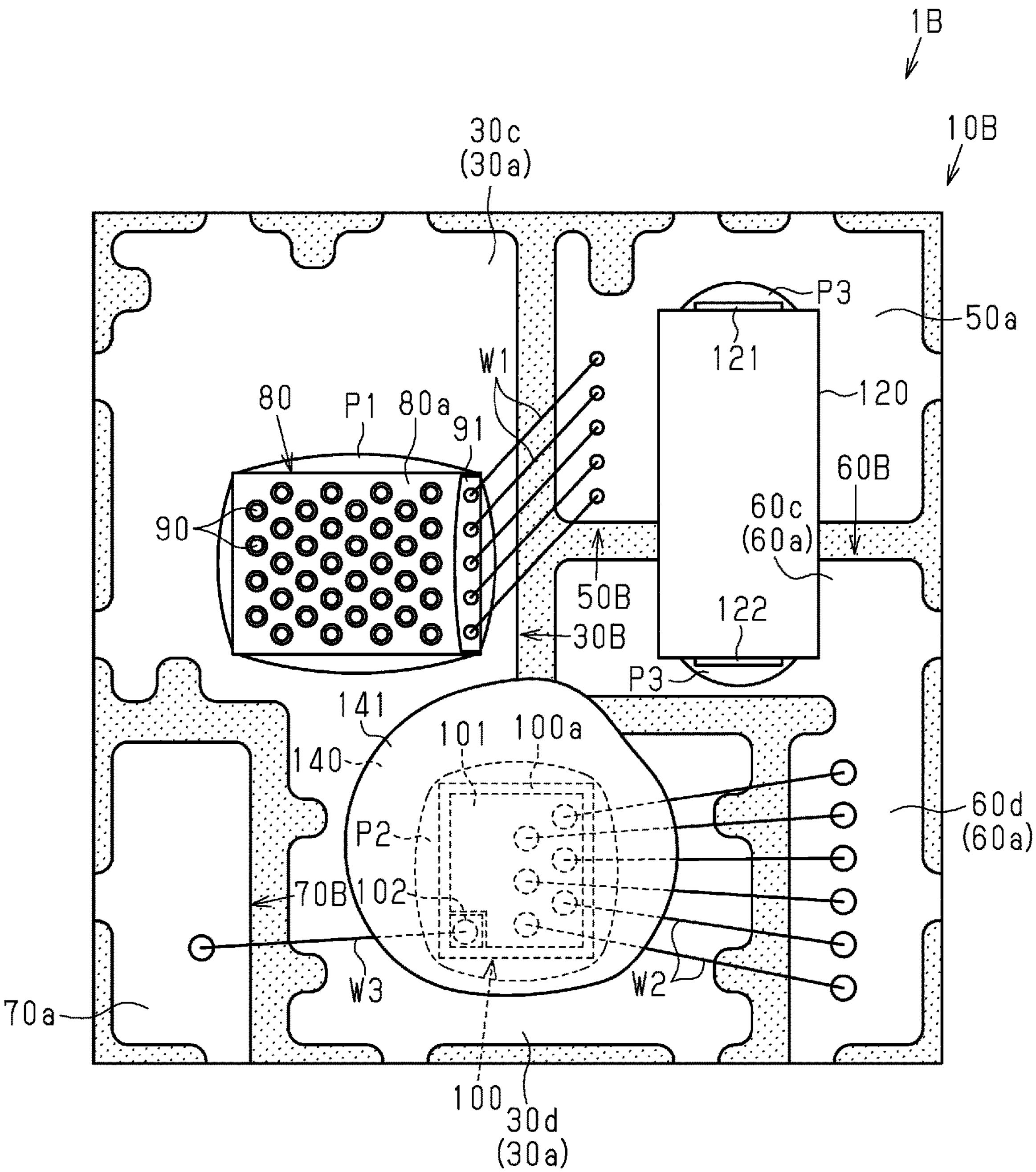
P1
80a
91
60B
13
90
60c
(60a)
50B
122
13
30B
P3
65
100
101
100a
13U
13
70B
70a
60d
(60a)
W3
102
P2
W2
13
13
30d
(30a)
13

Fig.35



1B
10B
30c
(30a)
P3
121
50a
W1
120
80
P1
80a
91
60B
90
60c
(60a)
50B
122
30B
P3
141
101
100a
140
60d
(60a)
P2
70B
102
W3
W2
70a
100
30d
(30a)

Fig.36

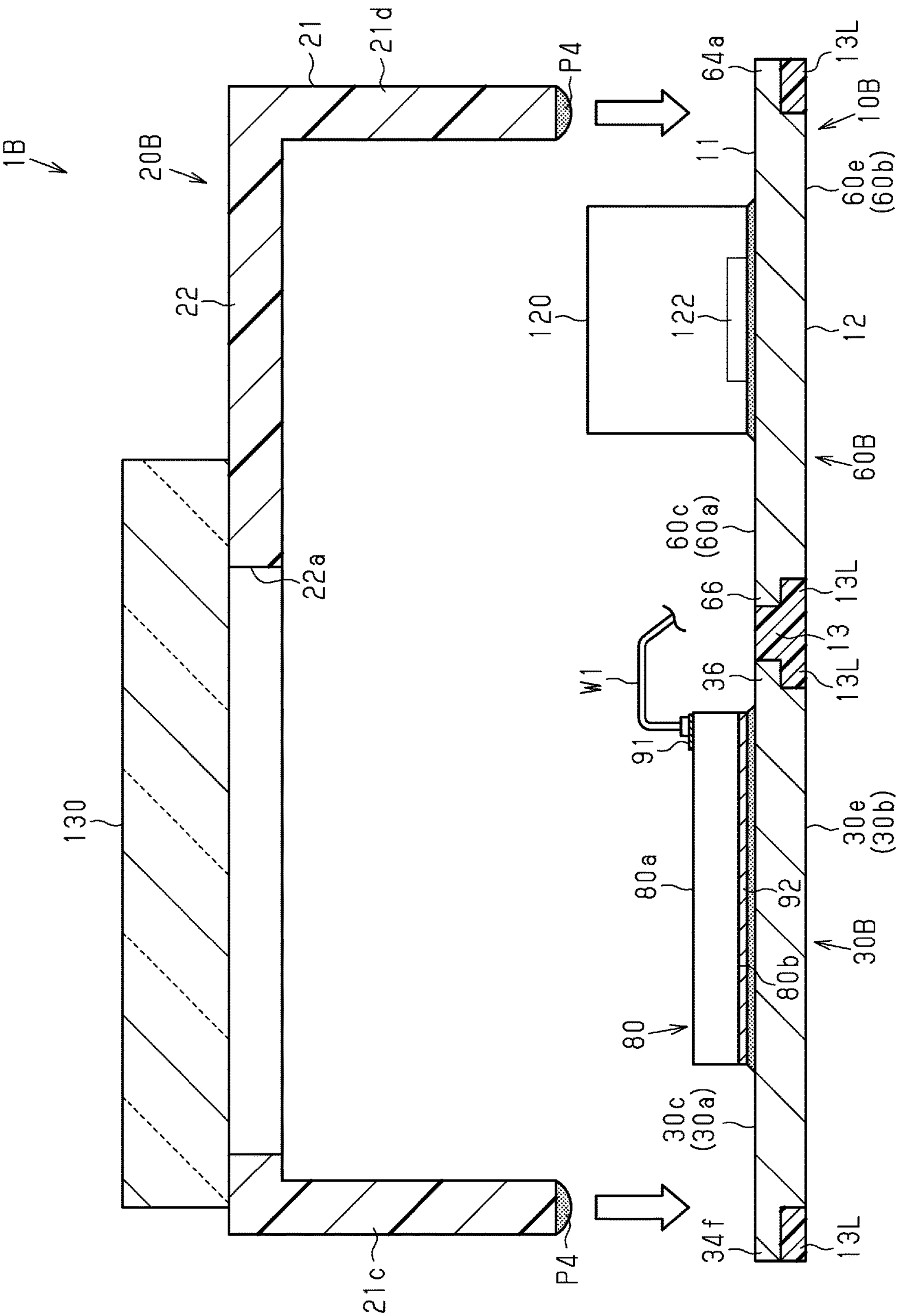

1B
20B
21
21d
P4
64a
13L
10B
11
60e
(60b)
22
120
122
12
60B
130
22a
60c
(60a)
66
13
13L
36
W1
91
80a
92
30e
(30b)
30B
80b
80
30c
(30a)
34f
21c
P4
13L

Fig.37

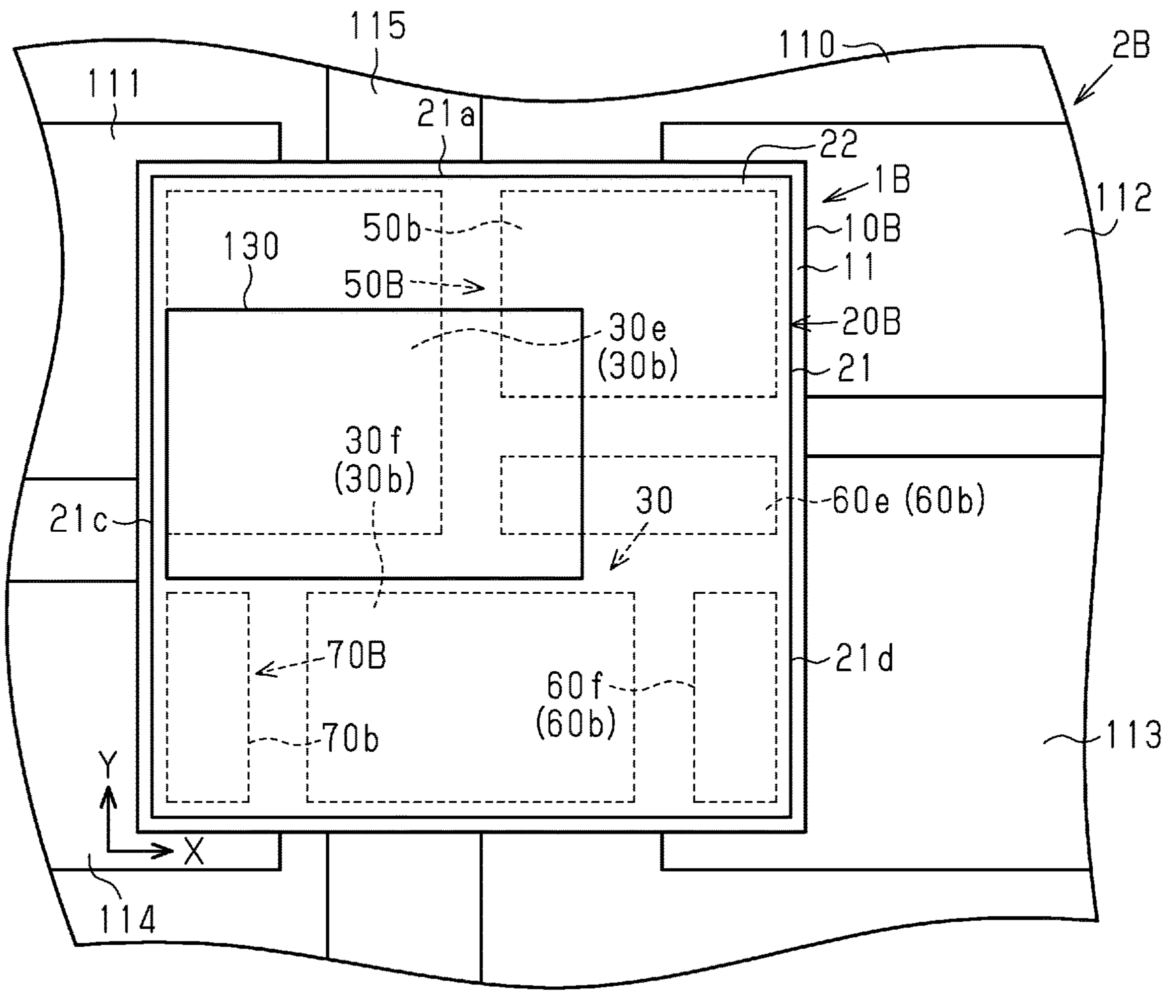

115
110
2B
111
21a
22
1B
112
10B
50b
130
11
50B
30e
20B
(30b)
21
30f
(30b)
30
60e (60b)
21c
70B
21d
60f
(60b)
70b
113
Y
X
114

Fig.38

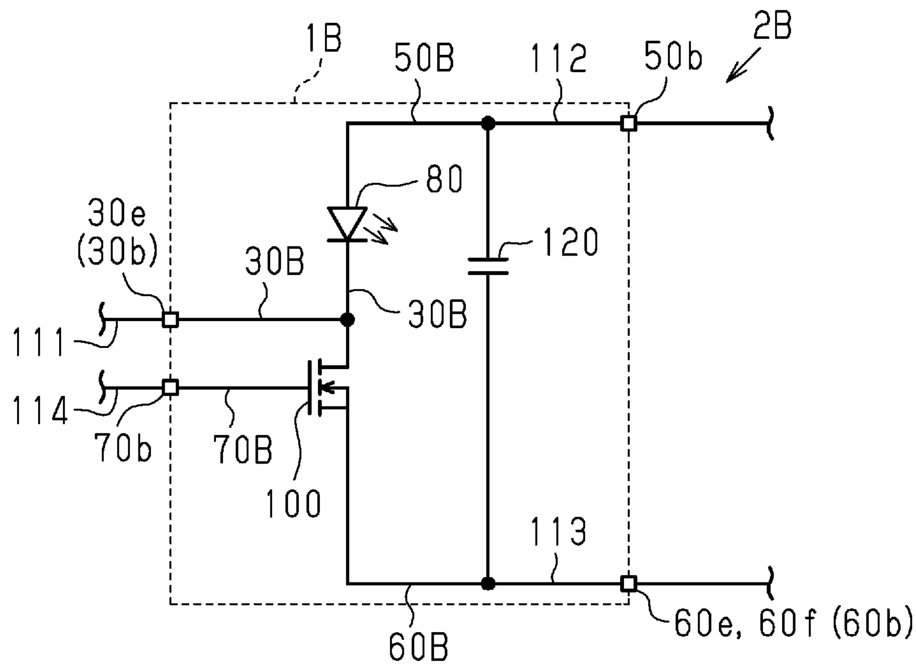

1B
50B
112
50b
2B
80
30e
(30b)
30B
120
111
30B
114
70b
70B
100
113
60B
60e, 60f (60b)

Fig.39

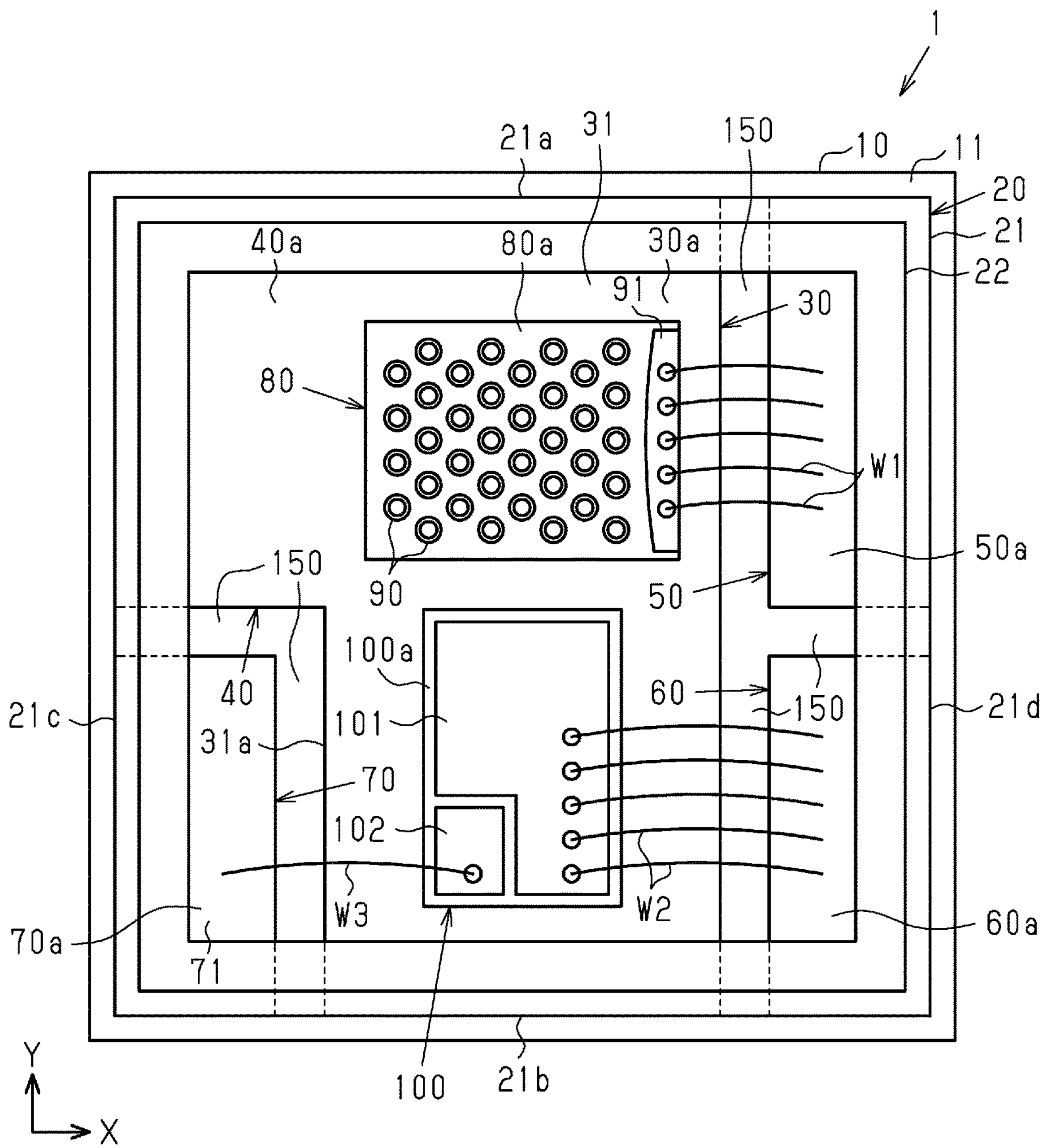

1
21a
31
150
10
11
20
21
22
40a
80a
30a
91
30
80
W1
50a
150
90
50
100a
60
150
40
101
21c
31a
70
21d
102
W3
W2
60a
70a
71
100
21b
Y
X

Fig.40

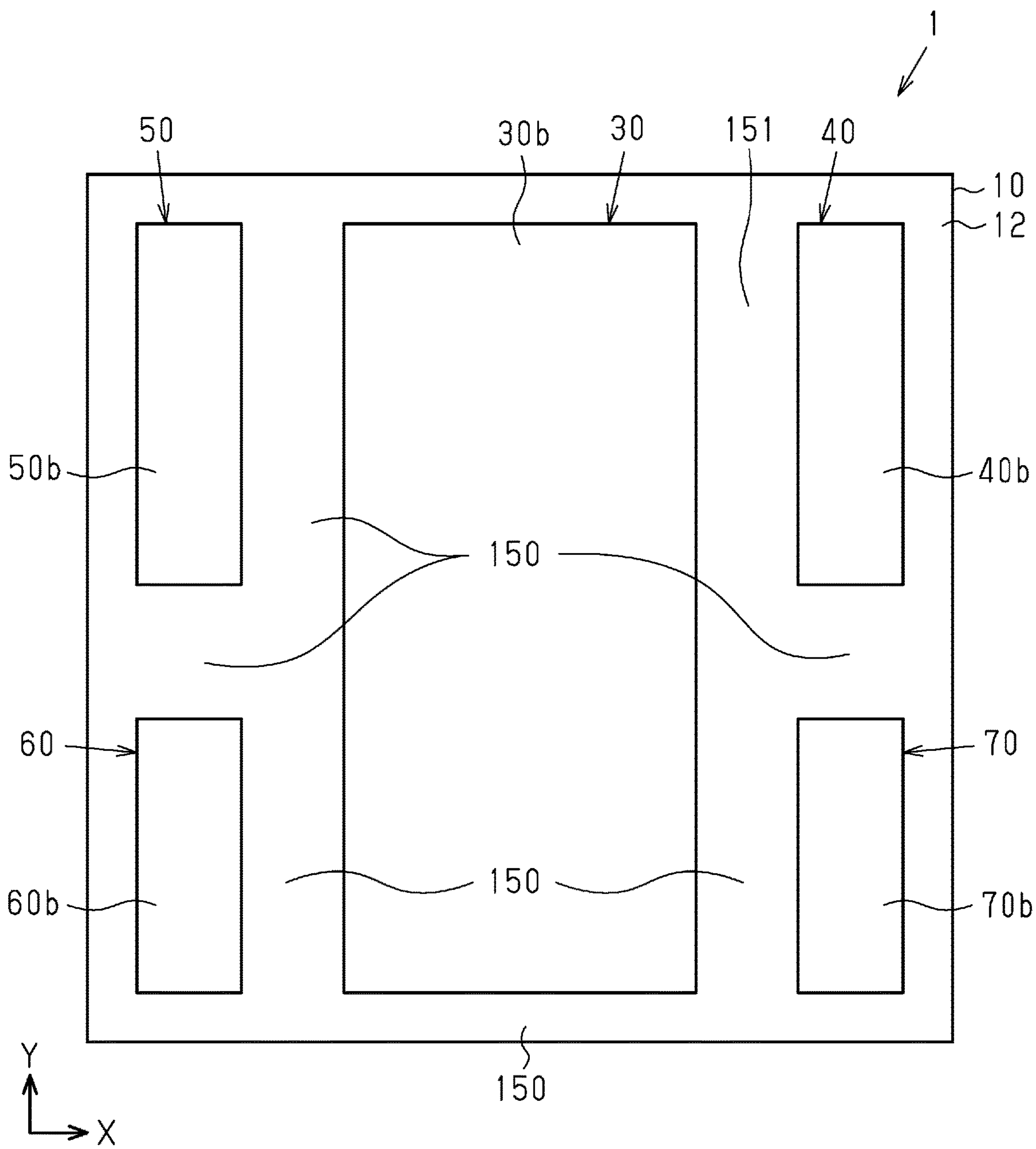


Fig.41

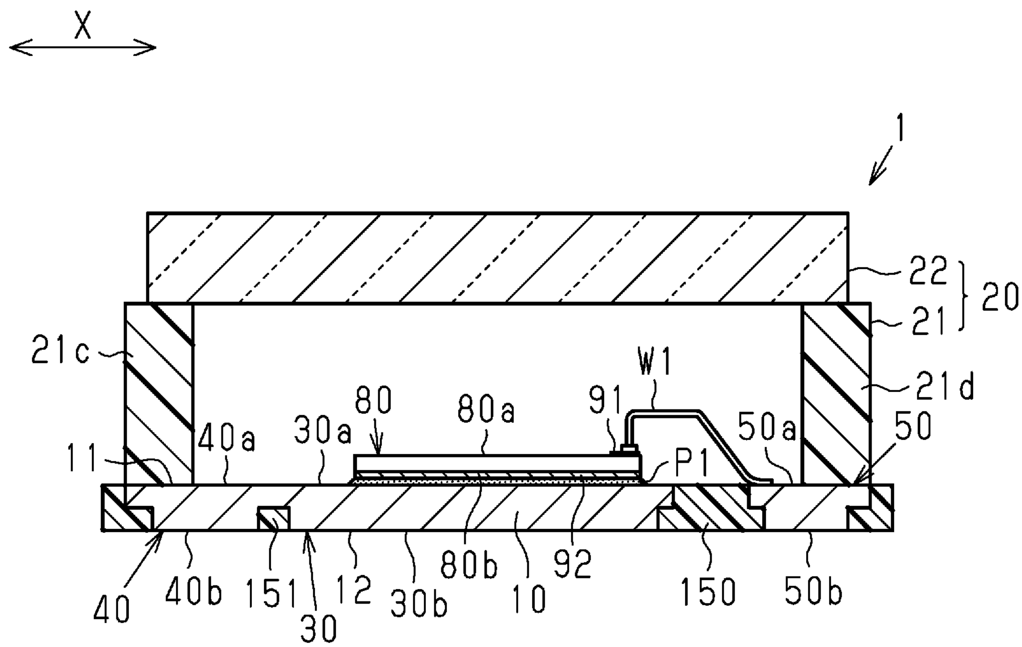

X
1
22
21
20
21c
21d
W1
80
80a
91
30a
40a
50a
50
11
P1
40
40b
151
12
30b
80b
10
92
150
50b
30

Fig.42

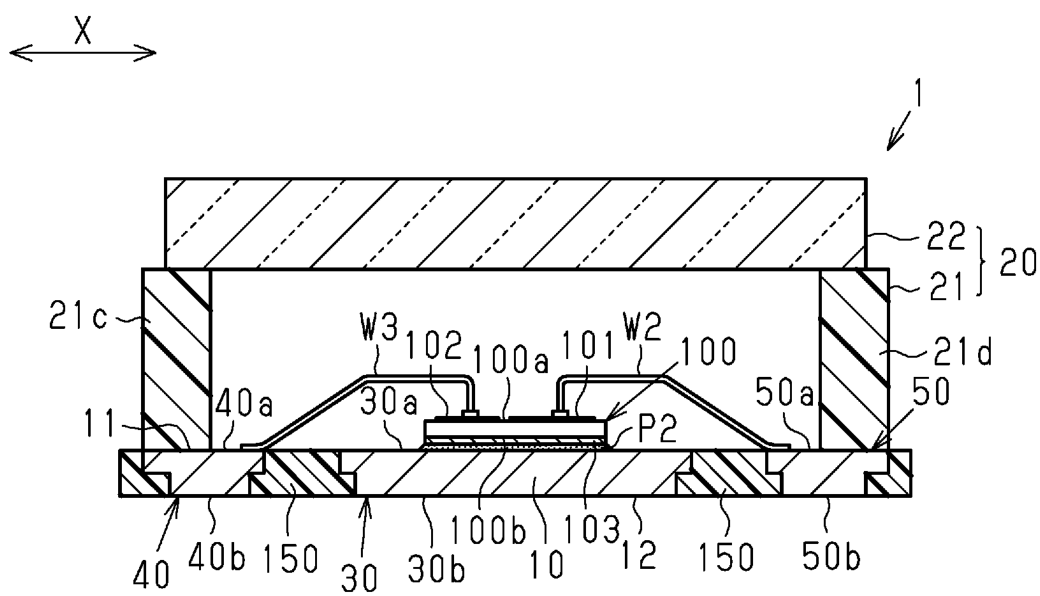

X
1
22
21
20
21c
21d
W3
102
100a
101
W2
100
30a
40a
50a
50
11
P2
40
40b
150
30
30b
100b
10
103
12
150
50b

Fig.43

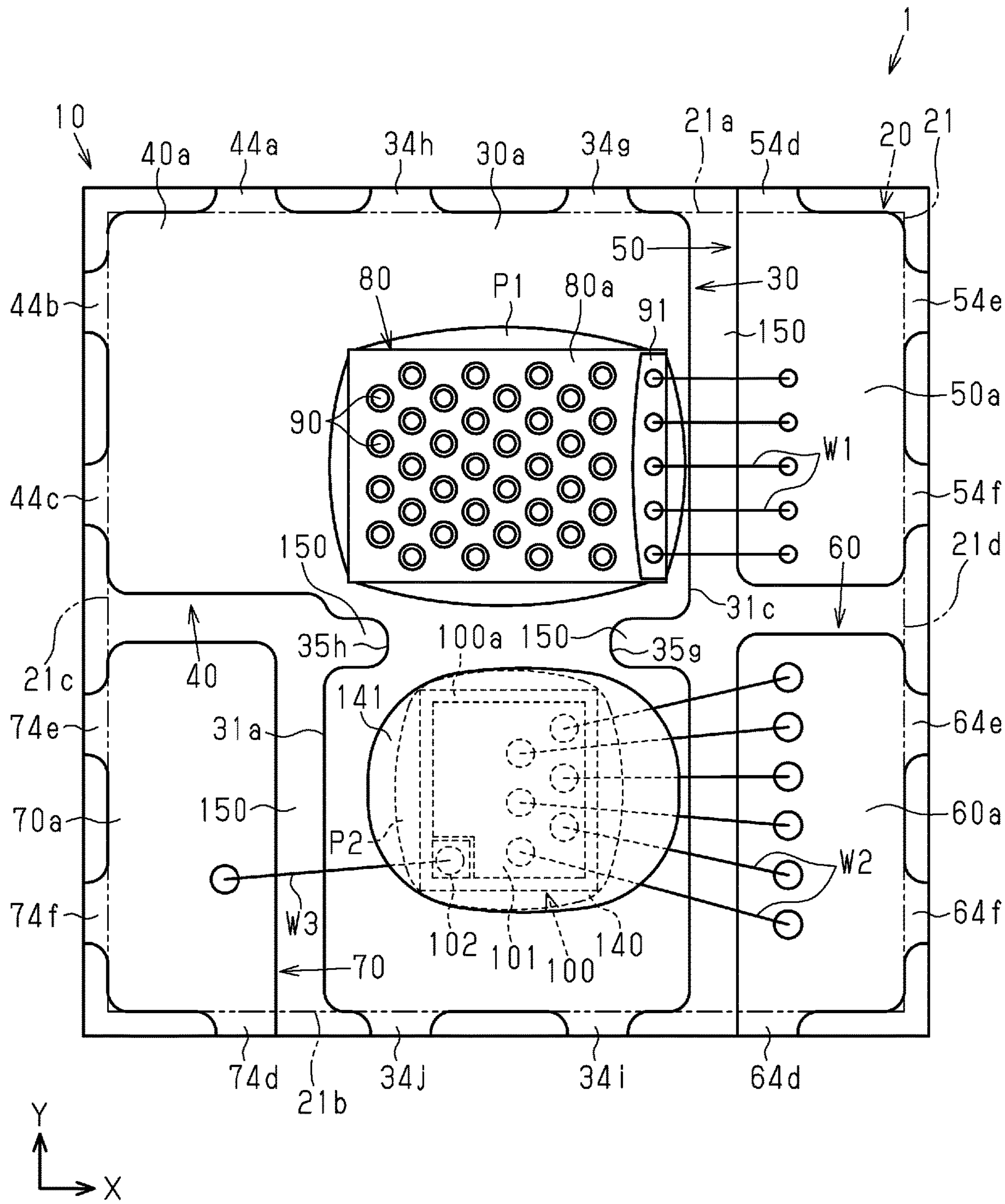

1
10
40a
44a
34h
30a
34g
21a
54d
20
21
50
30
80
P1
80a
91
150
44b
54e
90
50a
W1
44c
54f
150
60
21d
31c
35h
100a
150
35g
21c
40
141
74e
31a
64e
70a
150
60a
P2
W2
74f
W3
102
101
100
140
64f
70
74d
21b
34j
34i
64d
Y
X

Fig.44

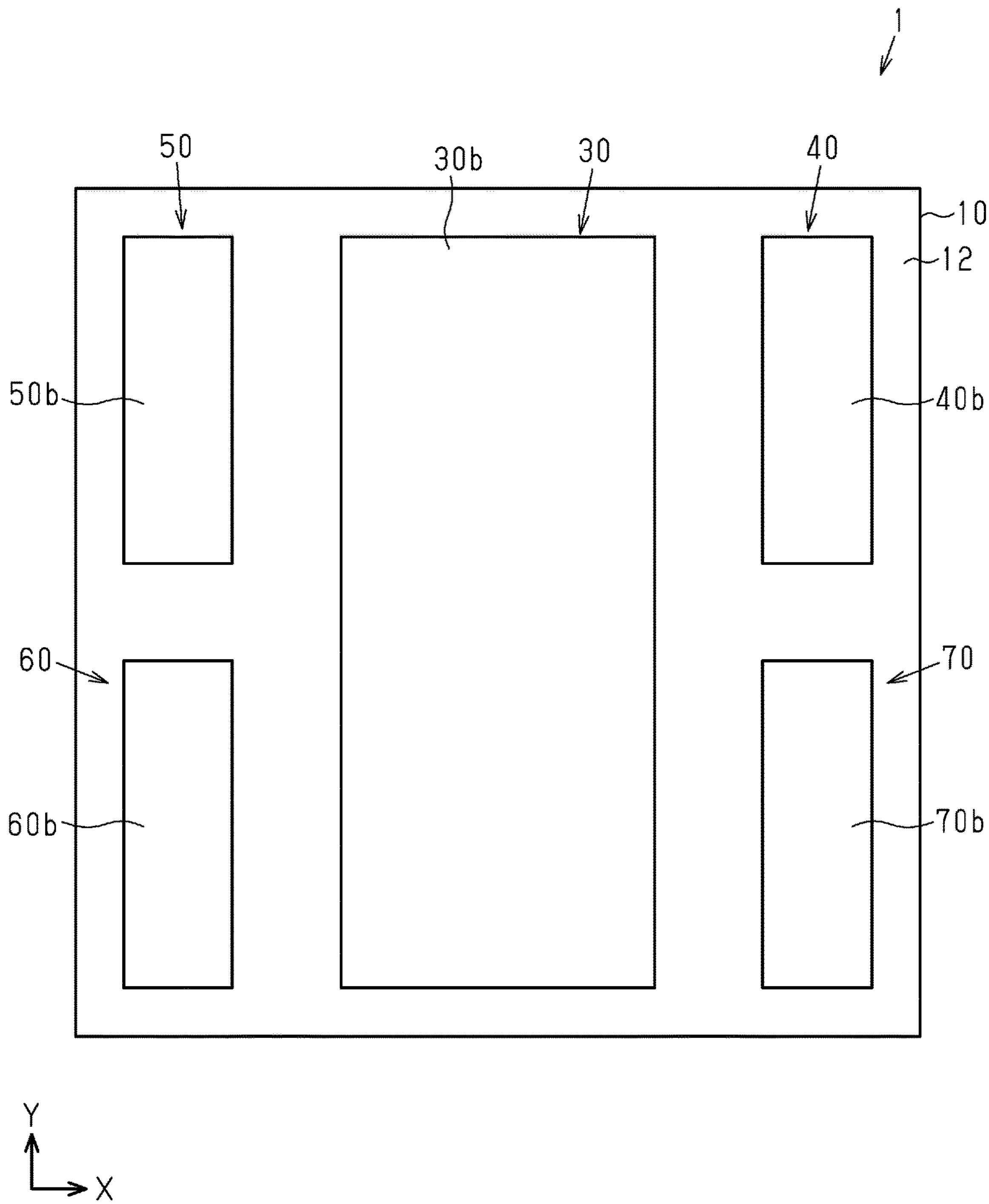


Fig.45

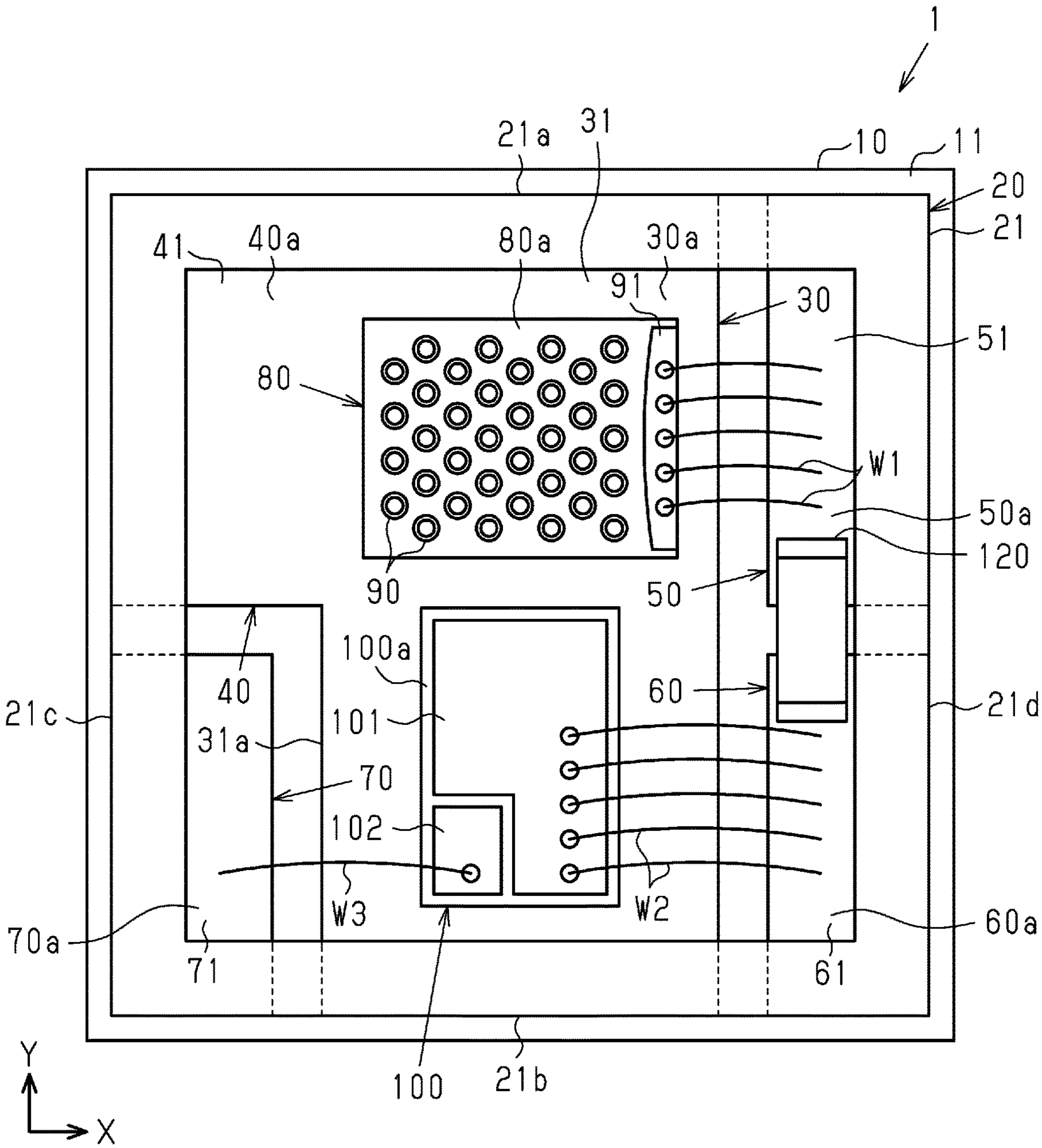

1
10
11
20
21
21a
31
30a
30
41
40a
80a
91
51
80
W1
50a
120
90
50
100a
40
60
21c
101
21d
31a
70
102
W3
W2
70a
60a
71
61
100
21b
Y
X

Fig.46

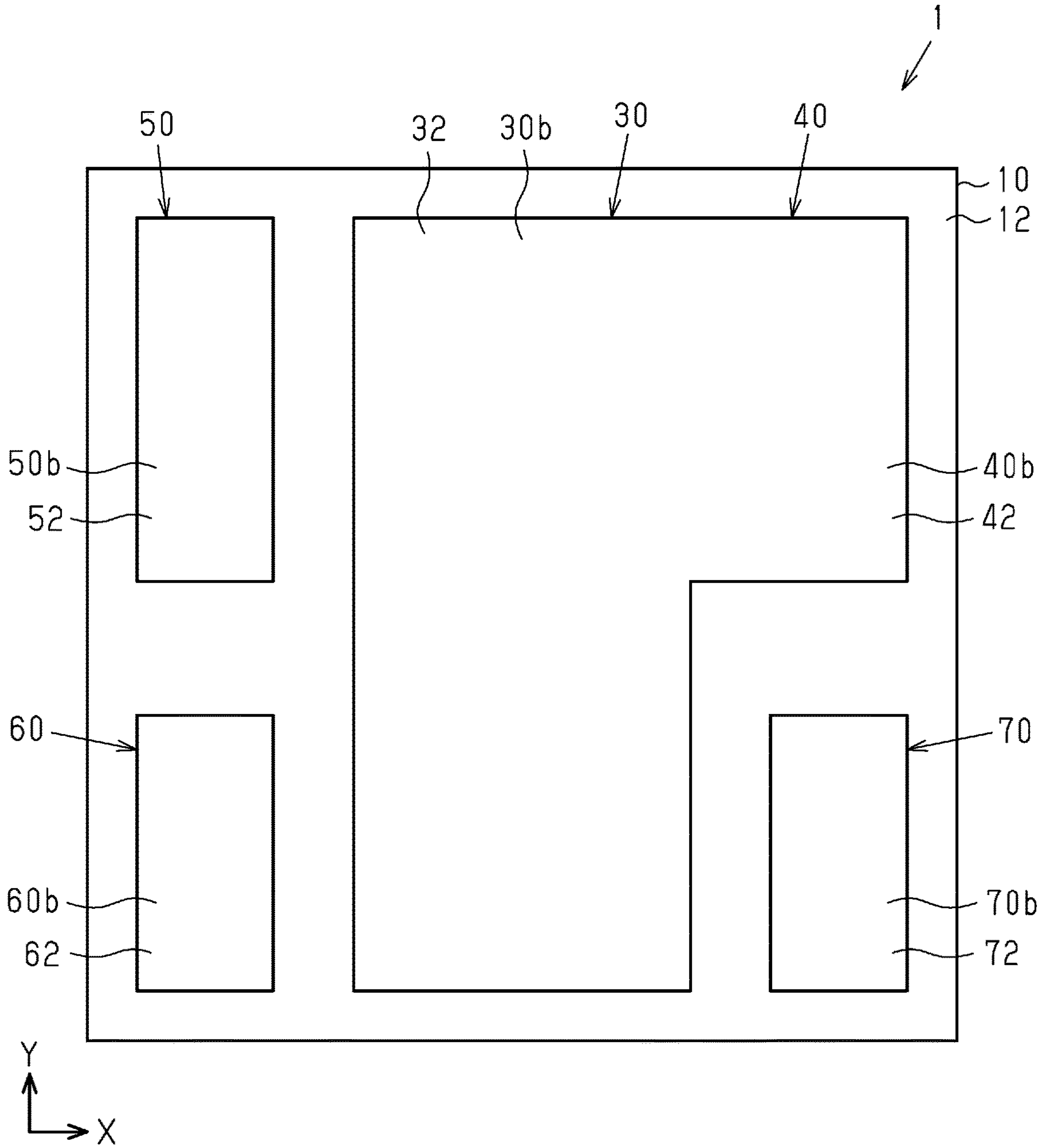


Fig.47

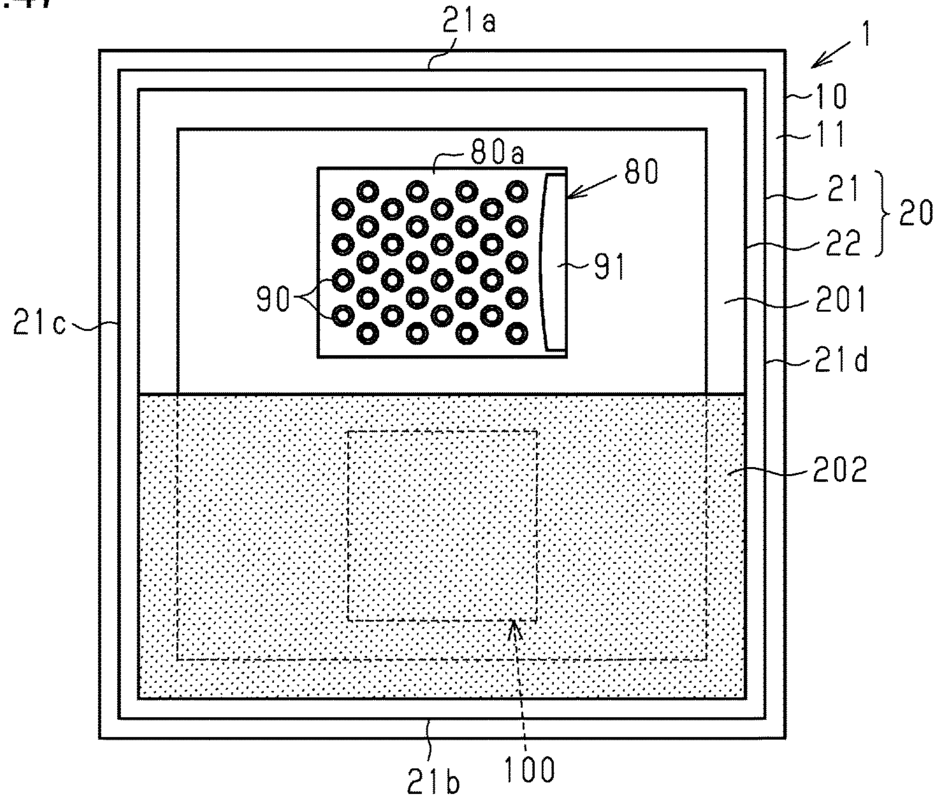

21a
1
10
11
80a
80
21
20
22
91
90
201
21c
21d
202
21b
100

Fig.48

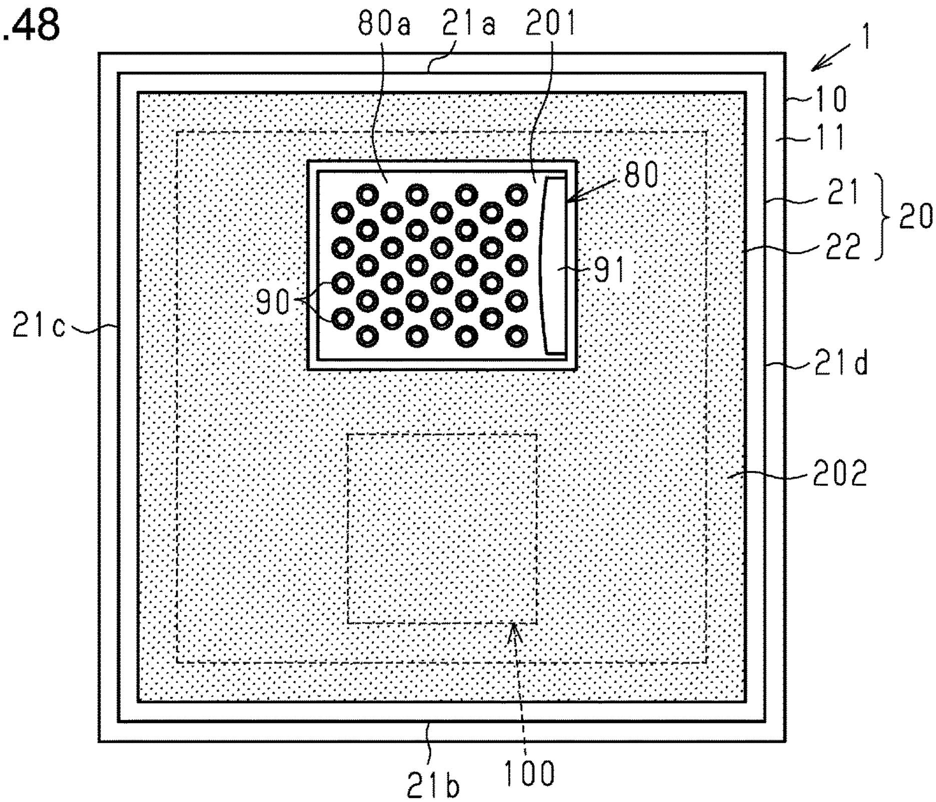
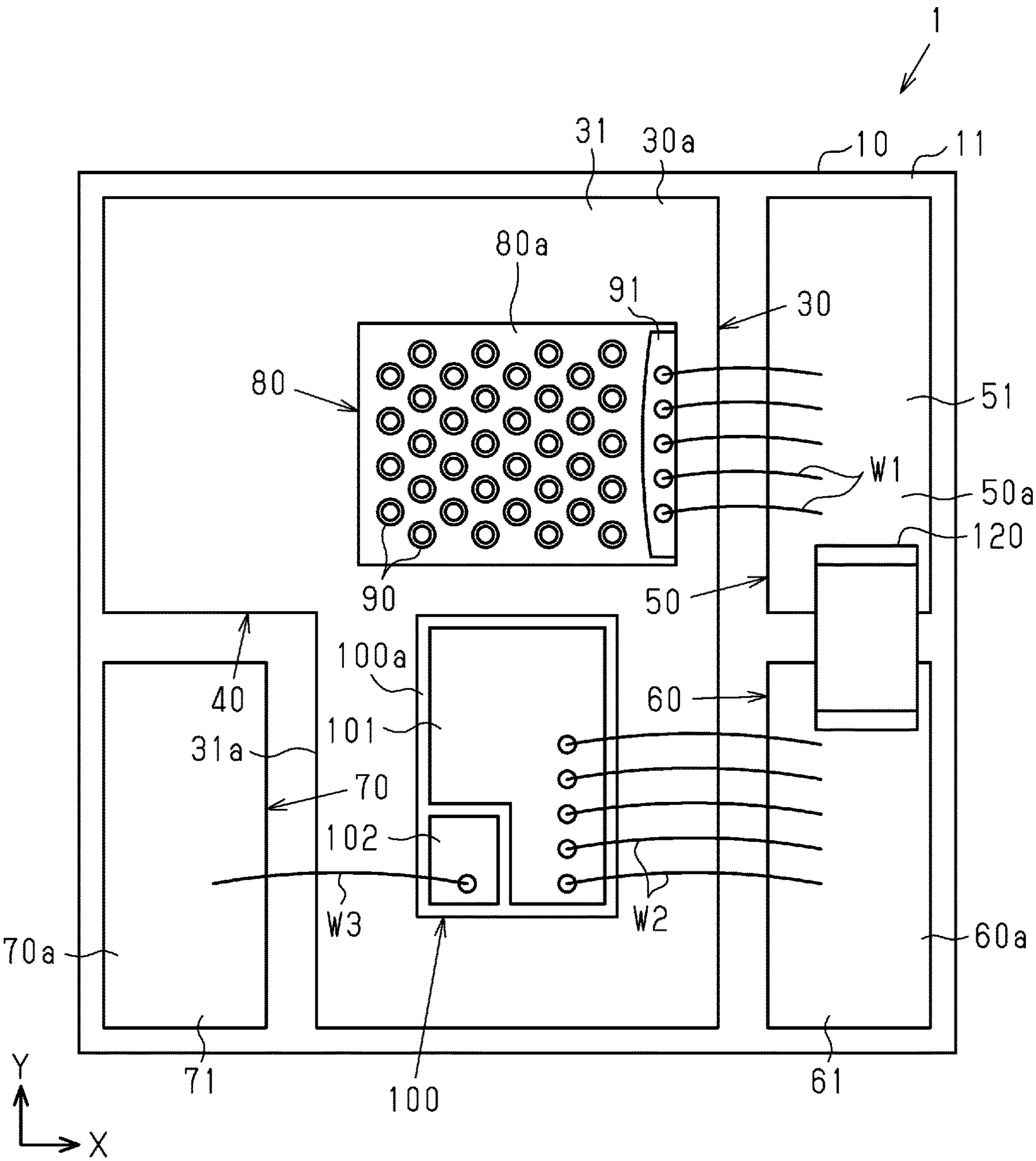
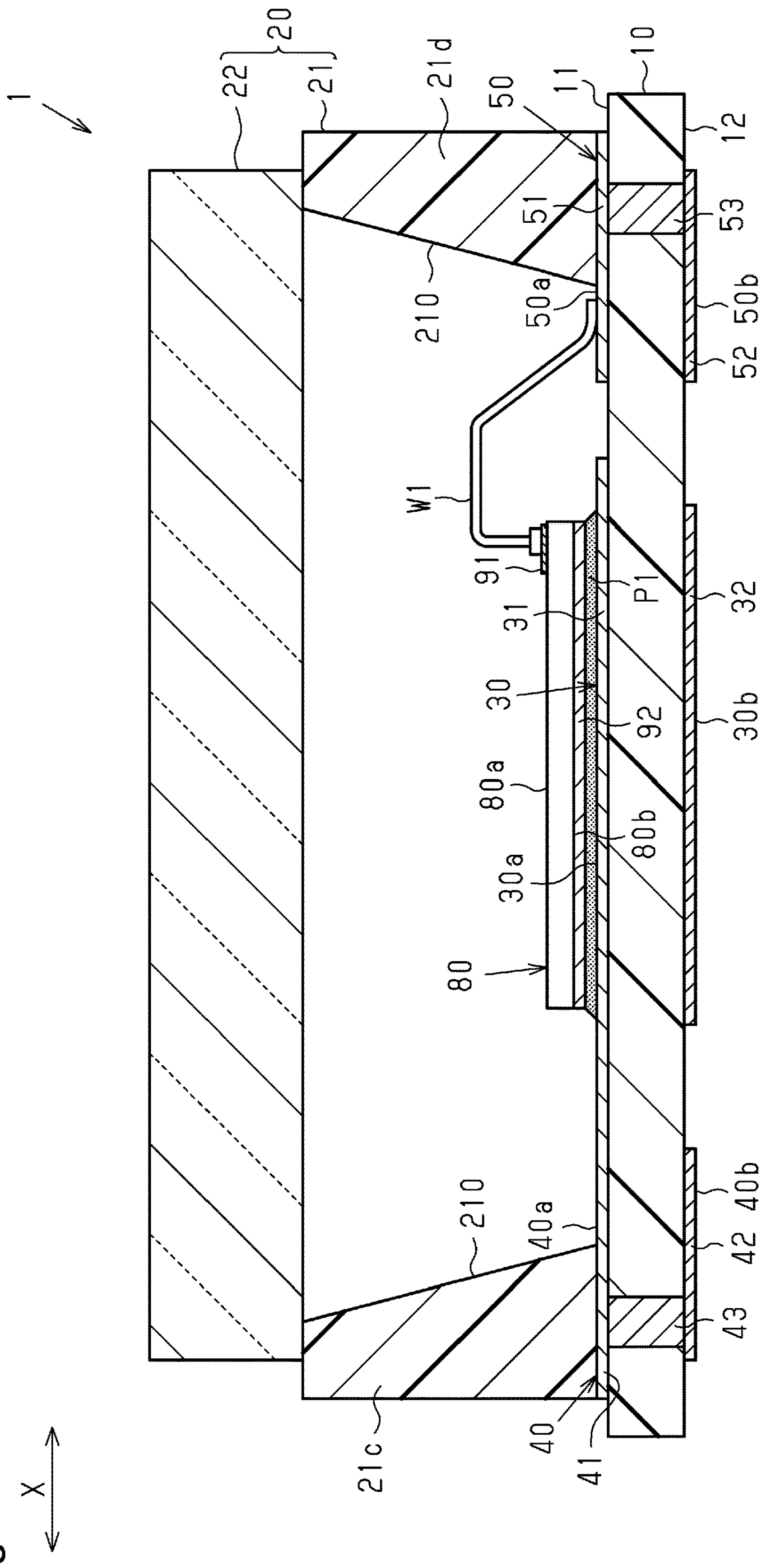

80a
21a
201
1
10
11
80
21
20
22
91
90
21c
21d
202
21b
100

Fig.49


1
31
30a
10
11
80a
91
30
80
51
W1
50a
120
90
50
100a
40
60
101
31a
70
102
W3
W2
70a
60a
71
100
61
Y
X



1
20
22
21
21d
50
11
10
12
51
53
210
50a
50b
52
W1
91
P1
32
31
30
92
30b
80a
80b
30a
80
210
40a
40b
42
43
21c
40
41
X


Fig.50

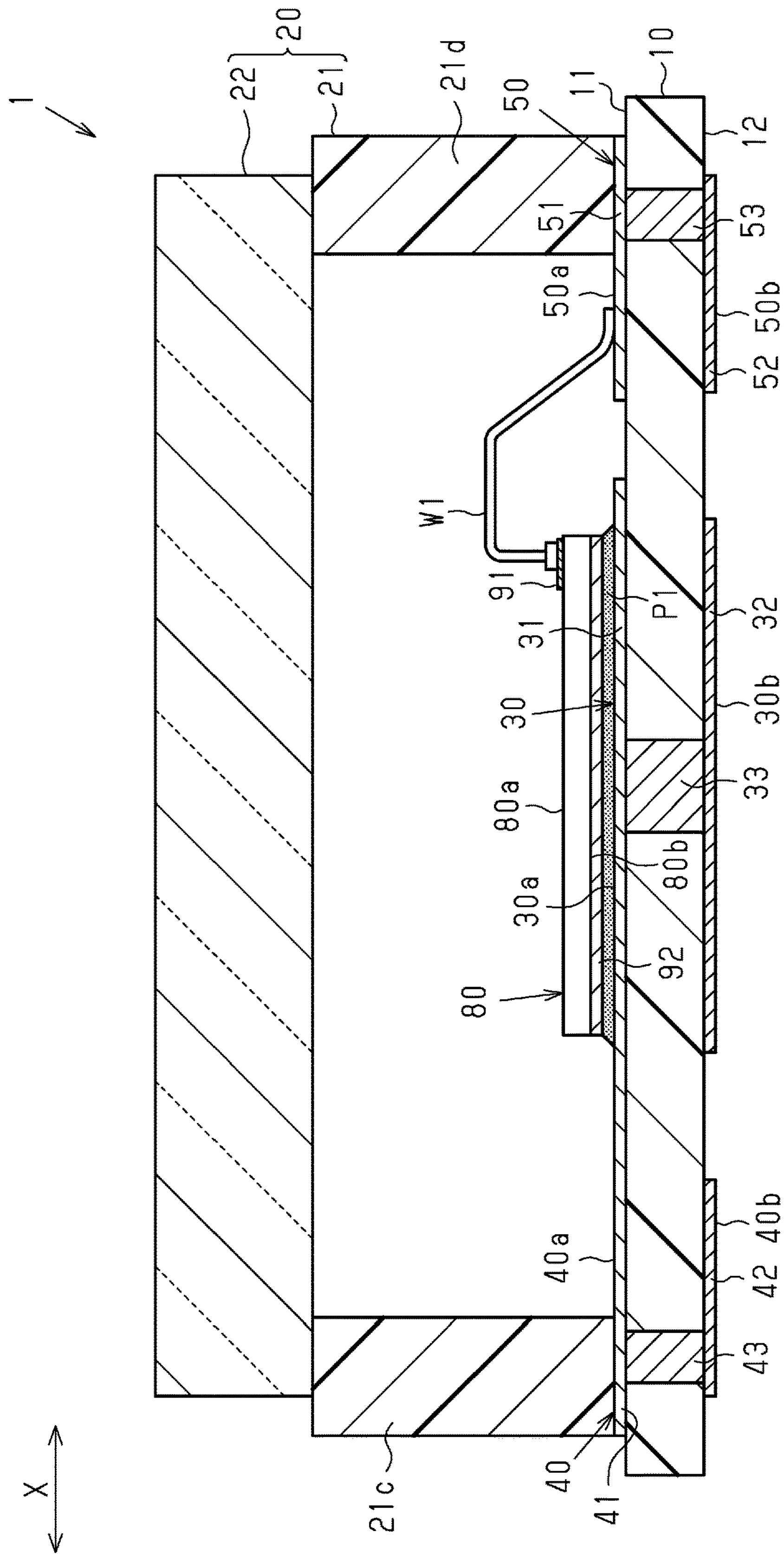

1
20
22
21
21d
50
11
10
12
51
53
50a
50b
52
W1
91
P1
32
31
30
30b
33
80a
80b
30a
92
80
40a
40b
42
43
40
41
21c
X

Fig.51

Fig.52

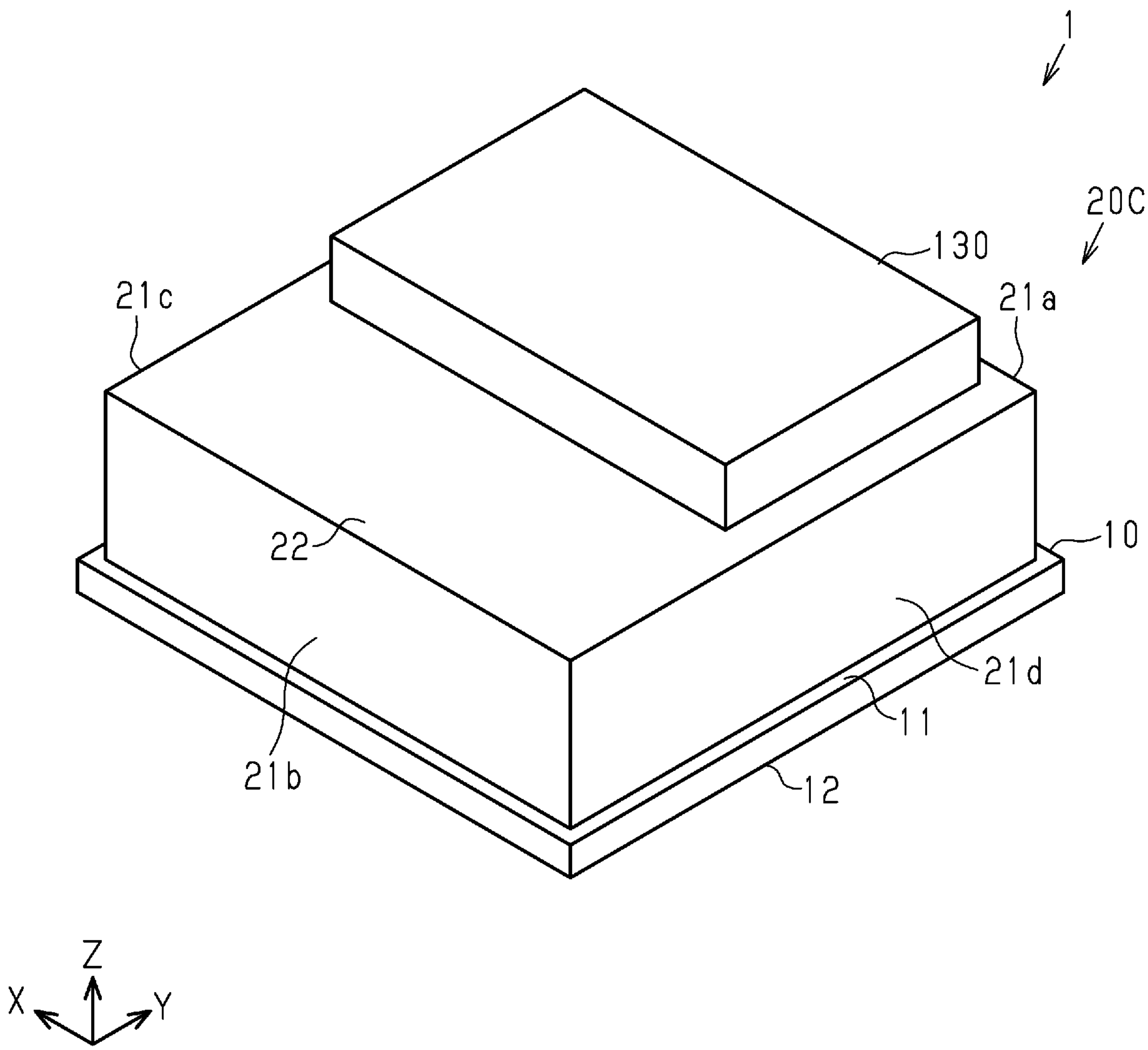
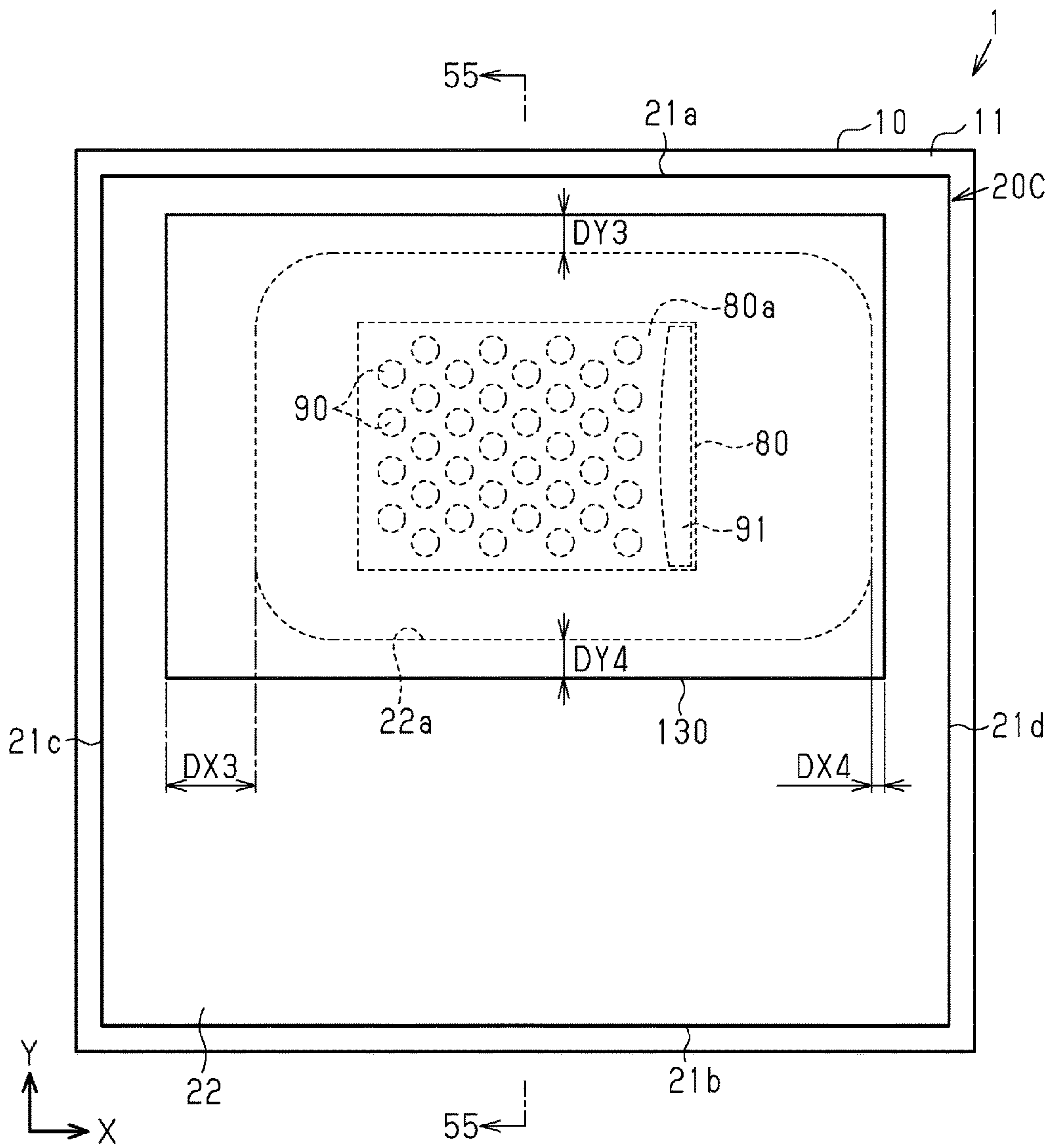


Fig.53


1
55
55
21a
10
11
20C
DY3
80a
90
80
91
DY4
22a
130
21c
21d
DX3
DX4
22
21b
Y
X

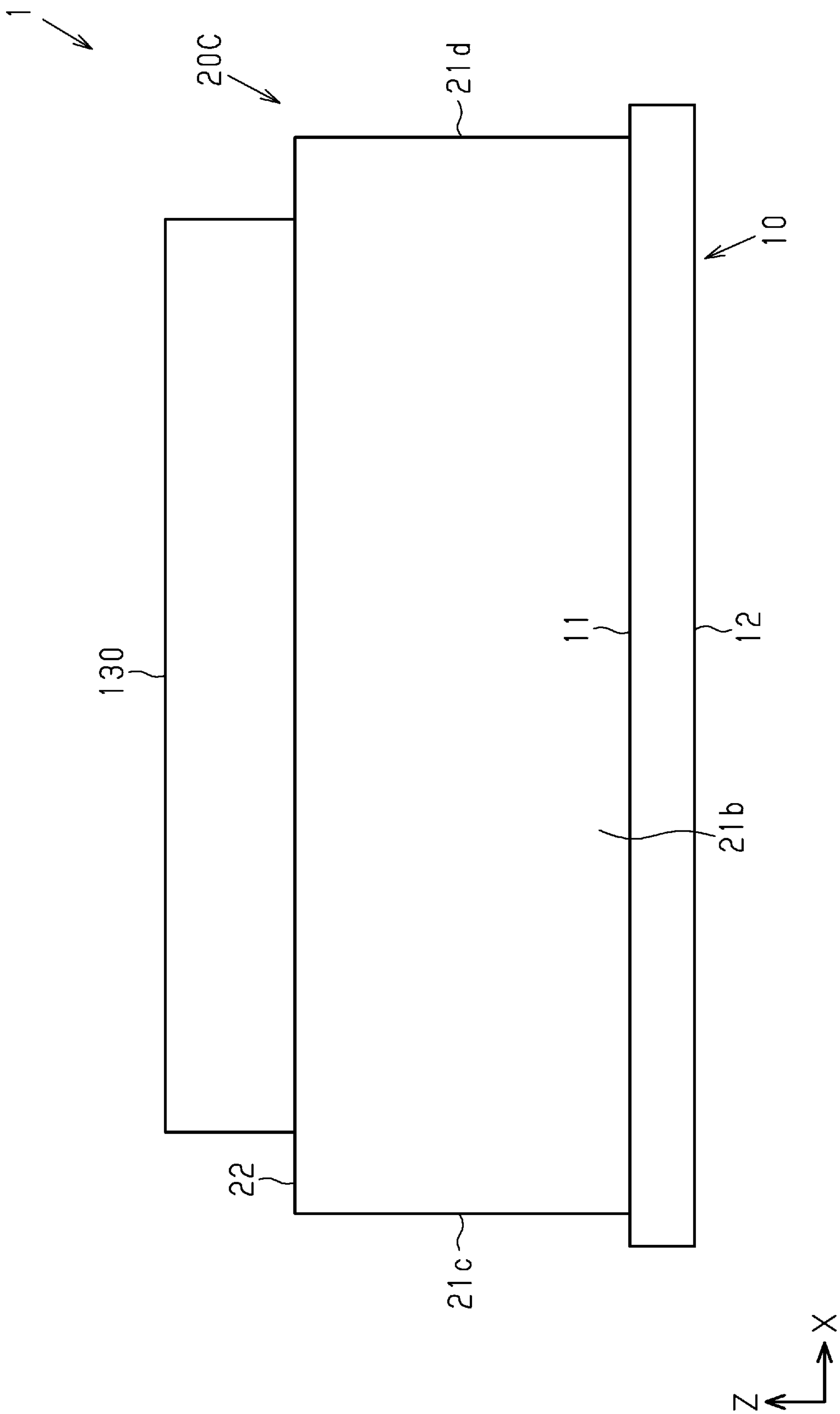



Fig.54

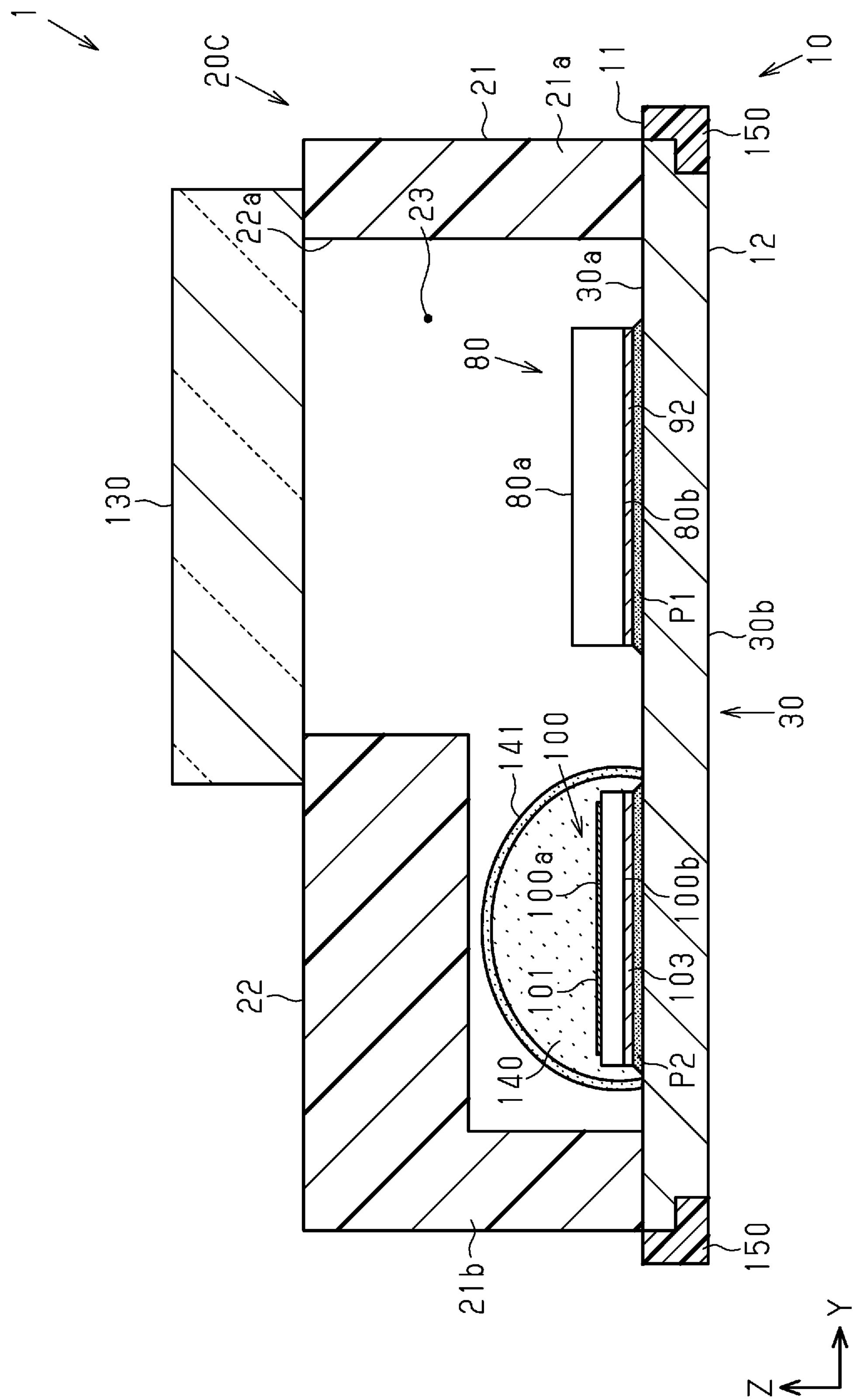

1
20C
21
21a
11
10
150
22a
23
12
30a
80
92
130
80a
80b
P1
30b
30
141
100
100a
100b
103
22
101
140
P2
150
21b
Y
Z

Fig.55

Fig.56

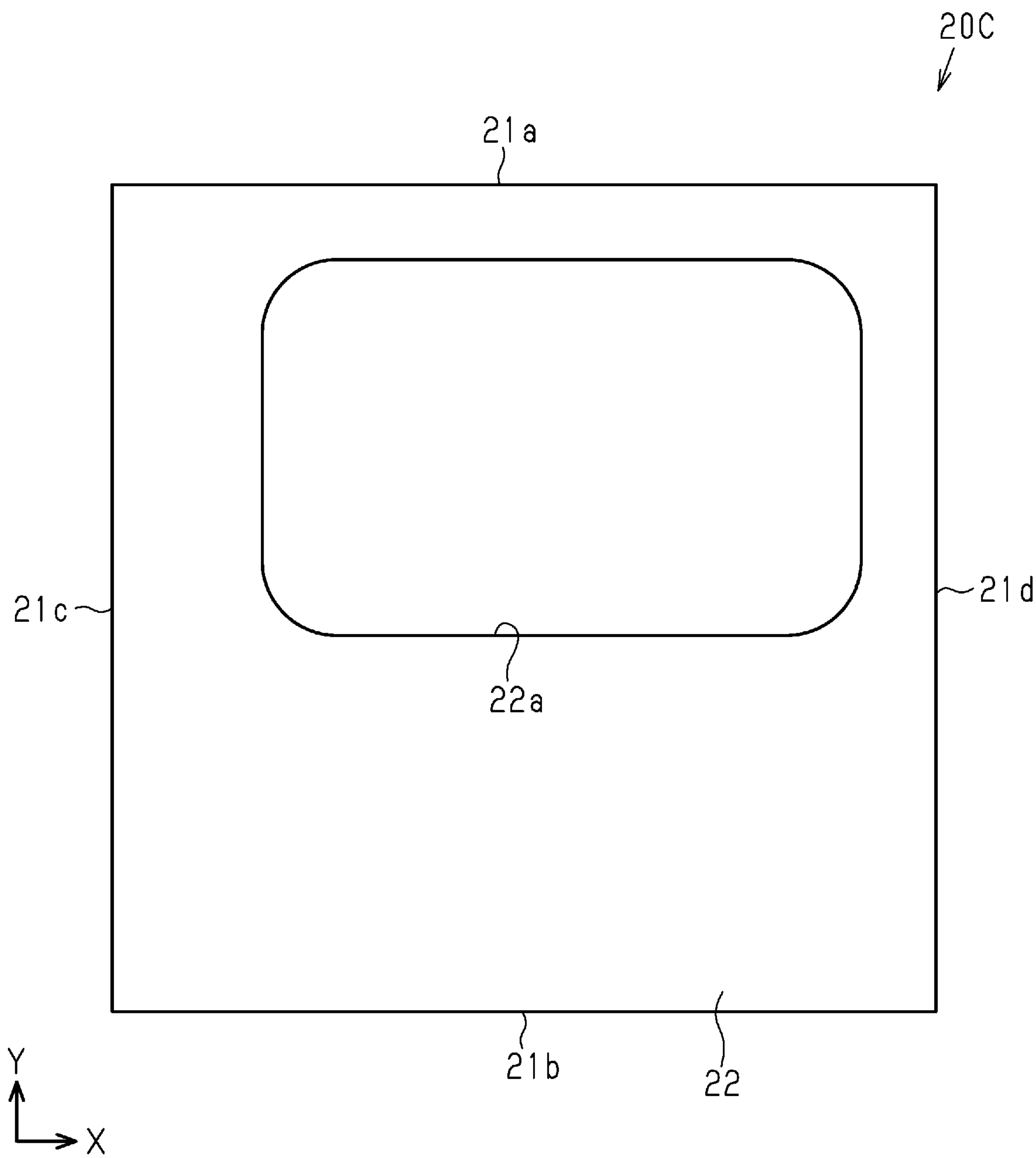
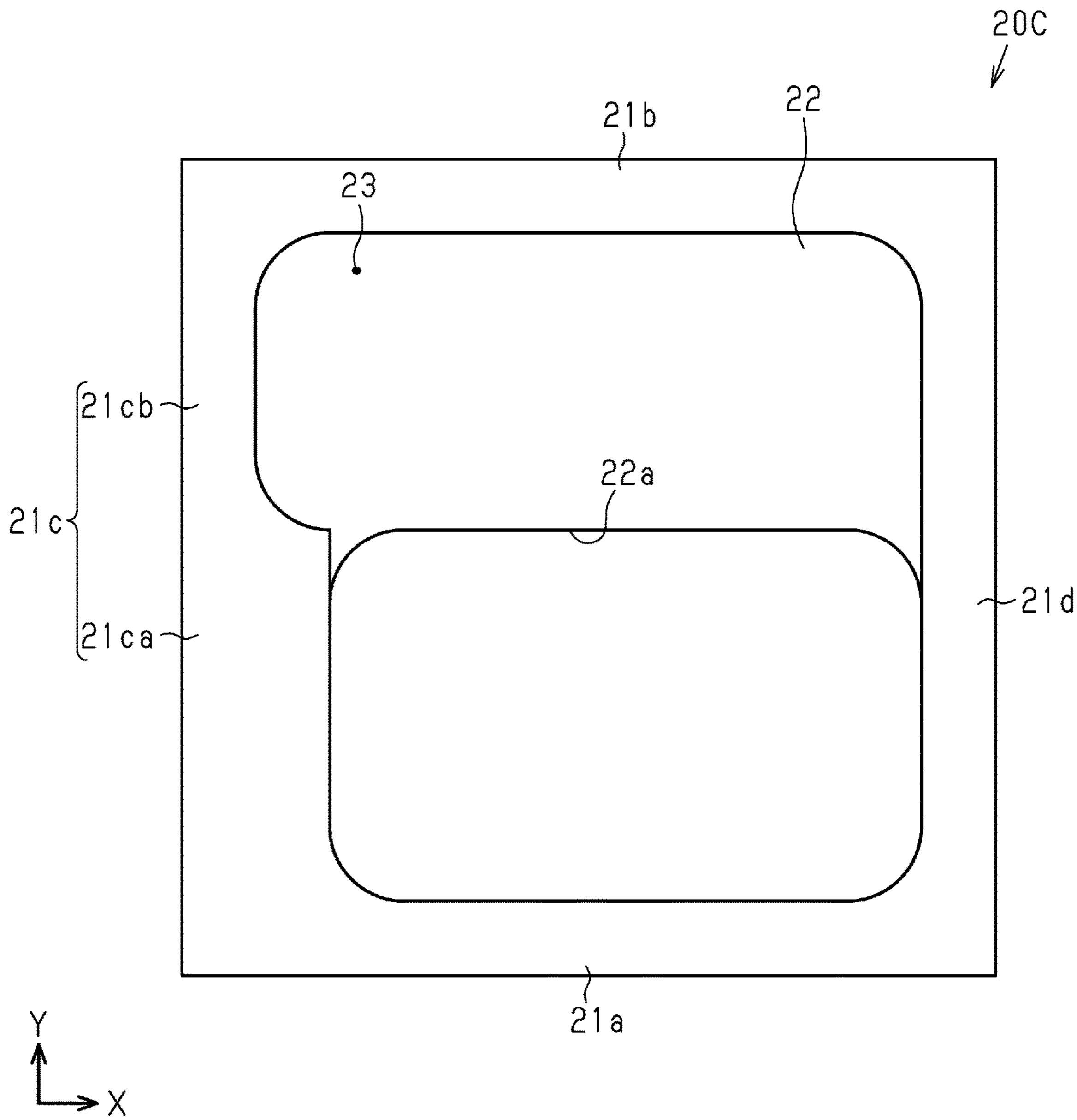


Fig.57



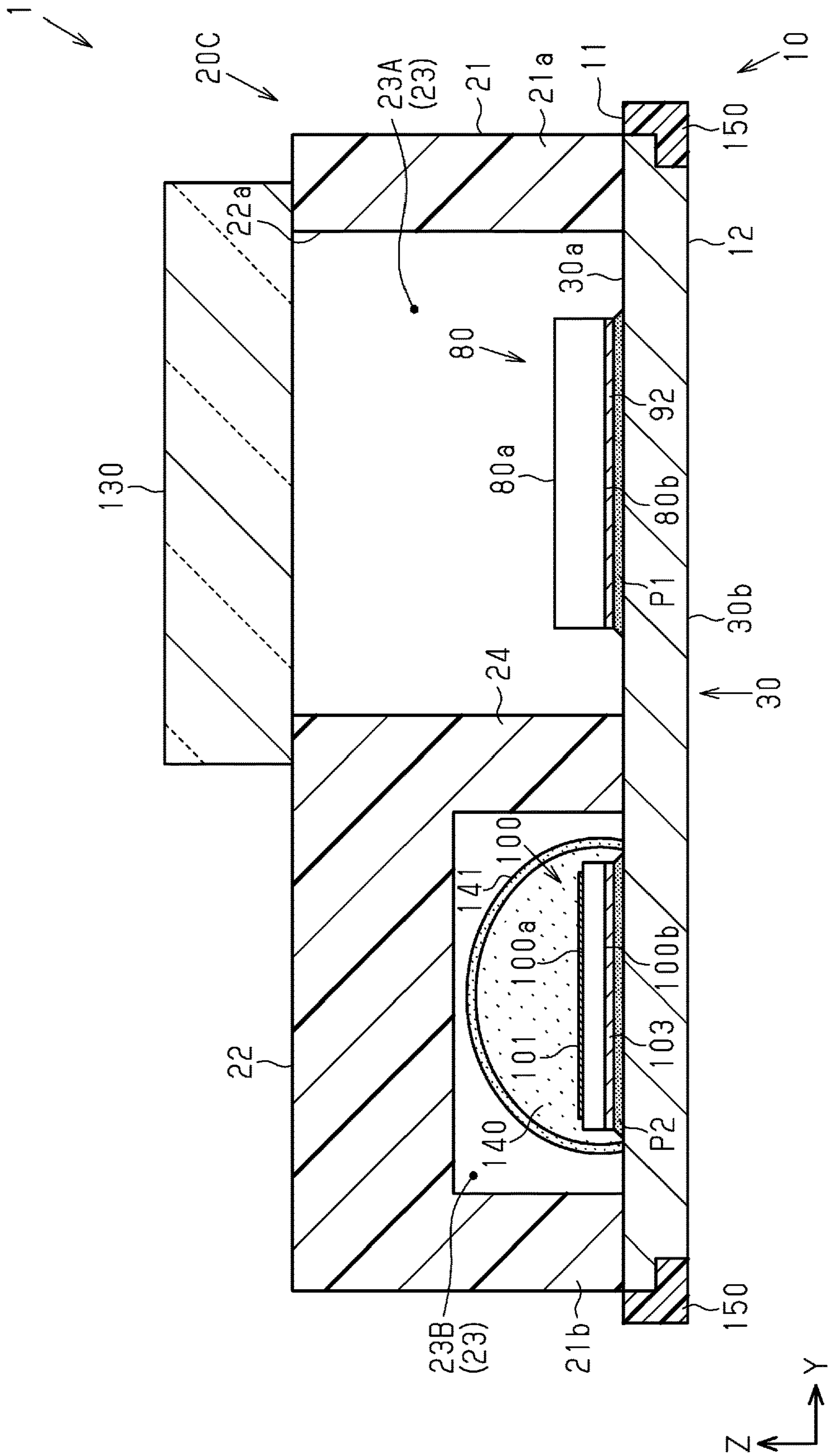
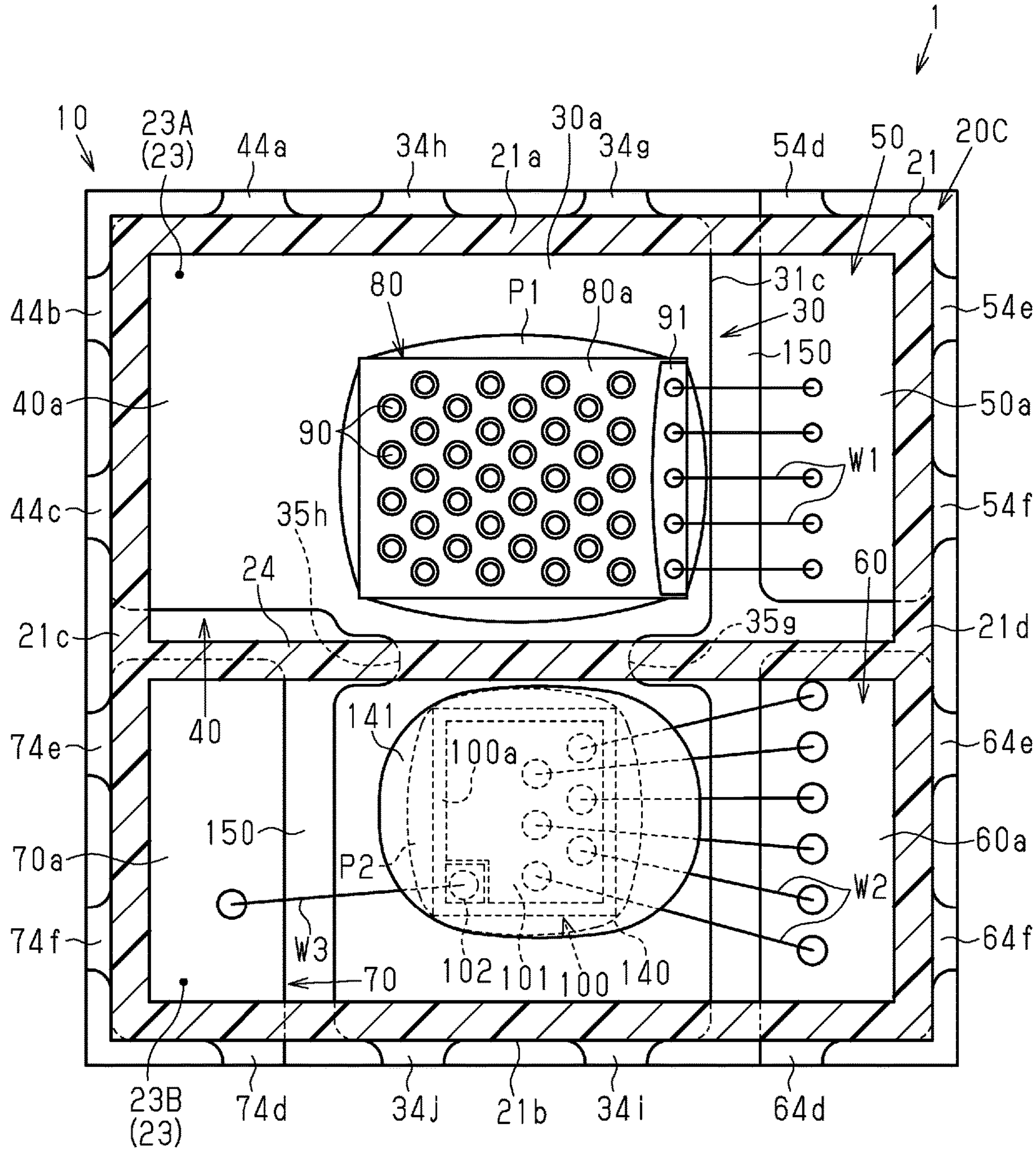

1
20C
23A
(23)
21
21a
11
10
150
22a
12
30a
80
92
130
80a
80b
P1
30b
24
30
100
141
100a
100b
101
103
22
140
P2
23B
(23)
21b
150
Y
Z

Fig.58

Fig.59


1
10
23A
(23)
44a
34h
21a
30a
34g
54d
50
21
20C
80
P1
80a
91
31c
30
150
44b
54e
40a
90
W1
50a
44c
35h
54f
24
60
35g
21c
21d
40
141
100a
74e
64e
150
70a
P2
60a
W2
W3
74f
64f
70
102
101
100
140
23B
(23)
74d
34j
21b
34i
64d

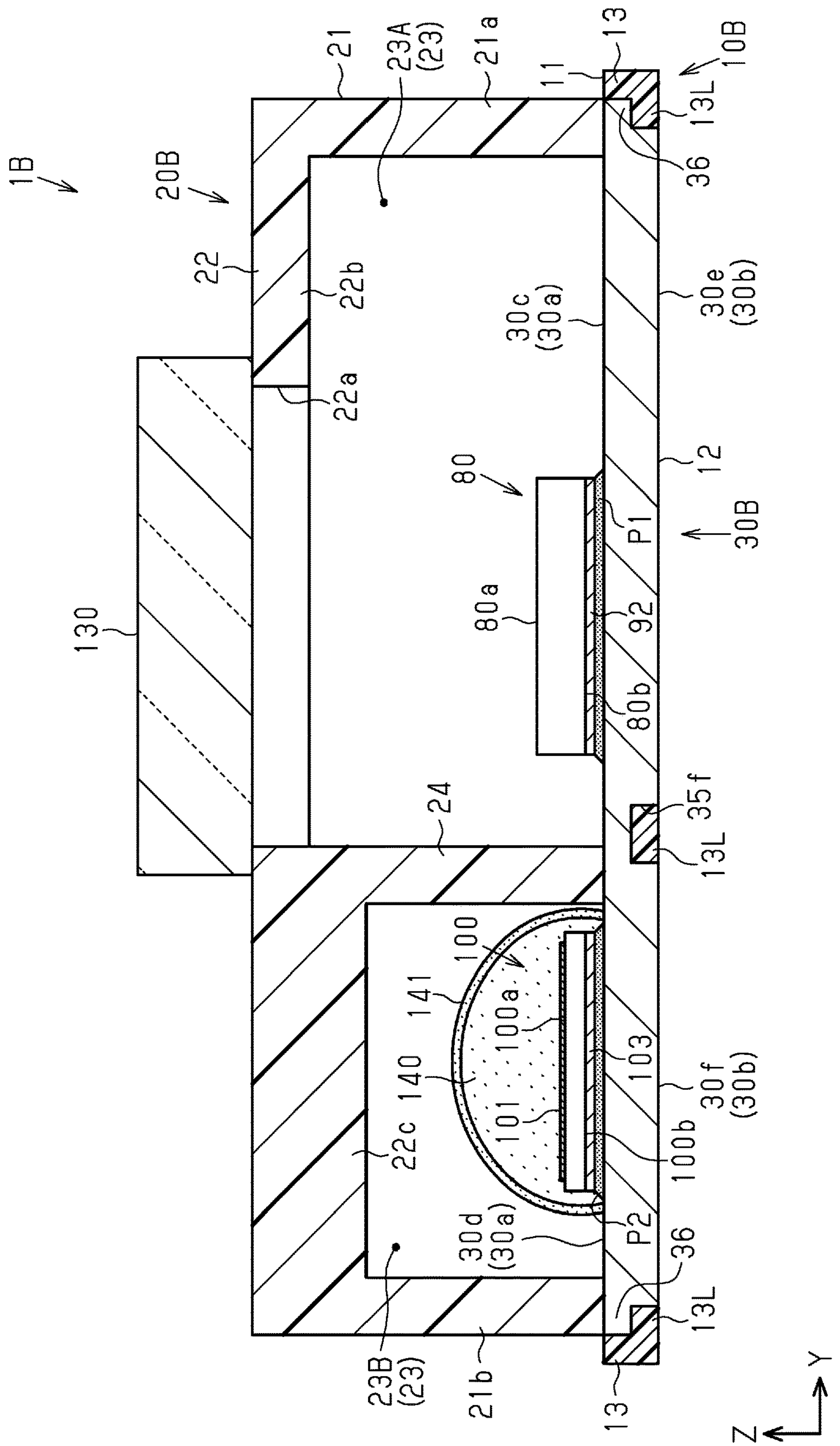
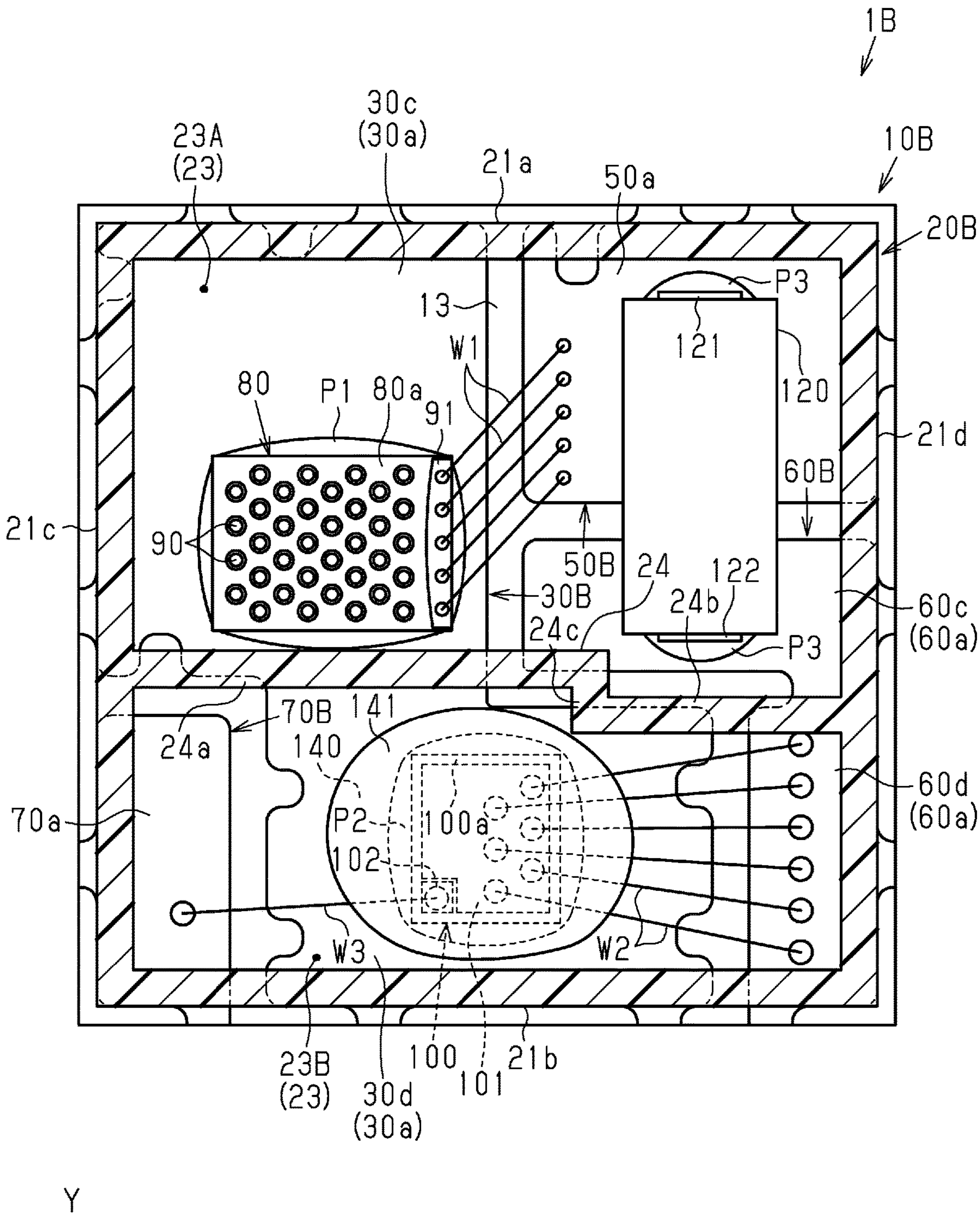

1B
20B
21
23A
(23)
21a
13
11
10B
13L
36
22
22b
30c
(30a)
30e
(30b)
22a
80
12
P1
30B
130
80a
92
80b
24
35f
13L
100
141
100a
140
103
30f
(30b)
101
22c
100b
30d
(30a)
P2
36
13L
23B
(23)
21b
13
Z
Y

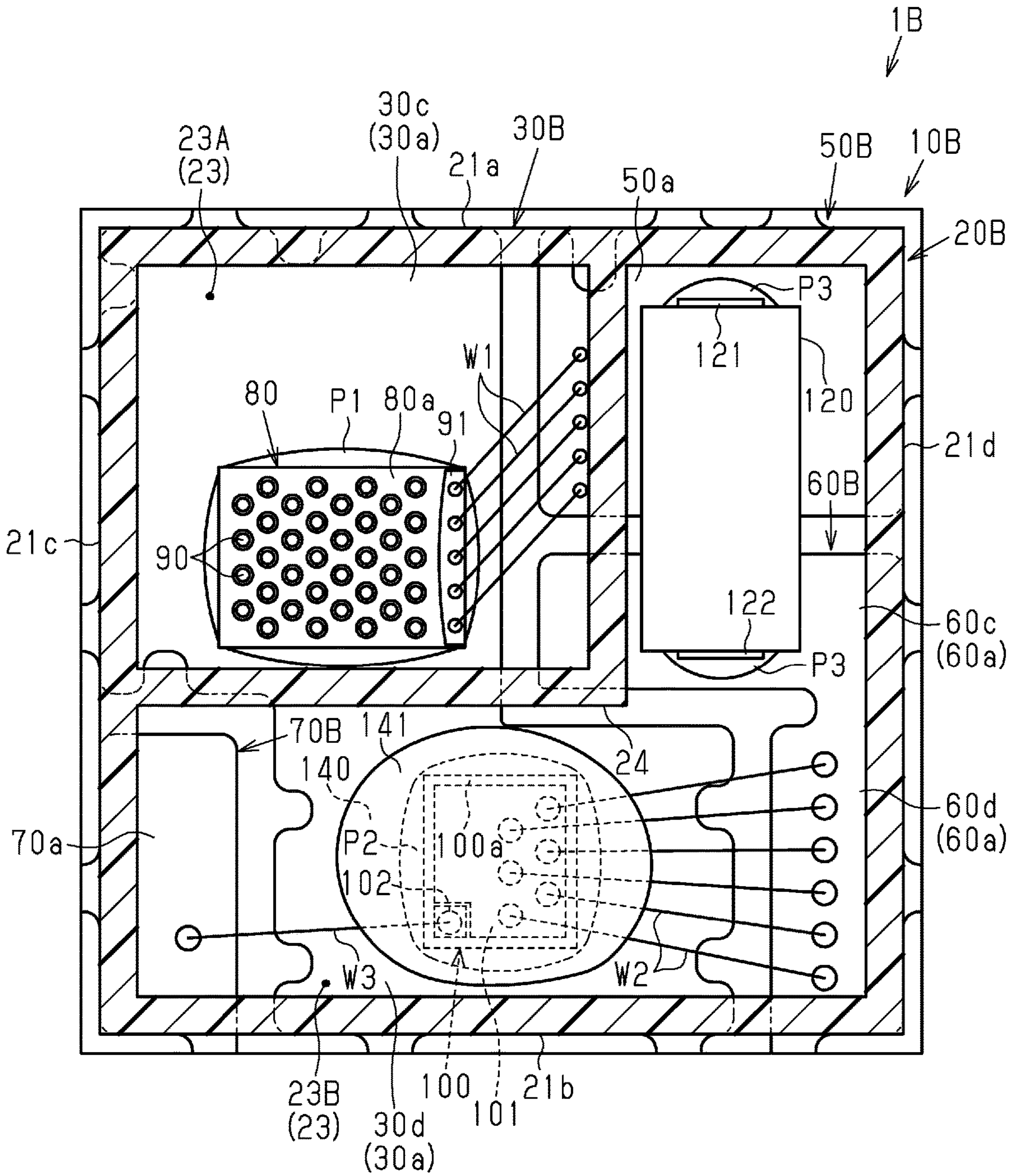
Fig.60

Fig.61



1B
10B
20B
30c
(30a)
23A
(23)
21a
50a
P3
121
120
13
W1
80
P1
80a
91
21d
60B
21c
90
50B
24
122
30B
24b
24c
60c
(60a)
P3
70B
141
24a
140
60d
(60a)
70a
P2
100a
102
W3
W2
23B
(23)
100
101
21b
30d
(30a)
Y
X

Fig.62


1B
10B
20B
50B
50a
30B
21a
30c
(30a)
23A
(23)
W1
80
P1
80a
91
90
21c
P3
121
120
21d
60B
122
P3
60c
(60a)
70B
141
140
24
P2
100a
102
60d
(60a)
70a
W3
W2
23B
(23)
30d
(30a)
100
101
21b

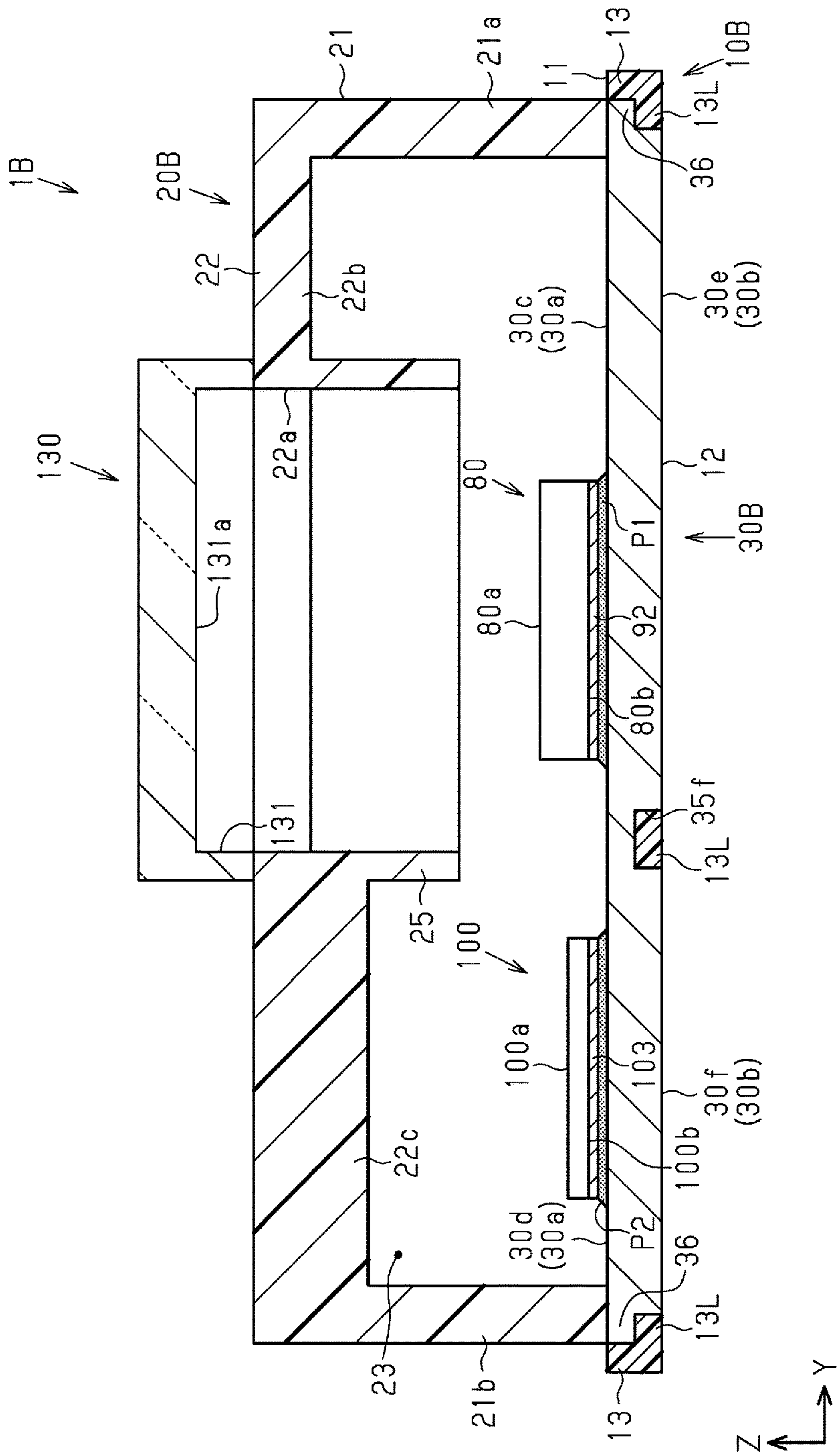

1B
20B
21
21a
13
11
10B
13L
36
22
22b
30c
(30a)
30e
(30b)
130
22a
80
12
131a
P1
30B
80a
92
80b
131
35f
13L
25
100
100a
103
30f
(30b)
22c
100b
30d
(30a)
P2
36
13L
23
21b
13
Z
Y

Fig.63

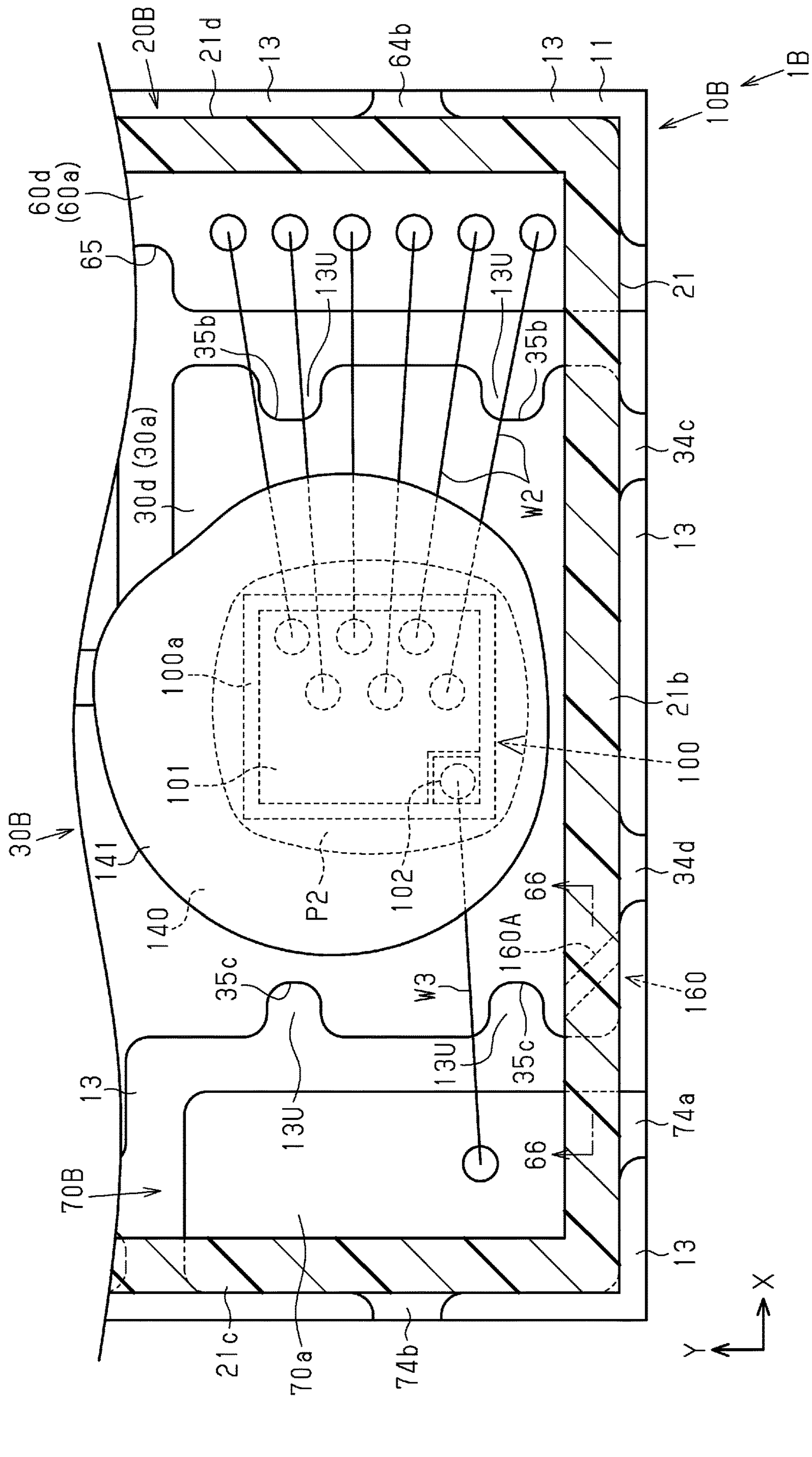

20B
21d
13
64b
13
11
1B
10B
60d
(60a)
65
13U
13U
21
35b
35b
34c
30d (30a)
W2
13
100a
21b
101
100
30B
141
102
P2
34d
66
140
160A
160
35c
W3
13U
35c
13
13U
74a
66
70B
13
21c
70a
74b
X
Y

Fig.64

Fig.65

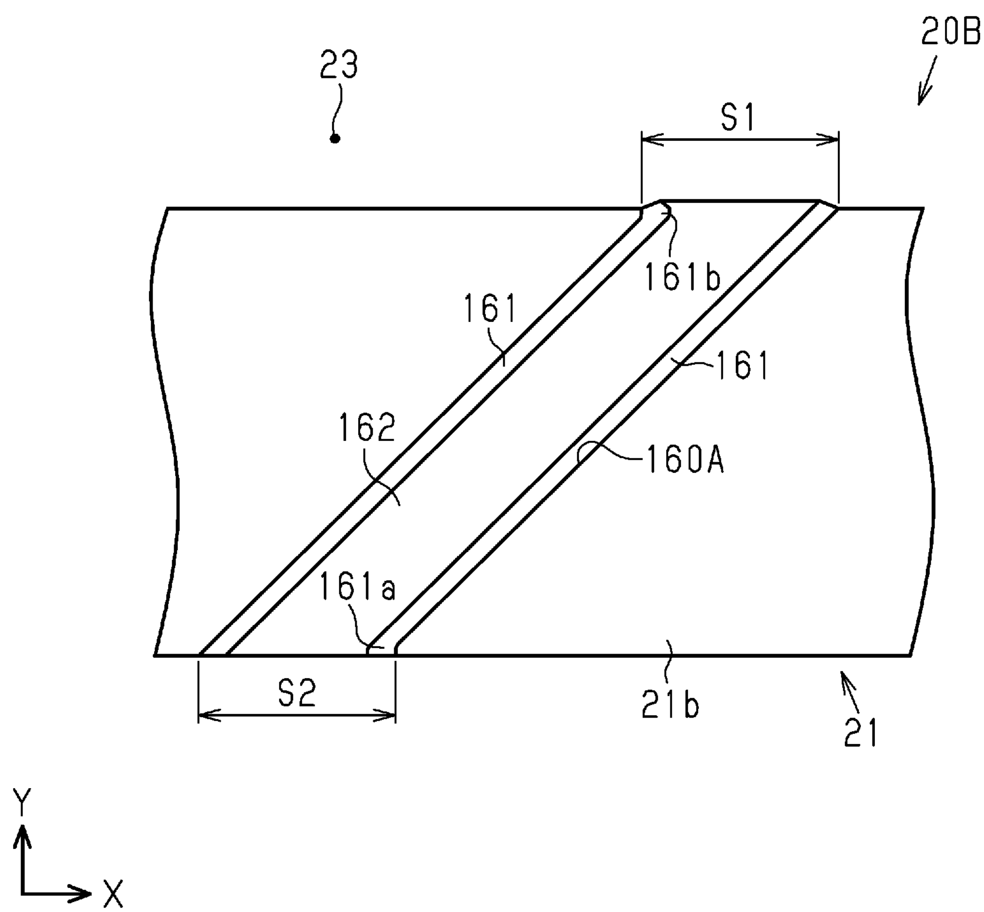


Fig.66

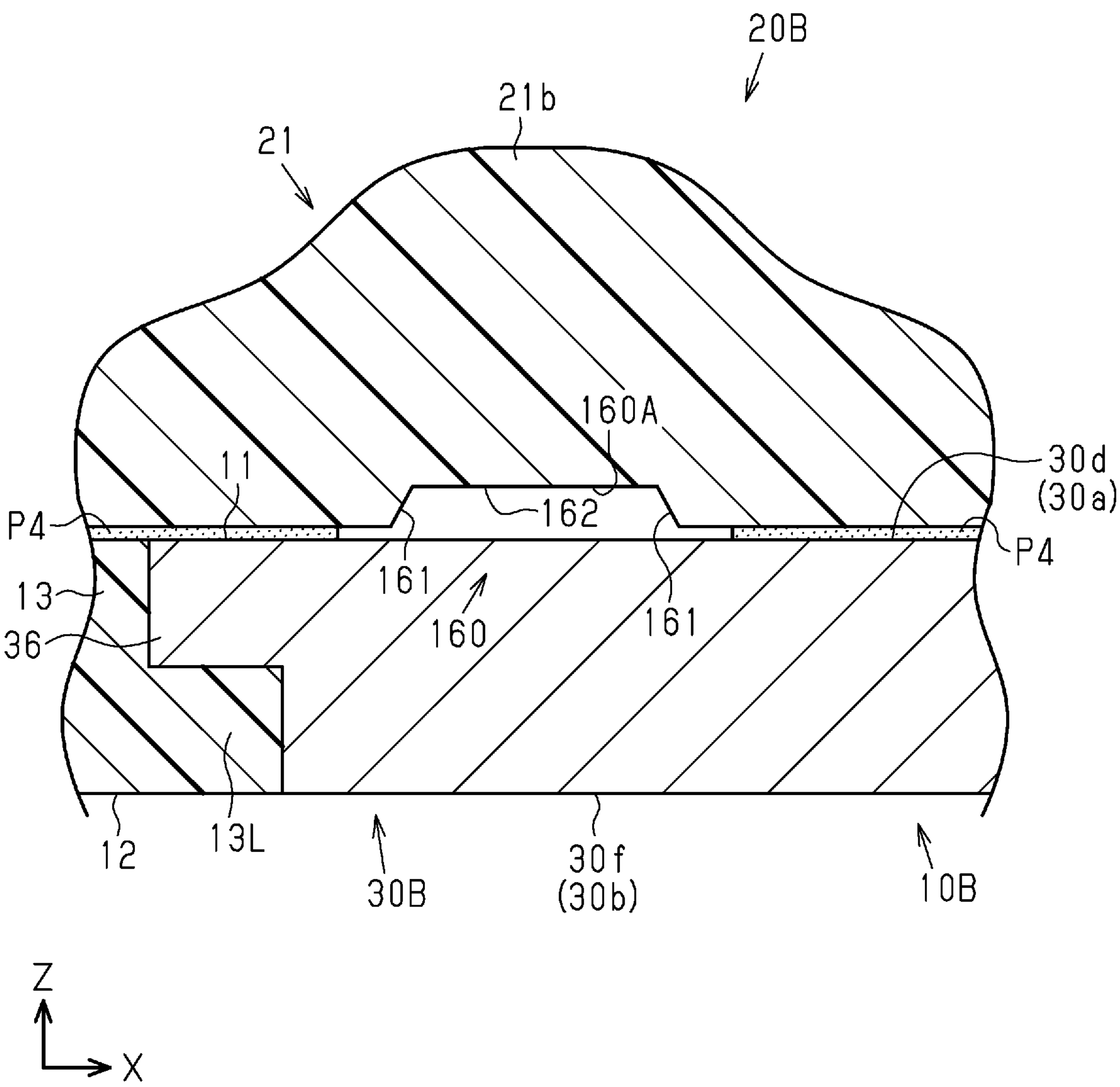

20B
21b
21
160A
30d
(30a)
11
P4
162
P4
13
161
36
160
161
12
13L
30B
30f
(30b)
10B
Z
X

Fig.67

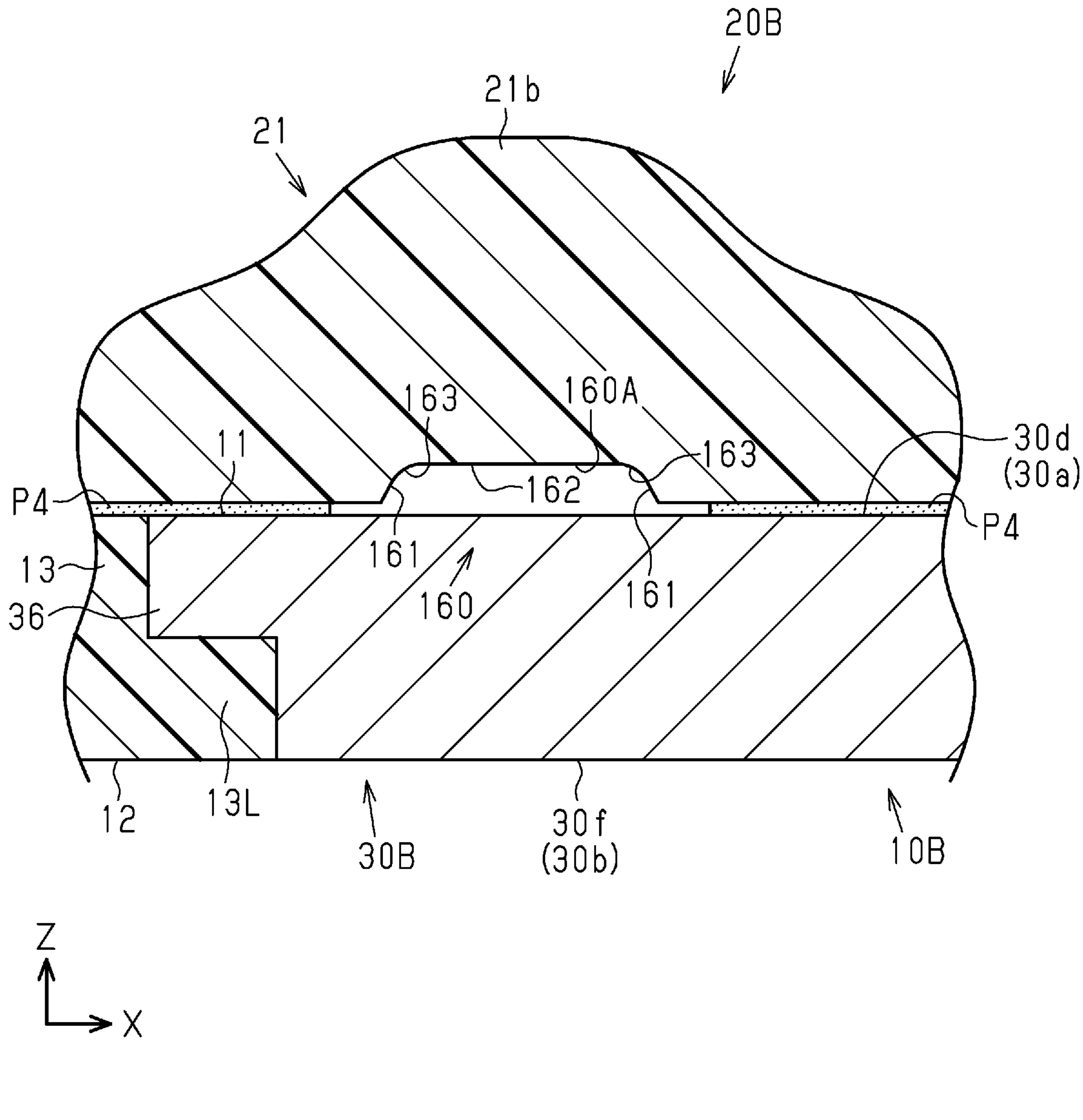

20B
21b
21
163
160A
11
163
30d
(30a)
P4
162
P4
161
13
161
36
160
12
13L
30B
30f
(30b)
10B
Z
X

Fig.68

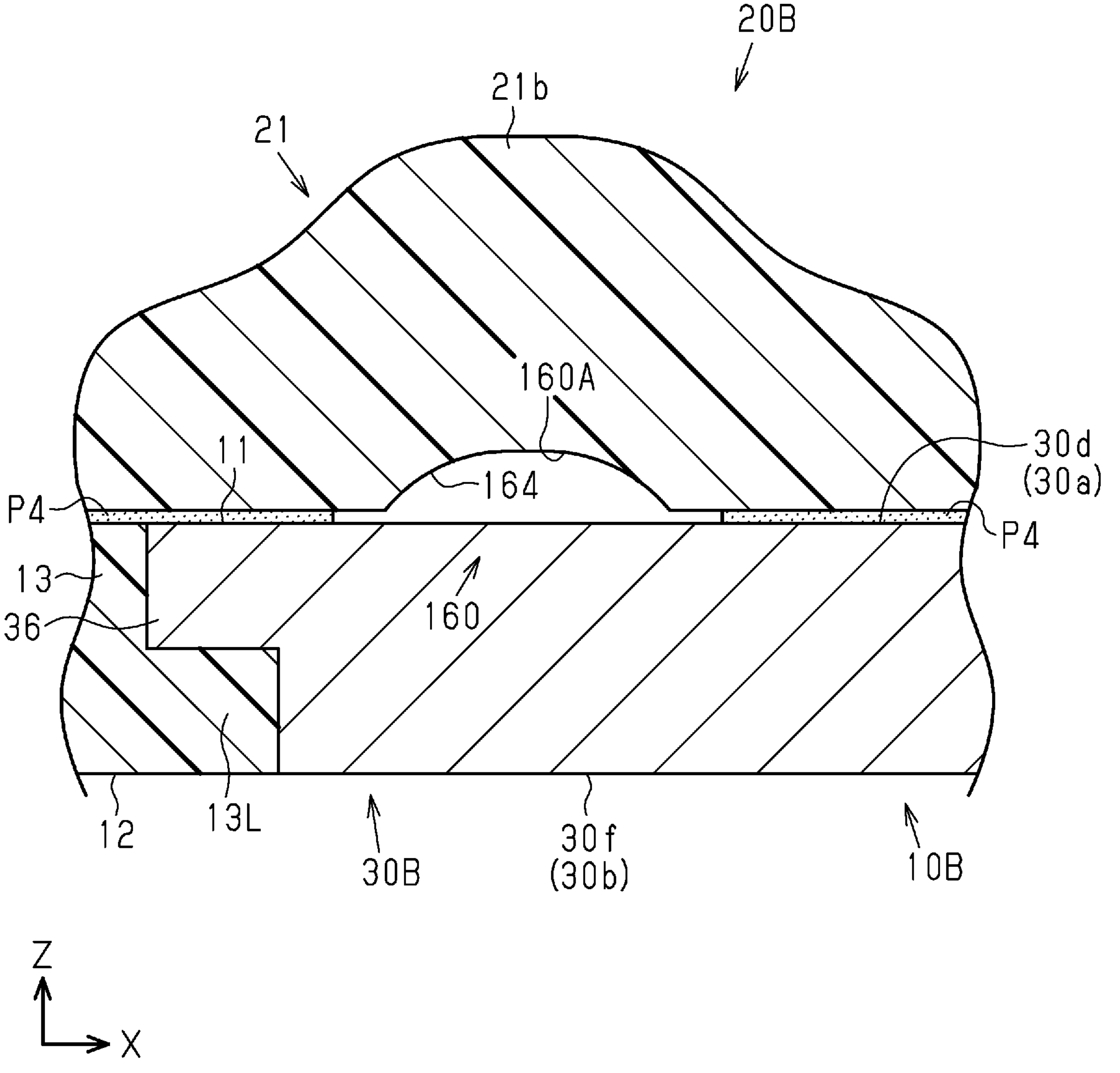

20B
21b
21
160A
30d
(30a)
11
164
P4
P4
13
160
36
12
13L
30B
30f
(30b)
10B
Z
X

Fig.69

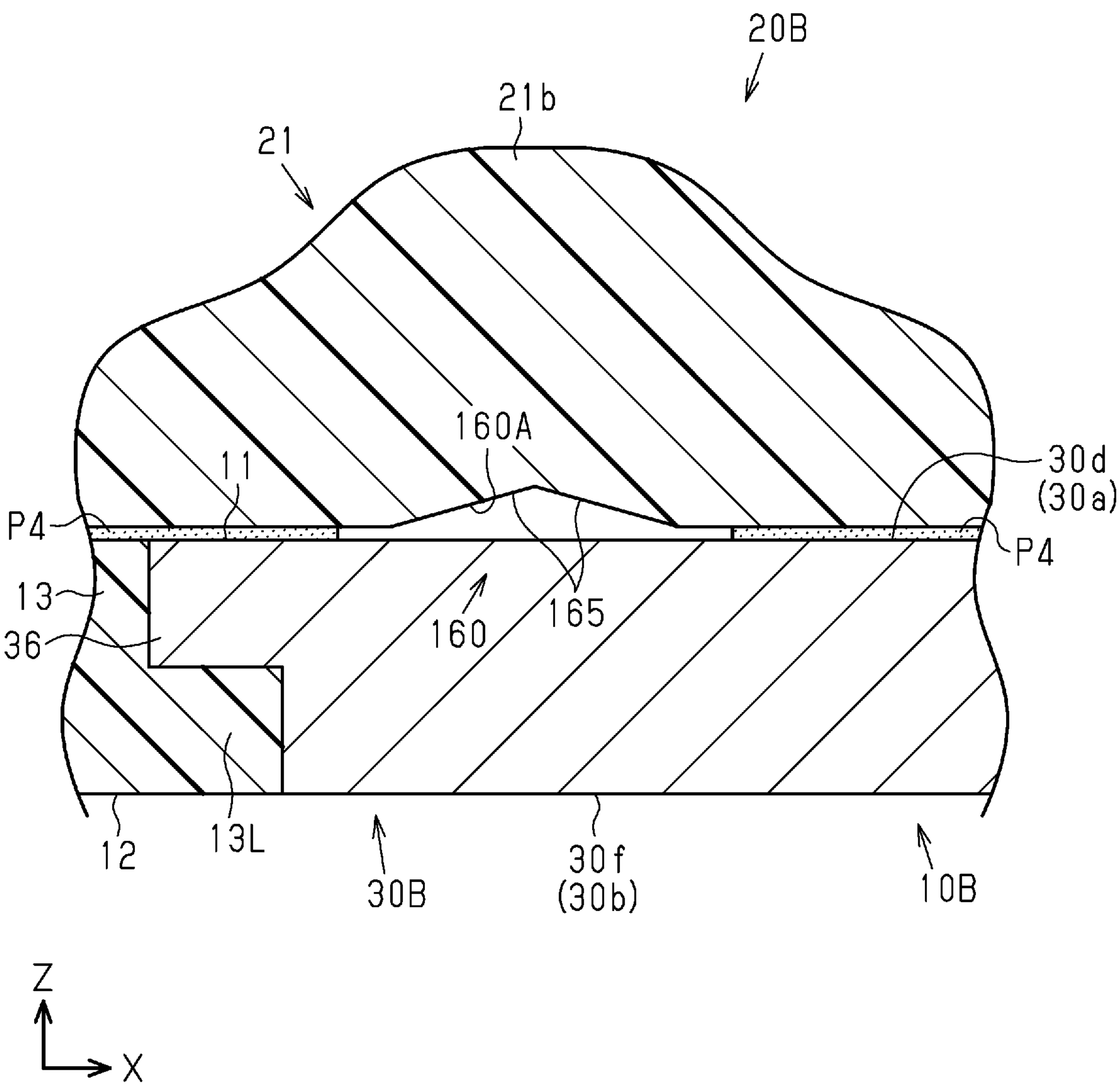

20B
21b
21
160A
11
30d
(30a)
P4
P4
13
36
160
165
12
13L
30B
30f
(30b)
10B
Z
X

Fig.70

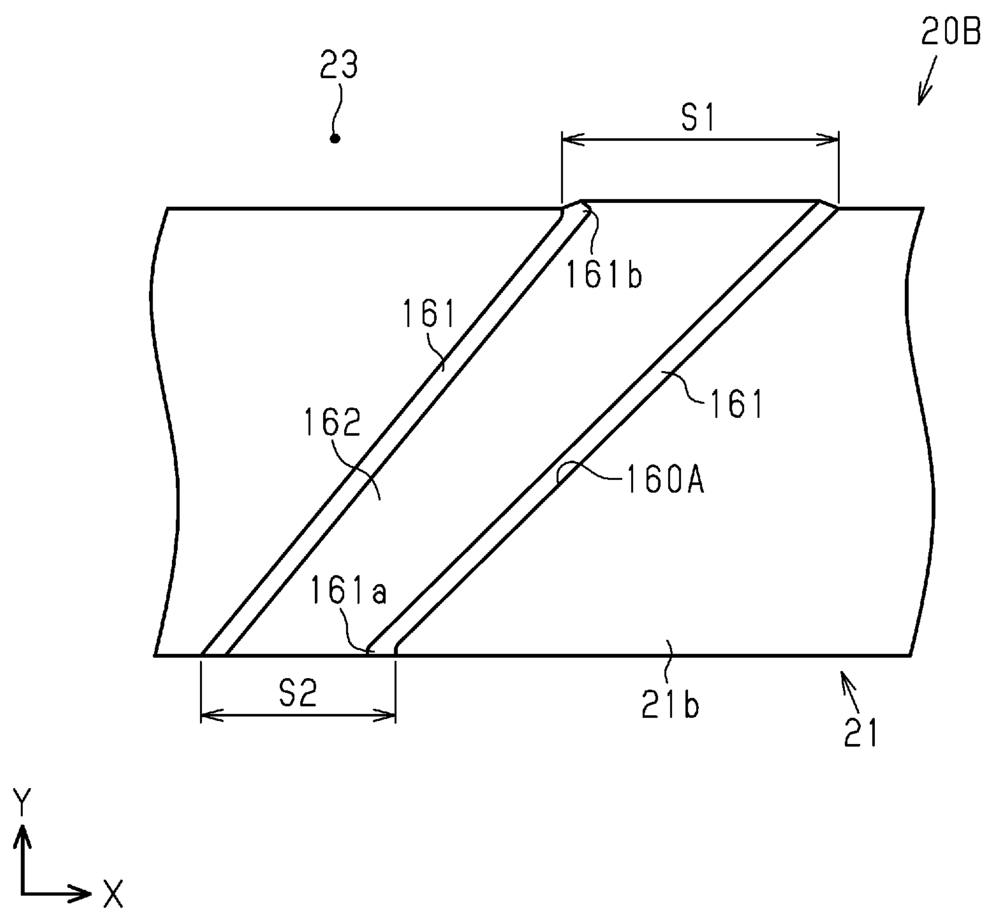


Fig.71

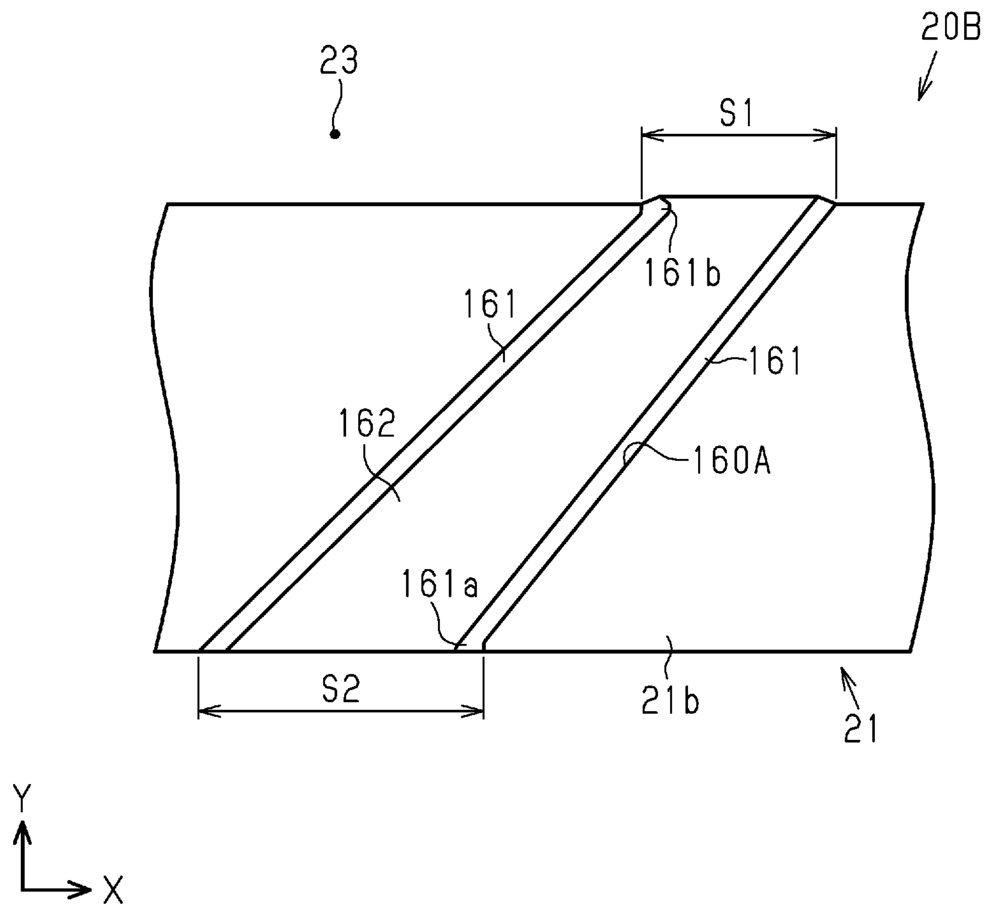


Fig.72

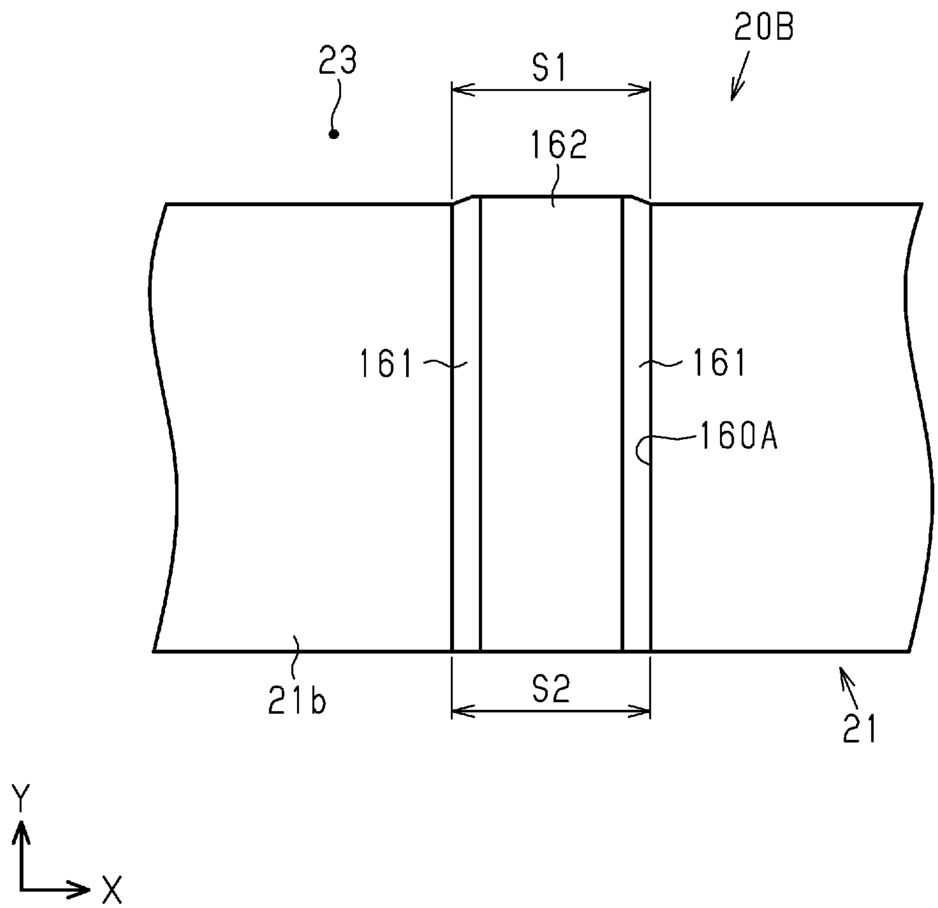


Fig.73

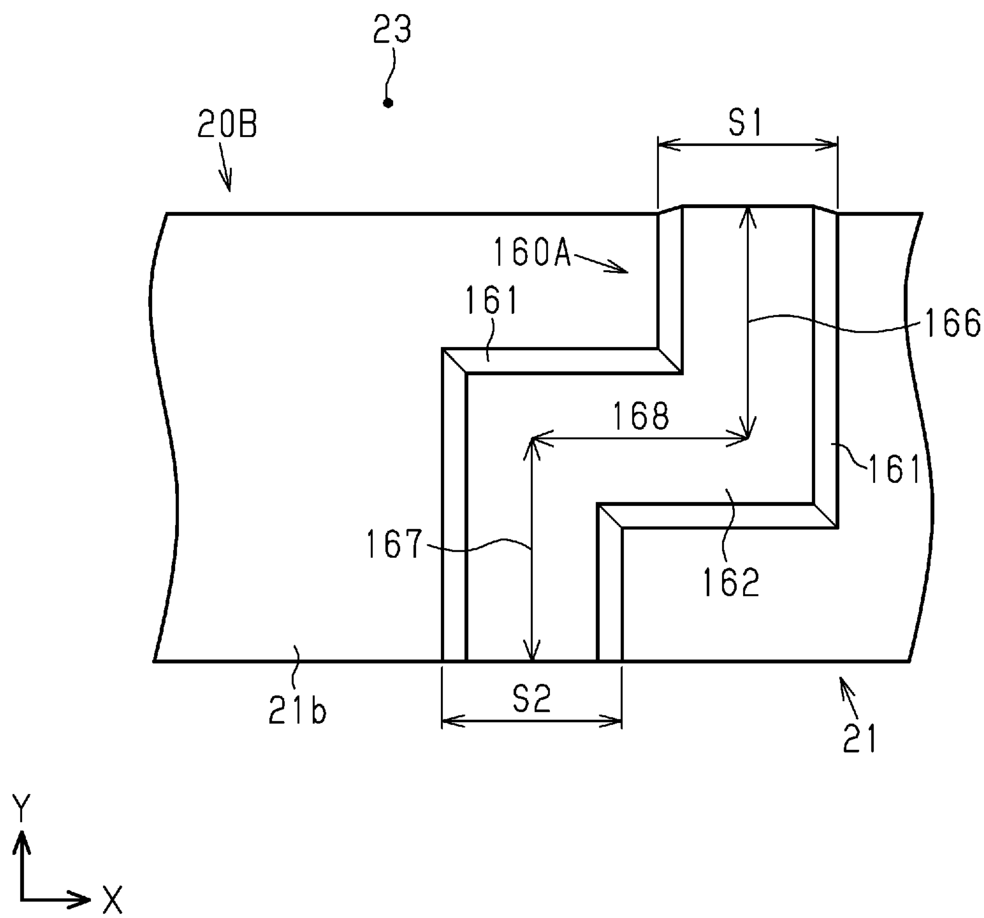
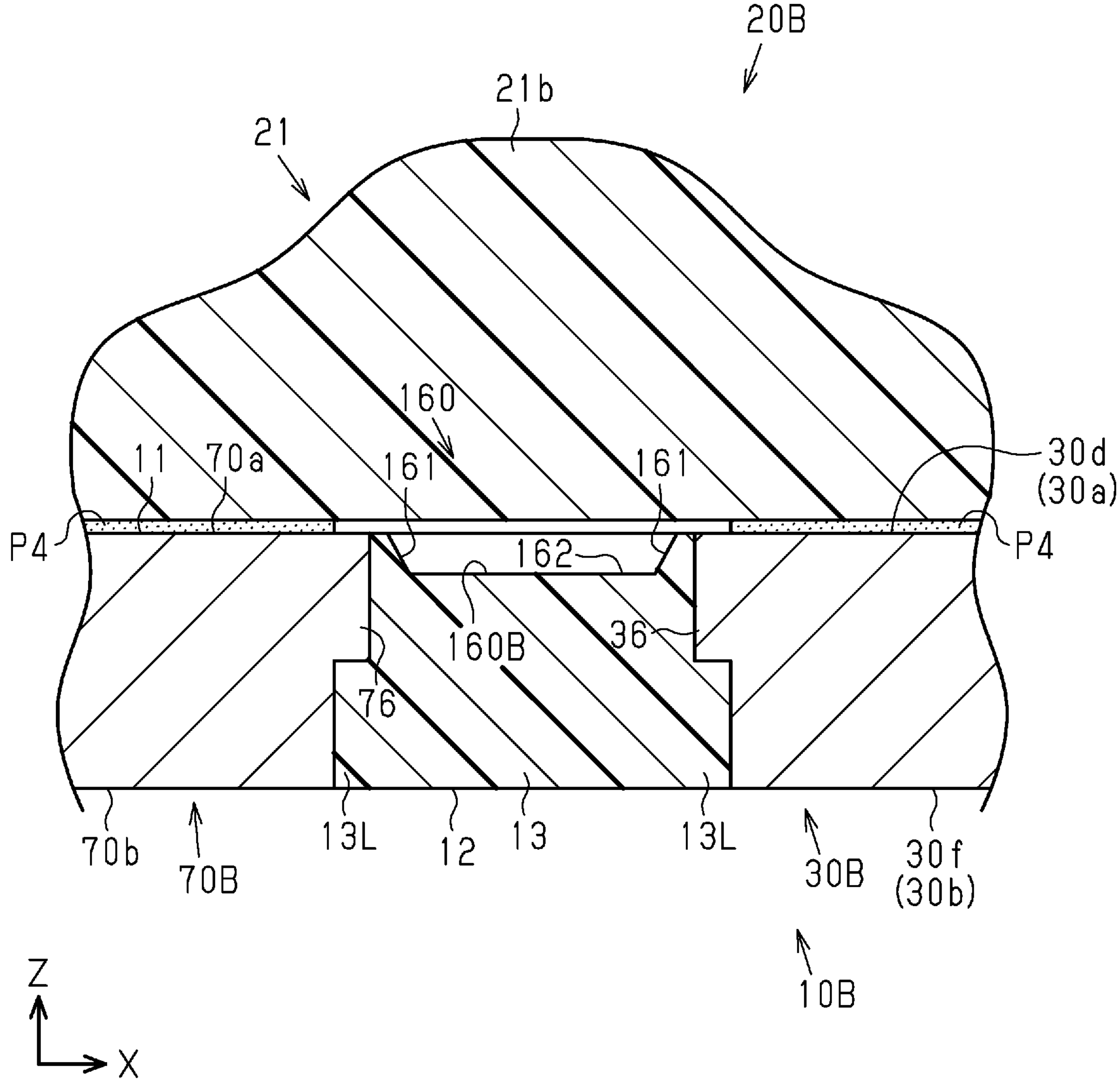


Fig.74



20B
21b
21
160
11
70a
161
161
30d
(30a)
P4
162
P4
160B
36
76
70b
70B
13L
12
13
13L
30B
30f
(30b)
10B
Z
X

Fig.75

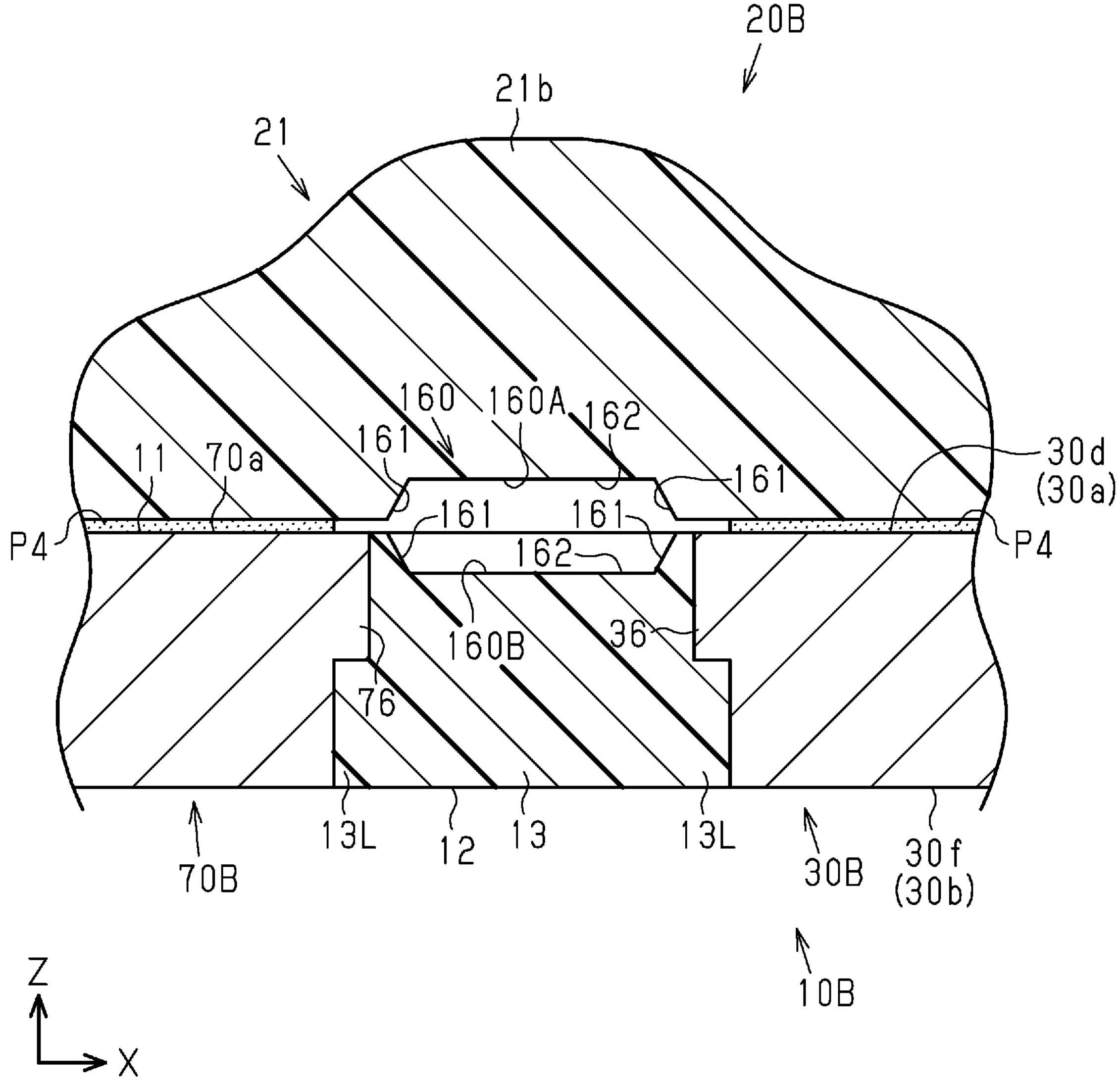

20B
21b
21
160
160A
162
161
30d
(30a)
11
70a
161
161
161
P4
P4
162
160B
36
76
13L
12
13
13L
70B
30B
30f
(30b)
10B
Z
X

Fig.76

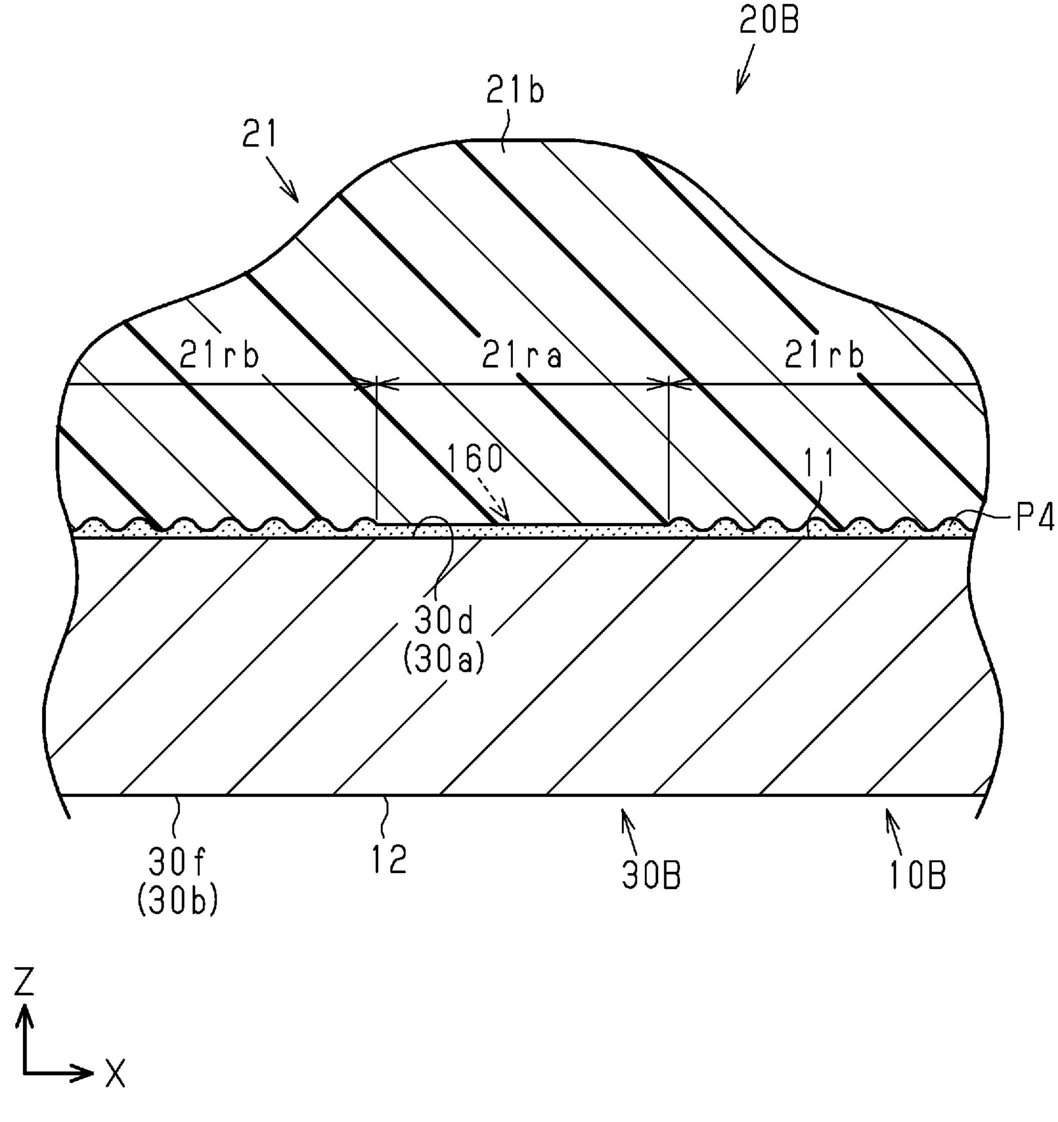


Fig.77

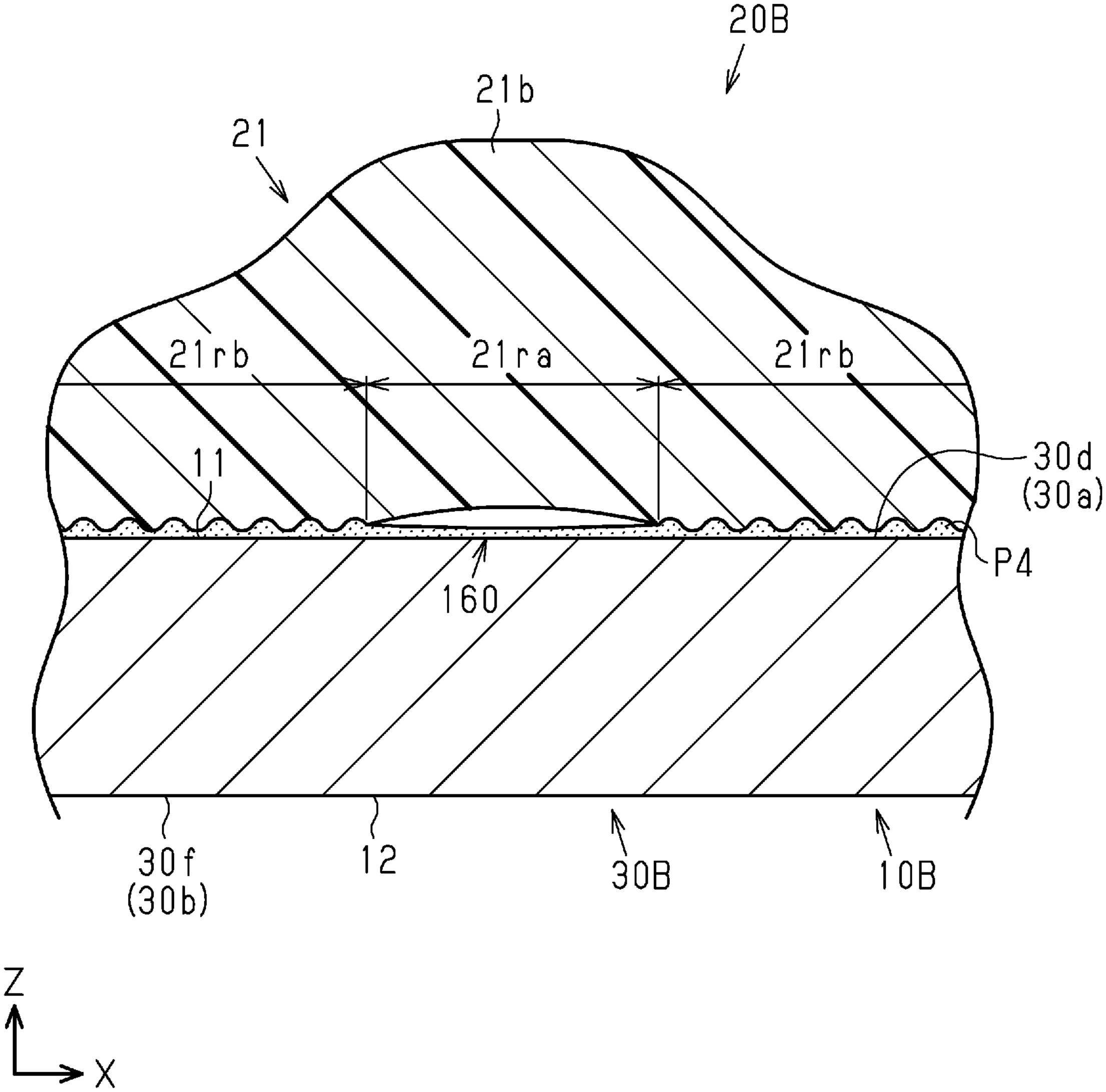
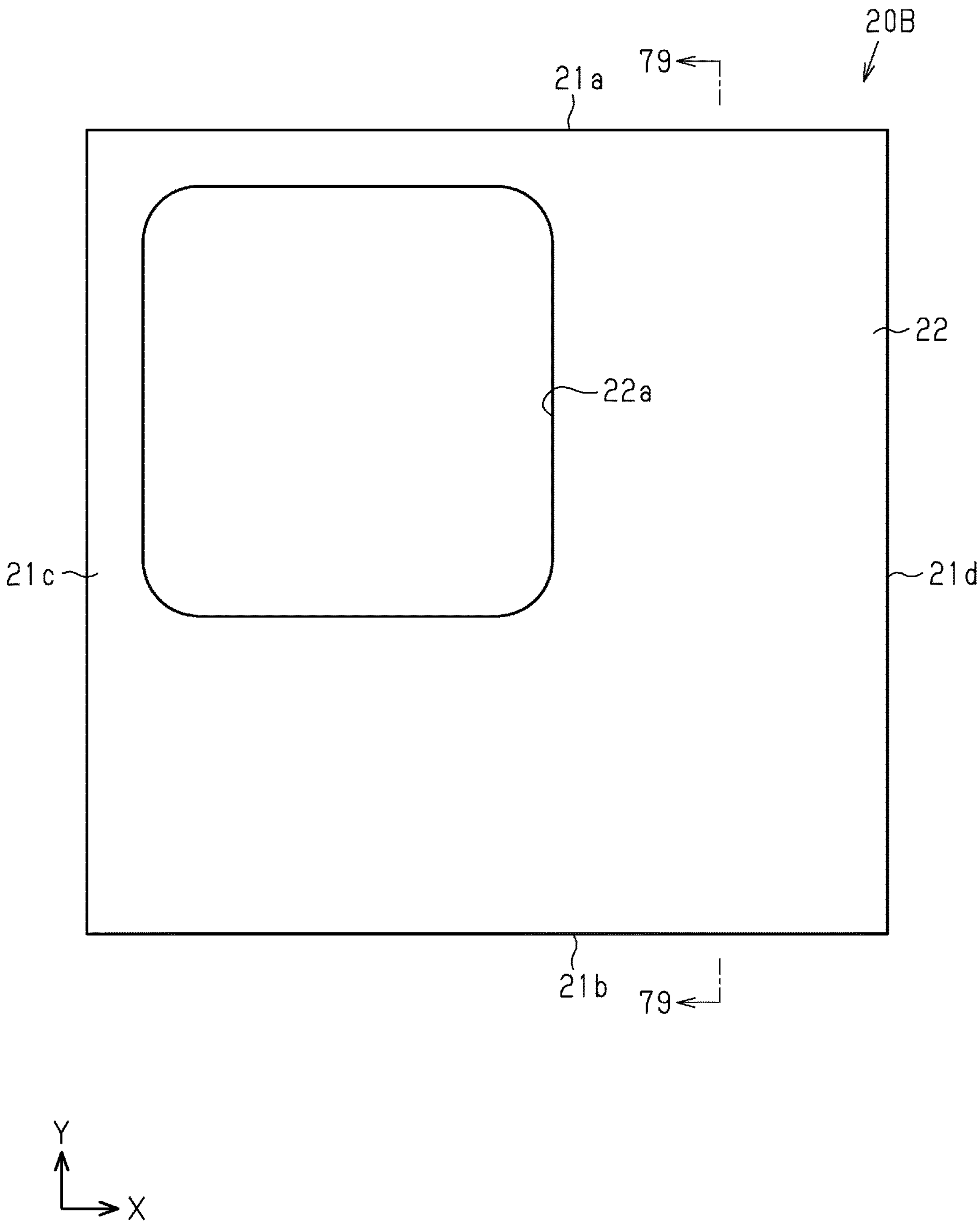


Fig.78



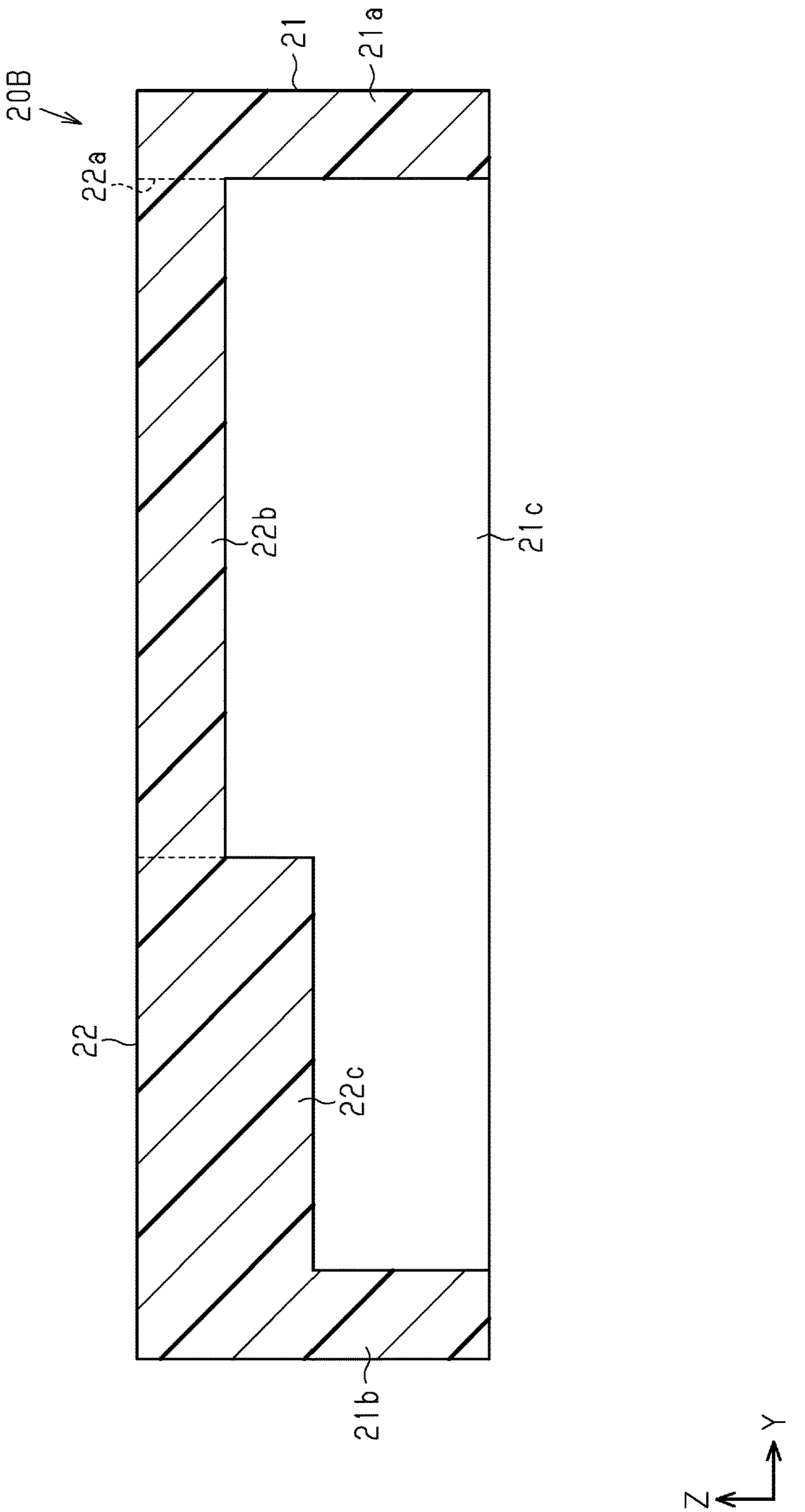
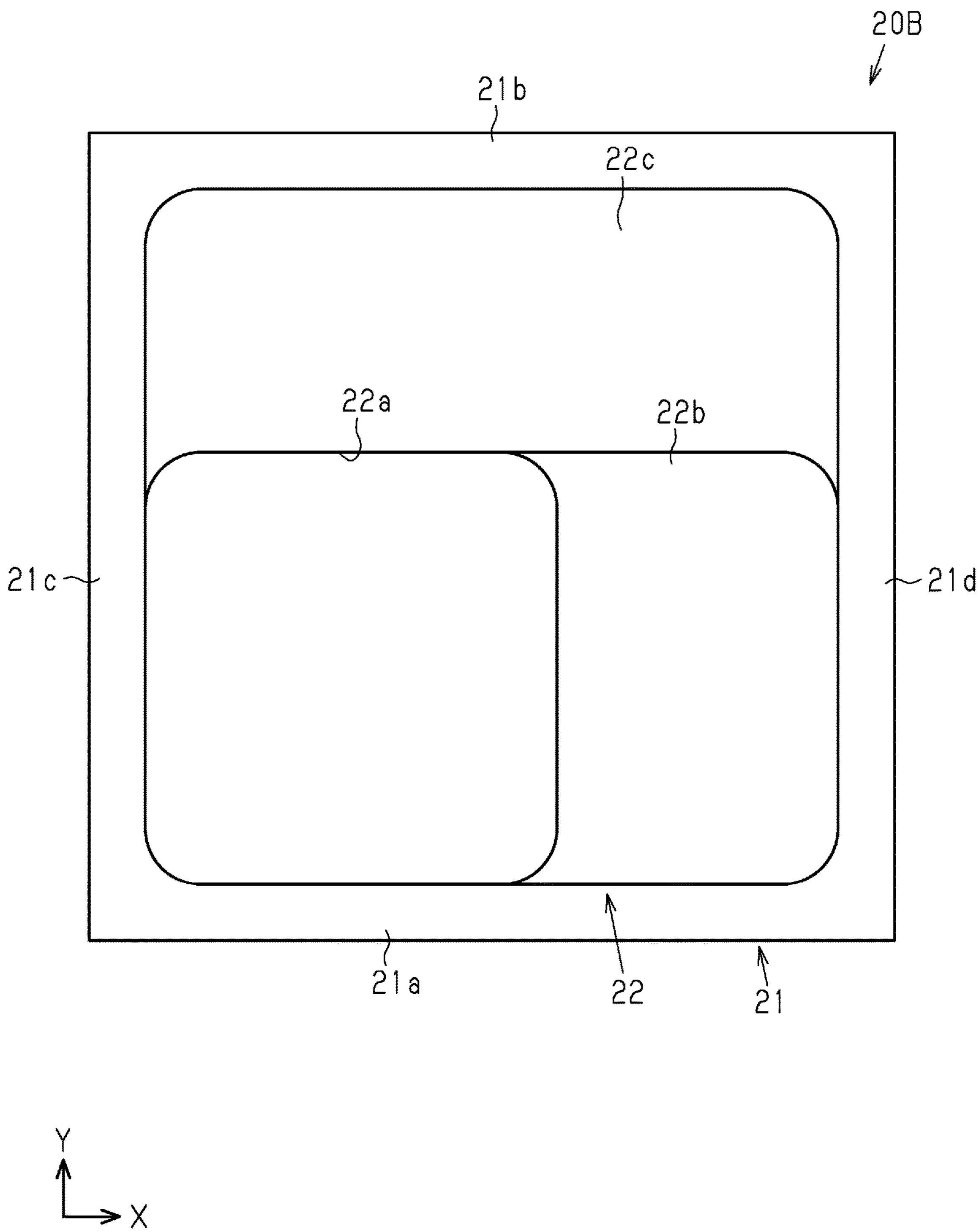
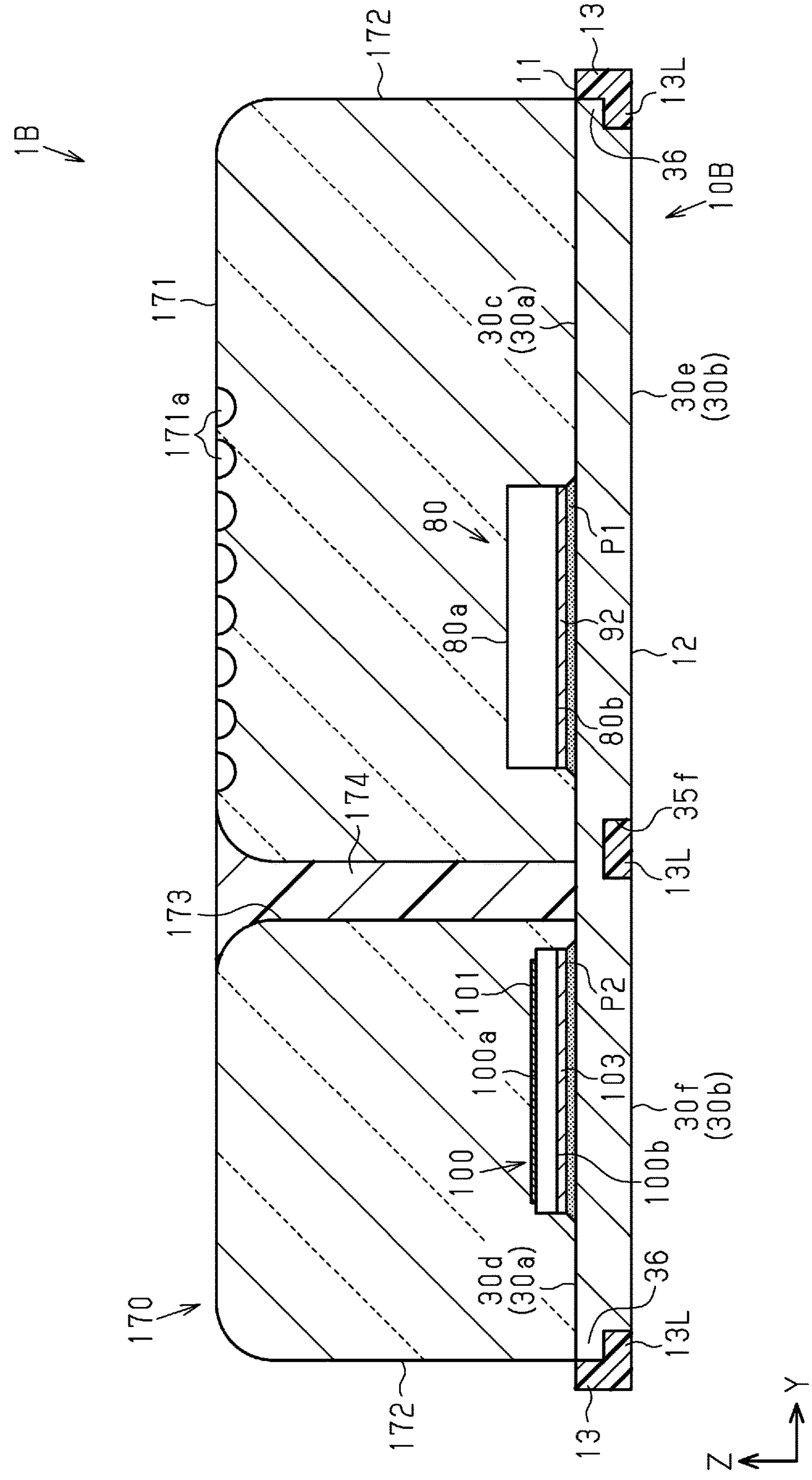



Fig.79

Fig.80

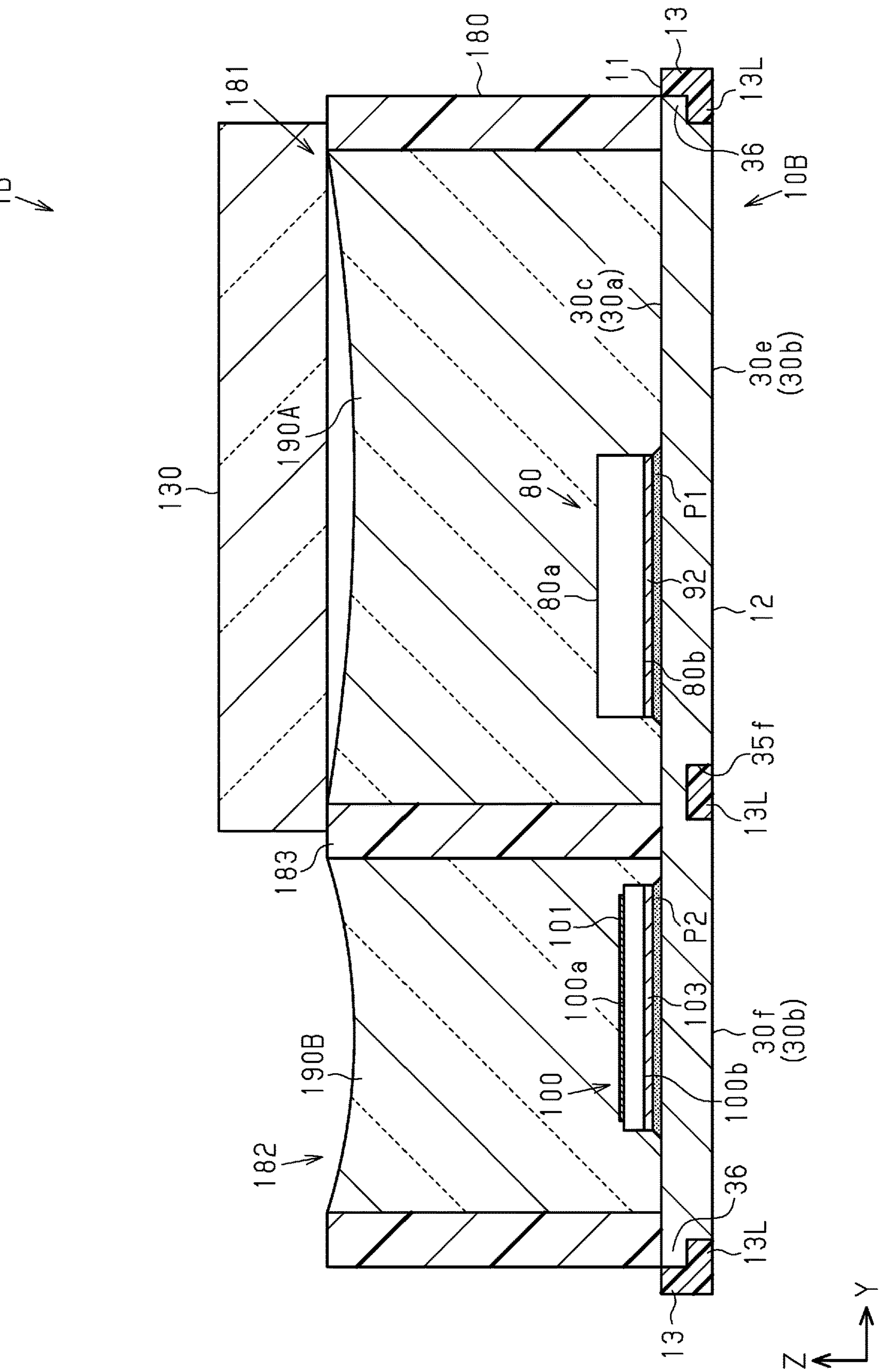




1B
170
171
171a
172
173
174
80
80a
80b
92
P1
100
100a
100b
101
103
P2
30c
(30a)
30d
(30a)
30e
(30b)
30f
(30b)
10B
11
12
13
13L
35f
36
Z
Y


Fig.81

Fig.82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181
13
11
13L
36
10B
130
190A
30c
(30a)
30e
(30b)
80
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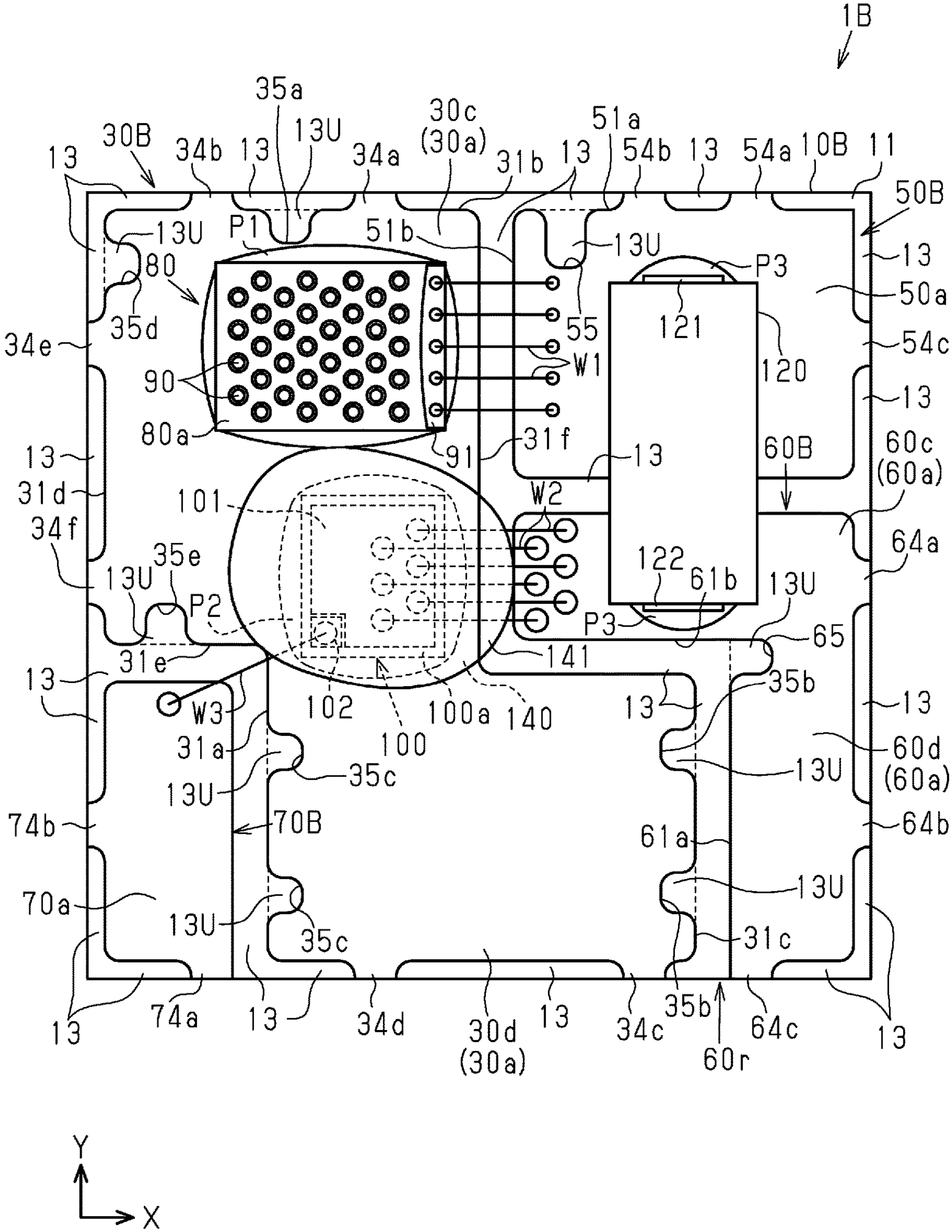
P1
92
12
80b
35f
13L
183
101
P2
100a
103
30f
(30b)
190B
100
100b
182
36
13L
13
Z
Y

Fig.83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13U
30c
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34b
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34a
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54a
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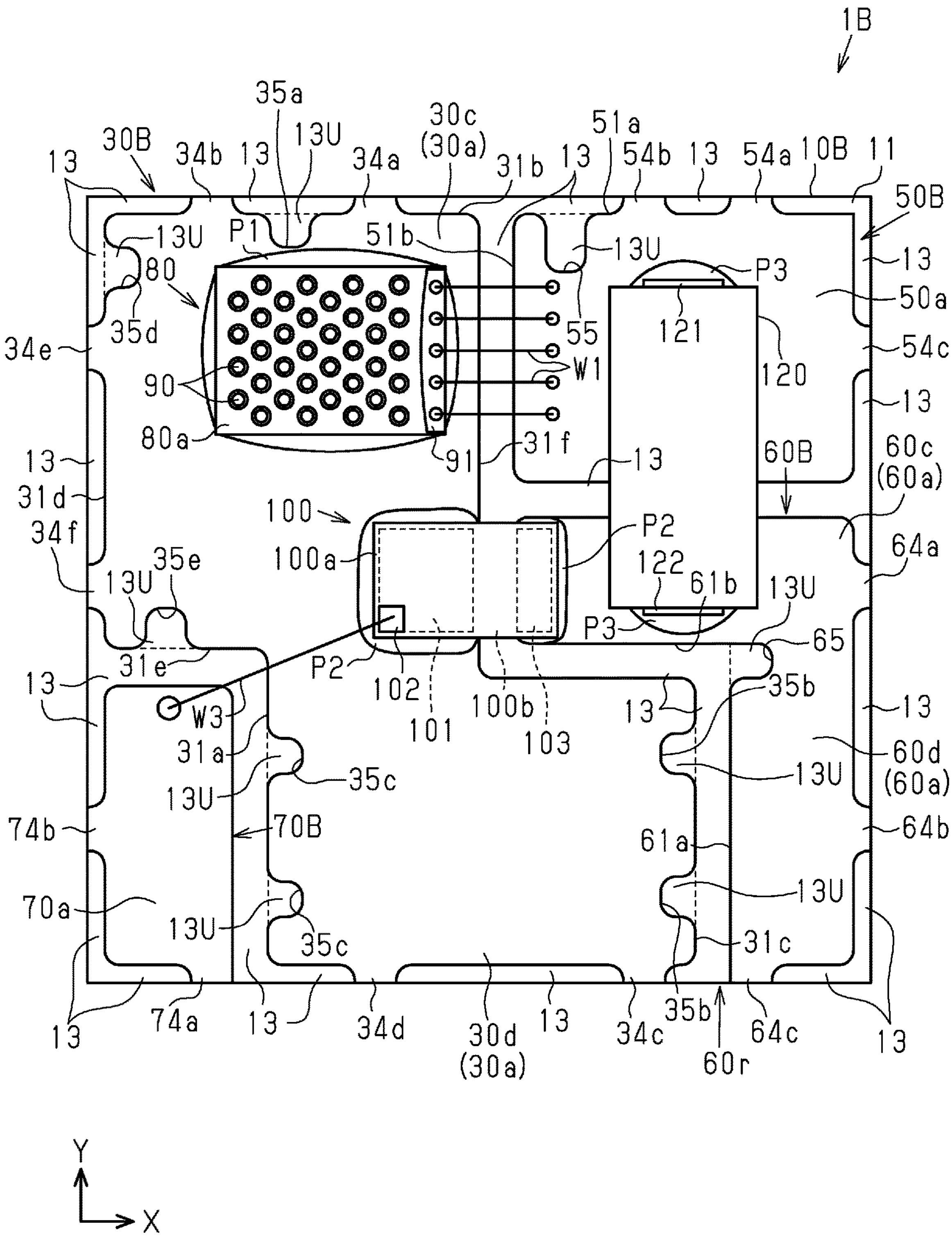
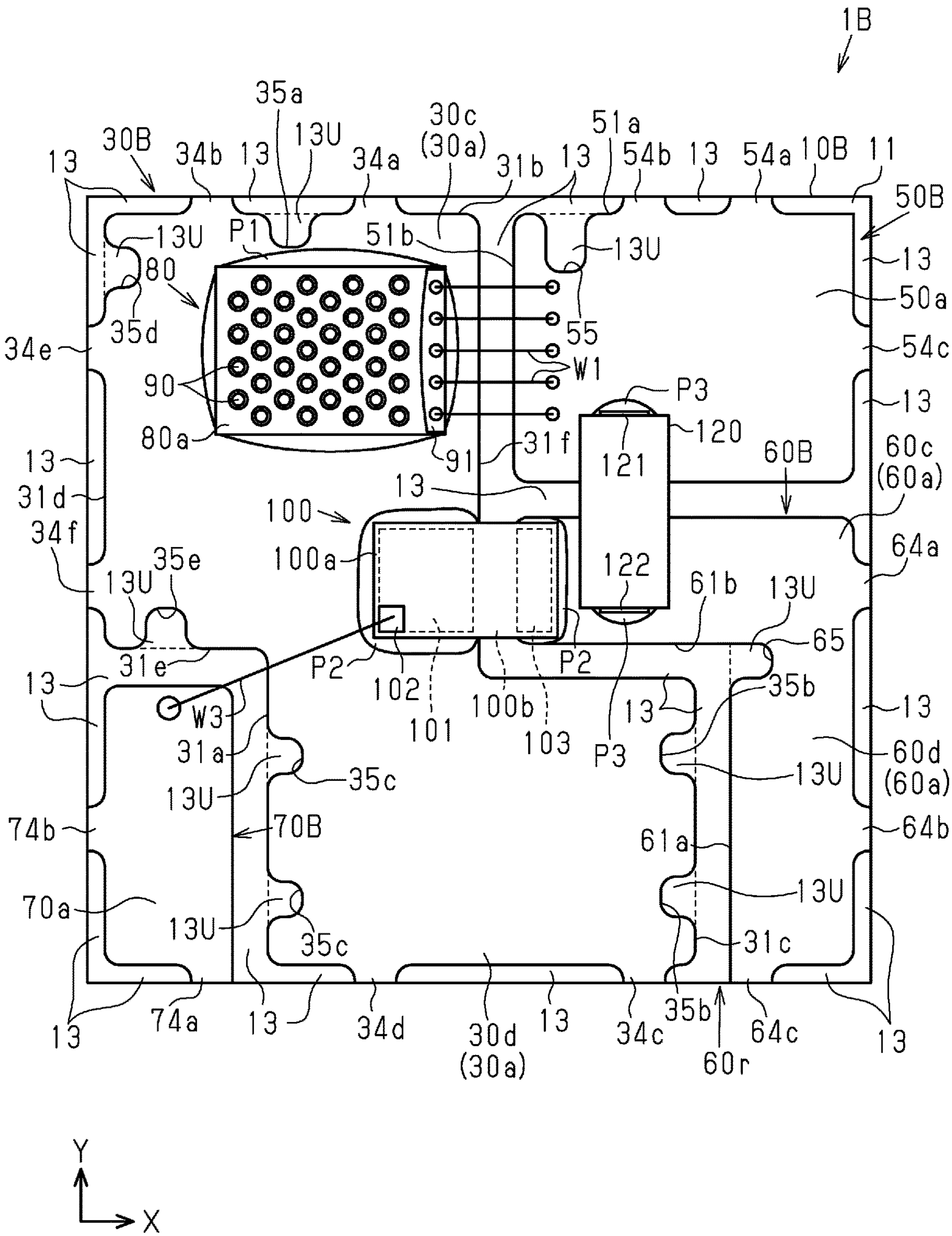
50B
51b
13U
P1
80
P3
35d
55
121
50a
34e
54c
W1
90
120
13
80a
91
31f
13
60B
60c
(60a)
31d
W2
101
64a
34f
35e
122
61b
13U
P2
P3
65
31e
141
35b
W3
102
100a
140
13
31a
100
13U
60d
(60a)
35c
13U
70B
74b
61a
64b
70a
13U
35c
31c
74a
34d
30d
(30a)
34c
35b
64c
60r
Y
X

Fig.84



1B
30B
35a
13U
34b
13
34a
30c
(30a)
31b
51a
54b
54a
10B
11
50B
13U
P1
51b
13U
P3
80
35d
55
121
50a
34e
90
W1
120
54c
80a
91
31f
13
60B
60c
(60a)
31d
100
P2
64a
100a
34f
35e
122
61b
13U
P3
65
31e
P2
102
101
100b
103
35b
W3
31a
13U
35c
60d
(60a)
74b
70B
61a
64b
70a
31c
74a
34d
30d
(30a)
34c
35b
60r
64c
Y
X

Fig.85


1B
13
30B
34b
13
35a
13U
34a
30c
(30a)
31b
13
51a
54b
13
54a
10B
11
50B
13U
P1
51b
13U
13
80
50a
35d
55
54c
34e
W1
90
P3
13
120
80a
121
31f
91
60B
60c
(60a)
13
13
31d
100
34f
100a
64a
35e
122
13U
61b
13U
65
31e
P2
P2
13
102
100b
35b
13
W3
101
103
13
13
31a
P3
13U
60d
(60a)
13U
35c
70B
74b
64b
61a
70a
13U
13U
35c
31c
13
74a
13
34d
30d
(30a)
13
34c
35b
64c
13
60r
Y
X

# SEMICONDUCTOR LIGHT EMITTING DEVICE

## TECHNICAL FIELD

The present disclosure relates to a semiconductor light emitting device.

## BACKGROUND ART

Patent Document 1 discloses a widely-known example of a semiconductor light emitting device including a semiconductor light emitting element as a light source. The semiconductor light emitting device disclosed in Patent Document **1** includes the semiconductor light emitting element and a substrate on which the semiconductor light emitting element is mounted.

## PRIOR ART DOCUMENT

### Patent Document

Patent Document 1: Japanese Laid-Open Patent Publication No. 2013-41866

## SUMMARY OF THE INVENTION

### Problems that the Invention is to Solve

For example, when the semiconductor light emitting device is used with an electronic component such as a capacitor or a switching element that drives the semiconductor light emitting element, the electronic component is arranged separately from the semiconductor light emitting device and electrically connected to the semiconductor light emitting device by a wire or the like. Such a structure has concern about parasitic capacitance caused by the wire or the like.

It is an objective of the present disclosure to provide a semiconductor light emitting device that electrically connects a semiconductor light emitting element to an electronic component while reducing parasitic capacitance.

### Means for Solving the Problems

To achieve the above objective, a semiconductor light emitting device includes a substrate, a common conductive portion formed on the substrate, a semiconductor light emitting element mounted on the common conductive portion, and an electronic component mounted on the common conductive portion and electrically connected to the semiconductor light emitting element by the common conductive portion.

This structure shortens the conductive path between the semiconductor light emitting element and the electronic component. As a result, parasitic capacitance caused by the conductive path between the semiconductor light emitting element and the electronic component is reduced. Thus, while reducing the parasitic capacitance, the semiconductor light emitting element and the electronic component are electrically connected.

### Effects of the Invention

With the semiconductor light emitting device described above, the semiconductor light emitting element is electrically connected to the electronic component while reducing parasitic capacitance.

## BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view showing an embodiment of a semiconductor light emitting device.

FIG. **2** is an exploded perspective view of the semiconductor light emitting device.

FIG. **3** is a front view of the semiconductor light emitting device.

FIG. **4** is a bottom view of the semiconductor light emitting device.

FIG. **5** is a cross-sectional view taken along line **5-5** in FIG. **3**.

FIG. **6** is a cross-sectional view taken along line **6-6** in FIG. **3**.

FIG. **7** is a perspective view showing a cross-sectional structure of a semiconductor light emitting element.

FIG. **8** is a partial cross-sectional view of the semiconductor light emitting element.

FIG. **9** is a front view showing part of an electronic apparatus.

FIG. **10** is a circuit diagram showing part of the electronic apparatus.

FIG. **11** is a perspective view showing a second embodiment of a semiconductor light emitting device.

FIG. **12** is a front view of the semiconductor light emitting device.

FIG. **13** is a side view of the semiconductor light emitting device.

FIG. **14** is a front view showing a frame of the semiconductor light emitting device that is cut along a plane orthogonal to a thickness-wise direction of the substrate.

FIG. **15** is a front view of the semiconductor light emitting device with a case removed.

FIG. **16** is a perspective view showing a bottom side of the semiconductor light emitting device.

FIG. **17** is a cross-sectional view taken along line **17-17** in FIG. **12**.

FIG. **18** is a cross-sectional view taken along line **18-18** in FIG. **12**.

FIG. **19** is a cross-sectional view taken along line **19-19** in FIG. **12**.

FIG. **20** is a cross-sectional view taken along line **20-20** in FIG. **15**.

FIG. **21** is a cross-sectional view taken along line **21-21** in FIG. **15**.

FIG. **22** is a cross-sectional view taken along line **22-22** in FIG. **15**.

FIG. **23** is a front view of the case.

FIG. **24** is a cross-sectional view taken along line **24-24** in FIG. **23**.

FIG. **25** is a cross-sectional view taken along line **25-25** in FIG. **23**.

FIG. **26** is a bottom view of the case.

FIG. **27** is a diagram showing an example of a step in a manufacturing method of the semiconductor light emitting device of the second embodiment.

FIG. **28** is a partial enlarged view of FIG. **27**.

FIG. **29** is a cross-sectional view taken along line **29-29** in FIG. **28**.

FIG. **30** is a diagram showing an example of a step in the manufacturing method of the semiconductor light emitting device.

FIG. **31** is a cross-sectional view taken along line **31-31** in FIG. **30**.

FIG. **32** is a diagram showing an example of a step in the manufacturing method of the semiconductor light emitting device.

FIG. **33** is a diagram showing an example of a step in the manufacturing method of the semiconductor light emitting device.

FIG. **34** is a diagram showing an example of a step in the manufacturing method of the semiconductor light emitting device.

FIG. **35** is a diagram showing an example of a step in the manufacturing method of the semiconductor light emitting device.

FIG. **36** is a diagram showing an example of a step in the manufacturing method of the semiconductor light emitting device.

FIG. **37** is a front view showing part of an electronic apparatus in the second embodiment.

FIG. **38** is a circuit diagram showing part of the electronic apparatus in the second embodiment.

FIG. **39** is a front view showing a semiconductor light emitting device in a modified example.

FIG. **40** is a bottom view showing the semiconductor light emitting device in the modified example.

FIG. **41** is a cross-sectional view showing the semiconductor light emitting device in the modified example.

FIG. **42** is a cross-sectional view showing the semiconductor light emitting device in the modified example.

FIG. **43** is a front view showing a semiconductor light emitting device in a modified example.

FIG. **44** is a bottom view showing the semiconductor light emitting device in a modified example.

FIG. **45** is a front view showing a semiconductor light emitting device in a modified example.

FIG. **46** is a bottom view showing the semiconductor light emitting device in a modified example.

FIG. **47** is a front view showing a semiconductor light emitting device in a modified example.

FIG. **48** is a front view showing a semiconductor light emitting device in a modified example.

FIG. **49** is a front view showing a semiconductor light emitting device in a modified example.

FIG. **50** is a cross-sectional view showing the semiconductor light emitting device in the modified example.

FIG. **51** is a cross-sectional view showing the semiconductor light emitting device in a modified example.

FIG. **52** is a perspective view showing the semiconductor light emitting device in a modified example.

FIG. **53** is a front view showing a semiconductor light emitting device in the modified example.

FIG. **54** is a side view showing a semiconductor light emitting device in the modified example.

FIG. **55** is a cross-sectional view taken along line **55-55** in FIG. **53**.

FIG. **56** is a front view showing a case of the semiconductor light emitting device shown in FIG. **52**.

FIG. **57** is a bottom view of the case shown in FIG. **56**.

FIG. **58** is a cross-sectional view showing the semiconductor light emitting device in a modified example.

FIG. **59** is a front view showing a frame of the semiconductor light emitting device of the modified example that is cut along a plane orthogonal to the thickness-wise direction of the substrate.

FIG. **60** is a cross-sectional view showing the semiconductor light emitting device in a modified example.

FIG. **61** is a front view showing a frame of the semiconductor light emitting device in the modified example that is cut along a plane orthogonal to the thickness-wise direction of the substrate.

FIG. **62** is a front view showing a frame of the semiconductor light emitting device in a modified example that is cut along a plane orthogonal to the thickness-wise direction of the substrate.

FIG. **63** is a cross-sectional view showing the semiconductor light emitting device in a modified example.

FIG. **64** is a partial enlarged front view showing a frame of the semiconductor light emitting device in a modified example that is cut along a plane orthogonal to the thickness-wise direction of the substrate.

FIG. **65** is a partial bottom view of the frame of the case.

FIG. **66** is a cross-sectional view taken along line **66-66** in FIG. **64**.

FIG. **67** is a cross-sectional view of a case and a substrate in a modified example of FIG. **66**.

FIG. **68** is a cross-sectional view of a case and a substrate in a modified example of FIG. **66**.

FIG. **69** is a cross-sectional view of a case and a substrate in a modified example of FIG. **66**.

FIG. **70** is a partial bottom view of the frame of a modified example of FIG. **65**.

FIG. **71** is a partial bottom view of the frame of a modified example of FIG. **65**.

FIG. **72** is a partial bottom view of the frame of a modified example of FIG. **65**.

FIG. **73** is a partial bottom view of the frame of a modified example of FIG. **65**.

FIG. **74** is a cross-sectional view of a case and a substrate in a modified example of FIG. **66**.

FIG. **75** is a cross-sectional view of a case and a substrate in a modified example of FIG. **66**.

FIG. **76** is a cross-sectional view of a case and a substrate in a modified example of FIG. **66**.

FIG. **77** is a cross-sectional view of the case and the substrate shown in FIG. **76** including a vent.

FIG. **78** is a front view of a case in a modified example.

FIG. **79** is a cross-sectional view taken along line **79-79** in FIG. **78**.

FIG. **80** is a bottom view of the case shown in FIG. **78**.

FIG. **81** is a cross-sectional view showing the semiconductor light emitting device in a modified example.

FIG. **82** is a cross-sectional view showing the semiconductor light emitting device in a modified example.

FIG. **83** is a front view showing a modified example of a semiconductor light emitting device with a case removed.

FIG. **84** is a front view showing a modified example of a semiconductor light emitting device with a case removed.

FIG. **85** is a front view showing a modified example of a semiconductor light emitting device with a case removed.

## MODES FOR CARRYING OUT THE INVENTION

### First Embodiment

An embodiment of a semiconductor light emitting device will be described below with reference to the drawings. The embodiments described below exemplify configurations and methods for embodying a technical concept and are not intended to limit the material, shape, structure, layout, dimensions, and the like of each component to those described below. The embodiments described below may undergo various modifications.

FIG. **1** is a perspective view showing a first embodiment of a semiconductor light emitting device **1**. FIG. **2** is an exploded perspective view showing the semiconductor light emitting device **1** of the first embodiment. FIG. **3** is a front view showing the semiconductor light emitting device **1** of the first embodiment. In FIG. **3**, a lower side of a cover **22** is indicated by solid lines.

As shown in FIG. **1**, the semiconductor light emitting device **1** is, for example, cuboid-shaped. In a plan view of the semiconductor light emitting device **1**, a direction extending along one side of the semiconductor light emitting device **1** is referred to as the X-direction, and a direction orthogonal to the X-direction is referred to as the Y-direction. In the present embodiment, the dimension of the semiconductor light emitting device **1** in the X-direction is approximately 3.5 mm. The dimension of the semiconductor light emitting device **1** in the Y-direction is approximately 3.5 mm. The dimensions of the semiconductor light emitting device **1** in the X-direction and the Y-direction may be changed in any manner.

As shown in FIGS. **1** to **3**, the semiconductor light emitting device **1** includes a substrate **10**, a case **20**, conductive portions **30**, **40**, **50**, **60**, and **70**, a semiconductor light emitting element **80**, and an electronic component **100**.

The substrate **10** has the shape of, for example, a square extending in the X-direction and the Y-direction. The X-direction and the Y-direction are two directions that are orthogonal to each other in a planar direction of the substrate **10**. The substrate **10** includes a substrate front surface **11** and a substrate back surface **12**.

In the present embodiment, the substrate **10** is formed from an insulative material. The substrate **10** may be, for example, a ceramic such as alumina or aluminum nitride, a silicon substrate, or a glass epoxy. For the sake of convenience, in the thickness-wise direction of the substrate **10**, a direction extending away from the substrate front surface **11** is referred to as "upward", and a direction extending toward the substrate front surface **11** is referred to as "downward."

The case **20** accommodates the semiconductor light emitting element **80** and the electronic component **100**. The case **20** is attached to the substrate **10**. The case **20** is, for example, hollow. However, alternatively, the case **20** may be filled with another member.

The case **20** includes a frame **21** that is open upward and the cover **22** that closes the opening of the frame **21**. The frame **21** is, for example, formed from a light-blocking material such as a colored resin. Light from the semiconductor light emitting element **80** is blocked by the frame **21**. The frame **21** has the shape of a square that is slightly smaller than the substrate **10**. As shown in FIG. **2**, the frame **21** includes opposite side walls in the Y-direction, namely, a first side wall **21***a* and a second side wall **21***b*, and opposite side walls in the X-direction, namely, a third side wall **21***c* and a fourth side wall **21***d*. The first side wall **21***a* and the second side wall **21***b* are opposed in the Y-direction. The third side wall **21***c* and the fourth side wall **21***d* are opposed in the X-direction.

The cover **22** is plate-shaped and is slightly smaller than the outer edges of the frame **21**. The cover **22** is formed from a transparent material, which is, for example, glass. The cover **22** is transmissive to light from the semiconductor light emitting element **80**.

As shown in FIGS. **2** and **3**, the conductive portions **30**, **40**, **50**, **60**, and **70** are formed on the substrate **10**. The conductive portions **30**, **40**, **50**, **60**, and **70** are formed from a conductive material. For example, Cu, Ni, Ti, or Au is appropriately selected. In addition, a surface layer formed from Sn may be formed on the conductive portions **30**, **40**, **50**, **60**, and **70**.

Each of the conductive portions **30**, **40**, **50**, **60**, and **70** includes, for example, a front surface conductive layer formed on the substrate front surface **11**, a back surface conductive layer formed on the substrate back surface **12**, a joint electrically connecting the front surface conductive layer to the back surface conductive layer.

As shown in FIGS. **3** to **5**, the conductive portions **30**, **40**, **50**, **60**, and **70** include a common conductive portion **30** that includes a common front surface conductive layer **31** formed on the substrate front surface **11**, a common back surface conductive layer **32** formed on the substrate back surface **12**, and a common joint **33** electrically connecting the common front surface conductive layer **31** to the common back surface conductive layer **32**. The common conductive portion **30** includes a common contact front surface **30***a* located on the substrate front surface **11** and a common contact back surface **30***b* located on the substrate back surface **12**. In the present embodiment, the common contact front surface **30***a* is a front surface of the common front surface conductive layer **31**, and the common contact back surface **30***b* is a back surface of the common back surface conductive layer **32**.

The common contact front surface **30***a* is located closer to a central part of the substrate front surface **11** in the X-direction than the third side wall **21***c* and the fourth side wall **21***d*. In the present embodiment, the common contact front surface **30***a* is located on the central part of the substrate front surface **11** in the X-direction. The common contact front surface **30***a* extends in the Y-direction. The common contact front surface **30***a* is rectangular so that the long sides extend in the Y-direction and the short sides extend in the X-direction. The common contact front surface **30***a* extends to a position that overlaps the frame **21**. As viewed from above, opposite ends of the common contact front surface **30***a* in the Y-direction coincide with an outer surface of the first side wall **21***a* and an outer surface of the second side wall **21***b*.

A connection conductive portion **40** and a control conductive portion **70** are located at one side of the common contact front surface **30***a* in the X-direction. An element conductive portion **50** and a drive conductive portion **60** are located at the other side of the common contact front surface **30***a* in the X-direction.

The connection conductive portion **40** includes a connection front surface conductive layer **41** formed on the substrate front surface **11**, a connection back surface conductive layer **42** formed on the substrate back surface **12**, and a connection joint **43** electrically connecting the connection front surface conductive layer **41** to the connection back surface conductive layer **42**.

The common front surface conductive layer **31** includes an end **31***a* in the X-direction located at a side opposite from the element conductive portion **50**. The connection front surface conductive layer **41** is a portion of the common front surface conductive layer **31** projecting from the end **31***a* in the X-direction. The connection front surface conductive layer **41** and the common front surface conductive layer **31** are formed integrally. Thus, the connection conductive portion **40** is electrically connected to the common conductive portion **30**.

The connection conductive portion **40** includes a connection contact front surface **40***a* located on the substrate front surface **11** and a connection contact back surface **40***b* located on the substrate back surface **12**. In the present embodiment, the connection contact front surface **40***a* is a front surface of the connection front surface conductive layer **41** and projects in the X-direction from one of the opposite ends of the common contact front surface **30***a* in the X-direction located at a side opposite from the element conductive portion **50**. The connection contact front surface **40***a* is continuous with the common contact front surface **30***a*. The connection contact back surface **40***b* is a back surface of the connection back surface conductive layer **42**.

The element conductive portion **50** includes the element front surface conductive layer **51** formed on the substrate front surface **11**, the element back surface conductive layer **52** formed on the substrate back surface **12**, and an element joint **53** electrically elementing the element front surface conductive layer **51** to the element back surface conductive layer **52**. The element conductive portion **50** includes an element contact front surface **50***a* located on the substrate front surface **11** and an element contact back surface **50***b* located on the substrate back surface **12**. In the present embodiment, the element contact front surface **50***a* is a front surface of the element front surface conductive layer **51**, and the element contact back surface **50***b* is a back surface of the element back surface conductive layer **52**.

As shown in FIGS. **3**, **4**, and **6**, the drive conductive portion **60** includes a drive front surface conductive layer **61** formed on the substrate front surface **11**, the drive back surface conductive layer **62** formed on the substrate back surface **12**, and a drive joint **63** electrically driving the drive front surface conductive layer **61** and the drive back surface conductive layer **62**. The drive conductive portion **60** includes a drive contact front surface **60***a* located on the substrate front surface **11** and a drive contact back surface **60***b* located on the substrate back surface **12**. In the present embodiment, the drive contact front surface **60***a* is a front surface of the drive front surface conductive layer **61**, and the drive contact back surface **60***b* is a back surface of the drive back surface conductive layer **62**.

The control conductive portion **70** includes a control front surface conductive layer **71** formed on the substrate front surface **11**, a control back surface conductive layer **72** formed on the substrate back surface **12**, and a control joint **73** electrically controlling the control front surface conductive layer **71** and the control back surface conductive layer **72**. The control conductive portion **70** includes a control contact front surface **70***a* located on the substrate front surface **11** and a control contact back surface **70***b* located on the substrate back surface **12**. In the present embodiment, the control contact front surface **70***a* is a front surface of the control front surface conductive layer **71**, and the control contact back surface **70***b* is a back surface of the control back surface conductive layer **72**.

The semiconductor light emitting element **80** is a light source of the semiconductor light emitting device **1** and emits light in a predetermined wavelength band. The semiconductor light emitting element **80** is not particularly limited to a specific configuration and is, for example, a semiconductor laser element or a light emitting diode (LED) element. In the present embodiment, the semiconductor light emitting element **80** is a semiconductor laser element. Particularly, a vertical cavity surface emitting laser (VCSEL) element is used. Light from the semiconductor light emitting element **80** is transmitted through the cover **22** and emitted to the outside.

As shown in FIGS. **7** and **8**, the present example of the semiconductor light emitting element **80** includes an element substrate **81**, a first semiconductor layer **82**, an active layer **83**, a second semiconductor layer **84**, a current narrow layer **85**, an insulation layer **86**, and a conductive layer **87**. Light emitting regions **90** are formed in the semiconductor light emitting element **80**. FIG. **8** shows an enlarged portion including one light emitting region **90**.

The element substrate **81** is formed of a semiconductor. The semiconductor forming the element substrate **81** is, for example, GaAs. The semiconductor forming the element substrate **81** may be other than GaAs.

The active layer **83** is formed of a compound semiconductor that limits light having, for example, a wavelength band of 980 nm (hereafter, denoted by "λa") through spontaneous emission and simulated emission. The active layer **83** is disposed between the first semiconductor layer **82** and the second semiconductor layer **84**. In the present embodiment, undoped GaAs well layers and undoped AlGaAs block layers (barrier layers) are alternately stacked to form a multilayer quantum well structure. For example, undoped $Al_{0.35}Ga_{0.65}As$ block layers and undoped GaAs well layers are alternately formed in two to six cycles of repetition.

The first semiconductor layer **82** is typically a distributed Bragg reflector (DBR) layer and is formed on the element substrate **81**. The first semiconductor layer **82** is formed of a semiconductor having a first conductive type. In the present example, the first conductive type is n-type. The first semiconductor layer **82** is formed as a DBR for efficiently reflecting the light emitted from the active layer **83**. More specifically, the active layer **83** is formed by stacking pairs of two AlGaAs layers having different reflectances and a thickness of λa/4. More specifically, the first semiconductor layer **82** is formed, for example, by alternately stacking n-type $Al_{0.16}Ga_{0.84}As$ layers having a thickness of 600 Å and a relatively low Al composition (low Al composition layers) and n-type $Al_{0.84}Ga_{0.16}As$ layers having a thickness of 700 Å and a relatively high Al composition (high Al composition layers) in cycles (for example, 20 cycles) of repetition. The n-type $Al_{0.16}Ga_{0.84}As$ layers and the n-type $Al_{0.84}Ga_{0.16}As$ layers are doped with, for example, an n-type impurity (e.g., Si) in concentration of $2\times10^{17}$ cm$^{-3}$ to $3\times10^{18}$ cm$^{-3}$ and$2\times^{17}$ cm$^{-3}$ to $3\times10^{18}$ cm$^{-3}$, respectively.

The second semiconductor layer **84** is typically a DBR layer and is formed of a semiconductor having a second conductive type. In the present example, the second conductive type is p-type. Other than the present embodiment, the first conductive type may be p-type, the second conductive type may be n-type. The first semiconductor layer **82** is disposed between the second semiconductor layer **84** and the element substrate **81**. The second semiconductor layer **84** is formed as a DBR for efficiently reflecting the light emitted from the active layer **83**. More specifically, the second semiconductor layer **84** is formed by stacking pairs of two AlGaAs layers having different reflectances and a thickness of λ/4. The second semiconductor layer **84** is formed, for example, by alternately stacking p-type $Al_{0.16}Ga_{0.84}As$ layers having relatively low Al composition (low Al composition layers) and p-type $Al_{0.84}Ga_{0.16}As$ layers having a relatively high Al composition (high Al composition layers) in cycles (for example, twenty cycles) of repetition.

The current narrow layer **85** is disposed in the second semiconductor layer **84**. The current narrow layer **85** is formed from, for example, an easily-oxidizable layer containing a large amount of Al. The current narrow layer **85** is formed by oxidizing the easily-oxidizable layer. However, the current narrow layer **85** does not necessarily have to be formed by oxidization and may be formed by using another process (e.g., ion implantation). The current narrow layer **85** has an opening **85***a*. Current flows through the opening **85***a*.

The insulation layer **86** is formed on the second semiconductor layer **84**. The insulation layer **86** is, for example, formed from $SiO_2$. The insulation layer **86** has an opening **86***a*.

The conductive layer **87** is formed on the insulation layer **86**. The conductive layer **87** is formed from a conductive material (e.g., metal). The conductive layer **87** is electrically connected to the second semiconductor layer **84** through the opening **86***a* in the insulation layer **86**. The conductive layer **87** has an opening **87***a*.

The light emitting region **90** is a region to which light from the active layer **83** is directly emitted or the light is reflected and then emitted. In the present example, the light emitting region **90** is annular in plan view but is not limited to a particular shape. The light emitting region **90** is formed by stacking the second semiconductor layer **84**, the current narrow layer **85**, the insulation layer **86**, and the conductive layer **87** and forming the opening **85***a* in the current narrow layer **85**, the opening **86***a* in the insulation layer **86**, and the opening **87***a* in the conductive layer **87** as described above. In the light emitting region **90**, the light from the active layer **83** is emitted through the opening **87***a* in the conductive layer **87**.

As shown in FIGS. **3** and **5**, the semiconductor light emitting element **80** includes an element upper surface **80***a* on which the light emitting regions **90** and an element upper surface electrode **91** are formed and an element lower surface **80***b* on which an element lower surface electrode **92** is formed. The element upper surface **80***a* is, for example, rectangular so that the long sides extend in the X-direction and the short sides extend in the Y-direction. The element upper surface electrode **91** is formed on an end of the element upper surface **80***a* in the X-direction. The element upper surface electrode **91** is located closer to the fourth side wall **21***d* than the third side wall **21***c*. The element upper surface electrode **91** extends in the Y-direction. The element upper surface electrode **91** is, for example, formed from metal and is electrically connected to the second semiconductor layer **84**. The element lower surface electrode **92** is, for example, formed on the entirety of the element lower surface **80***b* and is formed of, for example, metal. In the present embodiment, the semiconductor light emitting element **80** is a VCSEL element. The element upper surface electrode **91** is an anode electrode, and the element lower surface electrode **92** is a cathode electrode.

The electronic component **100** is, for example, used to drive the semiconductor light emitting element **80**. The electronic component **100** is, for example, a switching element and is an n-type metal-oxide-semiconductor field-effect transistor (MOSEFT) in the present embodiment.

As shown in FIGS. **3** and **6**, the electronic component **100** includes an upper surface **100***a* on which a first drive electrode **101** and a control electrode **102** are formed and a lower surface **100***b* on which a second drive electrode **103** is formed. In the present embodiment, the electronic component **100** is a MOSFET. The first drive electrode **101** is a source electrode, the second drive electrode **103** is a drain electrode, and the control electrode **102** is a gate electrode.

As shown in FIG. **3**, the upper surface **100***a* of the electronic component **100** is rectangular. As viewed from above, the control electrode **102** is formed on a lower left portion, and the remaining portion is the first drive electrode **101**. The first drive electrode **101** has a larger area than the control electrode **102**. The lower surface **100***b* is, for example, rectangular. The second drive electrode **103** is, for example, formed on the entire back surface of the electronic component **100** and is formed from, for example, metal.

The positional relationship among the conductive portions **30**, **40**, **50**, **60**, and **70**, the semiconductor light emitting element **80**, and the electronic component **100** will be described below in detail.

With reference to FIG. **3**, the layout at the side of the substrate front surface **11** will be described.

The semiconductor light emitting element **80** and the electronic component **100** are mounted on the common conductive portion **30** and are electrically connected to each other by the common conductive portion **30**. In the present embodiment, the semiconductor light emitting element **80** and the electronic component **100** are disposed on the common contact front surface **30***a*, which is formed of the front surface of the common front surface conductive layer **31**, and electrically connected by the common front surface conductive layer **31**.

More specifically, the common contact front surface **30***a* extends in the Y-direction, and the semiconductor light emitting element **80** and the electronic component **100** are arranged on the common contact front surface **30***a* in the Y-direction. In other words, in the present embodiment, the common contact front surface **30***a* extends in the arrangement direction of the semiconductor light emitting element **80** and the electronic component **100**. The Y-direction corresponds to the arrangement direction or the first direction. The X-direction corresponds to the second direction.

The semiconductor light emitting element **80** is located closer to the first side wall **21***a* than the electronic component **100**. The electronic component **100** is located closer to the second side wall **21***b* than the semiconductor light emitting element **80**. The semiconductor light emitting element **80** and the electronic component **100** are located at positions displaced from the central part of the substrate front surface **11** in the Y-direction, for example, at opposite sides of the central part in the Y-direction. In the illustrated example, the semiconductor light emitting element **80** is disposed on the common contact front surface **30***a* between the central part of the substrate front surface **11** in the Y-direction and the first side wall **21***a*. In the present example, the semiconductor light emitting element **80** is located closer to the central part of the substrate front surface **11** in the Y-direction than the first side wall **21***a*. The electronic component **100** is disposed between the central part of the substrate front surface **11** in the Y-direction and the second side wall **21***b*. In the present example, the electronic component **100** is located closer to the central part of the substrate front surface **11** in the Y-direction than the second side wall **21***b*. Thus, the semiconductor light emitting element **80** and the electronic component **100** are located at opposite sides of the central part of the substrate front surface **11** in the Y-direction and separated by a short distance in the Y-direction.

As shown in FIG. **5**, the element lower surface electrode **92**, which is formed on the element lower surface **80***b* of the semiconductor light emitting element **80**, is die-bonded to the common contact front surface **30***a* by a conductive bonding material P1 such as solder or a paste containing metal, for example, Ag. Thus, the element lower surface electrode **92** is bonded to the common conductive portion **30**.

As shown in FIG. **6**, the second drive electrode **103**, which is formed on the lower surface **100***b* of the electronic component **100**, is die-bonded to the common contact front surface **30***a* by a conductive bonding material P**2** such as solder or a paste containing metal, for example, Ag. Thus, the second drive electrode **103** is bonded to the common conductive portion **30**. The common conductive portion **30** electrically connects the element lower surface electrode **92** and the second drive electrode **103**.

As shown in FIG. **3**, the element conductive portion **50** is located on the substrate front surface **11** at one side of the common contact front surface **30***a* in the X-direction, and the connection conductive portion **40** is located at the other side of the common contact front surface **30**$a$ in the X-direction. In the present embodiment, the connection contact front surface **40**$a$ and the element contact front surface **50**$a$ are separately disposed at opposite sides of the semiconductor light emitting element **80** in the X-direction. The connection conductive portion **40** is located closer to the third side wall **21**$c$ than the semiconductor light emitting element **80**. The connection contact front surface **40**$a$ is located at an upper left portion of the substrate front surface **11**. The element conductive portion **50** is located closer to the fourth side wall **21**$d$ than the semiconductor light emitting element **80**. The element contact front surface **50**$a$ is located at an upper right portion of the substrate front surface **11**. The connection contact front surface **40**$a$, the element upper surface **80**$a$, and the element contact front surface **50**$a$ are arranged in the X-direction.

The connection contact front surface **40**$a$ is, for example, rectangular so that the long sides extend in the Y-direction and the short sides extend in the X-direction. The connection contact front surface **40**$a$ overlaps the frame **21** in plan view. An end of the connection contact front surface **40**$a$ in the long-side direction coincides with the outer surface of the first side wall **21**$a$ as viewed from above. An end of the connection contact front surface **40**$a$ in the short-side direction coincides with the outer surface of the third side wall **21**$c$ as viewed from above.

The element contact front surface **50**$a$ is, for example, rectangular so that the long sides extend the Y-direction and the short sides extend in the X-direction. The element contact front surface **50**$a$ overlaps the frame **21** in plan view. An end of the element contact front surface **50**$a$ in the long-side direction coincides with the outer surface of the first side wall **21**$a$ as viewed from above. An end of the element contact front surface **50**$a$ in the short-side direction coincides with the outer surface of the fourth side wall **21**$d$ as viewed from above.

The element contact front surface **50**$a$ is electrically connected to the element upper surface electrode **91** by wires W**1**. The wires W**1** are, for example, formed from metal such as Au and bonded to the element upper surface electrode **91** and the element contact front surface **50**$a$. The number of wires W**1** is not particularly limited. In the illustrated example, multiple wires W**1** (five wires W**1**) are arranged. In the illustrated example, the wires W**1** have a first bonding portion disposed on the element upper surface electrode **91** and a second bonding portion disposed on the element contact front surface **50**$a$.

In the illustrated example, the element upper surface electrode **91** is formed on one of the opposite ends of the element upper surface **80**$a$ in the X-direction that is located closer to the element conductive portion **50**. Thus, the wires W**1** are shortened.

The drive conductive portion **60** and the control conductive portion **70** are separately disposed at opposite sides of the common contact front surface **30**$a$ in the X-direction. The drive contact front surface **60**$a$ and the control contact front surface **70**$a$ are located at opposite sides of the electronic component **100** in the Y-direction. In the illustrated example, the drive conductive portion **60** is located closer to the fourth side wall **21**$d$ than the electronic component **100**. The drive contact front surface **60**$a$ is located at a lower right portion of the substrate front surface **11**. The control conductive portion **70** is located closer to the third side wall **21**$c$ than the electronic component **100**. The control contact front surface **70**$a$ is located at a lower left portion of the substrate front surface **11**. The control contact front surface **70**$a$, the upper surface **100**$a$ of the electronic component **100**, and the drive contact front surface **60**$a$ are arranged in the X-direction.

The element conductive portion **50** and the drive conductive portion **60** are located at the same side of the common contact front surface **30**$a$ in the Y-direction. In other words, the element contact front surface **50**$a$ is located at one of the opposite sides of the common contact front surface **30**$a$ in the X-direction where the drive contact front surface **60**$a$ is located.

The drive contact front surface **60**$a$ is, for example, rectangular so that the long sides extend the Y-direction and the short sides extend in the X-direction. The drive contact front surface **60**$a$ overlaps the frame **21**. An end of the drive contact front surface **60**$a$ in the long-side direction coincides with the outer surface of the second side wall **21**$b$ as viewed from above. An end of the drive contact front surface **60**$a$ in the short-side direction coincides with the outer surface of the fourth side wall **21**$d$ as viewed from above.

The drive contact front surface **60**$a$ is connected to the first drive electrode **101** by wires W**2**. Thus, the drive conductive portion **60** is electrically connected to the first drive electrode **101**. The wires W**2** are, for example, formed from metal such as Cu and bonded to the drive contact front surface **60**$a$ and the first drive electrode **101**. The number of wires W**2** is not particularly limited. In the illustrated example, multiple wires W**2** (five wires W**2**) are arranged. In the illustrated example, the wires W**2** have a first bonding portion disposed on the first drive electrode **101** and a second bonding portion disposed on the drive contact front surface **60**$a$.

In the illustrated example, the number of wires W**1** is equal to the number of wires W**2**. However, the number of wires W**1** may differ from the number of wires W**2**. For example, the number of wires W**1** may be greater than the number of wires W**2**. This configuration allows a larger amount of current to flow to the semiconductor light emitting element **80** than to the electronic component **100**.

The control contact front surface **70**$a$ is, for example, rectangular so that the long sides extend the Y-direction and the short sides extend in the X-direction. The control contact front surface **70**$a$ overlaps the frame **21**. An end of the control contact front surface **70**$a$ in the long-side direction coincides with the outer surface of the second side wall **21**$b$ as viewed from above. An end of the control contact front surface **70**$a$ in the short-side direction coincides with the outer surface of the third side wall **21**$c$.

The control contact front surface **70**$a$ is connected to the control electrode **102** by a wire W**3**. Thus, the control conductive portion **70** is electrically connected to the control electrode **102**. The wire W**3** is, for example, formed from metal such as Cu and bonded to the control contact front surface **70**$a$ and the control electrode **102**. The number of wires W**3** is not particularly limited. In the illustrated example, the number of wires W**3** is one. In the illustrated example, the wire W**3** has a first bonding portion disposed on the control electrode **102** and a second bonding portion disposed on the control contact front surface **70**$a$. The control electrode **102** is formed on one of the opposite ends of the upper surface **100**$a$ in the X-direction that is located closer to the control contact front surface **70**$a$. Thus, the wire W**3** is shortened.

In the illustrated example, the common contact front surface **30**$a$ is larger than other contact front surfaces **40**$a$, **50**$a$, **60**$a$, and **70**$a$. The element contact front surface **50**$a$ is larger than the drive contact front surface **60**$a$ and the control contact front surface **70**$a$. The connection contact front surface **40***a* is larger than the drive contact front surface **60***a* and the control contact front surface **70***a* and is larger than the element contact front surface **50***a*. The drive contact front surface **60***a* and the control contact front surface **70***a* have the same size.

The layout at the side of the substrate back surface **12** will be described with reference FIGS. **4** to **6**.

As shown in FIG. **4**, the common contact back surface **30***b* (common back surface conductive layer **32**) is formed on the central part of the substrate back surface **12**. The common contact back surface **30***b* and the common contact front surface **30***a* are formed on opposite sides. In the same manner as the common contact front surface **30***a*, the common contact back surface **30***b* is rectangular so that the short sides extend in the X-direction and the long sides extend in the Y-direction. As viewed from above, the common contact back surface **30***b* is smaller than the common contact front surface **30***a*. The dimension of the common contact back surface **30***b* in the X-direction is less than the dimension of the common contact front surface **30***a* in the X-direction. Opposite ends of the common contact back surface **30***b* in the X-direction are located closer to the center than opposite ends of the common contact front surface **30***a* in the X-direction.

The element contact back surface **50***b* and the drive contact back surface **60***b* are located at one side of the common contact back surface **30***b* in the X-direction. The connection contact back surface **40***b* and the control contact back surface **70***b* are located at the other side of the common contact back surface **30***b* in the X-direction.

The connection contact back surface **40***b* (connection back surface conductive layer **42**) is formed on the substrate back surface **12** at a position opposite to the connection contact front surface **40***a*. As shown in FIG. **4**, the connection contact back surface **40***b* is formed on an upper right portion of the substrate back surface **12**. The connection contact back surface **40***b* is rectangular so that the short sides extend in the X-direction and the long sides extend in the Y-direction. The connection contact back surface **40***b* is smaller than the connection contact front surface **40***a*. The connection contact back surface **40***b* and the common contact back surface **30***b* are separate and are not connected to each other.

As shown in FIG. **5**, as viewed from above (substrate front surface **11**), the left end of the connection contact back surface **40***b* is located closer to the center of the substrate **10** than the left end of the connection contact front surface **40***a*. The left end of the connection contact back surface **40***b* and the left end of the connection contact front surface **40***a* are ends of the connection contact back surface **40***b* and the connection contact front surface **40***a* in the X-direction that are located loser to the edge of the substrate **10**.

As shown in FIG. **4**, the element contact back surface **50***b* (element back surface conductive layer **52**) is formed on the substrate back surface **12** at a position opposite to the element contact front surface **50***a*. In the illustrated example, the element contact back surface **50***b* is formed on an upper left portion of the substrate back surface **12**. The element contact back surface **50***b* and the common contact back surface **30***b* are separate and are not connected to each other.

The element contact back surface **50***b* is rectangular so that the short sides extend in the X-direction and the long sides extend in the Y-direction. The element contact back surface **50***b* is smaller than the element contact front surface **50***a*.

As shown in FIG. **5**, as viewed from above, the left end of the element contact back surface **50***b* coincides with the left end of the element contact front surface **50***a*. The right end of the element contact back surface **50***b* is located closer to the center than the right end of the element contact front surface **50***a*. The left end of the element contact back surface **50***b* and the left end of the element contact front surface **50***a* are ends of the element contact back surface **50***b* and the element contact front surface **50***a* in the X-direction that are located closer to the center of the substrate **10**. The right end of the element contact back surface **50***b* and the right end of the element contact front surface **50***a* are ends of the element contact back surface **50***b* and the element contact front surface **50***a* in the X-direction that are located closer to the edge of the substrate **10**.

As shown in FIG. **4**, the drive contact back surface **60***b* (drive back surface conductive layer **62**) is formed on the substrate back surface **12** at a position opposite to the drive contact front surface **60***a*. In the illustrated example, the drive contact back surface **60***b* is formed on a lower left portion of the substrate back surface **12**. The drive contact back surface **60***b* is rectangular so that the short sides extend in the X-direction and the long sides extend in the Y-direction.

The drive contact back surface **60***b* and the common contact back surface **30***b* are separated from each other in the X-direction. The element contact back surface **50***b* and the drive contact back surface **60***b* are separated from each other in the Y-direction. The dimension of the drive contact back surface **60***b* in the Y-direction is less than the dimension of the element contact back surface **50***b* in the Y-direction. Distance **Y2** between the element contact back surface **50***b* and the drive contact back surface **60***b* is greater than distance **Y1** between the element contact front surface **50***a* and the drive contact front surface **60***a* (FIG. **3**).

As shown in FIG. **6**, as viewed from above, the left end of the drive contact back surface **60***b* coincides with the left end of the drive contact front surface **60***a*. The right end of the drive contact back surface **60***b* is located closer to the center than the right end of the drive contact front surface **60***a*. The left end of the drive contact back surface **60***b* and the left end of the drive contact front surface **60***a* are ends of the drive contact back surface **60***b* and the drive contact front surface **60***a* in the X-direction that are located closer to the center of the substrate **10**. The right end of the drive contact back surface **60***b* and the right end of the drive contact front surface **60***a* are ends of the drive contact back surface **60***b* and the drive contact front surface **60***a* in the X-direction that are located closer to the edge of the substrate **10**.

As shown in FIG. **4**, the control contact back surface **70***b* is formed on the substrate back surface **12** at a position opposite to the control contact front surface **70***a*. In the illustrated example, the control contact back surface **70***b* is formed on a lower right portion of the substrate back surface **12**. The control contact back surface **70***b* is rectangular so that the short sides extend in the X-direction and the long sides extend in the Y-direction. The control contact back surface **70***b* is smaller than the control contact front surface **70***a*.

The control contact back surface **70***b* and the common contact back surface **30***b* are separated in the X-direction. The connection contact back surface **40***b* and the control contact back surface **70***b* are separated from each other in the Y-direction. The dimension of the control contact back surface **70***b* in the Y-direction is less than the dimension of the connection contact back surface **40***b* in the Y-direction. Distance **Y4** between the connection contact back surface **40***b* and the control contact back surface **70***b* is greater than distance Y3 between the connection contact front surface **40**$a$ and the control contact front surface **70**$a$ (FIG. **3**).

As shown in FIG. **6**, as viewed from above, the right end of the control contact back surface **70**$b$ coincides with the right end of the control contact front surface **70**$a$. The left end of the control contact back surface **70**$b$ is located closer to the center than the left end of the control contact front surface **70**$a$. The right end of the control contact back surface **70**$b$ and the right end of the control contact front surface **70**$a$ are ends of the control contact back surface **70**$b$ and the control contact front surface **70**$a$ in the X-direction that are located closer to the center of the substrate **10**. The left end of the control contact back surface **70**$b$ and the left end of the control contact front surface **70**$a$ are ends of the control contact back surface **70**$b$ and the control contact front surface **70**$a$ in the X-direction that are located closer to the edge of the substrate **10**.

In the present embodiment, the common contact back surface **30**$b$ is larger than the connection contact back surface **40**$b$, the element contact back surface **50**$b$, the drive contact back surface **60**$b$, and the control contact back surface **70**$b$. The element contact back surface **50**$b$ and the connection contact back surface **40**$b$ are larger than the drive contact back surface **60**$b$ and the control contact back surface **70**$b$. The drive contact back surface **60**$b$ and the control contact back surface **70**$b$ are identical in shape.

The joints **33**, **43**, **53**, **63**, and **73** will now be described.

The joints **33**, **43**, **53**, **63**, and **73** form conductive paths in the thickness-wise direction of the substrate **10**. In the present embodiment, the joints **33**, **43**, **53**, **63**, and **73** extend through the substrate **10** in the thickness-wise direction.

As shown in FIGS. **3** and **4**, the common joint **33** is disposed between the common front surface conductive layer **31** and the common back surface conductive layer **32**. The common joint **33** is continuous with the common front surface conductive layer **31** and the common back surface conductive layer **32** to electrically connect the common front surface conductive layer **31** and the common back surface conductive layer **32**.

As shown in FIG. **3**, as viewed from above, the common joint **33** does not overlap the semiconductor light emitting element **80** and the electronic component **100**. For example, the common joint **33** is located between the semiconductor light emitting element **80** and the electronic component **100**. More specifically, the common joint **33** is located in the center of the substrate front surface **11**.

As shown in FIG. **5**, the connection joint **43** is disposed between the connection front surface conductive layer **41** and the connection back surface conductive layer **42**. The connection joint **43** is continuous with the connection front surface conductive layer **41** and the connection back surface conductive layer **42** to electrically connect the connection front surface conductive layer **41** and the connection back surface conductive layer **42**.

As shown in FIG. **3**, as viewed from above, the connection joint **43** overlaps the case **20**. For example, as viewed from above, the connection joint **43** overlaps the third side wall **21**$c$.

As shown in FIG. **5**, the element joint **53** is disposed between the element front surface conductive layer **51** and the element back surface conductive layer **52**. The element joint **53** is continuous with the element front surface conductive layer **51** and the element back surface conductive layer **52** to electrically connect the element front surface conductive layer **51** and the element back surface conductive layer **52**.

As shown in FIG. **3**, as viewed from above, the element joint **53** overlaps the case **20**. For example, as viewed from above, the element joint **53** overlaps the fourth side wall **21**$d$.

As shown in FIG. **6**, the drive joint **63** is disposed between the drive front surface conductive layer **61** and the drive back surface conductive layer **62**. The drive joint **63** is continuous with the drive front surface conductive layer **61** and the drive back surface conductive layer **62** to electrically connect the drive front surface conductive layer **61** and the drive back surface conductive layer **62**.

As shown in FIG. **3**, as viewed from above, the drive joint **63** overlaps the case **20**. For example, as viewed from above, the drive joint **63** overlaps the fourth side wall **21**$d$.

As shown in FIG. **6**, the control joint **73** is disposed between the control front surface conductive layer **71** and the control back surface conductive layer **72**. The control joint **73** is continuous with the control front surface conductive layer **71** and the control back surface conductive layer **72** to electrically connect the control front surface conductive layer **71** and the control back surface conductive layer **72**.

As shown in FIG. **3**, as viewed from above, the control joint **73** overlaps the case **20**. For example, as viewed from above, the control joint **73** overlaps the third side wall **21**$c$.

FIGS. **9** and **10** are respectively a plan view and a circuit diagram showing an example of an electronic apparatus **2** that uses the semiconductor light emitting device **1**. The electronic apparatus **2** is, for example, a sensor for measuring distance.

The electronic apparatus **2** includes the semiconductor light emitting device **1**, a circuit substrate **110** on which the semiconductor light emitting device **1** is mounted, and wiring patterns **111** to **114** formed on the circuit substrate **110**.

The wiring patterns **111** to **114** are separated from each other. The first wiring pattern **111** and the second wiring pattern **112** are arranged in the X-direction. The third wiring pattern **113** and the fourth wiring pattern **114** are arranged in the X-direction. The first wiring pattern **111** and the fourth wiring pattern **114** are arranged in the Y-direction. The second wiring pattern **112** and the third wiring pattern **113** are arranged in the Y-direction.

The connection back surface conductive layer **42** overlaps the first wiring pattern **111**. The connection contact back surface **40**$b$ and the first wiring pattern **111** are bonded by solder or the like. Thus, the first wiring pattern **111** is electrically connected to the element lower surface electrode **92** of the semiconductor light emitting element **80** and the second drive electrode **103** of the electronic component **100**.

The element back surface conductive layer **52** overlaps the second wiring pattern **112**. The element contact back surface **50**$b$ and the second wiring pattern **112** are bonded by solder or the like. Thus, the second wiring pattern **112** is electrically connected to the element upper surface electrode **91**, which is the anode electrode in the present embodiment.

The drive back surface conductive layer **62** overlaps the third wiring pattern **113**. The drive contact back surface **60**$b$ and the third wiring pattern **113** are bonded by solder or the like. Thus, the third wiring pattern **113** is electrically connected to the first drive electrode **101**, which is the source electrode in the present embodiment.

The control back surface conductive layer **72** overlaps the fourth wiring pattern **114**. The control contact back surface **70**$b$ and the fourth wiring pattern **114** are bonded by solder or the like. Thus, the fourth wiring pattern **114** is electrically connected to the control electrode **102**.

As described above, in the present embodiment, the contact back surfaces **40***b*, **50***b*, **60***b*, and **70***b* form external terminals of the semiconductor light emitting device **1**.

In the illustrated example, the common back surface conductive layer **32** (common contact back surface **30***b*) is mounted via solder or the like on a heat dissipation pattern **115** formed on the circuit substrate **110**. Heat of the semiconductor light emitting element **80** and the electronic component **100** is transmitted from the common contact back surface **30***b* to the circuit substrate **110**. This improves the heat dissipation efficiency of the semiconductor light emitting device **1**.

As shown in FIG. **9**, the electronic apparatus **2** includes capacitors **120**. The capacitors **120** are disposed so as to span between the second wiring pattern **112** and the third wiring pattern **113** and electrically connected to the second wiring pattern **112** and the third wiring pattern **113**. Thus, as shown in FIG. **10**, the capacitors **120** are connected in parallel to the semiconductor light emitting element **80** and the electronic component **100**, which are connected in series. This simplifies the layout of the wiring patterns that connect the capacitors **120**.

The operation of the present embodiment will now be described.

The semiconductor light emitting element **80** and the electronic component **100** are accommodated in the case **20**. The semiconductor light emitting element **80** and the electronic component **100** are mounted on the common conductive portion **30** formed on the substrate front surface **11**. Thus, the conductive path between the semiconductor light emitting element **80** and the electronic component **100** is shortened as compared to a structure in which the electronic component **100** is disposed outside the case **20**.

The semiconductor light emitting device **1** of the present embodiment has the following advantages.

(1-1) The semiconductor light emitting device **1** includes the substrate **10**, the common conductive portion **30** formed on the substrate **10**, and the semiconductor light emitting element **80** and the electronic component **100** that are mounted on the common conductive portion **30**. The semiconductor light emitting element **80** and the electronic component **100** are electrically connected by the common conductive portion **30**. This structure shortens the conductive path between the semiconductor light emitting element **80** and the electronic component **100**. As a result, parasitic capacitance caused by the conductive path between the semiconductor light emitting element **80** and the electronic component **100** is reduced. Thus, while reducing the parasitic capacitance, the semiconductor light emitting element **80** and the electronic component **100** are electrically connected.

(1-2) The element lower surface electrode **92** is formed on the element lower surface **80***b* of the semiconductor light emitting element **80**. The electronic component **100** is used to drive the semiconductor light emitting element **80** and includes the upper surface **100***a*, on which the first drive electrode **101** and the control electrode **102** are formed, and the lower surface **100***b*, on which the second drive electrode **103** is formed. The element lower surface electrode **92** and the second drive electrode **103** are bonded to the common conductive portion **30**. In this structure, the element lower surface electrode **92** and the second drive electrode **103** are electrically connected by the common conductive portion **30**.

(1-3) The semiconductor light emitting element **80** and the electronic component **100** are arranged in a predetermined direction. The common conductive portion **30** includes the common contact front surface **30***a* extending in the Y-direction, that is, the arrangement direction of the semiconductor light emitting element **80** and the electronic component **100**. The semiconductor light emitting element **80** and the electronic component **100** are disposed on the common contact front surface **30***a*. In this structure, the semiconductor light emitting element **80** and the electronic component **100** are disposed on the common conductive portion **30**.

The common contact front surface **30***a* is shaped to have a long side extending in the Y-direction and a short side extending in the X-direction. This structure provides spaces at opposite sides of the common contact front surface **30***a* in the X-direction, so that other conductive portions are disposed in the spaces.

(1-4) The drive conductive portion **60** and the control conductive portion **70** are formed on the substrate **10**. The drive conductive portion **60** includes the drive contact front surface **60***a* electrically connected to the first drive electrode **101**. The control conductive portion **70** includes the control contact front surface **70***a* electrically connected to the control electrode **102**. The drive conductive portion **60** and the control conductive portion **70** are separately disposed at opposite sides of the common contact front surface **30***a* in the X-direction. This structure ensures the areas of the drive contact front surface **60***a* and the control contact front surface **70***a*, while avoiding interference of the drive conductive portion **60** with the control conductive portion **70**.

(1-5) The drive contact front surface **60***a* and the control contact front surface **70***a* are located at opposite sides of the electronic component **100** in the X-direction. This shortens the wire W**3**, thereby reducing the parasitic capacitance caused by the wire W**3**.

(1-6) The element conductive portion **50** including the element contact front surface **50***a* is formed at one side of the common contact front surface **30***a* in the X-direction. The connection conductive portion **40** electrically connected to the common conductive portion **30** is formed at the other side of the common contact front surface **30***a* in the X-direction. The connection conductive portion **40** includes the connection contact front surface **40***a* projecting in the X-direction from one of the opposite ends of the common contact front surface **30***a* in the X-direction that is located at a side opposite from the element conductive portion **50**. In this structure, the connection conductive portion **40** is used for electrical connection with the common conductive portion **30** while avoiding interference with the element conductive portion **50**.

(1-7) The element conductive portion **50** and the drive conductive portion **60** are located at the same side of the common conductive portion **30** in the X-direction. In this structure, the element conductive portion **50** and the drive conductive portion **60** are located close to each other so that a component (e.g., a capacitor) connected to the element conductive portion **50** and the drive conductive portion **60** is readily arranged.

(1-8) The connection conductive portion **40** includes the connection contact back surface **40***b* located on the substrate back surface **12** at a position opposite to the connection contact front surface **40***a*. The element conductive portion **50** includes the element contact back surface **50***b* located on the substrate back surface **12** at a position opposite to the element contact front surface **50***a*. The drive conductive portion **60** includes the drive contact back surface **60***b* located on the substrate back surface **12** at a position opposite to the drive contact front surface **60***a*. The control conductive portion **70** includes the control contact back surface **70***b* located on the substrate back surface **12** at a position opposite to the control contact front surface **70***a*. In this structure, the contact back surfaces **40***b*, **50***b*, **60***b*, and **70***b* are used to ensure contact of the semiconductor light emitting device **1** with an external device.

(1-9) The element contact back surface **50***b* is larger than the drive contact back surface **60***b* and the control contact back surface **70***b*. This structure improves the heat dissipation efficiency of the element conductive portion **50**.

(1-10) The common conductive portion **30** includes the common contact back surface **30***b* located on the substrate back surface **12** at a position opposite to the common contact front surface **30***a*. In this structure, the common contact back surface **30***b* is used for heat dissipation. Thus, the heat dissipation efficiency of the common conductive portion **30** is improved.

(1-11) The common contact back surface **30***b* is separate from the connection contact back surface **40***b*. In this structure, either one or both of the common contact back surface **30***b* and the connection contact back surface **40***b* may be used when the semiconductor light emitting device **1** is mounted on the circuit substrate **110**. This increases the degree of freedom for designing the circuit substrate **110** on which the semiconductor light emitting device **1** is mounted.

(1-12) The substrate **10** is formed of an insulative material. The contact front surfaces **30***a*, **40***a*, **50***a*, **60***a*, and **70***a* are the front surfaces of the front surface conductive layers **31**, **41**, **51**, **61**, and **71** formed on the substrate front surface **11**. The contact back surfaces **30***b*, **40***b*, **50***b*, **60***b*, and **70***b* are the front surfaces of the back surface conductive layers **32**, **42**, **52**, **62**, and **72** formed on the substrate back surface **12**. The conductive portions **30**, **40**, **50**, **60**, and **70** include the joints **33**, **43**, **53**, **63**, and **73** connecting the front surface conductive layers **31**, **41**, **51**, **61**, and **71** to the back surface conductive layers **32**, **42**, **52**, **62**, and **72**. The joints **33**, **43**, **53**, **63**, and **73** are disposed below the case **20**. This structure limits interference of the joints **33**, **43**, **53**, **63**, and **73** with the wires W**1** to W**3**.

(1-13) The semiconductor light emitting device **1** includes the case **20** accommodating the semiconductor light emitting element **80** and the electronic component **100**. This structure protects the semiconductor light emitting element **80** and the electronic component **100**.

## Second Embodiment

A second embodiment of a semiconductor light emitting device 1B will now be described with reference to FIGS. **11** to **38**. In the following description, the same reference characters are given to those components that are the same as the corresponding components of the semiconductor light emitting device **1** of the first embodiment. Such components will not be described in detail. Those components having functions common to the corresponding components of the semiconductor light emitting device **1** of the first embodiment are denoted by the reference characters with a suffix "B." Such components may not be described in detail. In addition, the directions are referred to in the same manner as the first embodiment.

FIG. **11** is a perspective view showing the semiconductor light emitting device **1**B of the second embodiment. FIGS. **12** and **14** are front views of the semiconductor light emitting device **1**B. FIG. **13** is a side view of the semiconductor light emitting device **1**B. FIG. **15** is a bottom view of the semiconductor light emitting device **1**B. FIG. **14** shows the semiconductor light emitting device **1**B without showing the case **20**B.



The semiconductor light emitting device **1**B differs from the semiconductor light emitting device **1** of the first embodiment in the structure of the substrate **10**, in that the connection conductive portion **40** is omitted, in the shape of the conductive portions **30**, **50**, **60**, and **70**, and in the structure of the case **20**. The semiconductor light emitting device **1**B further includes a capacitor **120**. In the following description, the substrate of the present embodiment is referred to as the substrate **10**B. Conductive portions of the present embodiment are referred to as the conductive portions **30**B, **50**B, **60**B, and **70**B or the common conductive portion **30**B, the element conductive portion **50**B, the drive conductive portion **60**B, and the control conductive portion **70**B. The case of the present embodiment is referred to as the case **20**B. In the present embodiment, the semiconductor light emitting device **1**B has a dimension LX in the X-direction that is approximately 4.5 mm. The semiconductor light emitting device **1**B has a dimension LY in the Y-direction that is approximately 4.5 mm. The semiconductor light emitting device **1**B has a dimension LZ in a direction orthogonal to the X-direction and the Y-direction (hereafter, referred to as the Z-direction) that is approximately 1.83 mm. The Z-direction is also referred to as the thickness-wise direction of the substrate **10**B.

As shown in FIGS. **17** to **19**, the case **20**B accommodates the semiconductor light emitting element **80**, the electronic component **100**, and the capacitor **120**. In the same manner as the first embodiment, the case **20**B is attached to the substrate **10**B. The case **20**B is, for example, hollow. However, alternatively, the case **20**B may be filled with another member.

The case **20**B is box-shaped and has an open side in the Z-direction. In the present embodiment, the case **20**B includes the frame **21** and the cover **22** that are integrally formed as a single-piece component. The case **20**B is, for example, formed from a light-blocking material such as a colored resin. The case **20**B blocks light from the semiconductor light emitting element **80**. The frame **21** has the shape of a square that is slightly smaller than the substrate **10**B. The cover **22** is shaped to have the same size as the outer edges of the frame **21**.

As shown in FIG. **12**, the cover **22** includes an opening **22***a* through which light is transmitted from the semiconductor light emitting element **80**. The opening **22***a* is configured to expose at least the light emitting regions **90** of the semiconductor light emitting element **80** in the Z-direction. In the present embodiment, the opening **22***a* is configured to expose the entire element upper surface **80***a* of the semiconductor light emitting element **80** in the Z-direction. Therefore, in plan view, the opening **22***a* is larger than the element upper surface **80***a*. In plan view, the opening **22***a* is rectangular so that the long sides extend in the X-direction and the short sides extend in the Y-direction. As viewed in the Z-direction, the opening **22***a* is located in the cover **22** at a position toward the third side wall **21***c* in the X-direction and toward the first side wall **21***a* in the Y-direction.

As shown in FIGS. **12** and **13**, a plate-shaped light diffusion plate **130** is attached to the cover **22** so as to cover the opening **22***a* from a side opposite to the substrate **10**B in the Z-direction. In plan view, the light diffusion plate **130** is rectangular so that the long sides extend in the Y-direction and the short sides extend in the X-direction. For example, a light-transmissive resin material such as polycarbonate, polyester, or acrylic is selected for the light diffusion plate **130**. The size of the light diffusion plate **130** may be changed in any range capable of covering the entire opening **22***a* in the Z-direction. In the present embodiment, as viewed in the

Z-direction, the light diffusion plate **130** projects from the opening **22***a* toward the third side wall **21***c* in the X-direction by a projection distance DX**1** and projects from the opening **22***a* toward the fourth side wall **21***d* in the X-direction by a projection distance DX**2**. The projection distance DX**1** is less than the projection distance DX**2**. The light diffusion plate **130** projects from the opening **22***a* toward the first side wall **21***a* in the Y-direction by a projection distance DY**1** and projects from the opening **22***a* toward the second side wall **21***b* in the Y-direction by a projection distance DY**2**. The projection distance DX**1** is greater than the projection distance DY**2**. The projection distances DX**1** and DX**2** may be changed in any manner. In an example, the projection distance DX**1** may be greater than or equal to the projection distance DX**2**. At least one of the projection distances DX**1** and DX**2** may be less than or equal to the projection distances DY**1** and DY**2**. The light diffusion plate **130** may be sized to cover the entire cover **22** in the Z-direction.

As shown in FIGS. **14** and **15**, the substrate **10**B of the present embodiment is formed from a conductive material and is, for example, formed of a metal plate formed from Cu. In other words, the substrate **10**B is a lead frame. The substrate **10**B includes insulation portions **13** defining the conductive portions **30**B, **50**B, **60**B, and **70**B that are insulated from each other. The common conductive portion **30**B, the element conductive portion **50**B, the drive conductive portion **60**B, and the control conductive portion **70**B may be referred to as portions of the substrate **10**B that are insulated and separated from each other by the insulation portions **13**. The insulation portions **13** are formed from, for example, an epoxy resin. As shown in FIGS. **14** to **16**, the conductive portions **30**B, **50**B, **60**B, and **70**B are exposed in both the substrate front surface **11** and the substrate back surface **12** of the substrate **10**B.

A general structure of the conductive portions **30**B, **50**B, **60**B, and **70**B will now be described.

As shown in FIGS. **14** to **16**, the common conductive portion **30**B includes the common contact front surface **30***a* and the common contact back surface **30***b* facing in opposite directions in the Z-direction. In the present embodiment, the common contact front surface **30***a* is part of the substrate front surface **11**. The common contact back surface **30***b* is part of the substrate back surface **12**.

As shown in FIG. **14**, the common contact front surface **30***a* is located closer to the central part of the substrate front surface **11** in the X-direction than the third side wall **21***c* and the fourth side wall **21***d*. The common contact front surface **30***a* is formed from the upper side to the lower side of the substrate front surface **11**.

The layout of the conductive portions **50**B, **60**B, and **70**B on the common contact front surface **30***a* is the same as that of the first embodiment. That is, the control conductive portion **70**B is located at one side of the common contact front surface **30***a* in the X-direction. The element conductive portion **50**B and the drive conductive portion **60**B are located at the other side of the common contact front surface **30***a* in the X-direction. The element conductive portion **50**B and the drive conductive portion **60**B are separated in the Y-direction.

As shown in FIGS. **14** to **16**, the element conductive portion **50**B includes the element contact front surface **50***a* and the element contact back surface **50***b* facing in opposite directions in the Z-direction. In the present embodiment, the element contact front surface **50***a* is part of the substrate front surface **11**. The element contact back surface **50***b* is part of the substrate back surface **12**.

The drive conductive portion **60**B includes the drive contact front surface **60***a* and the drive contact back surface **60***b* facing in opposite directions in the Z-direction. In the present embodiment, the drive contact front surface **60***a* is part of the substrate front surface **11**. The drive contact back surface **60***b* is part of the substrate back surface **12**.

The control conductive portion **70**B includes the control contact front surface **70***a* and the control contact back surface **70***b* facing in opposite directions in the Z-direction. In the present embodiment, the control contact front surface **70***a* is part of the substrate front surface **11**. The control contact back surface **70***b* is part of the substrate back surface **12**.

The shapes of the conductive portions **30**B, **50**B, **60**B, and **70**B will now be described in detail with reference to FIGS. **14** to **22**.

As shown in FIGS. **14** and **15**, the common contact front surface **30***a* is generally crank-shaped in plan view. As shown in FIG. **15**, the insulation portions **13** are arranged around the common contact front surface **30***a*. The common contact front surface **30***a* includes a first common contact front surface portion **30***c* on which the semiconductor light emitting element **80** is mounted and a second common contact front surface portion **30***d* on which the electronic component **100** is mounted. The first common contact front surface portion **30***c* and the second common contact front surface portion **30***d* are integrally formed. The common contact front surface portion **30***c* and the second common contact front surface portion **30***d* are arranged in the Y-direction. As shown in FIG. **14**, the first common contact front surface portion **30***c* extends from the second common contact front surface portion **30***d* toward the first side wall **21***a* in the Y-direction. The first common contact front surface portion **30***c* is located closer to the first side wall **21***a* than the second common contact front surface portion **30***d* in the Y-direction. In other words, the second common contact front surface portion **30***d* is located closer to the second side wall **21***b* than the first common contact front surface portion **30***c* in the Y-direction.

The first common contact front surface portion **30***c* and the second common contact front surface portion **30***d* are located at different positions in the X-direction. The first common contact front surface portion **30***c* is located closer to the third side wall **21***c* than the second common contact front surface portion **30***d* in the X-direction. In other words, the second common contact front surface portion **30***d* is located closer to the fourth side wall **21***d* than the first common contact front surface portion **30***c* in the X-direction. Thus, the first common contact front surface portion **30***c* extends from the second common contact front surface portion **30***d* toward the third side wall **21***c* in the X-direction. More specifically, the first common contact front surface portion **30***c* includes a portion projecting in the X-direction toward the third side wall **21***c* from the end **31***a* of the second common contact front surface portion **30***d* located toward the third side wall **21***c*. The second common contact front surface portion **30***d* extends from the first common contact front surface portion **30***c* toward the fourth side wall **21***d* in the X-direction. More specifically, the second common contact front surface portion **30***d* includes a portion projecting in the X-direction toward the fourth side wall **21***d* from an end **31***f* of the first common contact front surface portion **30***c* located toward the fourth side wall **21***d*.

In plan view, the first common contact front surface portion **30***c* is rectangular so that the long sides extend in the Y-direction and the short sides extend in the X-direction. In plan view, the second common contact front surface portion

**30***d* is rectangular so that the long sides extend in the X-direction and the short sides extend in the Y-direction. The dimension of the first common contact front surface portion **30***c* in the X-direction is less than the dimensions of the second common contact front surface portion **30***d* in the X-direction and the Y-direction. The dimension of the first common contact front surface portion **30***c* in the Y-direction is greater than the dimensions of the second common contact front surface portion **30***d* in the X-direction and the Y-direction. The difference between the dimension of the first common contact front surface portion **30***c* in the Y-direction and the dimension of the second common contact front surface portion **30***d* in the Y-direction is greater than the difference between the dimension of the first common contact front surface portion **30***c* in the X-direction and the dimension of the second common contact front surface portion **30***d* in the X-direction. Therefore, the first common contact front surface portion **30***c* is larger than the second common contact front surface portion **30***d*.

As shown in FIG. **14**, the common contact front surface **30***a* extends in the Y-direction to a position that overlaps the frame **21**. As viewed in the Z-direction, opposite ends of the common contact front surface **30***a* in the Y-direction include portions projecting beyond the outer surface of the first side wall **21***a* and the outer surfaces of the second side wall **21***b* and the third side wall **21***c*.

More specifically, the end of the first common contact front surface portion **30***c* located toward the first side wall **21***a* in the Y-direction includes two projections **34***a* and **34***b*, and the end of the second common contact front surface portion **30***d* located toward the second side wall **21***b* includes two projections **34***c* and **34***d*. In addition, the end of the first common contact front surface portion **30***c* located toward the third side wall **21***c* in the X-direction includes two projections **34***e* and **34***f*.

As viewed in the Z-direction, the projections **34***a* and **34***b* project from the outer surface of the first side wall **21***a*. The projections **34***a* and **34***b* are separate from each other in the X-direction. The projection **34***a* is located closer to the element conductive portion **50**B (fourth side wall **21***d*) than the projection **34***b* in the X-direction. As viewed in the Z-direction, the projections **34***c* and **34***d* project from the outer surface of the second side wall **21***b*. The projections **34***c* and **34***d* are separate from each other in the X-direction. The projection **34***c* is located closer to the drive conductive portion **60**B (fourth side wall **21***d*) than the projection **34***d* in the X-direction. As viewed in the Z-direction, the projections **34***e* and **34***f* project from the outer surface of the third side wall **21***c*. The projections **34***e* and **34***f* are separate from each other in the Y-direction. The projection **34***e* is located closer to the first side wall **21***a* than the projection **34***f* in the Y-direction.

In the present embodiment, when a support lead supporting the common conductive portion **30**B is cut, residual portions of the lead frame are the projections **34***a* to **34***f*. The projections **34***a* to **34***f* are exposed from the side surface of the substrate **10**B. The projections **34***a* to **34***f* are also exposed from the substrate front surface **11**. The number of projections may be changed to any number. As viewed from above, connection portions between the common contact front surface **30***a* and the projections **34***a* to **34***f* each have a curved surface (refer to FIG. **15**). The number of projections may be changed to any number.

As shown in FIGS. **14** and **15**, the common conductive portion **30**B includes recesses that restrict movement of the common conductive portion **30**B. In the present embodiment, the common contact front surface **30***a* includes the recesses. In the illustrated example, the common contact front surface **30***a* includes a recess **35***a*, two recesses **35***b*, two recesses **35***c*, a recess **35***d*, and a recess **35***e*. The number of recesses may be changed to any number.

As shown in FIG. **14**, the first common contact front surface portion **30***c* includes an end **31***b* located toward the first side wall **21***a*. The recess **35***a* is recessed from the end **31***b* toward the second side wall **21***b* in the Y-direction. The bottom of the recess **35***a* is defined by a curved surface. In the present embodiment, in plan view, the recess **35***a* is concave and has a width that decreases toward the bottom. In the present embodiment, the recess **35***a* extends to the inner surface of the first side wall **21***a* as viewed in the Z-direction. As shown in FIG. **20**, the recess **35***a* is recessed in the Z-direction from the first common contact front surface portion **30***c* in a front layer part of the common conductive portion **30**B in the Z-direction. The recess **35***a* accommodates an insulation portion **13**. The insulation portion **13** accommodated in the recess **35***a* includes a front-half insulation portion **13**U that does not extend through the substrate **10**B in the Z-direction. The front-half insulation portion **13**U in the recess **35***a* is the portion surrounded by the recess **35***a* and the broken line shown in FIG. **15**. Part of the insulation portion **13** located above the front-half insulation portion **13**U of the recess **35***a* extends through the substrate **10**B in the Z-direction. The dimensions of the recess **35***a* in the X-direction and the Y-direction may be changed in any manner. In an example, as viewed in the Z-direction, the inner edge of the recess **35***a* may be located at an inner side of the inner surface of the first side wall **21***a* or at an outer side of the inner surface of the first side wall **21***a*.

As shown in FIG. **14**, the second common contact front surface portion **30***d* includes an end **31***c* located toward the fourth side wall **21***d* in the X-direction. The two recesses **35***b* are recessed from the end **31***c* in the X-direction. The bottom of each recess **35***b* is defined by a curved surface. In the present embodiment, in plan view, the recess **35***b* is concave and has a width that decreases toward the bottom. Although not illustrated, the two recesses **35***b* are recessed in the Z-direction from the second common contact front surface portion **30***d* in a front layer part of the common conductive portion **30**B in the Z-direction. The two recesses **35***b* accommodate an insulation portion **13**. The insulation portion **13** accommodated in the two recesses **35***b* includes front-half insulation portions **13**U that do not extend through the substrate **10**B in the Z-direction. The front-half insulation portion **13**U in each recess **35***b* is the portion shown in FIG. **15** surrounded by the recess **35***b* and the broken line and located at an inner side of a flange **36** (refer to FIG. **17**), which will be described later. Part of the insulation portion **13** located closer to the drive conductive portion **60**B than the front-half insulation portion **13**U in the recess **35***b* in the X-direction extends through the substrate **10**B in the Z-direction.

The two recesses **35***c* are recessed from the end **31***a* of the second common contact front surface portion **30***d* in the X-direction. The bottom of each recess **35***c* is defined by a curved surface. In the present embodiment, in plan view, the recess **35***c* is concave and has a width that decreases toward the bottom. Although not illustrated, in the same manner as the two recesses **35***b*, the two recesses **35***c* are recessed in the Z-direction from the second common contact front surface portion **30***d* in the front layer part of the common conductive portion **30**B in the Z-direction. The two recesses **35***c* accommodate an insulation portion **13**. The shape of the two recesses **35**$c$ in plan view is symmetric to the shape of the two recesses **35**$b$ in plan view.

The dimensions of the two recesses **35**$b$ and **35**$c$ in the X-direction and the Y-direction may be changed in any manner. In an example, at least one of the dimensions of the two recesses **35**$b$ in the X-direction and the Y-direction may differ from that of the recesses **35**$c$.

As shown in FIG. **14**, the first common contact front surface portion **30**$c$ includes an end **31**$d$ located toward the third side wall **21**$c$. The recess **35**$d$ is recessed from the end **31**$d$ toward the fourth side wall **21**$d$ in the Y-direction. The bottom of the recess **35**$d$ is defined by a curved surface. In the present embodiment, in plan view, the recess **35**$d$ is concave and has a width that decreases toward the bottom. In the present embodiment, the recess **35**$d$ extends to the inner surface of the third side wall **21**$c$ as viewed in the Z-direction. As shown in FIG. **20**, the recess **35**$d$ is recessed in the Z-direction from the first common contact front surface portion **30**$c$ in a front layer part of the common conductive portion **30**B in the Z-direction. The recess **35**$d$ accommodates an insulation portion **13**. The insulation portion **13** accommodated in the recess **35**$d$ includes a front-half insulation portion **13**U that does not extend through the substrate **10**B in the Z-direction. The front-half insulation portion **13**U in the recess **35**$d$ is the portion shown in FIG. **15** surrounded by the recess **35**$d$ and the broken line and located at the inner side of the flange **36** (refer to FIG. **17**). Part of the insulation portion **13** located at the left of the front-half insulation portion **13**U of the recess **35**$a$ extends through the substrate **10**B in the Z-direction. The dimensions of the recess **35**$d$ in the X-direction and the Y-direction may be changed in any manner. In an example, as viewed in the Z-direction, the inner edge of the recess **35**$d$ may be located at an inner side of the inner surface of the third side wall **21**$c$ or at an outer side of the inner surface of the third side wall **21**$c$.

As shown in FIG. **14**, the first common contact front surface portion **30**$c$ includes an end **31**$e$ located toward the second side wall **21**$b$. The recess **35**$e$ is recessed from the end **31**$e$ in the Y-direction. The bottom of the recess **35**$e$ is defined by a curved surface. In the present embodiment, in plan view, the recess **35**$e$ is concave and has a width that decreases toward the bottom. As shown in FIG. **21**, the recess **35**$e$ is recessed in the Z-direction from the first common contact front surface portion **30**$c$ in a front layer part of the common conductive portion **30**B in the Z-direction. The recess **35**$e$ accommodates an insulation portion **13**. The insulation portion **13** accommodated in the recess **35**$e$ includes a front-half insulation portion **13**U that does not extend through the substrate **10**B in the Z-direction. The front-half insulation portion **13**U in the recess **35**$e$ is the portion shown in FIG. **15** surrounded by the recess **35**$e$ and the broken line and located at the inner side of a flange **36** (refer to FIG. **19**). Part of the insulation portion **13** located below the front-half insulation portion **13**U of the recess **35**$e$ extends through the substrate **10**B in the Z-direction. The dimensions of the recess **35**$e$ in the X-direction and the Y-direction may be changed in any manner. In an example, as viewed in the Z-direction, the inner edge of the recess **35**$e$ may be located at an inner side of the inner surface of the third side wall **21**$c$ or at an outer side of the inner surface of the third side wall **21**$c$.

In plan view, the shape of the recesses **35**$a$ to **35**$e$ may be changed in any manner. In an example, in plan view, the recesses **35**$a$ to **35**$e$ are rectangular. In plan view, the recesses **35**$a$ to **35**$e$ may have different depths.

As shown in FIGS. **20** to **22**, in a cross-sectional view of the common conductive portion **30**B cut along a plane extending in the Z-direction, the flange **36** is formed in a peripheral edge of the common contact front surface **30**$a$. In other words, the common conductive portion **30**B includes a back layer part including the common contact back surface **30**$b$ and located closer to the common contact back surface **30**$b$ than the front layer part including the common contact front surface **30**$a$. The back layer part is recessed toward the front layer part in a direction orthogonal to the Z-direction. The recessed part includes an insulation portion **13**. The insulation portion **13** accommodated between the flange **36** and the common contact back surface **30**$b$ includes a back-half insulation portion **13**L that does not extend through the substrate **10**B in the Z-direction. The part of the insulation portion **13** located at an outer side of the flange **36** extends through the substrate **10**B in the Z-direction and is continuous with the back-half insulation portions **13**L of the flange **36**.

The peripheral edge of the common contact front surface **30**$a$ includes the projections **34**$a$ to **34**$f$ (refer to FIGS. **14** and **15**). The projections **34**$a$ to **34**$f$ extend from the flange **36** and have the same thickness as the flange **36**. That is, the insulation portion **13** is accommodated between the projections **34**$a$ to **34**$f$ and the common contact back surface **30**$b$, so that the projections **34**$a$ to **34**$f$ are not exposed from the substrate back surface **12**. The insulation portion **13** accommodated between the projections **34**$a$ to **34**$f$ and the common contact back surface **30**$b$ includes a back-half insulation portion **13**L that does not extend through the substrate **10**B in the Z-direction. As viewed in the Z-direction, the insulation portion **13** extends through the substrate **10**B and is continuous with the back-half insulation portions **13**L of the projections **34**$a$ to **34**$f$ at opposite sides of the projections **34**$a$ to **34**$f$ in a direction orthogonal to the direction in which the projections **34**$a$ to **34**$f$ extend. The common contact back surface **30**$b$ is exposed from the substrate back surface **12** as a part located at an inner side of the flange **36** of the common contact front surface **30**$a$. Although not illustrated, the recesses **35**$a$ to **35**$e$ extend to an inner side of the common contact front surface **30**$a$ beyond the peripheral edge of the common contact front surface **30**$a$ (flange **36**). That is, the recesses **35**$a$ to **35**$e$ partially overlap the common contact back surface **30**$b$ as viewed in the Z-direction. Thus, the front-half insulation portions **13**U in the recesses **35**$a$ to **35**$e$ partially overlap the common contact back surface **30**$b$ as viewed in the Z-direction.

As shown in FIGS. **14** and **15**, the element contact front surface **50**$a$ is located at a corner of the substrate **10**B located toward the first side wall **21**$a$ and the fourth side wall **21**$d$. The element contact front surface **50**$a$ is separate from the first common contact front surface portion **30**$c$ in the X-direction. The insulation portion **13** is arranged around the element contact front surface **50**$a$.

As shown in FIG. **14**, as viewed in the Y-direction, a portion of the element contact front surface **50**$a$ located toward the third side wall **21**$c$ overlaps a portion of the second common contact front surface portion **30**$d$ located toward the fourth side wall **21**$d$. The element contact front surface **50**$a$ is located in a recessed region formed by displacing the first common contact front surface portion **30**$c$ from the second common contact front surface portion **30**$d$ in the X-direction. The element contact front surface **50**$a$ faces the first common contact front surface portion **30**$c$ in the X-direction.

As shown in FIG. **15**, in plan view, the element contact front surface **50**$a$ is rectangular so that the long sides extend in the X-direction and the short sides extend in the Y-direction. The dimension of the element contact front surface **50***a* in the Y-direction is less than the dimension of the first common contact front surface portion **30***c* in the Y-direction. The dimension of the element contact front surface **50***a* in the X-direction is greater than the dimension of the first common contact front surface portion **30***c* in the X-direction.

As shown in FIG. **14**, as viewed in the Z-direction, an end of the element contact front surface **50***a* located toward the first side wall **21***a* extends to a position overlapping the first side wall **21***a* and includes portions projecting beyond the outer surface of the first side wall **21***a*. As viewed in the Z-direction, an end of the element contact front surface **50***a* located toward the fourth side wall **21***d* extends to a position overlapping the fourth side wall **21***d* and includes a portion projecting beyond the outer surface of the fourth side wall **21***d*.

More specifically, an end of the element contact front surface **50***a* in the Y-direction located toward the first side wall **21***a* includes projections **54***a* and **54***b*, and an end of the element contact front surface **50***a* in the X-direction located toward the fourth side wall **21***d* includes a projection **54***c*. As viewed in the Z-direction, the projections **54***a* and **54***b* project from the outer surface of the first side wall **21***a*. The projections **54***a* and **54***b* are separate from each other in the X-direction. The projection **54***a* is located closer to the fourth side wall **21***d* than the projection **54***b*. As viewed in the Z-direction, the projection **54***c* projects from the outer surface of the fourth side wall **21***d*. The projection **54***c* is formed on the central part of the element contact front surface **50***a* in the Y-direction. In the present embodiment, when a support lead supporting the element conductive portion **50**B is cut, residual portions of the lead frame are the projections **54***a* to **54***c*. The number of projections may be changed to any number. As viewed in the Z-direction, connection portions between the element contact front surface **50***a* and the projections **54***a* to **54***c* each have a curved surface (refer to FIG. **15**).

The element conductive portion **50**B includes a recess **55** that restricts movement of the element conductive portion **50**B. In the present embodiment, the recess **55** is formed in the element contact front surface **50***a*. The number of recesses may be changed in any manner.

As shown in FIG. **14**, the recess **55** is formed in the element contact front surface **50***a* at an end located toward the first side wall **21***a* and the third side wall **21***c*. The element contact front surface **50***a* has an end **51***a* located toward the first side wall **21***a*. The recess **55** is recessed from the end **51***a* toward the second side wall **21***b* in the Y-direction. The dimension of the recess **55** in the Y-direction (depth of the recess **55***a* in plan view) is greater (deeper) than the dimension of the recesses **35***a*, **35***d*, and **35***e* in the Y-direction and the dimension of the recesses **35***b* and **35***c* in the X-direction (depth of the recesses **35***a* to **35***e* in plan view). The bottom of the recess **55** is defined by a curved surface. In the present embodiment, in plan view, the recess **55** partially extends in the Y-direction with a fixed width and has a curved surface that reduces the width toward the bottom. In the present embodiment, as viewed in the Z-direction, the recess **55** extends inward beyond the inner surface of the first side wall **21***a*. The recess **55** is used as a mark of the position where the second bonding portions of the wires **W1** are formed.

As shown in FIG. **20**, the recess **55** is recessed in the Z-direction from the element contact front surface **50***a* in a front layer part of the element conductive portion **50**B in the Z-direction. The recess **55** accommodates in an insulation portion **13**. The insulation portion **13** accommodated in the recess **55** includes a front-half insulation portion **13**U that does not extend through the substrate **10**B in the Z-direction. The front-half insulation portion **13**U in the recess **55** is the portion shown in FIG. **15** surrounded by the recess **55** and the broken line, and located at an inner side of a flange **56** (refer to FIG. **20**), which will be described later. Part of the insulation portion **13** located above the front-half insulation portion **13**U of the recess **55** extends through the substrate **10**B in the Z-direction. In the present embodiment, the depth of the recess **55** in the Z-direction is equal to the depth of the recesses **35***a* to **35***e* in the Z-direction (refer to FIG. **15**). For example, when the difference between the depth of the recess **55** in the Z-direction and the depth of the recesses **35***a* to **35***e* in the Z-direction is within 5% of the depth of the recesses **35***a* to **35***e* in the Z-direction, the depth of the recess **55** in the Z-direction is considered to be equal to the depth of the recesses **35***a* to **35***e* in the Z-direction.

The dimensions of the recess **55** in the X-direction and the Y-direction, which are shown in FIG. **14**, may be changed in any manner. In an example, as viewed in the Z-direction, the inner edge of the recess **55** may be located at the same position as the inner surface of the first side wall **21***a* or at an outer side of the inner surface of the first side wall **21***a*. The direction of the recess **55** may be changed in any manner. In an example, the recess **55** may be recessed in the Y-direction toward the fourth side wall **21***d* from an end **51***b* of the element contact front surface **50***a* that is located toward the third side wall **21***c*.

As shown in FIG. **20**, in a cross-sectional view of the element conductive portion **50**B cut along a plane extending in the Z-direction, the flange **56** is formed in the peripheral edge of the element contact front surface **50***a*. In other words, the element conductive portion **50**B includes a back layer part including the element contact back surface **50***b* and located closer to the element contact back surface **50***b* than the element contact front surface **50***a* including the front layer part. The back layer part is recessed toward the front layer part in a direction orthogonal to the Z-direction. The recessed part accommodates an insulation portion **13**. The insulation portion **13** accommodated between the flange **56** and the element contact back surface **50***b* includes a back-half insulation portion **13**L that does not extend through the substrate **10**B in the Z-direction. The part of the insulation portion **13** located at an outer side of the flange **56** extends through the substrate **10**B in the Z-direction and is continuous with the back-half insulation portion **13**L of the flange **56**.

The peripheral edge of the element contact front surface **50***a* includes the projections **54***a* to **54***c* (refer to FIGS. **14** and **15**). The projections **54***a* to **54***c* are exposed in the substrate front surface **11** and side surfaces of the substrate **10**B. The projections **54***a* to **54***c* extend from the flange **56** and have the same thickness as the flange **56**. That is, the insulation portion **13** is accommodated between the projections **54***a* to **54***c* and the element contact back surface **50***b*, so that the projections **54***a* to **54***c* are not exposed from the substrate back surface **12**. The insulation portion **13** accommodated between the projections **54***a* to **54***c* and the element contact back surface **50***b* includes a back-half insulation portion **13**L that does not extend through the substrate **10**B in the Z-direction. As viewed in the Z-direction, the insulation portion **13** extends through the substrate **10**B and is continuous with the back-half insulation portions **13**L of the projections **54***a* to **54***c* at opposite sides of the projections **54***a* to **54***c* in a direction orthogonal to the direction in which the projections **54***a* to **54***c* extend. The element contact back surface **50***b* is exposed from the substrate back surface **12** as a part located at an inner side of the flange **56** of the element contact front surface **50***a*. Although not illustrated, the recess **55** extends to an inner side of the element contact front surface **50***a* beyond the peripheral edge of the element contact front surface **50***a* (flange **56**). That is, the recess **55** partially overlaps the element contact back surface **50***b* as viewed in the Z-direction. Thus, the front-half insulation portion **13**U in the recess **55** partially overlaps the element contact back surface **50***b* as viewed in the Z-direction.

As shown in FIGS. **14** and **15**, the drive contact front surface **60***a* is aligned with the element contact front surface **50***a* in the X-direction and is separate from the element contact front surface **50***a* in the Y-direction. As shown in FIG. **14**, the drive contact front surface **60***a* surrounds the second common contact front surface portion **30***d* from the fourth side wall **21***d* in the X-direction and the first side wall **21***a* in the Y-direction. In plan view, the drive contact front surface **60***a* is generally L-shaped. As shown in FIG. **15**, the insulation portion **13** is arranged around the drive contact front surface **60***a*.

The drive contact front surface **60***a* includes a first drive contact front surface portion **60***c* and a second drive contact front surface portion **60***d*. The first drive contact front surface portion **60***c* and the second drive contact front surface portion **60***d* are integrally formed. The first drive contact front surface portion **60***c* and the second drive contact front surface portion **60***d* are arranged in the Y-direction. As shown in FIG. **14**, the first drive contact front surface portion **60***c* is located closer to the first side wall **21***a* than the second drive contact front surface portion **60***d* in the Y-direction. In other words, the second drive contact front surface portion **60***d* is located closer to the second side wall **21***b* than the first drive contact front surface portion **60***c* in the Y-direction. That is, the first drive contact front surface portion **60***c* is located between the element contact front surface **50***a* and the second drive contact front surface portion **60***d* in the Y-direction. As viewed in the X-direction, the first drive contact front surface portion **60***c* overlaps the first common contact front surface portion **30***c*. The first drive contact front surface portion **60***c* is located closer to the first side wall **21***a* than the second common contact front surface portion **30***d*. As viewed in the X-direction, the second drive contact front surface portion **60***d* overlaps the second common contact front surface portion **30***d*.

As shown in FIG. **15**, the first drive contact front surface portion **60***c* extends in the X-direction, and the second drive contact front surface portion **60***d* extends in the Y-direction. In plan view, the first drive contact front surface portion **60***c* is rectangular so that the long sides extend in the X-direction and the short sides extend in the Y-direction. In plan view, the second drive contact front surface portion **60***d* is rectangular so that the long sides extend in the Y-direction and the short sides extend in the X-direction. As shown in FIG. **14**, the second drive contact front surface portion **60***d* extends toward the second side wall **21***b* from an end of the first drive contact front surface portion **60***c* that is located toward the fourth side wall **21***d*. Thus, the second drive contact front surface portion **60***d* is recessed from the first drive contact front surface portion **60***c* so as to have a smaller dimension than the first drive contact front surface portion **60***c* in the X-direction. More specifically, the drive contact front surface **60***a* includes a recessed region **60***r* defined by the first drive contact front surface portion **60***c* and the second drive contact front surface portion **60***d*. The recessed region **60***r* accommodates the second common contact front surface portion **30***d*. The second common contact front surface portion **30***d* faces the second drive contact front surface portion **60***d* in the X-direction. The part of the second common contact front surface portion **30***d* received in the recessed region **60***r* overlaps the first drive contact front surface portion **60***c* as viewed in the Y-direction.

The first drive contact front surface portion **60***c* faces the first common contact front surface portion **30***c* in the X-direction and faces the element contact front surface **50***a* in the Y-direction. The dimension of the first drive contact front surface portion **60***c* in the X-direction is equal to the dimension of the element contact front surface **50***a* in the X-direction. For example, when the difference between the dimension of the first drive contact front surface portion **60***c* in the X-direction and the dimension of the element contact front surface **50***a* in the X-direction is within 5% of the dimension of the element contact front surface **50***a* in the X-direction, the dimension of the first drive contact front surface portion **60***c* in the X-direction is considered to be equal to the dimension of the element contact front surface **50***a* in the X-direction.

As shown in FIG. **14**, as viewed in the Z-direction, an end of the drive contact front surface **60***a* located toward the fourth side wall **21***d* extends to a position overlapping the fourth side wall **21***d* and includes portions projecting beyond the outer surface of the fourth side wall **21***d*. As viewed in the Z-direction, an end of the drive contact front surface **60***a* located toward the second side wall **21***b* extends to a position overlapping the second side wall **21***b* and includes a portion projecting beyond the outer surface of the second side wall **21***b*.

More specifically, an end of the drive contact front surface **60***a* located toward the fourth side wall **21***d* in the X-direction includes projections **64***a* and **64***b*, and an end of the drive contact front surface **60***a* located toward the second side wall **21***b* in the Y-direction includes a projection **64***c*. As viewed in the Z-direction, the projections **64***a* and **64***b* project from the outer surface of the fourth side wall **21***d*. The projections **64***a* and **64***b* are separate from each other in the Y-direction. The projection **64***a* is located closer to the element contact front surface **50***a* than the projection **64***b*. In the present embodiment, the projection **64***a* is formed on the first drive contact front surface portion **60***c*, and the projection **64***b* is formed on the second drive contact front surface portion **60***d*. As viewed in the Z-direction, the projection **64***c* projects from the outer surface of the second side wall **21***b*. The projection **64***c* is formed on the second drive contact front surface portion **60***d*. The projection **64***c* is formed on an end of the second drive contact front surface portion **60***d* located toward the second common contact front surface portion **30***d* in the X-direction. In the present embodiment, when a support lead supporting the drive conductive portion **60**B is cut, residual portions of the lead frame are the projections **64***a* to **64***c*. The first drive contact front surface portion **60***c* includes the projection **64***a*. The second drive contact front surface portion **60***d* includes the projections **64***b* and **64***c*. The number of projections may be changed to any number. As viewed in the Z-direction, connection portions between the drive contact front surface **60***a* and the projections **64***a* to **64***c* each have a curved surface (refer to FIG. **15**).

The drive conductive portion **60**B includes a recess **65** that restricts movement of the drive conductive portion **60**B. In the present embodiment, the recess **65** is formed in the drive contact front surface **60***a*. The number of recesses may be changed in any manner.

The recess **65** is formed in an end of the second drive contact front surface portion **60***d* located toward the first drive contact front surface portion **60***c* in the Y-direction. The second drive contact front surface portion **60***d* includes an end **61***a* located toward the third side wall **21***c*. The recess **65** is recessed from the end **61***a* toward the fourth side wall **21***d*. The recess **65** is located adjacent to the first drive contact front surface portion **60***c* in the Y-direction. More specifically, the first drive contact front surface portion **60***c* includes an end **61***b* located toward the second side wall **21***b*, and the recess **65** is partially defined by the end **61***b*. In the present embodiment, the maximum width of the recess **65** is equal to a gap between the first drive contact front surface portion **60***c* and the second common contact front surface portion **30***d* in the Y-direction. For example, when the difference between the maximum width of the recess **65** and the gap is within 5% of the gap, the maximum width of the recess **65** is considered to be equal to the gap between the first drive contact front surface portion **60***c* and the second common contact front surface portion **30***d* in the Y-direction.

The recess **65** extends through the drive conductive portion **60**B in the Z-direction. The recess **65** accommodates an insulation portion **13**. In other words, the insulation portion **13** accommodated in the recess **65** extends through the substrate **10**B in the Z-direction, which is different from the recesses **35***a* to **35***e* and **55**.

As shown in FIG. **21**, in a cross-sectional view of the drive conductive portion **60**B cut along a plane extending in the Z-direction, a flange **66** is formed in the peripheral edge of the drive contact front surface **60***a*. In other words, the drive conductive portion **60**B includes a back layer part including the drive contact back surface **60***b* and located closer to the drive contact back surface **60***b* than a front layer part including the drive contact front surface **60***a*. The back layer part is recessed toward the front layer part in a direction orthogonal to the Z-direction. The recessed part accommodates an insulation portion **13**. The insulation portion **13** accommodated between the flange **66** and the drive contact back surface **60***b* includes a back-half insulation portion **13**L that does not extend through the substrate **10**B in the Z-direction. The part of the insulation portion **13** located at an outer side of the flange **66** extends through the substrate **10**B in the Z-direction. The peripheral edge of the drive contact front surface **60***a* includes the projections **64***a* to **64***c* (refer to FIGS. **14** and **15**). The projections **64***a* to **64***c* extend from the flange **66** and have the same thickness as the flange **66**. That is, the insulation portion **13** is accommodated between the projections **64***a* to **64***c* and the drive contact back surface **60***b*, so that the projections **64***a* to **64***c* are not exposed from the substrate back surface **12**. The insulation portion **13** accommodated between the projections **64***a* to **64***c* and the drive contact back surface **60***b* includes a back-half insulation portion **13**L that does not extend through the substrate **10**B in the Z-direction. As viewed in the Z-direction, the insulation portion **13** extends through the substrate **10**B and is continuous with the back-half insulation portion **13**L of the projections **64***a* to **64***c* at opposite sides of the projections **64***a* to **64***c* in a direction orthogonal to the direction in which the projections **64***a* to **64***c* extend. The drive contact back surface **60***b* is exposed from the substrate back surface **12** as a part located at an inner side of the flange **66** of the drive contact front surface **60***a*. Although not illustrated, the recess **65** extends to an inner side of the drive contact front surface **60***a* beyond the peripheral edge of the drive contact front surface **60***a* (flange **66**).

As shown in FIGS. **14** and **15**, the control contact front surface **70***a* is located at a corner of the substrate **10**B located toward the second side wall **21***b* and the third side wall **21***c*. The control contact front surface **70***a* is separate from the second common contact front surface portion **30***d* in the X-direction. The control contact front surface **70***a* is separate from the first common contact front surface portion **30***c* in the Y-direction. As viewed in the X-direction, the control contact front surface **70***a* overlaps the second common contact front surface portion **30***d*. That is, the control contact front surface **70***a* faces the second common contact front surface portion **30***d* in the X-direction. As viewed in the Y-direction, the control contact front surface **70***a* overlaps the first common contact front surface portion **30***c*. More specifically, in the Y-direction, the control contact front surface **70***a* faces the part of the first common contact front surface portion **30***c* projecting from the second common contact front surface portion **30***d* toward the third side wall **21***c*. Thus, the control contact front surface **70***a* is formed in a region surrounded by the first common contact front surface portion **30***c* and the second common contact front surface portion **30***d*. In other words, the control contact front surface **70***a* is located in a recessed region formed by displacing the first common contact front surface portion **30***c* from the second common contact front surface portion **30***d* in the X-direction.

As shown in FIGS. **14** and **15**, the second common contact front surface portion **30***d* is located between the drive conductive portion **60**B and the control conductive portion **70**B in the X-direction. In other words, the second common contact front surface portion **30***d* is located between the second drive contact front surface portion **60***d* and the control contact front surface **70***a* in the X-direction.

As shown in FIG. **15**, in plan view, the control contact front surface **70***a* is rectangular so that the long sides extend in the Y-direction and the short sides extend in the X-direction. The dimension of the control contact front surface **70***a* in the Y-direction is less than the dimension of the first common contact front surface portion **30***c* in the Y-direction. The dimension of the control contact front surface **70***a* in the X-direction is less than the dimension of the first common contact front surface portion **30***c* in the X-direction.

As viewed in the Z-direction, an end of the control contact front surface **70***a* located toward the second side wall **21***b* extends to a position overlapping the second side wall **21***b* and includes a portion projecting beyond the outer surface of the second side wall **21***b*. As viewed in the Z-direction, an end of the control contact front surface **70***a* located toward the third side wall **21***c* extends to a position overlapping the third side wall **21***c* and includes a portion projecting beyond the outer surface of the third side wall **21***c*.

More specifically, an end of the control contact front surface **70***a* located toward the second side wall **21***b* in the Y-direction includes a projection **74***a*, and an end of the control contact front surface **70***a* located toward the third side wall **21***c* in the X-direction includes a projection **74***b*. As viewed in the Z-direction, the projection **74***a* projects from the outer surface of the second side wall **21***b*. The projection **74***a* is formed on the central part of the control contact front surface **70***a* in the Y-direction. As viewed in the Z-direction, the projection **74***b* projects from the outer surface of the third side wall **21***c*. The projection **74***b* is formed on an end of the control contact front surface **70***a* located toward the second common contact front surface portion **30***d* in the X-direction. In the present embodiment, when a support lead supporting the control conductive portion **70**B is cut, residual portions of the lead frame are the projections **74***a* and **74***b*. The number of projections may be changed to any number. As viewed in the Z-direction, connection portions between the control contact front surface **70***a* and the projections **74***a* and **74***b* each have a curved surface (refer to FIG. **15**).

In a cross-sectional view of the control conductive portion **70**B cut along a plane extending in the Z-direction, a flange **76** (refer to FIG. **18**) is formed in the peripheral edge of the control contact front surface **70***a*. In other words, the control conductive portion **70**B includes a back layer part including the control contact back surface **70***b* and located closer to the control contact back surface **70***b* than a front layer part including the control contact front surface **70***a*, the back layer part is recessed toward the front layer part in the X-direction and the Y-direction. The recessed part accommodates an insulation portion **13**. The insulation portion **13** accommodated between the flange **76** of the control contact front surface **70***a* and the control contact back surface **70***b* includes a back-half insulation portion **13**L that does not extend through the substrate **10**B in the Z-direction. The part of the insulation portion **13** located at an outer side of the flange **76** extends through the substrate **10**B in the Z-direction. The peripheral edge of the control contact front surface **70***a* includes the projections **74***a* and **74***b*. The projections **74***a* and **74***b* extend from the flange **76** and have the same thickness as the flange **76**. That is, the insulation portion **13** is accommodated between the projections **74***a* and **74***b* and the control contact back surface **70***b*, so that the projections **74***a* and **74***b* are not exposed from the substrate back surface **12**. The insulation portion **13** accommodated between the projections **74***a* and **74***b* and the control contact back surface **70***b* includes a back-half insulation portion **13**L that does not extend through the substrate **10**B in the Z-direction. As viewed in the Z-direction, the insulation portion **13** extends through the substrate **10**B and is continuous with the back-half insulation portion **13**L of the projections **74***a* and **74***b* at opposite sides of the projections **74***a* and **74***b* in a direction orthogonal to the direction in which the projections **74***a* and **74***b* extend. The control contact back surface **70***b* is exposed from the substrate back surface **12** as a part located at an inner side of the flange **76**.

As shown in FIG. **15**, in the present embodiment, the common contact front surface **30***a* is larger than the other contact front surfaces **50***a*, **60***a*, and **70***a*. The first common contact front surface portion **30***c* is larger than the other contact front surfaces **50***a*, **60***a*, and **70***a*. The second common contact front surface portion **30***d* is larger than the control contact front surface **70***a*. The element contact front surface **50***a* is larger than the control contact front surface **70***a*. The drive contact front surface **60***a* is larger than the control contact front surface **70***a*.

The layout of the conductive portions **30**B, **50**B, **60**B, and **70**B at the side of the substrate back surface **12** will now be described.

As shown in FIG. **16**, the common contact back surface **30***b* includes a first common contact back surface portion **30***e* and a second common contact back surface portion **30***f* The first common contact back surface portion **30***e* is separate from the second common contact back surface portion **30***f* in the Y-direction. The insulation portion **13** is disposed between the first common contact back surface portion **30***e* and the second common contact back surface portion **30***f*.

More specifically, as shown in FIG. **16**, in the Y-direction, a part (indicated by broken lines) between the first common contact back surface portion **30***e* and the second common contact back surface portion **30***f* includes a recess **35***f* that is recessed from the substrate back surface **12** toward the substrate front surface **11**. The recess **35***f* accommodates an insulation portion **13**. The insulation portion **13** accommodated in the recess **35***f* includes a back-half insulation portion **13**L that does not extend through the substrate **10**B in the Z-direction. In other words, the back-half insulation portion **13**L separates the first common contact back surface portion **30***e* and the second common contact back surface portion **30***f* The back-half insulation portion **13**L extends through the common contact back surface **30***b* in the X-direction.

As shown in FIG. **16**, the insulation portions **13** arranged around the common conductive portion **30**B extend through the substrate **10**B. The back-half insulation portion **13**L arranged in the recess **35***f* is continuous with the insulation portions **13** located at left and right sides of the common conductive portion **30**B. More specifically, the back-half insulation portion **13**L arranged in the recess **65***a* connects the insulation portion **13** extending through the substrate **10**B and located at the left side of the second common contact back surface portion **30***f* and at the lower side of the part of the first common contact back surface portion **30***e* projecting leftward from the second common contact back surface portion **30***f* to the insulation portion **13** extending through the substrate **10**B and arranged at the right side of the first common contact back surface portion **30***e* and the upper side of the second common contact back surface portion **30***f*.

The first common contact back surface portion **30***e* is formed on the substrate back surface **12** at a position opposite to the first common contact front surface portion **30***c*. The first common contact back surface portion **30***e* is formed on an upper left portion as viewed from the substrate back surface **12**. In the same manner as the first common contact front surface portion **30***c*, the first common contact back surface portion **30***e* is rectangular so that the short sides extend in the X-direction and the long sides extend in the Y-direction. As viewed in the Z-direction, the first common contact back surface portion **30***e* is smaller than the first common contact front surface portion **30***c*.

The second common contact back surface portion **30***f* is formed on the substrate back surface **12** at a location opposite to the second common contact front surface portion **30***d*. The second common contact back surface portion **30***f* is formed on a central part of the substrate back surface **12** in the X-direction and a lower part of the substrate back surface **12** in the Y-direction. In the same manner as the second common contact front surface portion **30***d*, the second common contact back surface portion **30***f* is rectangular so that the long sides extend in the X-direction and the short sides extend in the Y-direction. As viewed in the Z-direction, the second common contact back surface portion **30***f* is smaller than the second common contact front surface portion **30***d*.

The element contact back surface **50***b* and the drive contact back surface **60***b* are located at the right side of the first common contact back surface portion **30***e*.

The element contact back surface **50***b* is formed on the substrate back surface **12** at a position opposite to the element contact front surface **50***a*. The element contact back surface **50***b* is formed on a left upper portion of the substrate back surface **12**. The element contact back surface **50***b* is separate from the common contact back surface **30***b* and is not continuous with the common contact back surface **30***b*. That is, the insulation portion **13** is disposed between the element contact back surface **50***b* and the common contact back surface **30***b*. The insulation portion **13** extends in the Y-direction.

The element contact back surface **50***b* is rectangular so that the long sides extend in the X-direction and the short sides extend in the Y-direction. As viewed in the Z-direction, the element contact back surface **50***b* is smaller than the element contact front surface **50***a*. The dimension of the element contact back surface **50***b* in the X-direction is equal to the dimension of the first common contact back surface portion **30***e* in the X-direction. When the difference between the dimension of the element contact back surface **50***b* in the X-direction and the dimension of the first common contact back surface portion **30***e* in the X-direction is within 5% of the dimension of the first common contact back surface portion **30***e* in the X-direction, the dimension of the element contact back surface **50***b* in the X-direction is equal to the dimension of the first common contact back surface portion **30***e* in the X-direction.

The drive contact back surface **60***b* includes a first drive contact back surface portion **60***e* and a second drive contact back surface portion **60***f*. The first drive contact back surface portion **60***e* is separate from the second drive contact back surface portion **60***f* in the Y-direction. The insulation portion **13** is disposed between the first drive contact back surface portion **60***e* and the second drive contact back surface portion **60***f*.

More specifically, as shown in FIG. **16**, the recess **65***a* is arranged in the drive conductive portion **60**B between the first drive contact back surface portion **60***e* and the second drive contact back surface portion **60***f* and is recessed from the substrate back surface **12** toward the substrate front surface **11**. The recess **65***a* does not extend through the substrate **10**B. The recess **65***a* accommodates a back-half insulation portion **13**L that does not extend through the substrate **10**B in the Z-direction. The back-half insulation portion **13**L extends through the drive contact back surface **60***b* in the X-direction. The recess **65** is continuous with the recess **65***a* in the Z-direction.

As shown in FIG. **16**, the insulation portions **13** arranged around the drive conductive portion **60**B extend through the substrate **10**B. The back-half insulation portion **13**L arranged in the recess **65***a* is continuous with the insulation portion **13** arranged at left and right sides of the drive conductive portion **60**B. More specifically, the back-half insulation portion **13**L arranged in the recess **65***a* connects the insulation portion **13** extending through the substrate **10**B and arranged at the left side of the second drive contact back surface portion **60***f* and at a lower side of the portion of the first drive contact back surface portion **60***e* projecting leftward from the second drive contact back surface portion **60***f* to the insulation portion **13** extending through the substrate **10**B and arranged at the right side of the first drive contact back surface portion **60***e* and the second drive contact back surface portion **60***f*.

The first drive contact back surface portion **60***e* is formed on the substrate back surface **12** at a position opposite to the first drive contact front surface portion **60***c*. The first drive contact back surface portion **60***e* is formed on a right central part of the substrate back surface **12**. The first drive contact back surface portion **60***e* is aligned with the element contact back surface **50***b* in the X-direction and is separate from the element contact back surface **50***b* in the Y-direction. The insulation portion **13** is disposed between the first drive contact back surface portion **60***e* and the element contact back surface **50***b*. The insulation portion **13** extends in the X-direction and is continuous with the insulation portion **13** that is disposed between the element contact back surface **50***b* and the common contact back surface **30***b*.

In the same manner as the first drive contact front surface portion **60***c*, the first drive contact back surface portion **60***e* is rectangular so that the long sides extend in the X-direction and the short sides extend in the Y-direction. As viewed in the Z-direction, the first drive contact back surface portion **60***e* is smaller than the first drive contact front surface portion **60***c*. The dimension of the first drive contact back surface portion **60***e* in the X-direction is equal to the dimension of the element contact back surface **50***b* in the X-direction.

The lower edge of the first drive contact back surface portion **60***e* is aligned with the lower edge of the first common contact back surface portion **30***e* in the Y-direction. Thus, the distance from the upper edge of the element contact back surface **50***b* to the lower edge of the first drive contact back surface portion **60***e* is equal to the dimension of the first common contact back surface portion **30***e* in the X-direction. When the difference between the distance from the upper edge of the element contact back surface **50***b* to the lower edge of the first drive contact back surface portion **60***e* and the dimension of the first common contact back surface portion **30***e* in the X-direction is within 5% of the dimension of the first common contact back surface portion **30***e* in the X-direction, the distance from the upper edge of the element contact back surface **50***b* to the lower edge of the first drive contact back surface portion **60***e* is considered to be equal to the dimension of the first common contact back surface portion **30***e* in the X-direction.

The second drive contact back surface portion **60***f* is formed on the substrate back surface **12** at a position opposite to the second drive contact front surface portion **60***d*. In the same manner as the second drive contact front surface portion **60***d*, the second drive contact back surface portion **60***f* is rectangular so that the short sides extend in the X-direction and the long sides extend in the Y-direction. The second drive contact back surface portion **60***f* is formed on a right lower portion of the substrate back surface **12**.

The dimension of the second drive contact back surface portion **60***f* in the Y-direction is equal to the dimension of the second common contact back surface portion **30***f* in the Y-direction. The dimension of the second drive contact back surface portion **60***f* in the X-direction is slightly greater than the dimension of the first drive contact back surface portion **60***e* in the Y-direction. For example, when the difference between the dimension of the second drive contact back surface portion **60***f* in the Y-direction and the dimension of the second common contact back surface portion **30***f* in the Y-direction is within 5% of the dimension of the second common contact back surface portion **30***f* in the Y-direction, the dimension of the second drive contact back surface portion **60***f* in the Y-direction is considered to be equal to the dimension of the second common contact back surface portion **30***f* in the Y-direction.

As shown in FIG. **16**, the right edge of the element contact back surface **50***b*, the right edge of the first drive contact back surface portion **60***e*, and the right edge of the second drive contact back surface portion **60***f* are aligned with each other.

The control contact back surface **70***b* is formed on a left lower portion of the substrate back surface **12**. The control contact back surface **70***b* is formed on the substrate back surface **12** at a position opposite to the control contact front surface **70***a*. In the present embodiment, the arrangement direction of the element conductive portion **50**B and the drive conductive portion **60**B, that is, the Y-direction, may be referred to as a third direction. The X-direction, which is orthogonal to the Y-direction as viewed in the Z-direction, may be referred to as a fourth direction. In this case, the second drive contact back surface portion **60***f* and the control contact back surface **70***b* are separately disposed at opposite sides of the second common contact back surface portion **30***f* in the fourth direction.

In the same manner as the control contact front surface **70***a*, the control contact back surface **70***b* is rectangular so that the short sides extend in the X-direction and the long sides extend in the Y-direction. The dimension of the control contact back surface **70***b* in the Y-direction is equal to the dimension of the second common contact back surface portion **30***f* in the Y-direction. The dimension of the control contact back surface **70***b* in the X-direction is equal to the dimension of the second drive contact back surface portion **60***f* in the X-direction. That is, the control contact back surface **70***b* is identical in shame to the second drive contact back surface portion **60***f*.

As shown in FIG. **16**, the second common contact back surface portion **30***f*, the second drive contact back surface portion **60***f*, and the control contact back surface **70***b* are aligned in the Y-direction and separate from each other in the X-direction. The lower edge of the second drive contact back surface portion **60***f*, the lower edge of the second common contact back surface portion **30***f*, and the lower edge of the control contact back surface **70***b* are aligned with each other. The upper edge of the second drive contact back surface portion **60***f*, the upper edge of the second common contact back surface portion **30***f*, and the upper edge of the control contact back surface **70***b* are aligned with each other. Thus, the second drive contact back surface portion **60***f* and the control contact back surface **70***b* are symmetrical about the second common contact back surface portion **30***f* at the left and right sides.

As shown in FIG. **16**, each insulation portion **13** has a constant width between ones of the contact back surfaces **30***b*, **50***b*, **60***b*, and **70***b* that are adjacent in the X-direction or the Y-direction. In addition, the dimension of the insulation portion **13** in the X-direction disposed between the first common contact back surface portion **30***e* and the element contact back surface **50***b* and the first drive contact back surface portion **60***e* in the X-direction is equal to the dimension of the insulation portion **13** in the Y-direction disposed between the element contact back surface **50***b* and the first drive contact back surface portion **60***e* in the Y-direction. In addition, the dimension of the insulation portion **13** in the X-direction disposed between the second common contact back surface portion **30***f* and the second drive contact back surface portion **60***f* in the X-direction is equal to the dimension of the insulation portion **13** in the X-direction disposed between the second common contact back surface portion **30***f* and the control contact back surface **70***b* in the X-direction.

For example, when the difference between the width of a predetermined insulation portion **13** and the width of another insulation portion **13** is within 10% of the width of the predetermined insulation portion **13**, the widths of the two insulation portions **13** are considered to be equal.

The positional relationship among the conductive portions **30**B, **50**B, **60**B, and **70**B, the semiconductor light emitting element **80**, and the electronic component **100** will be described below in detail.

As shown in FIG. **14**, in the same manner as the first embodiment, the semiconductor light emitting element **80** and the electronic component **100** are mounted on the common conductive portion **30**B and electrically connected by the common conductive portion **30**B. In the present embodiment, as viewed in the Z-direction, the semiconductor light emitting element **80** is larger than the electronic component **100**. More specifically, as viewed in the Z-direction, the semiconductor light emitting element **80** is rectangular and arranged so that the long sides extend in the X-direction and the short sides extend in the Y-direction. As viewed in the Z-direction, the electronic component **100** is rectangular and arranged so that the long sides extend in the Y-direction and the short sides extend in the X-direction. The dimension of the semiconductor light emitting element **80** in the X-direction is greater than the dimension of the electronic component **100** in the X-direction. The dimension of the semiconductor light emitting element **80** in the Y-direction is greater than the dimension of the electronic component **100** in the Y-direction.

In the present embodiment, the semiconductor light emitting element **80** and the electronic component **100** are disposed on the common contact front surface **30***a*. More specifically, the semiconductor light emitting element **80** is disposed on the first common contact front surface portion **30***c*, and the electronic component **100** is disposed on the second common contact front surface portion **30***d*. Thus, the semiconductor light emitting element **80**, which is larger than the electronic component **100**, is disposed on the first common contact front surface portion **30***c*, which is larger than the second common contact front surface portion **30***d*. This ensures the space for each of the semiconductor light emitting element **80** and the electronic component **100**. In addition, the dimension of the first common contact front surface portion **30***c* in the X-direction is greater than the dimension of the semiconductor light emitting element **80** in the X-direction. This allows for the arrangement of the semiconductor light emitting element **80** even when the semiconductor light emitting element **80** is increased in size from the illustrated one. The versatility for the semiconductor light emitting device **1**B is increased.

In addition, in the present embodiment, the semiconductor light emitting element **80** and the electronic component **100** are displaced from each other in the X-direction and separated from each other in the Y-direction on the common contact front surface **30***a*, which differs from the first embodiment.

In the present embodiment, the semiconductor light emitting element **80** is located closer to the third side wall **21***c* than the electronic component **100**. The electronic component **100** is located closer to the fourth side wall **21***d* than the semiconductor light emitting element **80**. In other words, when the electronic component **100** is located closer to the fourth side wall **21***d* than the semiconductor light emitting element **80** in the X-direction, the electronic component **100** and the semiconductor light emitting element **80** are arranged in the Y-direction. The semiconductor light emitting element **80** is displaced from the central part of the substrate front surface **11** in the X-direction and is, for example, located closer to the third side wall **21***c* than the central part. The electronic component **100** is located on the central part of the substrate front surface **11** in the X-direction. In the present embodiment, as viewed in the Y-direction, the semiconductor light emitting element **80** overlaps the electronic component **100**.

The semiconductor light emitting element **80** is located closer to the third side wall **21***c* than the first common contact front surface portion **30***c* in the X-direction. In the illustrated example, the semiconductor light emitting element **80** is located closer to the fourth side wall **21***d* than the recess **35***e* of the first common contact front surface portion

30*c*. As viewed in the Y-direction, the end of the semiconductor light emitting element **80** located toward the third side wall **21***c* overlaps the end of the control contact front surface **70***a* located toward the fourth side wall **21***d*. As viewed in the Z-direction, the semiconductor light emitting element **80** is arranged so as not to overlap the recess **35***f* of the common conductive portion **30**B (refer to FIG. **19**). In the present embodiment, as shown in FIG. **19**, the semiconductor light emitting element **80** and the electronic component **100** are located at opposite sides of the recess **35***f* This structure limits decreases in the heat dissipation efficiency of the semiconductor light emitting element **80**.

As viewed in the Y-direction, the electronic component **100** partially overlaps the first common contact front surface portion **30***c*. As viewed in the Y-direction, the electronic component **100** partially overlaps the first drive contact front surface portion **60***c*. As viewed in the Z-direction, the electronic component **100** is arranged not to overlap the recess **35***f*. In the present embodiment, as shown in FIG. **19**, the electronic component **100** and the semiconductor light emitting element **80** are located at opposite sides of the recess **35***f*. This structure limits decreases in the heat dissipation efficiency of the electronic component **100**.

The semiconductor light emitting element **80** is disposed on the first common contact front surface portion **30***c*, and the electronic component **100** is disposed on the second common contact front surface portion **30***d*. Thus, the semiconductor light emitting element **80** is located closer to the first side wall **21***a* than the electronic component **100**, and the electronic component **100** is located closer to the second side wall **21***b* than the semiconductor light emitting element **80**. The semiconductor light emitting element **80** and the electronic component **100** are displaced from the central part of the substrate front surface **11** in the Y-direction. In the illustrated example, the semiconductor light emitting element **80** is disposed on the first common contact front surface portion **30***c* between the central part of the substrate front surface **11** and the first side wall **21***a*. In the present example, the semiconductor light emitting element **80** is located closer to the central part of the substrate front surface **11** than the first side wall **21***a* in the Y-direction. In other words, the semiconductor light emitting element **80** is located at a position of the first common contact front surface portion **30***c* located toward the second common contact front surface portion **30***d* in the Y-direction. The phrase "position of the first common contact front surface portion **30***c* located toward the second common contact front surface portion **30***d* in the Y-direction" refers to a position toward one of the opposite ends of the first common contact front surface portion **30***c* in the Y-direction located toward the second common contact front surface portion **30***d*. The electronic component **100** is disposed on the second common contact front surface portion **30***d* between the central part of the substrate front surface **11** and the second side wall **21***b*. In the present example, the electronic component **100** is located closer to the central part of the substrate front surface **11** than the second side wall **21***b* in the Y-direction. In other words, the electronic component **100** is located at a position of the second common contact front surface portion **30***d* located toward the first common contact front surface portion **30***c* in the Y-direction. Thus, the semiconductor light emitting element **80** and the electronic component **100** are located at opposite sides of the central part of the substrate front surface **11** in the Y-direction and separated by a short distance in the Y-direction.

At least a portion of the semiconductor light emitting element **80** is disposed on the first common contact front surface portion **30***c* at a position closer to the electronic component **100** than the element contact front surface **50***a*. In the present embodiment, the semiconductor light emitting element **80** is located closer to the second side wall **21***b* than the element contact front surface **50***a* in the Y-direction. More specifically, the semiconductor light emitting element **80** is located closer to the second side wall **21***b* than the central part of the element contact front surface **50***a* in the Y-direction. A portion of the semiconductor light emitting element **80** is located closer to the second side wall **21***b* than the element contact front surface **50***a* in the Y-direction. As viewed in the X-direction, the semiconductor light emitting element **80** overlaps the element contact front surface **50***a* and the first drive contact front surface portion **60***c*. More specifically, in the present embodiment, the central part of the semiconductor light emitting element **80** in the Y-direction is located closer to the first drive contact front surface portion **60***c* than the insulation portion **13** that is disposed between the element contact front surface **50***a* and the first drive contact front surface portion **60***c*.

As shown in FIG. **17**, in the same manner as the first embodiment, the element lower surface electrode **92** of the semiconductor light emitting element **80** is die-bonded to the first common contact front surface portion **30***c* by the conductive bonding material P**1**. Thus, the element lower surface electrode **92** is bonded to the common conductive portion **30**B.

As shown in FIG. **18**, in the same manner as the first embodiment, the second drive electrode **103** of the electronic component **100** is die-bonded to the second common contact front surface portion **30***d* by the conductive bonding material P**2**. Thus, the second drive electrode **103** is bonded to the common conductive portion **30**B. The common conductive portion **30**B electrically connects the element lower surface electrode **92** and the second drive electrode **103**. In the present embodiment, the conductive bonding material P**2** is formed from an Ag paste containing a large amount of Ag. This improves the efficiency of heat dissipation from the electronic component **100** to the common conductive portion **30**B.

As shown in FIG. **15**, in the same manner as the first embodiment, the element upper surface electrode **91** of the semiconductor light emitting element **80** is connected to the element contact front surface **50***a* by the wires W**1**. In other words, each of the wires W**1** is connected to the element upper surface electrode **91** and the element contact front surface **50***a* through wire bonding. Thus, the element upper surface electrode **91** is electrically connected to the element conductive portion **50**B. The number of wires W**1** is not particularly limited. In the illustrated example, five wires W**1** are arranged. In the illustrated example, the wire W**1** has a first bonding portion disposed on the element upper surface electrode **91** and a second bonding portion disposed on the element contact front surface **50***a*. The first bonding portions of the five wires W**1** are arranged on the element upper surface electrode **91** and separated from each other in the Y-direction. The second bonding portions of the five wires W**1** are arranged on the element contact front surface **50***a* and separated from each other in the Y-direction. The second bonding portions are arranged on one of the opposite ends of the element contact front surface **50***a* in the X-direction located toward the third side wall **21***c*. The second bonding portions are disposed closer to the second side wall **21***b* than the recess **55** of the element contact front surface **50***a* in the Y-direction so as not to be disposed closer to the fourth side wall **21***d* than the recess **55** in the X-direction. In the illustrated example, the second bonding portions are disposed closer to the third side wall **21***c* than the central part of the recess **55** in the X-direction and overlap the recess **55** as viewed in the Y-direction.

As shown in FIGS. **14** and **15**, the first bonding portions and the second bonding portions of the five wires W**1** are located at different positions in the Y-direction and separate from each other in the X-direction. In the illustrated example, as viewed in the X-direction, one of the five first bonding portions located closest to the first side wall **21***a* overlaps one of the five second bonding portions located closest to the second side wall **21***b*. Thus, in plan view, the wires W**1** diagonally extend from the element upper surface electrode **91** toward the element contact front surface **50***a* so as to extend away from the electronic component **100**. In other words, in plan view, the wires W**1** diagonally extend toward the first side wall **21***a* from the first bonding portions toward the second bonding portions. In plan view, the wires W**1** extend parallel to each other.

In the illustrated example, the element upper surface electrode **91** is formed on one of the opposite ends of the element upper surface **80***a* in the X-direction that is located closer to the element conductive portion **50**B. Thus, the wires W**1** are shortened.

In the same manner as the first embodiment, the first drive electrode **101** of the electronic component **100** is connected to the drive contact front surface **60***a* by the wires W**2**. In other words, each of the wires W**2** is connected to the first drive electrode **101** and the drive contact front surface **60***a* through wire bonding. Thus, the first drive electrode **101** is electrically connected to the drive conductive portion **60**B. The number of wires W**2** is not particularly limited. In the illustrated example, six wires W**2** are arranged. In the illustrated example, the wire W**2** has a first bonding portion disposed on the first drive electrode **101** and a second bonding portion disposed on the drive contact front surface **60***a*. The second bonding portions of the six wires W**2** are separate from each other in the Y-direction.

The wires W**2** are connected to the second drive contact front surface portion **60***d*. In other words, the second bonding portions are disposed on the second drive contact front surface portion **60***d*. More specifically, the second bonding portions are located closer to the second side wall **21***b* than the recess **65** of the drive contact front surface **60***a* in the Y-direction. The second bonding portions are also located closer to the second common contact front surface portion **30***d* (third side wall **21***c*) than the central part of the second drive contact front surface portion **60***d* in the X-direction.

Among the wires W**2**, the farthest two wires W**2**, that is, the two wires W**2** located at opposite ends in the Y-direction, are connected to the first drive electrode **101** and the second drive contact front surface portion **60***d* so that the two wires W**2** are separated from each other more at the second drive contact front surface portion **60***d* than at the first drive electrode **101** in plan view. In other words, the distance in the Y-direction between the two of the wires W**2** located at opposite ends in the Y-direction is larger at the second drive contact front surface portion **60***d* than at the first drive electrode **101**. In the present embodiment, the wires W**2** are arranged so that the gap between adjacent ones of the wires W**2** gradually increases from the first bonding portions toward the second bonding portions in plan view. In an example, the distance between the second bonding portions of the two of the wires W**2** located at opposite ends in the Y-direction is greater than the dimension of the electronic component **100** in the Y-direction.

In the illustrated example, the wires W**2** are longer than the wires W**1**. The wires W**2** have a larger diameter than the wires W**1**. The number of wires W**2** is greater than the number of wires W**1**. This structure improves the heat dissipation efficiency of the electronic component **100**. However, alternatively, the diameter of the wires W**2** may be less than or equal to the diameter of the wires W**1**. The number of wires W**2** may be equal to the number of wires W**1**. The number of wires W**2** may be less than the number of wires W**1**.

In the same manner as the first embodiment, the control electrode **102** of the electronic component **100** is connected to the control contact front surface **70***a* by the wire W**3**. In other words, the wire W**3** is connected to the control electrode **102** and the control contact front surface **70***a* through wire bonding. The number of wires W**3** is not particularly limited. In the illustrated example, the number of wires W**3** is one. In the illustrated example, the wire W**3** has a first bonding portion disposed on the control electrode **102** and a second bonding portion disposed on the control contact front surface **70***a*. In the present embodiment, the control electrode **102** is formed on one of the opposite ends of the upper surface **100***a* in the X-direction that is located closer to the control contact front surface **70***a*. Thus, the wire W**3** is shortened.

The electronic component **100** is covered by an encapsulation resin **140**. The encapsulation resin **140** is covered by a coating agent **141** that inhibits sulfidation of the first drive electrode **101** and the second drive electrode **103** of the electronic component **100**. The encapsulation resin **140** and the coating agent **141** cover the first bonding portions of the wires W**2** connected to the first drive electrode **101** and the first bonding portion of the wire W**3** connected to the control electrode **102**. The encapsulation resin **140** is, for example, formed from a light-blocking resin material. In the present embodiment, an epoxy resin is used. Thus, the encapsulation resin **140** is a light-blocking resin (light-blocking member) that shields the electronic component **100** from external light. The coating agent **141** is, for example, formed from a fluoropolymer and glass.

As shown in FIGS. **14** and **15**, the coating agent **141** (encapsulation resin **140**) extends over the second common contact front surface portion **30***d*. A portion of the coating agent **141** (encapsulation resin **140**) covers the first common contact front surface portion **30***c* and the first drive contact front surface portion **60***c*. The coating agent **141** (encapsulation resin **140**) is arranged separately from the semiconductor light emitting element **80**. The electronic component **100** may be covered by the coating agent **141**, and the coating agent **141** may be covered by the encapsulation resin **140**.

In the present embodiment, the capacitor **120** is disposed so as to span between the element contact front surface **50***a* and the first drive contact front surface portion **60***c*. In an example, the capacitor **120** is a tantalum capacitor. Alternatively, the capacitor **120** may be a multilayer ceramic capacitor. The capacitor **120** is located closer to the fourth side wall **21***d* than the recess **55** of the element contact front surface **50***a* in the X-direction. Since the recess **55** is the mark for the second bonding portions of the wires W**1**, that is, the connection end of the element contact front surface **50***a*, the capacitor **120** is located closer to the fourth side wall **21***d* than the second bonding portions. The capacitor **120** is arranged so as not to extend toward the fourth side wall **21***d* beyond the recess **65** of the drive contact front surface **60***a* in the X-direction. The capacitor **120** includes a first electrode **121** and a second electrode **122** arranged in the Y-direction. In other words, as shown in FIGS. **14** and **15**, the capacitor **120** is arranged so that the long sides extend in the Y-direction and the short sides extend in the X-direction. In the present embodiment, the first electrode **121** is connected to the element contact front surface **50***a*, and the second electrode **122** is connected to the first drive contact front surface portion **60***c*. That is, the second bonding portions of the wires W**2**, which are connected to the second drive contact front surface portion **60***d*, are separated from the second electrode **122** by the recess **65**.

As shown in FIGS. **14** and **15**, the second electrode **122** of the capacitor **120** is die-bonded to the first drive contact front surface portion **60***cb* y a conductive bonding material P3 such as solder or a paste containing metal, for example Ag. In the same manner, the first electrode **121** of the capacitor **120** is die-bonded to the element contact front surface **50***a* by the conductive bonding material P3. As described above, the second bonding portions of the wires W**2** are separated from the second electrode **122** by the recess **65**. This limits spreading of the conductive bonding material P3 that bonds the second electrode **122** and the first drive contact front surface portion **60***c*, thereby limiting collection of the conductive bonding material P3 on the second bonding portions of the wires W**2**. The conductive bonding material P3 that bonds the element contact front surface **50***a* and the first electrode **121** has an extension extending from the first electrode **121**. The extension is disposed between the recess **55** and the capacitor **120** in the X-direction. That is, the recess **55** defines a range of the conductive bonding material P3 extending from the first electrode **121**. This allows for the manufacturing of the semiconductor light emitting device **1**B in which the conductive bonding material P3 are not in contact with the second bonding portions of the wires W**1**.

As shown in FIG. **19**, a height TM from the substrate front surface **11** to the upper surface **100***a* of the electronic component **100** is less than a height TV from the substrate front surface **11** to the element upper surface **80***a* of the semiconductor light emitting element **80**. In other words, the upper surface **100***a* of the electronic component **100** is located at a lower position (closer to the substrate front surface **11**) than the element upper surface **80***a* of the semiconductor light emitting element **80**. FIG. **19** does not show the encapsulation resin **140** and the coating agent **141**, which will be described later, for the sake of convenience. As shown in FIG. **18**, a height TC from the substrate front surface **11** to the upper surface of the capacitor **120** is greater than the height TV (refer to FIG. **19**) from the substrate front surface **11** to the element upper surface **80***a* of the semiconductor light emitting element **80**. The height TC is greater than the height TM (refer to FIG. **19**) from the substrate front surface **11** to the upper surface **100***a* of the electronic component **100**.

As shown in FIGS. **24** to **26**, the case **20**B includes an accommodation space **23** that accommodates the semiconductor light emitting element **80**, the electronic component **100**, and the capacitor **120**. The opening **22***a* connects the accommodation space **23** to the outside of the case **20**B. As shown in FIG. **23**, the opening **22***a* is located closer to the third side wall **21***c* than the central part of the cover **22** in the X-direction. Also, the opening **22***a* is located closer to the first side wall **21***a* than the central part of the cover **22** in the Y-direction. As shown in FIGS. **24** and **25**, the cover **22** has a thickness that varies in the Y-direction. More specifically, the cover **22** includes a first part **22***b* located at the same position as the opening **22***a* in the Y-direction and located closer to the first side wall **21***a* than the opening **22***a*. The cover **22** includes a second part **22***c* located closer to the second side wall **21***b* than the opening **22***a*. The thickness of the first part **22***b* is less than the thickness of the second part **22***c*. In other words, in the cover **22**, the thickness of the second part **22***c* is greater than the thickness of the first part **22***b*. As shown in FIG. **26**, in plan view, the accommodation space **23** defined by the inner surfaces of the side walls **21***a* to **21***d* is generally square. The opening **22***a* is located adjacent to the third side wall **21***c*.

As shown in FIG. **19**, the semiconductor light emitting element **80** is disposed at a position corresponding to the first part **22***b* of the cover **22**. The electronic component **100** is disposed at a position corresponding to the second part **22***c* of the cover **22**. The second part **22***c* of the cover **22** extends downward from an edge of the opening **22***a* located toward the second side wall **21***b*. Thus, when light from the semiconductor light emitting element **80** is reflected by the light diffusion plate **130**, irradiation of the electronic component **100** with the reflected light is limited.

Manufacturing Method

A method for manufacturing the semiconductor light emitting device **1**B will now be described with reference to FIGS. **27** to **36**. In FIGS. **28**, **30**, and **32**, a tetragon of double-dashed lines indicates the outline of each substrate **10**B.

The method for manufacturing the semiconductor light emitting device **1**B includes a step of preparing a lead frame **800** as shown in FIGS. **27** to **29**. The lead frame **800** is a metal plate formed from Cu. The lead frame **800** is formed, for example, by etching the Cu plate. The lead frame **800** includes conductive portions **830**B, **850**B, **860**B, and **870**B corresponding to the semiconductor light emitting devices **1**B. FIG. **28** shows the conductive portions **830**B, **850**B, **860**B, and **870**B corresponding to four semiconductor light emitting devices **1**B. The conductive portions **830**B, **850**B, **860**B, and **870**B are supported by support leads **880** of the lead frame **800**. The shape of the conductive portions **830**B, **850**B, **860**B, and **870**B conforms to the shape of the conductive portions **30**B, **50**B, **60**B, and **70**B shown in FIG. **14** except that the support leads **880** have not been cut. That is, a common conductive portion **830**B corresponds to the common conductive portion **30**B, an element conductive portion **850**B corresponds to the element conductive portion **50**B, a drive conductive portion **860**B corresponds to the drive conductive portion **60**B, and a control conductive portion **870**B corresponds to the control conductive portion **70**B. For example, in a cross-sectional view taken in the Z-direction, flanges are formed on the conductive portions **830**B, **850**B, **860**B, and **870**B, for example, by etching. In an example, as shown in FIG. **29**, the common conductive portion **830**B includes a flange **836**, and the drive conductive portion **860**B includes a flange **866**. As shown in FIG. **29**, the support lead **880** extends from the flange **866** and has the same thickness as the flange **866**.

As shown in FIGS. **30** and **31**, the method for manufacturing the semiconductor light emitting device **1**B includes a step of molding the lead frame **800** with an insulation portion **813**. The insulation portion **813** is formed from an electrically insulative resin material. In the present embodiment, an epoxy resin is used to form the insulation portion **813**. As indicated by the shading in FIG. **30**, the insulation portion **813** is formed to fill through holes formed in the lead frame **800**. As shown in FIG. **30**, the insulation portion **813** surrounds each of the conductive portions **830**B, **850**B, **860**B, and **870**B. Thus, in the part excluding the support leads **880**, the insulation portion **813** separates the conductive portions **830**B, **850**B, **860**B, and **870**B from each other. As shown in FIG. **31**, the insulation portion **813** is accommodated in the back side of the flange **866** and the support lead **880** to form a back-half insulation portion **813**L. In addition, the insulation portion **813** is accommodated in recesses of the conductive portions **830**B, **850**B, **860**B, and **870**B to form a front-half insulation portion **813**U. The recesses in the conductive portions **830**B, **850**B, **860**B, and **870**B correspond to the recesses **35***a* to **35***e*, **55**, and **65** (refer to FIG. **14**) of the conductive portions **30**B, **50**B, **60**B, and **70**B. The front-half insulation portion **813**U corresponds to the front-half insulation portion **13**U (refer to FIG. **14**) of the substrate **10**B.

As shown in FIG. **32**, the method for manufacturing the semiconductor light emitting device **1**B includes a step of mounting the semiconductor light emitting element **80** on the lead frame **800** and a step of mounting the electronic component **100** on the lead frame **800**. More specifically, the semiconductor light emitting element **80** is die-bonded to the common conductive portion **830**B by the conductive bonding material P**1**, and the electronic component **100** is die-bonded to the common conductive portion **830**B by the conductive bonding material P**2**. More specifically, the conductive bonding material P**1** is applied to a first common contact front surface portion **830***c* of the common conductive portion **830**B, and the conductive bonding material P**2** is applied to a second common contact front surface portion **830***d* of the common conductive portion **830**B. An Ag paste is used as the conductive bonding materials P**1** and P**2**. Then, the semiconductor light emitting element **80** is mounted on the conductive bonding material P**1**, and the electronic component **100** is mounted on the conductive bonding material P**2**. At this time, the element lower surface electrode **92** (refer to FIG. **17**) of the semiconductor light emitting element **80** is in contact with the conductive bonding material P**1**, and the first drive electrode **101** (refer to FIG. **18**) of the electronic component **100** is in contact with the conductive bonding material P**2**. The semiconductor light emitting element **80** is bonded to the first common contact front surface portion **830***cb* y the conductive bonding material P**1**, and the electronic component **100** is bonded to the second common contact front surface portion **830***d* by the conductive bonding material P**2**, for example, through a reflow process.

As shown in FIG. **32**, the method for manufacturing the semiconductor light emitting device **1**B includes a step of forming the wires W**1** to W**3**. In the present embodiment, a wire bonding machine is used to form the wires W**1** to W**3**. More specifically, the wire bonding machine forms first bonding portions on the element upper surface electrode **91** of the semiconductor light emitting element **80** and then moves to the element conductive portion **850**B to form second bonding portions on an element contact front surface **850***a*. As a result, the wires W**1** are formed. The second bonding portions of the wires W**1** are formed so as to be located at substantially the same position as a recess **855** formed in the element conductive portion **850**B in the X-direction. That is, the recess **855** is used as the mark of the position where the second bonding portions of the wires W**1** are formed. The wire bonding machine forms first bonding portions on the first drive electrode **101** of the electronic component **100** and then moves to the drive conductive portion **860**B to form second bonding portions on a second drive contact front surface portion **860***d*. As a result, the wires W**2** are formed. The wire bonding machine forms a first bonding portion on the control electrode **102** of the electronic component **100** and then moves to the control conductive portion **870**B to form a second bonding portion on a control contact front surface **870***a*. As a result, the wire W**3** is formed. The wires W**1** to W**3** may be formed in any order. The element contact front surface **850***a* corresponds to the element contact front surface **50***a* (refer to FIG. **14**) of the substrate **10**B. The second drive contact front surface portion **860***d* corresponds to the second drive contact front surface portion **60***d* (refer to FIG. **14**) of the substrate **10**B. The control contact front surface **870***a* corresponds to the control contact front surface **70***a* (refer to FIG. **14**) of the substrate **10**B.

As shown in FIG. **33**, the method for manufacturing the semiconductor light emitting device **1**B includes a step of forming the substrate **10**B. More specifically, for example, a dicing blade is used to cut the insulation portion **813** and the support lead **880** of the lead frame **800** along the double-dashed lines shown in FIG. **32**. In the present embodiment, the dicing blade cuts the lead frame **800** from a front surface **801** toward a back surface **802** (refer to FIG. **31**). Accordingly, the projections **34***a* to **34***f*, **54***a* to **54***c*, **64***a* to **64***c*, **74***a*, and **74***b* are formed. At this time, since the support lead **880** is located closer to the front surface **801** than the back surface **802** and the insulation portion **813** is disposed at the side of the support lead **880** located toward the back surface **802**, even when a burr is produced on the support lead **880** when cut by the dicing blade, protrusion of the burr toward the back surface **802** is limited.

As shown in FIG. **34**, the method for manufacturing the semiconductor light emitting device **1**B includes a step of mounting the capacitor **120**. More specifically, the capacitor **120** is disposed across the element conductive portion **50**B and the drive conductive portion **60**B and die-bonded to the element conductive portion **50**B and the drive conductive portion **60**B by the conductive bonding material P3. More specifically, the conductive bonding material P3 is applied to the element contact front surface **50***a* of the element conductive portion **50**B and the first drive contact front surface portion **60***c* of the drive conductive portion **60**B. For example, an Ag paste is used as the conductive bonding material P**3**. Then, the capacitor **120** is mounted on the conductive bonding material P**3**. At this time, the first electrode **121** of the capacitor **120** is in contact with the conductive bonding material P3 that is applied to the element conductive portion **50**B. The second electrode **122** is in contact with the conductive bonding material P3 that is applied to the drive conductive portion **60**B. The capacitor **120** is bonded to the element contact front surface **50***a* and the first drive contact front surface portion **60***cb* y the conductive bonding material P3m for example, through a reflow process. In this step, the application of the conductive bonding material P3 and the mounting of the capacitor **120** are performed so that an extension of the conductive bonding material P3 extending from the first electrode **121** is located between the recess **55** and the first electrode **121**.

As shown in FIG. **35**, the method for manufacturing the semiconductor light emitting device **1**B includes a step of covering the electronic component **100** with the encapsulation resin **140** and the coating agent **141**. In the present embodiment, the encapsulation resin **140** is applied to cover the electronic component **100** and the first bonding portions of the wires W**2** and W**3**, and then the encapsulation resin **140** is cured by application of heat or ultraviolet irradiation. Then, the coating agent **141** is applied to cover the encapsulation resin **140**. A light-blocking resin material is used as the encapsulation resin **140**. In the present embodiment, an epoxy resin is used. The coating agent **141** is formed from a fluoropolymer and glass.

As shown in FIG. **36**, the method for manufacturing the semiconductor light emitting device **1**B includes a step of attaching the case **20**B provided with the light diffusion plate

**130** to the substrate **10**B. In the present embodiment, a light-blocking adhesive agent P**4** is applied to end surfaces of the side walls **21***a* to **21***d* of the case **20**B, and the case **20**B is attached to the substrate front surface **11** of the substrate **10**B. An electrically insulative material is used as the adhesive agent P**4**. In the present embodiment, an adhesive containing a black epoxy resin as a main component is used as the adhesive agent P**4** Through the steps described above, the semiconductor light emitting device **1**B is manufactured.

Electronic Apparatus Using Semiconductor Light Emitting Device

FIGS. **37** and **38** are respectively a plan view and a circuit diagram showing an example of an electronic apparatus **2**B that uses the semiconductor light emitting device **1**B. The electronic apparatus **2**B is, for example, a sensor for measuring distance.

The electronic apparatus **2**B includes the semiconductor light emitting device **1**B, a circuit substrate **110** on which the semiconductor light emitting device **1**B is mounted, and wiring patterns **111** to **114** formed on the circuit substrate **110**. The layout of the wiring patterns **111** to **114** is the same as the layout of the wiring patterns **111** to **114** (refer to FIGS. **9** and **10**) of the first embodiment.

The common conductive portion **30**B partially overlaps the first wiring pattern **111**. The first common contact back surface portion **30***e* and the first wiring pattern **111** are bonded by solder or the like. Thus, the first wiring pattern **111** is electrically connected to the element lower surface electrode **92**, that is, the cathode electrode of the semiconductor light emitting element **80**, and the second drive electrode **103**, that is, the drain electrode of the electronic component **100**.

The element conductive portion **50**B overlaps the second wiring pattern **112**. The second wiring pattern **112** of the present embodiment has a smaller width than the second wiring pattern **112** of the first embodiment. The element contact back surface **50***b* and the second wiring pattern **112** are bonded by solder or the like. Thus, the second wiring pattern **112** is electrically connected to the element upper surface electrode **91**, that is, the anode electrode of the semiconductor light emitting element **80**, and the first electrode **121** of the capacitor **120**.

The drive conductive portion **60**B overlaps the third wiring pattern **113**. The third wiring pattern **113** of the present embodiment has a larger width than the third wiring pattern **113** of the first embodiment. The first drive contact back surface portion **60***e* and the second drive contact back surface portion **60***f* are bonded to the third wiring pattern **113** by solder or the like. Thus, the third wiring pattern **113** is electrically connected to the first drive electrode **101**, that is, the source electrode of the electronic component **100**, and the second electrode **122** of the capacitor **120**. The width of the third wiring pattern **113** may be the same as the width of the third wiring pattern **113** of the first embodiment. In this case, the third wiring pattern **113** is bonded to the second drive contact back surface portion **60***f* by solder or the like. The third wiring pattern **113** may be bonded to the first drive contact back surface portion **60***e* instead of the second drive contact back surface portion **60***f* by solder or the like.

The control conductive portion **70**B overlaps the fourth wiring pattern **114**. The control contact back surface **70***b* and the fourth wiring pattern **114** are bonded by solder or the like. Thus, the fourth wiring pattern **114** is electrically connected to the control electrode **102** of the electronic component **100**.

As described above, in the present embodiment, the contact back surfaces **30***b*, **50***b*, **60***b*, and **70***b* form external terminals of the semiconductor light emitting device **1**B.

In the illustrated example, part of the first common contact back surface portion **30***e* and the second common contact back surface portion **30***f* are mounted via solder or the like on the heat dissipation pattern **115** formed on the circuit substrate **110**. The heat dissipation pattern **115** of the present embodiment has a smaller width than the heat dissipation pattern **115** of the first embodiment. The heat dissipation pattern **115** is not bonded to the element contact back surface **50***b* and the first drive contact back surface portion **60***e*. Therefore, heat of the semiconductor light emitting element **80** and the electronic component **100** is transmitted from the common contact back surface portions **30***e* and **30***f* to the circuit substrate **110**. This improves the heat dissipation efficiency of the semiconductor light emitting device **1**B.

In the present embodiment, the semiconductor light emitting device **1**B incorporates the capacitor **120**. As described above, the capacitor **120** is electrically connected to the third wiring pattern **113** and the fourth wiring pattern **114**. As shown in FIG. **38**, the capacitor **120** is connected in parallel to the semiconductor light emitting element **80** and the electronic component **100**, which are connected in series.

The semiconductor light emitting device **1**B of the present embodiment has the following advantages in addition to the advantages of (1-1) to (1-11) and (1-13) of the first embodiment.

(2-1) The common conductive portion **30**B, the element conductive portion **50**B, the drive conductive portion **60**B, and the control conductive portion **70**B are formed of a lead frame and are exposed in the substrate front surface **11** and the substrate back surface **12**. In this structure, each of the common conductive portion **30**B, the element conductive portion **50**B, the drive conductive portion **60**B, and the control conductive portion **70**B has a large volume, so that the heat dissipation efficiency of each of the common conductive portion **30**B, the element conductive portion **50**B, the drive conductive portion **60**B, and the control conductive portion **70**B is improved.

(2-2) The common contact front surface **30***a* includes the first common contact front surface portion **30***c* and the second common contact front surface portion **30***d*. The second common contact front surface portion **30***d* is disposed between the drive conductive portion **60**B and the control conductive portion **70**B and extends toward the fourth side wall **21***d* beyond the first common contact front surface portion **30***c* in the X-direction. The first common contact front surface portion **30***c* extends from the second common contact front surface portion **30***d* in the Y-direction. The semiconductor light emitting element **80** is located at a position of the first common contact front surface portion **30***c* located toward the second common contact front surface portion **30***d*. This structure shortens the conductive path between the semiconductor light emitting element **80** and the electronic component **100**. As a result, parasitic capacitance caused by the conductive path between the semiconductor light emitting element **80** and the electronic component **100** is reduced.

(2-3) The electronic component **100** is located at a position of the second common contact front surface portion **30***d* located toward the first common contact front surface portion **30***c*. This structure further shortens the conductive path between the semiconductor light emitting element **80** and the electronic component **100**. As a result, parasitic capacitance caused by the conductive path between the semiconductor light emitting element **80** and the electronic component **100** is further reduced.

(2-4) The common contact front surface **30***a* is larger than the drive contact front surface **60***a* and the control contact front surface **70***a*. This structure improves the heat dissipation efficiency of the common conductive portion **30**B.

(2-5) As viewed in the X-direction, a portion of the semiconductor light emitting element **80** is disposed on the first common contact front surface portion **30***ca*t a position closer to the electronic component **100** than the element contact front surface **50***a*. The element upper surface electrode **91** of the semiconductor light emitting element **80** is connected to the element contact front surface **50***a* by multiple wires W**1**. The wires W**1** diagonally extend from the element upper surface electrode **91** toward the element contact front surface **50***a* so as to extend away from the electronic component **100**. With this structure, even when the semiconductor light emitting element **80** and the element contact front surface **50***a* are located at different positions in the Y-direction, the multiple wires W**1** allow the element upper surface electrode **91** to be readily electrically connected to the element conductive portion **50**B.

(2-6) The drive contact front surface **60***a* includes the first drive contact front surface portion **60***c* and the second drive contact front surface portion **60***d*. The first drive contact front surface portion **60***c* extends in the X-direction. The second drive contact front surface portion **60***d* extends in the Y-direction. In this structure, the second electrode **122** of the capacitor **120** is readily connected to the first drive contact front surface portion **60***c*, and the wires W**2** are readily connected to the second drive contact front surface portion **60***d*.

In addition, the drive contact front surface **60***a* includes the recessed region **60***r* that is recessed from the first drive contact front surface portion **60***c* so that the second drive contact front surface portion **60***d* has a smaller dimension in the X-direction than the first drive contact front surface portion **60***c*. The second common contact front surface portion **30***d* is accommodated in the recessed region **60***r*. In this structure, the dimension of the second common contact front surface portion **30***d* in the X-direction is increased, thereby improving the heat dissipation efficiency of the electronic component **100**.

(2-7) In plan view, the wires W**2** are arranged in the Y-direction. In this structure, the second drive contact front surface portion **60***d* extends in the Y-direction to ensure a space for forming the second bonding portion of each wire W**2**. Thus, the second bonding portion of each wire W**2** is readily formed.

(2-8) Inductance between the first drive electrode **101** and the second drive contact front surface portion **60***d* is reduced in accordance with an increase in the distance between the two wires W**2** that are configured to be the combination of the furthermost ones of the wires W**2**. In this regard, in the semiconductor light emitting device **1**B, in plan view, the distance between the furthermost ones of the wires W**2** in the Y-direction increases from the first drive electrode **101** of the electronic component **100** toward the second drive contact front surface portion **60***d*. Thus, the two wires W**2** that are configured to be the combination of the furthermost ones of the wires W**2** are separated by a great distance, so that the inductance between the first drive electrode **101** and the second drive contact front surface portion **60***d* is reduced.

(2-9) The first common contact back surface portion **30***e* is larger than the element contact back surface **50***b*, the drive contact back surface **60***b*, and the control contact back surface **70***b*. This structure improves the heat dissipation efficiency of the semiconductor light emitting element **80**.

(2-10) The first common contact back surface portion **30***e* is larger than the second common contact back surface portion **30***f*. This structure improves the heat dissipation efficiency of the semiconductor light emitting element **80**.

(2-11) The first common contact front surface portion **30***c* extends from the second common contact front surface portion **30***d* toward the third side wall **21***c* in the X-direction. In this structure, the dimension of the first common contact front surface portion **30***c* in the X-direction is increased, thereby improving the heat dissipation efficiency of the semiconductor light emitting element **80**.

(2-12) The common conductive portion **30**B includes the recess **35***a*, the two recesses **35***b*, the two recesses **35***c*, the recess **35***d*, and the recess **35***e*. Each of the recesses **35***a* to **35***e* accommodates the insulation portion **13**. This structure resists separation of the insulation portion **13** from the common conductive portion **30**B. In addition, the recesses **35***a* to **35***e* are arranged in the common contact front surface **30***a* and accommodate the front-half insulation portions **13**U, which do not extend through in the Z-direction. In this structure, as viewed in the Z-direction, the front-half insulation portions **13**U accommodated in the recesses **35***a* to **35***e* overlaps the common contact back surface **30***b*, thereby restricting movement of the common conductive portion **30**B toward the case **20**B from the insulation portion **13** in the Z-direction.

(2-13) The element conductive portion **50**B includes the recess **55**. The recess **55** accommodates the insulation portion **13**. This structure resists separation of the insulation portion **13** from the element conductive portion **50**B. In addition, the recess **55** is arranged in the element contact front surface **50***a* and accommodates the front-half insulation portion **13**U, which does not extend through in the Z-direction. In this structure, as viewed in the Z-direction, the front-half insulation portion **13**U accommodated in the recess **55** overlaps the element contact back surface **50***b*, thereby restricting movement of the element conductive portion **50**B toward the case **20**B from the insulation portion **13** in the Z-direction.

(2-14) The drive conductive portion **60**B includes the recess **65**. The recess **65** accommodates the insulation portion **13**. This structure resists separation of the insulation portion **13** from the drive conductive portion **60**B. In addition, the recess **65** is arranged in the drive contact front surface **60***a* and accommodates the front-half insulation portion **13**U, which does not extend through in the Z-direction. In this structure, as viewed in the Z-direction, the front-half insulation portion **13**U accommodated in the recess **65** overlaps the drive contact back surface **60***b*, thereby restricting movement of the drive conductive portion **60**B toward the case **20**B from the insulation portion **13** in the Z-direction.

(2-15) The recess **65** is disposed between the first drive contact front surface portion **60***c* and the second drive contact front surface portion **60***d*. This structure limits entrance of the conductive bonding material P**3**, which is used to connect the capacitor **120** to the first drive contact front surface portion **60***c*, into the second drive contact front surface portion **60***d*. Thus, contact of the conductive bonding material P3 with the second bonding portions of the wires W**2** is hindered.

(2-16) The recess **35***f* is disposed between the first common contact back surface portion **30***e* and the second common contact back surface portion **30***f*. The recess **35***f* accommodates the back-half insulation portion **13**L. The back-half insulation portion **13**L accommodated in the recess **35***f* extends through the common contact back surface **30***b* in the X-direction and is continuous with the insulation portions **13** that are located at opposite sides of the common contact back surface **30***b* and extend through the substrate **10**B. In this structure, the back-half insulation portion **13**L in the recess **35***f* integrates the insulation portions **13** located at opposite sides of the common contact back surface **30***b*. This improves the strength of the insulation portions **13** located around the common conductive portion **30**B. In addition, the back-half insulation portion **13**L in the recess **35***f* supports the common conductive portion **30**B in the Z-direction and increases the area of contact of the insulation portion **13** with the common conductive portion **30**B. This restricts movement of the common conductive portion **30**B toward the substrate back surface **12** in the Z-direction.

(2-17) The recess **65***a* is disposed between the first drive contact back surface portion **60***e* and the second drive contact back surface portion **60***f*. The recess **65***a* accommodates the back-half insulation portion **13**L. The back-half insulation portion **13**L accommodated in the recess **65***a* extends through the drive contact back surface **60***b* in the X-direction and is continuous with the insulation portions **13** that are located at opposite sides of the drive contact back surface **60***b* and extend through the substrate **10**B. In this structure, the back-half insulation portion **13**L in the recess **65***a* integrates the insulation portions **13** located at opposite sides of the drive contact back surface **60***b*. This improves the strength of the insulation portions **13** located around the drive conductive portion **60**B. In addition, the back-half insulation portion **13**L in the recess **65***a* supports the drive conductive portion **60**B in the Z-direction and increases the area of contact of the insulation portion **13** with the drive conductive portion **60**B. This restricts movement of the drive conductive portion **60**B toward the substrate back surface **12** in the Z-direction.

(2-18) The common conductive portion **30**B, the element conductive portion **50**B, the drive conductive portion **60**B, and the control conductive portion **70**B include the flanges **36**, **56**, **66**, and **76**, respectively. The insulation portion **13** (back-half insulation portion **13**L) is accommodated between the substrate back surface **12** and each of the flanges **36**, **56**, **66**, and **76**. This structure restricts movement of the common conductive portion **30**B, the element conductive portion **50**B, the drive conductive portion **60**B, and the control conductive portion **70**B toward a side of the insulation portion **13** opposite from the case **20**B in the Z-direction.

(2-19) The electronic component **100** is covered by the light-blocking resin material (encapsulation resin **140**). In this structure, when the light from the semiconductor light emitting element **80** is reflected and emitted toward the electronic component **100** by the light diffusion plate **130** and the like, the encapsulation resin **140** hinders the reflected light from reaching the electronic component **100**. Thus, erroneous actuation of the electronic component **100** is limited.

(2-20) The conductive bonding material P**2**, which is an Ag paste connecting the electronic component **100** to the common conductive portion **30**B, contains a large amount of Ag. This improves the efficiency of heat dissipation from the electronic component **100** to the common conductive portion **30**B. However, the sulfidation resistance of the conductive bonding material P**2** is decreased. In this regard, in the present embodiment, the encapsulation resin **140** is covered by the coating agent **141**, which inhibits sulfidation. This structure inhibits sulfidation of the conductive bonding material P**2**, the second drive electrode **103**, the control electrode **102**, and the first drive electrode **101**. Thus, the sulfidation resistance of the conductive bonding material P**2**, the second drive electrode **103**, the control electrode **102**, and the first drive electrode **101** is improved.

(2-21) The semiconductor light emitting device **1**B includes the capacitor **120**. This structure eliminates the need for arranging the capacitor **120** outside the semiconductor light emitting device **1**B or reduces the number of capacitors **120** arranged outside the semiconductor light emitting device **1**B, thereby saving the space in the electronic apparatus **2**B.

(2-22) The semiconductor light emitting element **80** is located closer to the third side wall **21***c* than the electronic component **100** in the X-direction. The capacitor **120** is disposed at a side of the semiconductor light emitting element **80** toward the fourth side wall **21***d* in the X-direction. The electronic component **100** is disposed on the central part of the substrate **10**B in the X-direction. The capacitor **120** is located closer to the first side wall **21***a* than the electronic component **100** in the Y-direction. This structure ensures the space for the capacitor **120** in the accommodation space **23** defined by the case **20**B and the substrate **10**B without immoderate extension of one of the wires W**2** and W**3** relative to the other wires W**2** and W**3**.

(2-23) The cover **22** of the case **20**B formed from a light-blocking material includes the opening **22***a* in the part facing the semiconductor light emitting element **80** in the Z-direction. The light diffusion plate **130** is attached to the cover **22** so as to cover the opening **22***a*. In this structure, excluding the part of the cover **22** facing the semiconductor light emitting element **80** in the Z-direction, the cover **22** is shielded from light, so that irradiation of the electronic component **100** with light is limited. Thus, erroneous actuation of the electronic component **100** caused by light irradiation is limited.

(2-24) The common conductive portion **30**B includes the projections **34***a* to **34***f*. This structure increases the volume of the common conductive portion **30**B, thereby improving the heat dissipation efficiency of the semiconductor light emitting element **80** and the electronic component **100**.

(2-25) The element conductive portion **50**B includes the projections **54***a* to **54***c*. This structure increases the volume of the element conductive portion **50**B, thereby improving the heat dissipation efficiency of the semiconductor light emitting device **1**B. The drive conductive portion **60**B includes the projections **64***a* to **64***c*. This structure increases the volume of the drive conductive portion **60**B, thereby improving the heat dissipation efficiency of the semiconductor light emitting device **1**B. The control conductive portion **70**B includes the projections **74***a* and **74***b*. This structure increases the volume of the control conductive portion **70**B, thereby improving the heat dissipation efficiency of the semiconductor light emitting device **1**B.

(2-26) The height TM of the electronic component **100** from the substrate front surface **11** is less than the height TV of the semiconductor light emitting element **80** from the substrate front surface **11**. In this structure, when light from the semiconductor light emitting element **80** is reflected by the light diffusion plate **130** or the like, irradiation of the electronic component **100** with the light is limited. Thus, erroneous actuation of the electronic component **100** cause by irradiation of the electronic component **100** with light is limited.

## Modified Examples

The embodiments exemplify, without any intention to limit, applicable forms of a semiconductor light emitting device according to the present disclosure. The semiconductor light emitting device according to the present disclosure may be applicable to forms differing from the above embodiments. In an example of such a form, a portion of the configurations of the above embodiments is replaced, changed, or omitted, or a further configuration is added to the above embodiments. The modified examples described below may be combined with one another as long as there is no technical inconsistency. In the following modified examples, the same reference characters are given to those parts that are the same as the corresponding parts of the above embodiments. Such parts will not be described in detail.

In the first embodiment, as shown in FIGS. **39** to **42**, the substrate **10** may be formed of a conductive material such as Cu. In this form, the substrate **10** includes insulation portions **150** that separate the substrate **10** into the conductive portions **30**, **50**, **60**, and **70**. The insulation portions **150** are formed from an insulative material such as an epoxy resin. The insulation portions **150** have steps having, for example, a wide part and a narrow part. The conductive portions **30**, **50**, **60**, and **70** are portions of the substrate **10** insulated and separated from each other by the insulation portions **150**.

The connection conductive portion **40** is electrically connected to the common conductive portion **30** and is insulated from the other conductive portions **50**, **60**, and **70**. More specifically, the insulation portion **150** is disposed between the connection conductive portion **40** and the other conductive portions **50**, **60**, and **70** and extends through the substrate **10** in the thickness-wise direction. In addition, as shown in FIGS. **40** and **41**, a half insulation portion **151** that does not extend through the substrate **10** in the thickness-wise direction is disposed between the common conductive portion **30** and the connection conductive portion **40**. The half insulation portion **151** is formed in the substrate back surface **12** and is not formed in the substrate front surface **11**. This allows the common conductive portion **30** to be electrically connected to the connection conductive portion **40**.

As described above, in a form in which the conductive portions **30**, **40**, **50**, **60**, and **70** are formed by portions of the substrate **10**, the common contact front surface **30***a*, the connection contact front surface **40***a*, the element contact front surface **50***a*, the drive contact front surface **60***a*, and the control contact front surface **70***a* are formed in the substrate front surface **11**. The common contact front surface **30***a*, the element contact front surface **50***a*, the drive contact front surface **60***a*, and the control contact front surface **70***a* are separated from each other by the insulation portions **150**. The connection contact front surface **40***a* is separated from the element contact front surface **50***a*, the drive contact front surface **60***a*, and the control contact front surface **70***a*. However, since the half insulation portion **151** is not formed in the substrate front surface **11**, the connection contact front surface **40***a* is continuous with the common contact front surface **30***a*.

In the same manner, the common contact back surface **30***b*, the connection contact back surface **40***b*, the element contact back surface **50***b*, the drive contact back surface **60***b*, and the control contact back surface **70***b* are formed in the substrate back surface **12**. The common contact back surface **30***b*, the element contact back surface **50***b*, the drive contact back surface **60***b*, and the control contact back surface **70***b* are separated from each other by the insulation portions **150**. The common contact back surface **30***b*, the connection contact back surface **40***b*, the element contact back surface **50***b*, the drive contact back surface **60***b*, and the control contact back surface **70***b* are separated from each other by the insulation portions **150** and the half insulation portion **151**.



In the modified example of the first embodiment shown in FIGS. **39** to **42**, the conductive portions **30**, **40**, **50**, **60**, and **70** may include projections and recesses in the same manner as the conductive portions **30**B, **50**B, **60**B, and **70**B of the second embodiment.

FIG. **43** shows an example of a structure of the conductive portions **30**, **40**, **50**, **60**, and **70** that include projections and recesses. In FIG. **43**, for the sake of convenience, the outer surface of the frame **21** of the case **20** is indicated by double-dashed lines.

As shown in FIG. **43**, the projections **34***g* to **34***j* are formed on the common contact front surface **30***a* of the common conductive portion **30**. The projections **34***g* and **34***h* are formed on an end of the common contact front surface **30***a* located toward the first side wall **21***a*. The projections **34***g* and **34***h* are separate from each other in the X-direction. The projection **34***g* is located closer to the element conductive portion **50** than the projection **34***h*. The projections **34***g* and **34***h* project from the outer surface of the first side wall **21***a* as viewed in the Z-direction. The projections **34***i* and **34***j* are formed on an end of the common contact front surface **30***a* located toward the second side wall **21***b*. The projections **34***i* and **34***j* are separate from each other in the X-direction. The projection **34***i* is located closer to the drive conductive portion **60** than the projection **34***j*. The projections **34***i* and **34***j* project from the outer surface of the second side wall **21***b* as viewed in the Z-direction. When a support lead supporting the common conductive portion **30** is cut, residual portions of the lead frame are the projections **34***g* to **34***j*. The number of projections may be changed to any number. As viewed in the Z-direction, connection portions between the common contact front surface **30***a* and the projections **34***g* to **34***j* each have a curved surface.

In addition, the common contact front surface **30***a* includes recesses **35***g* and **35***h*. In the Y-direction, the recesses **35***g* and **35***h* are formed between the semiconductor light emitting element **80** and the electronic component **100**. The recess **35***g* is formed in an end of the common contact front surface **30***a* located toward the fourth side wall **21***d*. The recess **35***h* is formed in an end of the common contact front surface **30***a* located toward the third side wall **21***c*. The recess **35***g* is recessed toward the third side wall **21***c* from the end **31***c* of the common contact front surface **30***a* located toward the fourth side wall **21***d* in the X-direction. The bottom of the recess **35***g* is defined by a curved surface. In the illustrated example, in plan view, the recess **35***g* partially extends in the X-direction with a fixed width and has a curved surface that reduces the width toward the bottom. The recess **35***h* is recessed toward the fourth side wall **21***d* from the end **31***a* of the common contact front surface **30***a* located toward the third side wall **21***c* in the X-direction. In the illustrated example, the shape of the recess **35***h* in plan view is symmetric to the shape of the recess **35***g* in plan view. The recesses **35***g* and **35***h* accommodate the insulation portions **150**.

The connection contact front surface **40***a* of the connection conductive portion **40** includes projections **44***a* to **44***c*. The projection **44***a* is formed in an end of the connection contact front surface **40***a* located toward the first side wall **21***a*. The projections **44***b* and **44***c* are formed in an end of the connection contact front surface **40***a* located toward the third side wall **21***c*. The projections **44***b* and **44***c*are separate from each other in the Y-direction. The projection **44***b* is located closer to the first side wall **21***a* than the projection **44***c*. The projection **44***a* projects from the outer surface of the first side wall **21***a* as viewed in the Z-direction. The projections **44***b* and **44***c* project from the outer surface of the third side wall **21***c* as viewed in the Z-direction. When a support lead supporting the connection conductive portion **40** is cut, residual portions of the lead frame are the projections **44***a* to **44***c*. The number of projections may be changed to any number. As viewed in the Z-direction, connection portions between the connection contact front surface **40***a* and the projections **44***a* to **44***c* each have a curved surface.

The element contact front surface **50***a* of the element conductive portion **50** includes projections **54***d* to **54***f* The projection **54***d* is formed in an end of the element contact front surface **50***a* located toward the first side wall **21***a*. The projections **54***e* and **54***f* are formed in an end of the element contact front surface **50***a* located toward the fourth side wall **21***d*. The projections **54***e* and **54***f* are separate from each other in the Y-direction. The projection **54***e* is located closer to the first side wall **21***a* than the projection **54***f* The projection **54***d* projects from the outer surface of the first side wall **21***a* as viewed in the Z-direction. As viewed in the Z-direction, the projections **54***e* and **54***f* project from the outer surface of the fourth side wall **21***d*. When a support lead supporting the element conductive portion **50** is cut, residual portions of the lead frame are the projections **54***d* to **54***f* The number of projections may be changed to any number. As viewed in the Z-direction, connection portions between the element contact front surface **50***a* and the projections **54***d* to **54***f* each have a curved surface.

The drive contact front surface **60***a* of the drive conductive portion **60** includes projections **64***d* to **64***f* The projection **64***d* is formed in an end of the drive contact front surface **60***a* located toward the second side wall **21***b*. The projections **64***e* and **64***f* are formed in an end of the drive contact front surface **60***a* located toward the fourth side wall **21***d*. The projections **64***e* and **64***f* are separate from each other in the Y-direction. The projection **64***e* is located closer to the first side wall **21***a* than the projection **64***f* As viewed in the Z-direction, the projection **64***d* projects from the outer surface of the second side wall **21***b*. As viewed in the Z-direction, the projections **64***e* and **64***f* project from the outer surface of the fourth side wall **21***d*. When a supporting lead supporting the drive conductive portion **60** is cut, residual portions of the lead frame are the projections **64***d* to **64***f* The number of projections may be changed to any number. As viewed in the Z-direction, connection portions between the drive contact front surface **60***a* and the projections **64***d* to **64***f* each have a curved surface.

The control contact front surface **70***a* of the control conductive portion **70** includes projections **74***d* to **74***f* The projection **74***d* is formed on an end of the control contact front surface **70***a* located toward the second side wall **21***b*. The projections **74***e* and **74***f* are formed on an end of the control contact front surface **70***a* located toward the third side wall **21***c*. The projections **74***e* and **74***f* are separate from each other in the Y-direction. The projection **74***e* is located closer to the first side wall **21***a* than the projection **74***f* As viewed in the Z-direction, the projection **74***d* projects from the outer surface of the second side wall **21***b*. The projections **74***e* and **74***f* project from the outer surface of the third side wall **21***c* as viewed in the Z-direction. When a support lead supporting the control conductive portion **70** is cut, residual portions of the lead frame are the projections **74***d* to **74***f*. The number of projections may be changed to any number. As viewed in the Z-direction, connection portions between the control contact front surface **70***a* and the projections **74***d* to **74***f* each have a curved surface. When the projections and the recesses are arranged as described above, the structure of the conductive portions of the substrate back surface **12** is the same as the structure of the conductive portions shown in FIG. **40**.

The structure described above increases the volume of each of the conductive portions **30**, **40**, **50**, **60**, and **70**, thereby improving the heat dissipation efficiency of the semiconductor light emitting device **1**.

The common conductive portion **30** includes the recesses **35***g* and **35***h*. The recesses **35***g* and **35***h* accommodate the insulation portions **150**. This /resists/is resistant to/hinders/ separation of the insulation portions **150** from the common conductive portion **30**. In addition, the common contact front surface **30***a* includes the recesses **35***g* and **35***h*. The recesses **35***g* and **35***h* accommodate the insulation portions **150** that do not extend through the Z-direction. This structure restricts movement of the common conductive portion **30** toward the case **20** from the insulation portions **150** in the Z-direction.

In the semiconductor light emitting device **1** shown in FIG. **43**, the connection conductive portion **40**, the element conductive portion **50**, the drive conductive portion **60**, and the control conductive portion **70** do not include a recess. However, at least one of the connection conductive portion **40**, the element conductive portion **50**, the drive conductive portion **60**, and the control conductive portion **70** may include a recess. The recess has the same structure as the recesses **35***g* and **35***h* of the common conductive portion **30**. Although not illustrated, at least one of the conductive portions **30**, **40**, **50**, **60**, and **70** may include a flange in the same manner as the flanges **36**, **56**, **66**, and **76** of the conductive portions **30**B, **50**B, **60**B, and **70**B in the second embodiment.

In the first embodiment, the size of each of the connection contact back surface **40***b*, the element contact back surface **50***b*, the drive contact back surface **60***b*, and the control contact back surface **70***b* may be changed in any manner. In an example, as shown in FIG. **44**, the area of the connection contact back surface **40***b* may be equal to the area of the control contact back surface **70***b*. The area of the element contact back surface **50***b* may be equal to the area of the drive contact back surface **60***b*.

In the first embodiment, as shown in FIG. **45**, the capacitor **120** may be disposed in the case **20**. The capacitor **120** is disposed so as to span between the element contact front surface **50***a* and the drive contact front surface **60***a*. This structure allows for further reduction in the space.

In the first embodiment, the semiconductor light emitting element **80** and the electronic component **100** are disposed in the central part in the X-direction. However, the semiconductor light emitting element **80** and the electronic component **100** may be disposed at one side in the X-direction. For example, the semiconductor light emitting element **80** and the electronic component **100** may be located toward the third side wall **21***c* from the central part. This structure allows for increase in the space for the capacitor **120**.

In each embodiment, the electronic component **100** is not limited to a MOSFET and may be another switching element, such as a bipolar transistor. For example, when the electronic component **100** is a bipolar transistor, one of the first drive electrode **101** and the second drive electrode **103** corresponds to a collector electrode, the other one of the first drive electrode **101** and the second drive electrode **103** corresponds to an emitter electrode, and the control electrode **102** corresponds to a base electrode.

Alternatively, the electronic component **100** may be an integrated circuit (IC) instead of a switching element. Further, the electronic component **100** is not limited to an active element such as a switching element and may be a passive element such as a capacitor. The electronic component **100** does not have to be used to drive the semiconductor light emitting element **80**.

In each element, the arrangement direction of the semiconductor light emitting element **80** and the electronic component **100** may be changed in any manner. For example, the semiconductor light emitting element **80** and the electronic component **100** may be arranged in the X-direction or in a direction that intersects the X-direction and the Y-direction. The common contact front surface **30***a* only needs to extend in the arrangement direction of the semiconductor light emitting element **80** and the electronic component **100** so that the semiconductor light emitting element **80** and the electronic component **100** are disposed on the common contact front surface **30***a*.

In the first embodiment, the layout of the conductive portions **30**, **40**, **50**, **60**, and **70** may be changed in any manner. For example, the drive conductive portion **60** and the control conductive portion **70** may be located at the same side of the common contact front surface **30***a* in the X-direction. At least one of the drive conductive portion **60** and the control conductive portion **70** may be separated from the common contact front surface **30***a* in the Y-direction. The same applies to the connection conductive portion **40** and the element conductive portion **50**. The control conductive portion **70** and the element conductive portion **50** may be located at the same side.

In the second embodiment, the layout of the conductive portions **30**B, **50**B, **60**B, and **70**B may be changed in any manner. For example, the drive conductive portion **60**B and the control conductive portion **70**B may be located at the same side of the common contact front surface **30***a* in the X-direction. In this case, the first common contact front surface portion **30***c* may be aligned with the second common contact front surface portion **30***d* in the X-direction. At least one of the drive conductive portion **60**B and the control conductive portion **70** may be separated from the common contact front surface **30***a* in the Y-direction. The same applies to the element conductive portion **50**B. The control conductive portion **70**B and the element conductive portion **50**B may be located at the same side.

In the first embodiment, the shape of the contact front surfaces **30***a*, **40***a*, **50***a*, **60***a*, and **70***a* may be changed in any manner. For example, the contact front surfaces **30***a*, **40***a*, **50***a*, **60***a*, and **70***a* may have the same size or different sizes. At least one of the contact front surfaces **30***a*, **40***a*, **50***a*, **60***a*, and **70***a* may be elliptical or circular.

In the first embodiment, as shown in FIG. **46**, the common back surface conductive layer **32** may be continuous with the connection back surface conductive layer **42**, and the common contact back surface **30***b* may be continuous with the connection contact back surface **40***b*.

In the first embodiment, the shape of the contact back surfaces **30***b*, **40***b*, **50***b*, **60***b*, and **70***b* may be changed in any manner. For example, the common contact back surface **30***b* may be smaller than one of the connection contact back surface **40***b* and the element contact back surface **50***b*. The drive contact back surface **60***b* and the control contact back surface **70***b* may be larger than the connection contact back surface **40***b* and the element contact back surface **50***b*. The contact back surfaces **30***b*, **40***b*, **50***b*, **60***b*, and **70***b* may have the same size or different sizes. At least one of the contact back surfaces **30***b*, **40***b*, **50***b*, **60***b*, and **70***b* may be elliptical or circular.

In the first embodiment, the connection conductive portion **40** may be omitted. Even in this case, the common contact back surface **30***b* may be used to ensure contact of the semiconductor light emitting device **1** with an external device.

In each embodiment, the common contact back surface **30***b* is not necessary. That is, the common back surface conductive layer **32** may be omitted.

In the first embodiment, as shown in FIG. **47**, the cover **22** may include a transmissive portion **201** and a light-blocking portion **202**. The transmissive portion **201** is disposed above the semiconductor light emitting element **80** and is transmissive to light from the semiconductor light emitting element **80**. The light-blocking portion **202** is disposed above the electronic component **100** and blocks light. This structure limits irradiation of the electronic component **100** with light, thereby limiting erroneous actuation of the electronic component **100** caused by the light irradiation.

The transmissive portion **201** and the light-blocking portion **202** may have any shape. For example, as shown in FIG. **48**, the transmissive portion **201** may be formed at only a portion overlapping the semiconductor light emitting element **80**, and the remaining portion may be the light-blocking portion **202**.

In the first embodiment, the cover **22** may diffuse light from the semiconductor light emitting element **80**.

In each embodiment, the cover **22** may be omitted, and only the frame **21** may be used.

In the first embodiment, as shown in FIG. **49**, the case **20** may be omitted. In this case, the capacitor **120** may be disposed in the space used for the frame **21**.

In the first embodiment, as shown in FIG. **50**, the frame **21** may include inclined surfaces **210** that are inclined to have an opening that widens upward.

In the first embodiment, as shown in FIG. **51**, the common joint **33** may be formed at a position overlapping the semiconductor light emitting element **80**. The common joint **33** may be formed at a position overlapping the electronic component **100**.

In the first embodiment, as shown in FIGS. **52** to **55**, the semiconductor light emitting device **1** may include a case **20**C having the same structure as the case **20**B of the second embodiment instead of the case **20**. In this case, the semiconductor light emitting device **1** may further include the light diffusion plate **130**. FIGS. **56** and **57** show only the case **20**C. As shown in FIG. **53**, the case **20**C is smaller than the case **20**B and includes the opening **22***a* formed in a position different from that of the case **20**B. As shown in FIGS. **53** and **56**, the opening **22***a* of the case **20**C is formed in the cover **22** at a position toward the first side wall **21***a* and slightly toward the fourth side wall **21***d* from the central part in the X-direction. The opening **22***a* shown in FIGS. **53** and **56** has, for example, the same size as the opening **22***a* in the second embodiment. The dimension of the case **20**C in the X-direction is less than the dimension of the case **20**B in the X-direction. Therefore, the dimension of the opening **22***a* in the X-direction is greater than or equal to two-thirds of the dimension of the case **20**C in the X-direction. In an example, in FIG. **56**, the dimensions of the case **20**C in the X-direction and the Y-direction are each approximately 3.3 mm. The dimension of the opening **22***a* in the X-direction is approximately 2.4 mm. In FIG. **54**, the dimension of the case **20**C in the Z-direction is, for example, equal to the dimension of the case **20**B in the Z-direction. The light diffusion plate **130** shown in FIGS. **53** and **54** has the same size as the light diffusion plate **130** of the second embodiment. In this modified example, the size of the light diffusion plate **130** may be changed in a range capable of covering the entirety of the opening **22***a*. In an example, the light diffusion plate **130** may be sized to cover the entirety of the cover **22**.

As shown in FIG. **53**, the light diffusion plate **130** is located in the central part of the cover **22** of the case **20**C in the X-direction. Therefore, the light diffusion plate **130** is not aligned with the opening **22***a* in the X-direction. More specifically, in this modified example, as viewed in the Z-direction, the light diffusion plate **130** projects from the opening **22***a* toward the third side wall **21***c* in the X-direction by a projection distance DX**3** and projects from the opening **22***a* toward the fourth side wall **21***d* in the X-direction by a projection distance DX**4**. The projection distance DX**3** is greater than the projection distance DX**4**. The light diffusion plate **130** projects from the opening **22***a* toward the first side wall **21***a* in the Y-direction by a projection distance DY**3** and projects from the opening **22***a* toward the second side wall **21***b* in the Y-direction by a projection distance DY**4**. The projection distance DX**3** is greater than the projection distance DY**4**. The projection distance DY**3** is equal to the projection distance DY**4**. The projection distance DX**4** is less than the projection distances DY**3** and DY**4**. Therefore, the light diffusion plate **130** is located toward the third side wall **21***c* relative to the opening **22***a*.

As shown in FIG. **57**, as viewed in the Z-direction, the shape of the accommodation space **23** defined by the inner surfaces of the side walls **21***a* to **21***d* is different from a square. In other words, the frame **21** has portions having different widths. More specifically, the third side wall **21***c* includes a first part **21***ca* corresponding to the opening **22***a* in the Y-direction and a second part **21***cb* located closer to the second side wall **21***b* than the first part **21***ca*. The width of the first part **21***ca* is greater than the width of the second part **21***cb*. The width of the second part **21***cb* is equal to each of the side walls **21***a*, **21***b*, **21***d*. The term "width" refers to a dimension of each of the side walls **21***a* to **21***d* in a direction orthogonal to the direction in which the side walls **21***a* to **21***d* extend as viewed in the Z-direction.

As viewed in the Z-direction, the dimension of the part of the accommodation space **23** located toward the second side wall **21***b* from the part having the opening **22***a* is greater in the X-direction than the dimension of the part of the accommodation space **23** having the opening **22***a*. As viewed in the Z-direction, the dimension of the part of the accommodation space **23** located toward the second side wall **21***b* from the part having the opening **22***a* in the X-direction is specified by the distance between the inner surface of the second part **21***cb* and the inner surface of the fourth side wall **21***d* in the X-direction. The dimension of the part of the accommodation space **23** having the opening **22***a* in the X-direction is specified by the distance between the inner surface of the first part **21***ca* and the inner surface of the fourth side wall **21***d* in the X-direction as viewed in the Z-direction.

As shown in FIG. **55**, the opening **22***a* is formed in a part of the cover **22** facing the semiconductor light emitting element **80**. The electronic component **100** is located closer to the second side wall **21***b* than the opening **22***a*. That is, the electronic component **100** is located in the accommodation space **23** at a side having the larger dimension in the X-direction. This ensures the space for the wires W**2** and W**3** (for example, refer to FIG. **43**).

In the modified examples of the first embodiment shown in FIGS. **52** to **57** and the second embodiment, the cases **20**B and **20**C only need to include a transmissive portion that is transmissive to light from the semiconductor light emitting element **80**. Therefore, the cases **20**B and **20**C may include a light-transmissive material instead of the opening **22***a*. In an example, the cases **20**B and **20**C are formed by molding a light-transmissive resin material and a light-blocking resin material in two colors.

In the first embodiment, the contact front surfaces **30***a*, **40***a*, **50***a*, **60***a*, and **70***a* may be identical in shape to the contact back surfaces **30***b*, **40***b*, **50***b*, **60***b*, and **70***b*.

In the second embodiment, the contact front surfaces **30***a*, **50***a*, **60***a*, and **70***a* may be identical in shape to the contact back surfaces **30***b*, **50***b*, **60***b*, and **70***b*.

In the first embodiment, the arrangement positions of the joints **43**, **53**, **63**, and **73** are not limited to positions overlapping the case **20** and may be changed in any manner.

In each embodiment, the shape of the element upper surface **80***a* and the element lower surface **80***b* may be changed in any manner. Also, the position and the shape of the light emitting regions **90** and the element upper surface electrode **91** may be changed in any manner.

In each embodiment, the shape of the upper surface **100***a* and the lower surface **100***b* is not limited to a rectangle and may be changed in any manner. Also, the position and the shape of the first drive electrode **101** and the control electrode **102** may be changed in any manner.

In each embodiment, the shape of the substrates **10** and **10**B is not limited to a square and may be changed in any manner. For example, the substrates **10** and **10**B may be a rectangle so that one side is longer than the other side.

In each embodiment, the element lower surface electrode **92** may be formed on part of the element lower surface **80***b*. Also, the second drive electrode **103** may be formed on part of the lower surface **100***b*.

In each embodiment, the electronic apparatuses **2** and **2**B may have any specific structure. For example, the capacitor **120** may be omitted from the electronic apparatus **2**. The capacitor **120** may be added to the electronic apparatus **2**B. The electronic apparatuses **2** and **2**B may include a light receiving element mounted on the circuit substrate **110**.

In each embodiment, the wiring patterns **111** to **114** may have any specific layout. The heat dissipation pattern **115** may be omitted.

In each embodiment, the semiconductor light emitting devices **1** and **1**B may further include a partition wall that separates the semiconductor light emitting element **80** from the electronic component **100**.

In a first example, as shown in FIGS. **58** and **59**, the semiconductor light emitting device **1** including the case **20**C includes a partition wall **24**. The partition wall **24** is formed from a light-blocking material. In an example, the partition wall **24** and the case **20**B are integrally formed from the same resin material. The partition wall **24** extends downward from the cover **22** of the case **20**B. In the illustrated example, the partition wall **24** is in contact with the substrate front surface **11**. The partition wall **24** is disposed between the semiconductor light emitting element **80** and the electronic component **100** in the Y-direction and separates the accommodation space **23** into a first accommodation space **23**A accommodating the semiconductor light emitting element **80** and a second accommodation space **23**B accommodating the electronic component **100**. As shown in FIG. **59**, the partition wall **24** connects the third side wall **21***c* and the fourth side wall **21***d*. That is, the first accommodation space **23**A is defined by the first side wall **21***a*, the third side wall **21***c*, the fourth side wall **21***d*, and the partition wall **24**. The second accommodation space **23**B is defined by the second side wall **21***b*, the third side wall **21***c*, the fourth side wall **21***d*, and the partition wall **24**. In the illustrated example, the partition wall **24** extends in the

X-direction in plan view. The partition wall **24** overlaps the recesses **35***e* and **35***f* in the common conductive portion **30** in the Z-direction.

In a second example, as shown in FIGS. **60** and **61**, the semiconductor light emitting device **1**B of the second embodiment includes a partition wall **24**. The partition wall **24** extends downward from the cover **22** of the case **20**B. In the illustrated example, the partition wall **24** is in contact with the substrate front surface **11**. The partition wall **24** is disposed between the semiconductor light emitting element **80** and the electronic component **100** in the Y-direction and separates the accommodation space **23** into a first accommodation space **23**A accommodating the semiconductor light emitting element **80** and the capacitor **120** and a second accommodation space **23**B accommodating the electronic component **100**. As shown in FIG. **61**, the partition wall **24** connects the third side wall **21***c* and the fourth side wall **21***d*. That is, the first accommodation space **23**A is defined by the first side wall **21***a*, the third side wall **21***c*, the fourth side wall **21***d*, and the partition wall **24**. The second accommodation space **23**B is defined by the second side wall **21***b*, the third side wall **21***c*, the fourth side wall **21***d*, and the partition wall **24**.

As shown in FIG. **61**, the partition wall **24** includes a first part **24***a* extending from the third side wall **21***c* in the X-direction, a second part **24***b* extending from the fourth side wall **21***d* in the X-direction, and a step **24***c* formed between the first part **24***a* and the second part **24***b* in the X-direction. That is, the first part **24***a* and the second part **24***b* are located at different positions in the Y-direction. In the illustrated example, the second part **24***b* is located closer to the second side wall **21***b* than the first part **24***a*.

The first part **24***a* separates the semiconductor light emitting element **80** and the electronic component **100**. The second part **24***b* separates the capacitor **120** and the wires **W2**. The step **24***c* is located between the semiconductor light emitting element **80** and the capacitor **120** in the X-direction. That is, the step **24***c* and the second part **24***b* extends the dimension of the part of the first accommodation space **23**A accommodating the capacitor **120** in the Y-direction. Thus, the capacitor **120** is accommodated in the first accommodation space **23**A. The step **24***c* is located between the electronic component **100** and the capacitor **120** in the X-direction. This ensures the space for the encapsulation resin **140** and the coating agent **141** covering the electronic component **100**. The shape of the step **24***c* in plan view may be changed in any manner. In an example, the step **24***c* may diagonally extend toward the second side wall **21***b* from the first part **24***a* to the fourth side wall **21***d*. This ensures the space for the encapsulation resin **140** and the coating agent **141** covering the electronic component **100**.

In a third example, in the semiconductor light emitting device **1**B of the second example, the shape of the partition wall **24** may be changed as shown in FIG. **62**. More specifically, the partition wall **24** is formed so that the semiconductor light emitting element **80** is disposed in the first accommodation space **23**A and that the electronic component **100** and the capacitor **120** are disposed in the second accommodation space **23**B. That is, in plan view, the partition wall **24** is L-shaped and connects the third side wall **21***c* and the first side wall **21***a*. That is, the first accommodation space **23**A is defined by the first side wall **21***a*, the third side wall **21***c*, and the partition wall **24**. The second accommodation space **23**B is L-shaped and defined by the first to fourth side walls **21***a* to **21***d* and the partition wall **24**.

In the second embodiment, as shown in FIG. **63**, a light-blocking wall **25** may extend downward from the edge of the opening **22***a* in the case **20**B. In FIG. **63**, the light-blocking wall **25** extends along the entire edge of the opening **22***a*. In this structure, when light from the semiconductor light emitting element **80** is reflected by the light diffusion plate **130**, irradiation of the electronic component **100** with the reflected light is limited. Thus, erroneous actuation of the electronic component **100** caused by light of the semiconductor light emitting element **80** is limited. The dimension of the light-blocking wall **25** in the Z-direction may be changed in any range that limits irradiation of the electronic component **100** with the light reflected from the light diffusion plate **130**. The light-blocking wall **25** is not limited to the structure in which the light-blocking wall **25** extends along the entire edge of the opening **22***a* and may be formed along a portion of the edge of the opening **22***a*. The light-blocking wall **25** may be formed along the opening **22***a* at least at a part between the semiconductor light emitting element **80** and the electronic component **100**. In FIG. **63**, the light diffusion plate **130** may be plate-shaped as shown in FIG. **19**. Although the encapsulation resin **140** and the coating agent **141** are omitted from FIG. **63**, the encapsulation resin **140** and the coating agent **141** may be added to cover the electronic component **100**.

In the second embodiment, the shape of the light diffusion plate **130** is not limited to being plate-shaped and may be changed in any manner. In an example, as shown in FIG. **63**, the light diffusion plate **130** includes a recess **131** that is rectangular in cross-sectional view. In the illustrated example, the recess **131** includes a flat surface orthogonal to the Z-direction, defining a bottom **131***a*. Since light from the semiconductor light emitting element **80** is reflected at the bottom of the recess **131** in the light diffusion plate **130**, the inner surface defining the opening **22***a* is used as a light-blocking wall. This limits irradiation of the electronic component **100** with the reflected light from the light diffusion plate **130**. As shown in FIG. **63**, the case **20**B may include the combination of the light diffusion plate **130** having the recess **131** and the light-blocking wall **25**. In this case, the irradiation of the electronic component **100** with the reflected light from the light diffusion plate **130** is further limited.

Also, in the first embodiment, when the semiconductor light emitting device **1** includes the case **20**C shown in FIG. **56**, the case **20**C may include at least one of the light-blocking wall **25** and the light diffusion plate **130** having the recess **131**.

In each embodiment, a vent may be arranged between the substrate **10** (**10**B) and the case **20** (**20**B, **20**C) to connect the accommodation space of the case **20** (**20**B, **20**C) and the outside of the case **20** (**20**B, **20**C). A vent **160** will be described using the case **20**B of the semiconductor light emitting device **1**B in the second embodiment.

In an example, as shown in FIG. **64**, the second side wall **21***b* of the frame **21** of the case **20**B includes a side wall recess **160**A. The side wall recess **160**A extends from the inner surface to the outer surface of the second side wall **21***b*. The vent **160** is defined by the side wall recess **160**A and the substrate front surface **11** of the substrate **10**B. The vent **160** extends from the inner surface to the outer surface of the second side wall **21***b* in the Y-direction diagonally from the fourth side wall **21***d* toward the third side wall **21***c* in the X-direction. As shown in FIG. **65**, the width of the vent **160** (side wall recess **160**A) is constant. The vent **160** (side wall recess **160**A) has a first opening region **S1** at the accommodation space **23** and a second opening region **S2** at the outside of the case **20**B. The first opening region **S1**and the second opening region **S2** have the same width. The width of the vent **160** (side wall recess **160**A) refers to a width of the vent **160** in a direction orthogonal to the direction in which the vent **160** extends in plan view. The first opening region S**1**is also referred to as an inner opening region that is open in the inner surface of the side wall (second side wall **21***b*) of the case **20**B. The second opening region S**2** is also referred to as an outer opening region that is open in the outer surface of the side wall (second side wall **21***b*) of the case **20**B.

As shown in FIG. **66**, the side wall recess **160**A is formed by recessing in the Z-direction from an end surface of the second side wall **21***b* that faces the substrate front surface **11** in the Z-direction. As shown in FIGS. **65** and **65**, the side wall recess **160**A is defined by two side walls **161** separated from each other and a top wall **162** connecting the side walls **161**. In the illustrated example, the side walls **161** include tapered inclined surfaces that become closer to each other toward the top wall **162**. The top wall **162** includes a flat surface that is orthogonal to the Z-direction.

The side wall recess **160**A further includes a first side end surface **161***a* and a second side end surface **161***b*. The first side end surface **161***a* is located between the outer surface of the second side wall **21***b* and one of the side walls **161** and includes a bulged surface as viewed in the Z-direction. Another bulged surface that is similar to the first side end surface **161***a* may be located between the outer surface of the second side wall **21***b* and the other one of the side walls **161**. The second side end surface **16** lb is located between the inner surface of the second side wall **21***b* and the other one of the side walls **161** and includes a bulged surface as viewed in the Z-direction. Another bulged surface that is similar to the second side end surface **161***b* may be located between the inner surface of the second side wall **21***b* and the one of the side walls **161**.

The periphery of the vent **160** is partially free of the adhesive agent P**4**, which fixes the case **20**B to the substrate front surface **11**. Avoidance of entrance of the adhesive agent P**4** into the vent **160** maintains ventilation characteristics. The number of vents **160** may be changed in any manner. For example, the frame **21** may include a plurality of vents **160**.

In a side view of the case **20**B, the shape of the vent **160** (side wall recess **160**A) may be changed in any manner.

In a first example, as shown in FIG. **67**, the side wall recess **160**A further includes two curves **163** disposed between the top wall **162** and the side walls **161**. The curves **163** connect the side walls **161** to the top wall **162**. Alternatively, the side walls **161** of the vent **160** may be curved instead of the curves **163**.

In a second example, as shown in FIG. **68**, the side wall recess **160**A includes a curved surface **164** instead of the side walls **161** and the top wall **162**. The curved surface **164** is recessed to be further from the substrate front surface **11** as the curved surface **164** extends toward a width-wise central part of the vent **160**.

In a third example, as shown in FIG. **69**, the side wall recess **160**A is a V-shaped groove. More specifically, the vent **160** includes two inclined surfaces **165** that are inclined further from the substrate front surface **11** as the inclined surfaces **165** extend toward the width-wise central part of the vent **160**. The inclined surfaces **165** are connected each other at the width-wise central part of the vent **160**.

As viewed in the Z-direction, the shape of the vent **160** (side wall recess **160**A) may be changed in any manner.

In a first example, as shown in FIG. **70**, the vent **160** (side wall recess **160**A) has a structure in which the first opening region S**1** has a greater width than the second opening region S**2**. In other words, the vent **160** (side wall recess **160**A) has a structure in which the second opening region S**2** has a greater width than the first opening region S**1**. In the illustrated example, the width of the vent **160** (side wall recess **160**A) decreases from the first opening region S**1** toward the second opening region S**2**. That is, as viewed in the Z-direction, the vent **160** (side wall recess **160**A) is tapered to become smaller from the first opening region S**1** toward the second opening region S**2**. In this structure, the width of the second opening region S**2** is decreased to further hinder unintentional entrance of an object.

In a second example, as shown in FIG. **71**, the vent **160** (side wall recess **160**A) has a structure in which the first opening region S**1** has a smaller width than the second opening region S**2**. In the illustrated example, the width of the vent **160** (side wall recess **160**A) decreases from the second opening region S**2** toward the first opening region S**1**. That is, as viewed in the Z-direction, the vent **160** (side wall recess **160**A) is tapered to become smaller from the second opening region S**2** toward the first opening region S**1**. In this structure, the width of the first opening region S**1** is decreased to further hinder unintentional entrance of an object.

In a third example, as shown in FIG. **72**, the side wall recess **160**A linearly extends in the Y-direction as viewed in the Z-direction. Thus, the vent **160** is linearly formed in the Y-direction. In the illustrated example, the width of the vent **160** (side wall recess **160**A) is constant. The width of the first opening region S**1** is equal to the width of the second opening region S**2**. In the side wall recess **160**A shown in FIG. **72**, the width of the vent **160** (side wall recess **160**A) may be changed. In an example, the width of the vent **160** (side wall recess **160**A) may decrease or increase from the first opening region S**1** toward the second opening region S**2**.

In a fourth example, as shown in FIG. **73**, as viewed in the Z-direction, the side wall recess **160**A is labyrinth-shaped (crank-shaped). More specifically, the side wall recess **160**A includes a first recess **166** including the first opening region S**1**, a second recess **167** including the second opening region S**2**, and a third recess **168** connecting the first recess **166** to the second recess **167**. In the illustrated example, as viewed in the Z-direction, the first recess **166** and the second recess **167** each extend in the Y-direction. As viewed in the Z-direction, the first recess **166** and the second recess **167** are located at different positions in the X-direction. The third recess **168** extends in a direction intersecting the first recess **166** and the second recess **167**. In the illustrated example, the third recess **168** extends in the X-direction. As described above, the vent **160** is labyrinth-shaped (crank-shaped). In other words, the vent **160** includes a first vent defined by the first recess **166** and the substrate front surface **11**, a third vent defined by the second recess **167** and the substrate front surface **11**, and a second vent defined by the third recess **168** and the substrate front surface **11**. More specifically, as a labyrinth structure, the vent **160** includes the first vent extending from the inner surface toward the outer surface of the side wall (second side wall **21***b*) of the case **20**B, the second vent connected to the first vent and extending in a direction that intersects the extension direction of the first vent, and the third vent connected to the second vent and extending from the inner surface toward the outer surface of the side wall (second side wall **21***b*). This structure further hinders unintentional entrance of an object.

In FIGS. **64** to **73**, the vent **160** is defined by the side wall recess **160**A in the case **20**B and the substrate front surface **11**. However, there is no limitation to such structures. For example, as in a first example and a second example described below, the structure of the vent **160** may be changed.

In the first example, as shown in FIG. **74**, the substrate **10**B includes a substrate recess **160**B. The vent **160** is defined by the substrate recess **160**B and an end surface of the frame **21**. In the illustrated example, the vent **160** is defined by the substrate recess **160**B and the end surface of the second side wall **21***b*.

The substrate recess **160**B is recessed from the substrate front surface **11** toward the substrate back surface **12**. In other words, the substrate recess **160**B is recessed in the substrate front surface **11** to be separate from the frame **21** in the Z-direction. In the illustrated example, the substrate recess **160**B is formed in the insulation portion **13**. In the illustrated example, the shape of the substrate recess **160**B is symmetric to the shape of the side wall recess **160**A shown in FIG. **66**. That is, the substrate recess **160**B is defined by the side walls **161** separated from each other and the top wall **162** connecting the side walls **161**. The side walls **161** include tapered inclined surfaces that become closer to each other toward the top wall **162**. The shape of the substrate recess **160**B is not limited to the above structure and may be changed to the side wall recesses **160**A shown in FIGS. **67** to **73**.

In a second example, as shown in FIG. **75**, the case **20**B includes the side wall recess **160**A, and the substrate **10**B includes the substrate recess **160**B. The vent **160** is defined by the side wall recess **160**A and the substrate recess **160**B. The side wall recess **160**A is symmetrical to the substrate recess **160**B. The shape of the side wall recess **160**A and the shape of the substrate recess **160**B are not limited to those shown in FIG. **75** and may be changed to the shape of the side wall recesses **160**A shown in FIGS. **67** to **73**. In this case, in a side view of the case **20**B, the side wall recess **160**A does not have to be symmetrical to the substrate recess **160**B.

The structure of the vent **160** is not limited to those shown in FIGS. **64** to **75** and may be a structure obtained by using the difference in the adhesiveness of the case **20**B to the substrate **10**B. More specifically, as shown in FIGS. **76** and **77**, the case **20**B and the substrate **10**B do not include the side wall recess **160**A and the substrate recess **160**B. The case **20**B shown in FIGS. **76** and **77** has a structure such that an end surface of the frame **21** in the Z-direction is separated into a first region **21***ra* and a second region **21***rb* in a peripheral direction of the frame **21**. The first region **21***ra* is substantially smaller than the second region **21***rb*. The surface roughness of the first region **21***ra* differs from the surface roughness of the second region **21***rb*, so that the first region **21***ra* and the second region **21***rb* differ from each other in bonding strength to the adhesive agent P4. For example, the surface roughness (Ra) of the first region **21***ra* is approximately greater than or equal to 0.01 μm and less than or equal to 0.1 μm, and the surface roughness (Ra) of the second region **21***rb* is approximately greater than or equal to 1.0 μm and less than or equal to 20 μm. The second region **21***rb* has a greater roughness than the first region **21***ra*. The thickness of the adhesive agent P4 is, for example, approximately greater than or equal to 15 μm and less than or equal to 40 μm. Examples of a process for forming the second region **21***rb* as described above include a mechanical process such as sandblasting and a chemical process such as one using a chemical solution. An example of such processes is a chemical process that applies a chemical such as a stripping agent to part of the end surface of the frame **21**.

As shown in FIGS. **76** and **77**, the vent **160** is the part defined between the adhesive agent P4 and the first region **21***ra* of the end surface of the frame **21** in the Z-direction. However, during general transportation, storage, and use of the semiconductor light emitting device **1**B, the case **20**B is bonded to the substrate **10**B by the adhesive agent P**4** in the first region **21***ra*. In this state, the vent **160** is not a clear hole that connects the accommodation space **23** to the outside.

FIG. **77** schematically shows that the internal pressure of the accommodation space **23** is increased in a process for mounting the semiconductor light emitting device **1**B using, for example, a reflow oven. An increase in the internal pressure of the accommodation space **23** produces force that separates the case **20**B from the adhesive agent P**4**. The force locally separates the bonded portion in the first region **21***ra*, which is set to have a relatively weak bonding force. As a result, the vent **160** has the form of a gap and connects the accommodation space **23** to the outside. To facilitate understanding, FIG. **77** shows a form in which the vent **160** is a clear gap. However, an actual vent **160** may be any vent that releases gas from the accommodation space **23** to the outside. When the first region **21***ra* is slightly separated from the adhesive agent P**4**, the gas is guided from the accommodation space **23** to the outside. When the internal pressure of the accommodation space **23** is decreased as a result of the ventilation, the first region **21***ra* is again in contact with the adhesive agent P**4**.

This structure increases the reliability of the semiconductor light emitting device **1**B. In addition, the vent **160** including the first region **21***ra* is closed in general use. That is, the first region **21***ra* is in contact with the adhesive agent P**4**. This further ensures the hindrance of unintentional entrance of an object such as moisture. In addition, after the ventilation shown in FIG. **77** is performed, the vent **160** may be again closed. This hinders unintentional entrance of an object even in subsequent use.

The formation position of the vent **160** and the number of vents **160** may be changed in any manner.

When the vent **160** described above is applied to the first embodiment, the side wall recess **160**A is arranged in the end surface of the frame **21** that faces the substrate front surface **11** in the Z-direction. The side wall recess **160**A is formed from the inner surface to the outer surface of a side wall of the frame **21**. The side wall may be at least one of the side walls **21***a* to **21***d* of the frame **21**.

The vent **160** described above may be applied to the semiconductor light emitting devices **1** and **1**B having the partition wall **24** in each embodiment. In this case, the vent **160** includes a first vent that connects the first accommodation space **23**A to the outside of the cases **20** and **20**B and a second vent that connects the second accommodation space **23**B to the outside of the cases **20** and **20**B. The first vent is, for example, arranged in one of the side walls **21***a* to **21***d* defining the first accommodation space **23**A. The second vent is, for example, arranged in one of the side walls **21***a* to **21***d* defining the second accommodation space **23**B.

In the second embodiment, the opening **22***a* in the case **20**B may be changed in any manner. In an example, as shown in FIGS. **78** to **80**, the shape of the opening **22***a* may be a square in plan view. In the illustrated example, the square opening **22***a* is obtained by extending the dimension of the opening **22***a* in the Y-direction in the second embodiment. In this case, as shown in FIGS. **79** and **80**, in the accommodation space **23**, the first part **22***b*, which is part of the cover **22** located at the same position as the opening **22***a* in the Y-direction, has a smaller thickness than the second part **22***c*, which is part of the cover **22** located closer to the second side wall **21***b* than the opening **22***a*. In other words, the second part **22***c* has a larger thickness than the first part **22***b*. As shown in FIG. **80**, the first part **22***b* is located adjacent to the opening **22***a* in the X-direction and extends from the opening **22***a* to the fourth side wall **21***d*. The second part **22***c* extends from the third side wall **21***c* to the fourth side wall **21***d* in the X-direction. The shape of the opening **22***a* in plan view is not limited to an equiangular parallelogram such as a square or a rectangle and may be a circle, an ellipse, or an oval.

In each embodiment, the structure for accommodating the semiconductor light emitting element **80** and the electronic component **100** may be changed in any manner. In an example, in the semiconductor light emitting device **1**, the semiconductor light emitting element **80** and the electronic component **100** may be encapsulated by an encapsulation resin instead of the case **20**. In the semiconductor light emitting device **1**B, the semiconductor light emitting element **80**, the electronic component **100**, and the capacitor **120** may be encapsulated by an encapsulation resin instead of the case **20**B.

In a specific example, as shown in FIG. **81**, the semiconductor light emitting device **1**B includes a light-transmissive encapsulation resin **170** that encapsulates the semiconductor light emitting element **80**, the electronic component **100**, and the capacitor **120**. The encapsulation resin **170** is rectangular-box-shaped. The encapsulation resin **170** is formed from an electrically insulative resin material. The encapsulation resin **170** includes an upper surface **171** and side surfaces **172**. Recesses **171***a* are formed in part of the upper surface **171** facing the semiconductor light emitting element **80** in the Z-direction. The recesses **171***a* are, for example, separated from each other in the X-direction and the Y-direction. This allows for diffusion of light from the semiconductor light emitting element **80**.

A cutaway portion **173** is formed in the encapsulation resin **170** between the semiconductor light emitting element **80** and the electronic component **100** in the Y-direction. The cutaway portion **173** includes a light-blocking wall **174**. In an example, the light-blocking wall **174** is formed by filling the cutaway portion **173** with a light-blocking material. An example of the light-blocking material is a colored resin material. This structure limits irradiation of the electronic component **100** with light from the semiconductor light emitting element **80**, thereby limiting erroneous actuation of the electronic component **100**. As viewed in the Z-direction, the shape of each of the cutaway portion **173** and the light-blocking wall **174** is, for example, the same as the shape of the partition wall **24** shown in FIG. **61** or **62**. Instead of forming the light-blocking wall **174** by filling the cutaway portion **173** with a light-blocking material, for example, the light-blocking wall **174** may be plate-shaped in advance and inserted into the cutaway portion **173**.

In another example, as shown in FIG. **82**, the semiconductor light emitting device **1**B includes a frame **180**. The frame **180** separates the semiconductor light emitting element **80** and the electronic component **100**. The frame **180** is, for example, formed from a light-blocking material. In an example, the frame **180** is formed from a colored resin material. The frame **180** includes a partition wall **183** that separates a first receptacle **181** accommodating the semiconductor light emitting element **80** and a second receptacle **182** accommodating the electronic component **100**. In the illustrated example, the partition wall **183** and the frame **180** are integrally formed from the same material. The first receptacle **181** is filled with a first encapsulation resin **190**A. The second receptacle **182** is filled with a second encapsulation resin **190**B. In the illustrated example, the encapsulation resins **190**A and **190**B are both transmissive to light. The material of the first encapsulation resin **190**A is the same as the material of the second encapsulation resin **190**B. The material of the first encapsulation resin **190**A may differ from the material of the second encapsulation resin **190**B. In an example, the second encapsulation resin **190**B may be filled with a light-blocking material. In an example, a light-blocking resin material is used as the second encapsulation resin **190**B. As viewed in the Z-direction, the shape of the partition wall **183** is, for example, the same as the shape of the partition wall **24** shown in FIG. **61** or **62**. The first encapsulation resin **190**A may be omitted.

In the semiconductor light emitting device **1**B, the light diffusion plate **130** is attached to the frame **180** to cover the first receptacle **181** in the Z-direction. The light diffusion plate **130** is supported by the partition wall **183** and part of the frame **180**. In this structure, the partition wall **183** limits irradiation of the electronic component **100** with light from the semiconductor light emitting element **80**, thereby limiting erroneous actuation of the electronic component **100**.

In the second embodiment, at least one of the encapsulation resin **140** and the coating agent **141** that covers the electronic component **100** may be omitted (for example, refer to FIG. **63**).

In the first embodiment, at least one of the encapsulation resin **140** and the coating agent **141** that covers the electronic component **100** may be added.

In the second embodiment, the coating agent **141** may be applied to only the portion of the conductive bonding material P**2**, which connects the electronic component **100** to the second common contact front surface portion **30***d*, extending from the electronic component **100**. Thus, sulfidation of the conductive bonding material P**2** is inhibited. Also, in the first embodiment, the coating agent **141** may be applied to only the portion of the conductive bonding material P**2** extending from the electronic component **100**.

In the second embodiment, one of the two recesses **35***b* may be omitted from the common conductive portion **30**B. Also, one of the two recesses **35***c* may be omitted from the common conductive portion **30**B. Also, at least one of the recess **35***a*, the two recesses **35***b*, the two recesses **35***c*, the recess **35***d*, and the recess **35***e* may be omitted from the common conductive portion **30**B.

In the second embodiment, the control conductive portion **70**B may include at least one recess. The recess has the same structure as, for example, the recess **35***a*.

In the second embodiment, the first common contact back surface portion **30***e* may be continuous with the second common contact back surface portion **30***f*. That is, the recess **35***f* may be omitted from the common conductive portion **30**B. This structure increases the size of the common contact back surface **30***b*, thereby improving the heat dissipation efficiency of the semiconductor light emitting element **80** and the electronic component **100**.

In the second embodiment, the common conductive portion **30**B may include a groove arranged between the semiconductor light emitting element **80** and the electronic component **100** and recessed from the common contact front surface **30***a* toward the substrate back surface **12** in the Z-direction. In an example, the groove extends through the common conductive portion **30**B in the X-direction. The groove is free of the insulation portion **13**. In this case, the recess **35***f* may be omitted from the common conductive portion **30**B. This structure hinders the encapsulation resin **140** and the coating agent **141** from entering the light emitting regions **90** of the semiconductor light emitting element **80**. The groove does not have to extend through the common conductive portion **30**B in the X-direction. The number of grooves may be changed. The common conductive portion **30**B may include a plurality of grooves.

In the second embodiment, the first drive contact back surface portion **60***e* may be continuous with the second drive contact back surface portion **60***f*. That is, the recess **65***a* may be omitted from the drive conductive portion **60**B.

In the second embodiment, the position of the semiconductor light emitting element **80** relative to the first common contact front surface portion **30***c* in the Y-direction may be changed in any manner. In an example, the semiconductor light emitting element **80** may be disposed on the central part of the first common contact front surface portion **30***c* in the Y-direction or a part of the first common contact front surface portion **30***c* located toward the first side wall **21***a*. The semiconductor light emitting element **80** may be disposed on the first common contact front surface portion **30***c* so as not to extend to the element contact front surface **50***a* in the Y-direction.

In the second embodiment, the position of the semiconductor light emitting element **80** relative to the first common contact front surface portion **30***c* in the X-direction may be changed in any manner. In an example, the semiconductor light emitting element **80** may be disposed on the central part of the first common contact front surface portion **30***c* in the X-direction or a part of the first common contact front surface portion **30***c* located toward the third side wall **21***c*.

In the second embodiment, the position of the electronic component **100** relative to the second common contact front surface portion **30***d* in the Y-direction may be changed in any manner. In an example, the electronic component **100** may be disposed on the central part of the second common contact front surface portion **30***d* in the Y-direction or a part of the second common contact front surface portion **30***d* located toward the second side wall **21***b*.

In the second embodiment, the position of the electronic component **100** relative to the second common contact front surface portion **30***d* in the X-direction may be changed in any manner. In an example, the electronic component **100** may be disposed on a part of the second common contact front surface portion **30***d* located toward the second drive contact front surface portion **60***d* or a part of the second common contact front surface portion **30***d* located toward the control contact front surface **70***a*.

In the second embodiment, the semiconductor light emitting element **80** and the electronic component **100** may be aligned with each other in the X-direction and separated from each other in the Y-direction.

In the second embodiment, as viewed in the X-direction, the capacitor **120** may overlap the electronic component **100**.

In the second embodiment, the number of capacitors **120** incorporated in the semiconductor light emitting device **1**B may be changed in any manner. For example, the semiconductor light emitting device **1**B may include a plurality of capacitors **120**.

In the second embodiment, the shape of the contact front surfaces **30***a*, **50***a*, **60***a*, and **70***a* may be changed in any manner. For example, the common contact front surface **30***a* may be smaller than the element contact front surface **50***a*. The drive contact front surface **60***a* and the control contact front surface **70***a* may be larger than the element contact front surface **50***a*. The contact front surfaces **30***a*, **50***a*, **60***a*, and **70***a* may all have the same size. Alternatively, some of the contact front surfaces **30***a*, **50***a*, **60***a*, and **70***a* may have the same size, while the remaining contact front surfaces differ from each other in size. At least one of the contact front surfaces **30***a*, **50***a*, **60***a*, and **70***a* may be elliptical or circular.

In the second embodiment, the size of the first common contact front surface portion **30***c* may be changed in any manner. In an example, the first common contact front surface portion **30***c* may be smaller than or equal to the drive contact front surface **60***a*. In another example, the first common contact front surface portion **30***c* may be smaller than or equal to at least one of the element contact front surface **50***a* and the control contact front surface **70***a*.

In the second embodiment, the size of the second common contact front surface portion **30***d* may be changed in any manner. In an example, the second common contact front surface portion **30***d* may be smaller than or equal to the drive contact front surface **60***a*. In another example, the second common contact front surface portion **30***d* may be smaller than or equal to at least one of the element contact front surface **50***a* and the control contact front surface **70***a*.

In the second embodiment, the size of the first drive contact front surface portion **60***c* may be changed in any manner. In an example, the first drive contact front surface portion **60***c* may be larger than or equal to the element contact front surface **50***a*. In another example, the first drive contact front surface portion **60***c* may be smaller than or equal to the control contact front surface **70***a*.

In the second embodiment, the size of the second drive contact front surface portion **60***d* may be changed in any manner. In an example, the second drive contact front surface portion **60***d* may be larger than or equal to the element contact front surface **50***a*. In another example, the second drive contact front surface portion **60***d* may be smaller than or equal to the control contact front surface **70***a*.

In the second embodiment, the shape of the contact back surfaces **30***b*, **50***b*, **60***b*, and **70***b* may be changed in any manner. For example, the common contact back surface **30***b* may be smaller than the element contact back surface **50***b*. The drive contact back surface **60***b* and the control contact back surface **70***b* may be larger than the element contact back surface **50***b*. The contact back surfaces **30***b*, **50***b*, **60***b*, and **70***b* may all have the same size. Alternatively, some of the contact back surfaces **30***b*, **50***b*, **60***b*, and **70***b* may have the same size while the remaining contact back surfaces differ from each other in size. At least one of the contact back surfaces **30***b*, **50***b*, **60***b*, and **70***b* may be elliptical or circular.

In the second embodiment, the size of the first common contact back surface portion **30***e* may be changed in any manner. In an example, the first common contact back surface portion **30***e* may be smaller than or equal to the drive contact back surface **60***b*. In another example, the first common contact back surface portion **30***e* may be smaller than or equal to at least one of the element contact back surface **50***b* and the control contact back surface **70***b*.

In the second embodiment, the size of the second common contact back surface portion **30***f* may be changed in any manner. In an example, the second common contact back surface portion **30***f* may be smaller than or equal to the drive contact back surface **60***b*. In another example, the second common contact back surface portion **30***f* may be smaller than or equal to at least one of the element contact back surface **50***b* and the control contact back surface **70***b*.

In the second embodiment, the size of the first drive contact back surface portion **60***e* may be changed in any manner. In an example, the first drive contact back surface portion **60***e* may be larger than or equal to the element contact back surface **50***b*. In another example, the first drive contact back surface portion **60**e may be smaller than or equal to the control contact back surface **70**b.

In the second embodiment, the size of the second drive contact back surface portion **60**f may be changed in any manner. In an example, the second drive contact back surface portion **60**f may be larger than or equal to the element contact back surface **50**b. In another example, the second drive contact back surface portion **60**f may be smaller than or equal to the control contact back surface **70**b.

In the second embodiment, the flange **36** may be omitted from the common conductive portion **30**B. The flange **56** may be omitted from the element conductive portion **50**B. The flange **66** may be omitted from the drive conductive portion **60**B. The flange **76** may be omitted from the control conductive portion **70**B.

In the second embodiment, the light diffusion plate **130** may be omitted from the semiconductor light emitting device **1**B.

In the second embodiment, the height of the semiconductor light emitting element **80** from the substrate front surface **11** and the height of the electronic component **100** from the substrate front surface **11** may be changed in any manner. In an example, the height of the electronic component **100** from the substrate front surface **11** may be greater than or equal to the height of the semiconductor light emitting element **80** from the substrate front surface **11**.

In the second embodiment, the position of the flange **36** relative to the common conductive portion **30**B in the Z-direction may be changed in any manner. In an example, the flange **36** may be formed toward the common contact back surface **30**b rather than the common contact front surface **30**a. The flange **36** may be flush with the common contact back surface **30**b.

In the second embodiment, the position of the flange **56** relative to the element conductive portion **50**B in the Z-direction may be changed in any manner. In an example, the flange **56** may be formed toward the element contact back surface **50**b rather than the element contact front surface **50**a. The flange **56** may be flush with the element contact back surface **50**b.

In the second embodiment, the position of the flange **66** relative to the drive conductive portion **60**B in the Z-direction may be changed in any manner. In an example, the flange **66** may be formed toward the drive contact back surface **60**b rather than the drive contact front surface **60**a. The flange **66** may be flush with the drive contact back surface **60**b.

In the second embodiment, the number of capacitors **120** may be changed in any manner. In an example, the semiconductor light emitting device **1**B includes two capacitors **120**.

In the second embodiment, the position of the flange **76** relative to the control conductive portion **70**B in the Z-direction may be changed in any manner. In an example, the flange **76** may be formed toward the control contact back surface **70**b rather than the control contact front surface **70**a. The flange **76** may be flush with the control contact back surface **70**b.

In the second embodiment, the structure of the substrate **10**B may be changed in any manner. In an example, the semiconductor light emitting device **1**B may include a substrate formed from an insulative material like the substrate **10** of the first embodiment instead of a substrate formed of a lead frame. In this case, the substrate may be, for example, a ceramic such as alumina or aluminum nitride, a silicon substrate, or a glass epoxy. The substrate includes a common conductive portion **30**B, an element conductive portion **50**B, a drive conductive portion **60**B, and a control conductive portion **70**B that include, for example, a front surface conductive layers formed on the substrate front surface, back surface conductive layers formed on the substrate back surface, and joints electrically connecting the front surface conductive layers to the back surface conductive layers.

In each embodiment, the positions of the semiconductor light emitting element **80** and the electronic component **100** on the common contact front surface **30**a of the common conductive portion **30**B may be changed in any manner. In an example, as shown in FIG. **83**, the semiconductor light emitting element **80** and the electronic component **100** may be disposed on the first common contact front surface portion **30**c of the common conductive portion **30**B. In the illustrated example, as viewed in the X-direction, the semiconductor light emitting element **80** and the electronic component **100** overlap with the capacitor **120**.

The semiconductor light emitting element **80** is disposed on the first common contact front surface portion **30**c close to the end **31**b in the Y-direction. The semiconductor light emitting element **80** is disposed on a part of the first common contact front surface portion **30**c opposed to the element conductive portion **50**B in the X-direction. In the illustrated example, the semiconductor light emitting element **80** is disposed on the first common contact front surface portion **30**c closer to the end **31**b in the Y-direction than the insulation portion **13** that is located between the element conductive portion **50**B and the drive conductive portion **60**B. In this case, as viewed from above, the wires W1 extend in the X-direction. The semiconductor light emitting element **80** is also disposed on the first common contact front surface portion **30**c toward the element conductive portion **50**B in the X-direction.

The electronic component **100** is disposed on an end of the first common contact front surface portion **30**c located close to the second common contact front surface portion **30**d in the Y-direction. More specifically, the electronic component **100** is disposed on a part of the first common contact front surface portion **30**c opposed to the first drive contact front surface portion **60**c of the drive conductive portion **60**B in the X-direction. In this case, the wires W**2** are connected to the first drive contact front surface portion **60**c. The electronic component **100** is also disposed on the first common contact front surface portion **30**c toward the drive conductive portion **60**B in the Y-direction. The wire W3 is connected to one of the opposite ends of the control conductive portion **70**B in the Y-direction that is located closer to the first common contact front surface portion **30**c.

As viewed in the X-direction, the semiconductor light emitting element **80** may partially extend closer to toward the end **31**b of the common conductive portion **30**B than the capacitor **120**. The electronic component **100** may partially extend from the first common contact front surface portion **30**c to the second common contact front surface portion **30**d in the Y-direction.

The structure shown in FIG. **83** shortens the conductive path between the semiconductor light emitting element **80** and the electronic component **100** and the conductive path between the electronic component **100** and the capacitor **120**. Accordingly, parasitic capacitance caused by the conductive path between the semiconductor light emitting element **80** and the electronic component **100** and parasitic capacitance caused by the conductive path between the electronic component **100** and the capacitor **120** are decreased.

In each electronic component, the structure of the electronic component **100** may be changed in any manner. In an example, as shown in FIG. **84**, the first drive electrode **101** and the second drive electrode **103** of the electronic component **100** are formed on the lower surface **100***b* of the electronic component **100**. The control electrode **102** is formed on the upper surface **100***a* of the electronic component **100**. The first drive electrode **101** and the second drive electrode **103** of the electronic component **100** are aligned with each other in the Y-direction and separated from each other in the X-direction. FIG. **84** does not show the encapsulation resin **140** and the coating agent **141** for the sake of convenience.

As shown in FIG. **84**, the electronic component **100** may be flip-chip-mounted on the common contact front surface **30***a* of the common conductive portion **30**B and the drive contact front surface **60***a* of the drive conductive portion **60**B. In this case, the electronic component **100** extends over the insulation portion **13** located between the common conductive portion **30**B and the drive conductive portion **60**B. The first drive electrode **101** faces the first common contact front surface portion **30***c* of the common conductive portion **30**B in the Z-direction. The second drive electrode **103** faces the first drive contact front surface portion **60***c* of the drive conductive portion **60**B in the Z-direction.

More specifically, in the Z-direction, the first drive electrode **101** is bonded to part of the first common contact front surface portion **30***c* opposed to the first drive contact front surface portion **60***c* of the drive conductive portion **60**B in the X-direction. The second drive electrode **103** is bonded to one of the opposite ends of the first drive contact front surface portion **60***c* in the X-direction located closer to the first common contact front surface portion **30***c*.

When the electronic component **100** is disposed in this manner, the semiconductor light emitting element **80** is located on the common contact front surface **30***a* close to the end **31***b* in the Y-direction. More specifically, the semiconductor light emitting element **80** is disposed on a part of the first common contact front surface portion **30***c* opposed to the element conductive portion **50**B in the X-direction.

The structure shown in FIG. **84** shortens the conductive path between the semiconductor light emitting element **80** and the electronic component **100** and the conductive path between the electronic component **100** and the capacitor **120**. Accordingly, parasitic capacitance caused by the conductive path between the semiconductor light emitting element **80** and the electronic component **100** and parasitic capacitance caused by the conductive path between the electronic component **100** and the capacitor **120** are decreased. In addition, when the electronic component **100** is flip-chip-mounted on the common contact front surface **30***a* and the drive contact front surface **60***a*, parasitic capacitance between the electronic component **100** and the drive contact front surface **60***a* is decreased as compared to when the first drive electrode **101** is connected to the drive contact front surface **60***a* by the wires W**2**.

In the modified example shown in FIG. **84**, the capacity of the capacitor **120** may be reduced. Accordingly, as shown in FIG. **85**, the capacitor **120** may be located closer to the semiconductor light emitting element **80** and the electronic component **100**. More specifically, the first electrode **121** of the capacitor **120** is located adjacent to one of the wires W**1** of the semiconductor light emitting element **80** located closer to the drive conductive portion **60**B in the X-direction. The second electrode **122** of the capacitor **120** is located adjacent to the electronic component **100** in the X-direction. FIG. **85** does not show the encapsulation resin **140** and the coating agent **141** for the sake of convenience.

The structure shown in FIG. **85** shortens the conductive path between the semiconductor light emitting element **80** and the electronic component **100**, the conductive path between the semiconductor light emitting element **80** and the capacitor **120**, and the conductive path the electronic component **100** and the capacitor **120**. Accordingly, parasitic capacitance caused by the conductive path between the semiconductor light emitting element **80** and the electronic component **100**, parasitic capacitance caused by the conductive path between the semiconductor light emitting element **80** and the capacitor **120**, and parasitic capacitance caused by the conductive path between the electronic component **100** and the capacitor **120** are decreased.

In the modified examples shown in FIGS. **83** to **85**, the semiconductor light emitting element **80** may include a first electrode **91** and a second electrode **92** formed on the lower surface **80***b* of the semiconductor light emitting element **80**. The first electrode **91** and the second electrode **92** are, for example, aligned with each other in the Y-direction and separated from each other in the X-direction. In this case, the semiconductor light emitting element **80** is flip-chip-mounted on the common contact front surface **30***a* of the common conductive portion **30**B and the element contact front surface **50***a* of the element conductive portion **50**B. Therefore, the semiconductor light emitting element **80** extends over the insulation portion **13** located between the common conductive portion **30**B and the element conductive portion **50**B. The first electrode **91** faces the element contact front surface **50***a* of the element conductive portion **50**B in the Z-direction. The second electrode **92** faces the first common contact front surface portion **30***c* in the Z-direction.

## CLAUSES

The technical ideas obtainable from the above embodiments and modified examples are described below.

### Clause A1

A semiconductor light emitting device, including

a substrate formed from an insulative material;

a common front surface conductive layer formed on a front surface of the substrate;

a semiconductor light emitting element disposed on a common contact front surface formed of a front surface of the common front surface conductive layer, and

an electronic component disposed on the common contact front surface and electrically connected to the semiconductor light emitting element by the common front surface conductive layer.

### Clause A2

A semiconductor light emitting device, including

a substrate formed from a conductive material;

a common conductive portion formed of a portion of the substrate and including a common contact front surface formed of a front surface of the substrate;

a semiconductor light emitting element disposed on the common contact front surface; and

an electronic component disposed on the common contact front surface and electrically connected to the semiconductor light emitting element by the common conductive portion.

Clause A3

An electronic apparatus on which a semiconductor light emitting device is mounted.

Clause B1

The semiconductor light emitting device, in which

the substrate includes a substrate contact surface that is in contact with a side wall of the frame,

the substrate contact surface includes a substrate recess recessed downward from the substrate contact surface and formed from an inner surface to an outer surface of the side wall of the case, and

the vent is defined by the side wall of the frame and the substrate recess.

Clause B2

The semiconductor light emitting device, in which

a side wall of the frame includes a side wall contact surface that is in contact with the substrate,

the side wall contact surface includes a recess upwardly recessed from the side wall contact surface and formed from an inner surface to an outer surface of the side wall,

the substrate includes a substrate contact surface that is in contact with the side wall of the frame,

the substrate contact surface includes a substrate recess downwardly recessed from the substrate contact surface and formed from the inner surface to the outer surface of the side wall of the frame, and

the vent is defined by the side wall recess and the substrate recess.

Clause B3

The semiconductor light emitting device according to any one of clauses B1 and B2, in which the vent has a labyrinth structure.

Clause B4

The semiconductor light emitting device according to clause B3, in which as the labyrinth structure, the vent includes a first vent extending from an inner surface toward an outer surface of the side wall of the frame, a second vent connected to the first vent and extending in a direction that intersects an extension direction of the first vent, and a third vent connected to the second vent and extending from the inner surface toward the outer surface.

Clause B5

The semiconductor light emitting device according to any one of clauses B1 and B2, in which as viewed in a direction orthogonal to the planar direction, the vent extends in a direction inclined from a direction orthogonal to an extension direction of the side wall of the frame.

Clause B6

The semiconductor light emitting device according to any one of clauses B1 and B2, in which as viewed in the planar direction, the vent extends in a direction in which the side wall of the frame extends.

Clause B7

The semiconductor light emitting device according to any one of clauses B1 and B2, in which

the side wall recess includes an inner opening region that is open in an inner surface of the side wall of the frame,

the side wall recess includes an outer opening region that is open in an outer surface of the side wall of the frame, and

the inner opening region is smaller than the outer opening region.

Clause B8

The semiconductor light emitting device according to any one of clauses B1 and B2, in which

the side wall recess includes an inner opening region that is open in an inner surface of the side wall of the frame,

the side wall recess includes an outer opening region that is open in an outer surface of the side wall of the frame, and

the inner opening region is larger than the outer opening region.

Clause B9

The semiconductor light emitting device according to any one of clauses B1 and B2, in which

the side wall recess includes an inner opening region that is open in an inner surface of the side wall of the frame,

the side wall recess includes an outer opening region that is open in an outer surface of the side wall of the frame, and

the inner opening region is equal to the outer opening region.

Clause B10

The semiconductor light emitting device, in which

the frame is disposed inward from a peripheral edge of the substrate front surface,

as viewed in a direction orthogonal to the planar direction of the substrate, at least one of the common conductive portion, the control conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion includes a projection extending outward from the frame.

Clause C1

The semiconductor light emitting device, in which

the electronic component has a lower surface on which a first drive electrode and a second drive electrode are formed,

the second drive electrode is bonded to the common conductive portion, and the substrate includes a drive conductive portion including a drive contact front surface that is opposed and bonded to the first drive electrode.

Clause C2

The semiconductor light emitting device according to clause C1, in which the semiconductor light emitting element and the electronic component are arranged in a predetermined direction,

the common conductive portion includes a common contact front surface extending in an arrangement direction of the semiconductor light emitting element and the electronic component, in a planar direction of the substrate, when the arrangement direction of the semiconductor light emitting element and the electronic component is referred to as a first direction, and a direction orthogonal to the first direction is referred to as a second direction, the common contact front surface includes a first common contact front surface portion and a second common contact front surface portion that are arranged in the first direction,

the drive contact front surface is located adjacent to the first contact front surface, and

the semiconductor light emitting element and the electronic component are disposed on the first common contact front surface.

Clause D1. A semiconductor light emitting device, comprising:

a substrate;

a common conductive portion formed on the substrate;

a semiconductor light emitting element mounted on the common conductive portion; and

an electronic component mounted on the common conductive portion and electrically connected to the semiconductor light emitting element by the common conductive portion.

Clause D2. The semiconductor light emitting device according to clause D1, wherein the electronic component is configured to drive the semiconductor light emitting element.

Clause D3. The semiconductor light emitting device according to clause D2, wherein

the semiconductor light emitting element has a lower surface on which an element lower surface electrode is formed,

the electronic component has an upper surface on which a first drive electrode and a control electrode are formed and a lower surface on which a second drive electrode is formed, and

the element lower surface electrode and the second drive electrode are bonded to the common conductive portion.

Clause D4. The semiconductor light emitting device according to clause D3, wherein

the semiconductor light emitting element and the electronic component are arranged in a predetermined direction,

the common conductive portion includes a common contact front surface extending in an arrangement direction of the semiconductor light emitting element and the electronic component, and

the semiconductor light emitting element and the electronic component are disposed on the common contact front surface.

Clause D5. The semiconductor light emitting device according to clause D4, wherein

the substrate includes a drive conductive portion and a control conductive portion,

the drive conductive portion includes a drive contact front surface electrically connected to the first drive electrode by a wire,

the control conductive portion includes a control contact front surface electrically connected to the control electrode by a wire, and

in a planar direction of the substrate, when the arrangement direction is referred to as a first direction, and a direction orthogonal to the first direction is referred to as a second direction, the drive conductive portion and the control conductive portion are separately disposed at opposite sides of the common contact front surface in the second direction.

Clause D6. The semiconductor light emitting device according to clause D5, wherein the drive contact front surface and the control contact front surface are separately disposed at opposite sides of the electronic component in the second direction.

Clause D7. The semiconductor light emitting device according to clause D5 or D6, wherein the common contact front surface is larger than the drive contact front surface and the control contact front surface.

Clause D8. The semiconductor light emitting device according to clause D6 or D7, wherein

the semiconductor light emitting element includes an upper surface on which an element upper surface electrode is formed,

an element conductive portion is formed at one side of the common contact front surface in the second direction,

the element conductive portion includes an element contact front surface electrically connected to the element upper surface electrode,

a connection conductive portion is formed at the other side of the common contact front surface in the second direction and is electrically connected to the common conductive portion, and

the connection conductive portion includes a connection contact front surface projecting in the second direction from one of opposite ends of the common contact front surface in the second direction that is located at a side opposite from the element conductive portion.

Clause D9. The semiconductor light emitting device according to clause D8, wherein the element conductive portion and the drive conductive portion are disposed at the same side of the common contact front surface in the second direction.

Clause D10. The semiconductor light emitting device according to clause D9, further comprising a capacitor disposed so as to span between the element contact front surface and the drive contact front surface.

Clause D11. The semiconductor light emitting device according to any one of clauses D8 to D10, wherein

the connection conductive portion includes a connection contact back surface disposed on a back surface of the substrate at a position opposite from the connection contact front surface,

the element conductive portion includes an element contact back surface disposed on the back surface of the substrate at a position opposite from the element contact front surface,

the drive conductive portion includes a drive contact back surface disposed on the back surface of the substrate at a position opposite from the drive contact front surface, and

the control conductive portion includes a control contact back surface disposed on the back surface of the substrate at a position opposite from the control contact front surface.

Clause D12. The semiconductor light emitting device according to clause D11, wherein the element contact back surface is larger than the drive contact back surface and the control contact back surface.

Clause D13. The semiconductor light emitting device according to clause D11 or D12, wherein the common conductive portion includes a common contact back surface disposed on the back surface of the substrate at a position opposite from the common contact front surface.

Clause D14. The semiconductor light emitting device according to clause D13, wherein the common contact back surface is separate from the connection contact back surface.

Clause D15. The semiconductor light emitting device according to clause D13 or D14, wherein the common contact back surface is larger than the connection contact back surface, the element contact back surface, the drive contact back surface, and the control contact back surface.

Clause D16. The semiconductor light emitting device according to any one of clauses D11 to D15, wherein

the substrate is formed from an insulative material,

each of the common contact front surface, the connection contact front surface, the element contact front surface, the drive contact front surface, and the control contact front surface is a surface of a conductive layer formed on a front surface of the substrate, and

each of the connection contact back surface, the element contact back surface, the drive contact back surface, and the control contact back surface is a surface of a conductive layer formed on a back surface of the substrate.

Clause D17. The semiconductor light emitting device according to any one of clauses D11 to D15, wherein

the substrate is formed of a conductive material,

the common conductive portion, the connection conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion are portions of the substrate that are insulated and separated from each other by an insulation portion,

the common contact front surface, the connection contact front surface, the element contact front surface, the drive contact front surface, and the control contact front surface are formed of a front surface of the substrate, and

the connection contact back surface, the element contact back surface, the drive contact back surface, and the control contact back surface are formed of a back surface of the substrate.

Clause D18. The semiconductor light emitting device according to clause D17, wherein

at least one of the common conductive portion, the connection conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion has a peripheral edge including a recess recessed in a direction orthogonal to the planar direction of the substrate, and

the insulation portion is disposed in the recess.

Clause D19. The semiconductor light emitting device according to clause D18, wherein at least one of the common contact front surface, the connection contact front surface, the element contact front surface, the drive contact front surface, and the control contact front surface has a peripheral edge including the recess.

Clause D20. The semiconductor light emitting device according to any one of clauses D17 to D19, wherein

at least one of the common conductive portion, the connection conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion has a peripheral edge including a flange,

the substrate includes a substrate back surface located opposite from a surface on which the semiconductor light emitting element is mounted, and

the insulation portion is disposed between the flange and the substrate back surface in a direction orthogonal to the planar direction of the substrate.

Clause D21. The semiconductor light emitting device according to any one of clauses D5 to D7, wherein

the common contact front surface includes a first common contact front surface portion on which the semiconductor light emitting element is mounted and a second common contact front surface portion on which the electronic component is mounted,

the second common contact front surface portion is disposed between the drive conductive portion and the control conductive portion in the second direction and extends more than the first common contact front surface portion toward one side in the second direction,

the first common contact front surface portion extends from the second common contact front surface portion in the first direction, and

the semiconductor light emitting element is disposed on the first common contact front surface portion toward the second common contact front surface portion.

Clause D22. The semiconductor light emitting device according to clause D21, wherein

the semiconductor light emitting element has an upper surface on which an element upper surface electrode is formed,

an element conductive portion is formed at one side of the common contact front surface in the second direction,

the element conductive portion includes an element contact front surface electrically connected to the element upper surface electrode,

the element conductive portion is disposed at one side of the first common contact front surface in the second direction,

at least a portion of the semiconductor light emitting element is disposed on the first common contact front surface portion at a position closer to the electronic component than the element contact front surface,

the element upper surface electrode and the element contact front surface are connected by a wire, and

in plan view, the wire diagonally extends from the element upper surface electrode toward the element contact front surface so as to extend away from the electronic component.

Clause D23. The semiconductor light emitting device according to clause D22, wherein the semiconductor light emitting element and the electronic component are arranged in the first direction so that the electronic component is displaced from the semiconductor light emitting element toward one side in the second direction.

Clause D24. The semiconductor light emitting device according to clause D22 or D23, further comprising a capacitor disposed so as to span between the element contact front surface and the drive contact front surface, wherein

the drive contact front surface includes a first drive contact front surface portion on which the capacitor is disposed and a second drive contact front surface portion electrically connected to the first drive electrode by a wire,

the first drive contact front surface portion is located closer to the element contact front surface than the second drive contact front surface portion,

the first drive contact front surface portion extends in the second direction, and

the second drive contact front surface portion extends in the first direction.

Clause D25. The semiconductor light emitting device according to clause D24, wherein

the second drive contact front surface portion is recessed relative to the first drive contact front surface portion so as to have a smaller dimension than the first drive contact front surface portion in the second direction,

the first common contact front surface portion is opposed to the first drive contact front surface portion in the second direction,

the second common contact front surface portion is disposed in a recess region defined by the first drive contact front surface portion and the second drive contact front surface portion, and

the second common contact front surface portion is opposed to the second drive contact front surface portion in the second direction.

Clause D26. The semiconductor light emitting device according to clause D24 or D25, wherein

the wire connecting the first drive electrode and the second drive contact front surface portion includes multiple wires, and

the wires are arranged in the first direction as viewed in a direction orthogonal to the planar direction of the substrate.

Clause D27. The semiconductor light emitting device according to clause D26, wherein among the wires, the farthest two wires are connected to the first drive electrode and the second drive contact front surface portion so that the two wires are separated from each other more at the second drive contact front surface portion than at the first drive electrode as viewed in the direction orthogonal to the planar direction of the substrate.

Clause D28 The semiconductor light emitting device according to any one of clauses D24 to D27, wherein

the semiconductor light emitting element and the electronic component are arranged in the first direction so that the electronic component is displaced from the semiconductor light emitting element toward one side in the second direction, and

the capacitor is disposed at one side of the semiconductor light emitting element in the second direction.

Clause D29. The semiconductor light emitting device according to clause D28, wherein the capacitor is disposed at one side of the electronic component in the first direction without overlapping the electronic component as viewed in the second direction.

Clause D30. The semiconductor light emitting device according to any one of clauses D22 to D29, wherein

the first common contact front surface portion extends more than the second common contact front surface portion toward a side opposite from the one side in the second direction, and

the control contact front surface is formed in a region surrounded by the first common contact front surface portion and the second common contact front surface portion.

Clause D31. The semiconductor light emitting device according to any one of clauses D22 to D30, wherein

the common conductive portion includes a common contact back surface disposed on a back surface of the substrate at a position opposite from the common contact front surface,

the element conductive portion includes an element contact back surface disposed on the back surface of the substrate at a position opposite from the element contact front surface,

the drive conductive portion includes a drive contact back surface disposed on the back surface of the substrate at a position opposite from the drive contact front surface, and

the control conductive portion includes a control contact back surface disposed on the back surface of the substrate at a position opposite from the control contact front surface.

Clause D32. The semiconductor light emitting device according to clause D31, wherein the element contact back surface is equal to the drive contact back surface and the control contact back surface.

Clause D33. The semiconductor light emitting device according to clause D31 or D32, wherein

the common contact back surface includes a first common contact back surface portion and a second common contact back surface portion,

the first common contact back surface portion is disposed on the back surface of the substrate at a position opposite from the first common contact front surface portion,

the second common contact back surface portion is disposed on the back surface of the substrate at a position opposite from the second common contact front surface portion, and

the first common contact back surface portion is separate from the second common contact back surface portion.

Clause D34. The semiconductor light emitting device according to any one of clauses D31 to D33, wherein

the common contact back surface includes a first common contact back surface portion and a second common contact back surface portion,

the first common contact back surface portion is disposed on the back surface of the substrate at a position opposite from the first common contact front surface portion,

the second common contact back surface portion is disposed on the back surface of the substrate at a position opposite from the second common contact front surface portion, and

the first common contact back surface portion is larger than the element contact back surface, the drive contact back surface, and the control contact back surface.

Clause D35. The semiconductor light emitting device according to any one of clauses D31 to D34, further comprising a capacitor disposed so as to span between the element contact front surface and the drive contact front surface, wherein

the drive contact front surface includes a first drive contact front surface portion on which the capacitor is disposed and a second drive contact front surface portion electrically connected to the first drive electrode by a wire,

the drive contact back surface includes a first drive contact back surface portion and a second drive contact back surface portion,

the first drive contact back surface portion is disposed on the back surface of the substrate at a position opposite from the first drive contact front surface portion,

the second drive contact back surface portion is disposed on the back surface of the substrate at a position opposite from the second drive contact front surface portion, and

the first drive contact back surface portion is separate from the second drive contact back surface portion.

Clause D36. The semiconductor light emitting device according to clause D35, wherein

the common contact back surface includes a first common contact back surface portion and a second common contact back surface portion, the first common contact back surface portion is disposed on the back surface of the substrate at a position opposite from the first common contact front surface portion,

the second common contact back surface portion is disposed on the back surface of the substrate at a position opposite from the second common contact front surface portion, and

in the planar direction of the substrate, when an arrangement direction of the element conductive portion and the drive conductive portion is referred to as a third direction and a direction orthogonal to the third direction is referred to as a fourth direction, the second drive contact back surface portion and the control contact back surface are separately disposed at opposite sides of the second common contact back surface portion in the fourth direction.

Clause D37. The semiconductor light emitting device according to any one of clauses D31, 32, and 34, wherein

the substrate is formed from a conductive material,

the common conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion are portions of the substrate that are insulated and separated from each other by an insulation portion,

the common contact front surface, the element contact front surface, the drive contact front surface, and the control contact front surface are formed of a front surface of the substrate, and

the element contact back surface, the drive contact back surface, and the control contact back surface are formed of a back surface of the substrate.

Clause D38. The semiconductor light emitting device according to clause D37, wherein

at least one of the common conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion includes a peripheral edge including a recess recessed in a direction orthogonal to the planar direction of the substrate, and

the insulation portion is accommodated in the recess.

Clause D39. The semiconductor light emitting device according to clause D38, wherein at least one of the common contact front surface, the element contact front surface, the drive contact front surface, and the control contact front surface includes a peripheral edge including the recess.

Clause D40. The semiconductor light emitting device according to clause D39, further comprising a capacitor disposed so as to span between the element contact front surface and the drive contact front surface, wherein

the drive contact front surface includes a first drive contact front surface portion on which the capacitor is disposed and a second drive contact front surface portion electrically connected to the first drive electrode by the wire, and

the recess is disposed in the drive contact front surface between the first drive contact front surface portion and the second drive contact front surface portion.

Clause D41. The semiconductor light emitting device according to any one of clauses D37 to D40, wherein

at least one of the common conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion includes a peripheral edge including a flange,

the substrate includes a substrate back surface located opposite from a surface on which the semiconductor light emitting element is mounted, and

the insulation portion is accommodated between the flange and the substrate back surface in a direction orthogonal to the planar direction of the substrate.

Clause D42. The semiconductor light emitting device according to any one of clauses D1 to D41, wherein the electronic component is covered by a light-blocking resin material.

Clause D43. The semiconductor light emitting device according to any one of clauses D1 to D42, wherein

the electronic component is bonded to the common conductive portion by a conductive bonding material, and

at least a portion of the conductive bonding material exposed from the electronic component is covered by an anti-sulfidation coating agent.

Clause D44. The semiconductor light emitting device according to any one of clauses D1 to D43, further comprising a case accommodating the semiconductor light emitting element and the electronic component.

Clause D45. The semiconductor light emitting device according to clause D44, wherein

the semiconductor light emitting element and the electronic component are arranged in a predetermined direction,

the case includes opposite side walls in an arrangement direction of the semiconductor light emitting element and the electronic component, defining a first side wall and a second side wall,

the semiconductor light emitting element is located closer to the first side wall than the electronic component, and

the electronic component is located closer to the second side wall than the semiconductor light emitting element.

Clause D46. The semiconductor light emitting device according to clause D44 or D45, wherein

the case includes a frame and a cover,

the frame is formed from a light-blocking material and has an upward opening, and

the cover closes the opening of the frame.

Clause D47. The semiconductor light emitting device according to clause D46, wherein the cover is transmissive to light from the semiconductor light emitting element.

Clause D48. The semiconductor light emitting device according to clause D47, wherein

the cover includes a transmissive portion and a light-blocking portion,

the transmissive portion is disposed above the semiconductor light emitting element and transmissive to light from the semiconductor light emitting element, and

the light-blocking portion is disposed above the electronic component and blocks light.

Clause D49. The semiconductor light emitting device according to clause D48, wherein the cover diffuses light from the semiconductor light emitting element.

Clause D50. The semiconductor light emitting device according to clause D44, wherein

the case is box-shaped and includes a cover and a side wall,

the cover is formed from a light-blocking material and opposed to the substrate in a direction orthogonal to the planar direction of the substrate,

the side wall is formed from a light-blocking material and extends downward from an edge of the cover,

the cover includes an opening in a position opposed to the semiconductor light emitting element,

the opening extends through the cover in the direction orthogonal to the planar direction, a light diffusion plate is attached to the cover at a side opposite from the semiconductor light emitting element in the direction orthogonal to the planar direction to cover the opening, and

the light diffusion plate transmits and diffuses light from the semiconductor light emitting element.

Clause D51. The semiconductor light emitting device according to clause D50, wherein the light diffusion plate includes a recess recessed from the opening in the direction orthogonal to the planar direction in a direction away from the semiconductor light emitting element.

Clause D52. The semiconductor light emitting device according to any one of clauses D1 to D51, further comprising a light-blocking partition wall that separates the semiconductor light emitting element and the electronic component.

Clause D53. The semiconductor light emitting device according to any one of clauses D44 to D50, further comprising a light-blocking partition wall that separates the semiconductor light emitting element and the electronic component,

wherein the partition wall is arranged on the case.

Clause D54. The semiconductor light emitting device according to clause D53, wherein

the case is box-shaped and includes a cover and a side wall,

the cover is opposed to the substrate in a direction orthogonal to the planar direction of the substrate,

the side wall extends downward from an edge of the cover, and

the partition wall is disposed so that the cover extends downward.

Clause D55. The semiconductor light emitting device according to any one of clauses D44 to D51, D53, and D54, wherein

the semiconductor light emitting element is accommodated in an accommodation space defined by the substrate and the case, and

a vent is provided between the substrate and the case to connect the accommodation space to the outside.

Clause D56. The semiconductor light emitting device according to clause D55, wherein

the side wall of the case includes an end surface opposed to the substrate in the direction orthogonal to the planar direction of the substrate,

the end surface includes a side wall recess recessed from the end surface in a direction away from the substrate and formed from an inner surface to an outer surface of the side wall of the case, and

the vent is defined by the substrate and the side wall recess.

Clause D57. The semiconductor light emitting device according to clause D55, wherein

the substrate includes a substrate front surface opposed to the side wall of the case in the direction orthogonal to the planar direction of the substrate,

the substrate front surface includes a substrate recess recessed from the substrate front surface in a direction away from the case and formed from an inner surface to an outer surface of the side wall of the case, and

the vent is defined by the side wall of the case and the substrate recess.

Clause D58. The semiconductor light emitting device according to clause D55, wherein

the side wall of the case includes an end surface opposed to the substrate in the direction orthogonal to the planar direction of the substrate,

the end surface includes a side wall recess recessed from the end surface in a direction away from the substrate and formed from an inner surface to an outer surface of the side wall,

the substrate includes a substrate front surface opposed to the side wall of the case in a direction orthogonal to the planar direction of the substrate.

the substrate front surface includes a substrate recess recessed from the substrate front surface in a direction away from the case and formed from the inner surface to the outer surface of the side wall of the case, and

the vent is defined by the side wall recess and the substrate recess.

Clause D59. The semiconductor light emitting device according to any one of clauses D55 to D58, wherein the vent has a labyrinth structure.

Clause D60. The semiconductor light emitting device according to clause D59, wherein as the labyrinth structure, the vent includes a first vent extending from an inner surface toward an outer surface of the side wall of the case, a second vent connected to the first vent and extending in a direction that intersects an extension direction of the first vent, and a third vent connected to the second vent and extending from the inner surface toward the outer surface.

Clause D61. The semiconductor light emitting device according to any one of clauses D55 to D58, wherein as viewed in the direction orthogonal to the planar direction of the substrate, the vent extends in a direction inclined from a direction orthogonal to an extension direction of the side wall of the case.

Clause D62. The semiconductor light emitting device according to any one of clauses D55 to D58, wherein, as viewed in the direction orthogonal to the planar direction of the substrate, the vent extends in a direction orthogonal to an extension direction of the side wall of the case.

Clause D63. The semiconductor light emitting device according to any one of clauses D55 to D58, wherein

the vent includes an inner opening region that is open in an inner surface of the side wall of the case and an outer opening region that is open in an outer surface of the side wall of the case, and

the inner opening region is smaller than the outer opening region.

Clause D64. The semiconductor light emitting device according to any one of clauses D55 to D58, wherein

the vent includes an inner opening region that is open in an inner surface of the side wall of the case and an outer opening region that is open in an outer surface of the side wall of the case, and

the inner opening region is larger than the outer opening region.

Clause D65. The semiconductor light emitting device according to any one of clauses D55 to D58, wherein

the vent includes an inner opening region that is open in an inner surface of the side wall of the case and an outer opening region that is open in an outer surface of the side wall of the case, and

the inner opening region is equal to the outer opening region.

Clause D66. The semiconductor light emitting device according to clause D46, wherein

the semiconductor light emitting element is accommodated in an accommodation space defined by the substrate and the case, and

a vent is provided between the substrate and a side wall of the frame to connect the accommodation space to the outside.

Clause D67. The semiconductor light emitting device according to clause D66, wherein

the side wall of the frame includes an end surface opposed to the substrate in the direction orthogonal to the planar direction of the substrate,

the end surface includes a side wall recess recessed from the end surface in a direction away from the substrate and formed from an inner surface to an outer surface of the side wall, and

the vent is defined by the substrate and the side wall recess.

Clause D68. The semiconductor light emitting device according to any one of clauses D44 to D51 and D53 to D67, wherein

the substrate includes a drive conductive portion, a control conductive portion, and an element conductive portion,

the drive conductive portion includes a drive contact front surface electrically connected to a first drive electrode of the electronic component by a wire,

the control conductive portion includes a control contact front surface electrically connected to a control electrode of the electronic component by a wire,

the element conductive portion includes an element contact front surface electrically connected to an element upper surface electrode of the electronic component by a wire,

the substrate includes a substrate front surface opposed to a side wall of the case in a direction orthogonal to the planar direction of the substrate,

the side wall of the case is disposed inward from a peripheral edge of the substrate front surface, and

at least one of the common conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion includes a projection extending outward from the side wall of the case.

Clause D69. The semiconductor light emitting device according to any one of clauses D1 to D68, wherein an upper surface of the electronic component is located at a lower position than an upper surface of the semiconductor light emitting element.

Clause D70. The semiconductor light emitting device according to any one of clauses D1 to D43, further comprising a light-transmissive encapsulation resin encapsulating the semiconductor light emitting element and the electronic component,

wherein a light-blocking partition wall is disposed in the encapsulation resin between the semiconductor light emitting element and the electronic component.

Clause D71. The semiconductor light emitting device according to any one of clauses D1 to D43, wherein

the semiconductor light emitting element and the electronic component are accommodated in a frame having an upward opening,

the frame is attached to the substrate,

the frame includes a partition wall that separates the semiconductor light emitting element from the electronic component, and

the frame includes a first receptacle accommodating the semiconductor light emitting element and a second receptacle accommodating the electronic component that are separated by the partition wall.

Clause D72. The semiconductor light emitting device according to clause D71, further comprising a light-transmissive light diffusion plate covering the first receptacle from above.

Clause D73, The semiconductor light emitting device according to clause D71 or D72, wherein

the semiconductor light emitting element is encapsulated by a light-transmissive first encapsulation resin in the first receptacle, and

the electronic component is encapsulated by a second encapsulation resin in the second receptacle.

Clause D74. The semiconductor light emitting device according to clause D73, wherein the first encapsulation resin and the second encapsulation resin are formed from different materials.

Clause D75. The semiconductor light emitting device according to clause D74, wherein the second encapsulation resin is formed from a light-blocking material.

Clause D76. The semiconductor light emitting device according to any one of clauses D1 to D75, wherein the semiconductor light emitting element includes a semiconductor laser element.

Clause D77. The semiconductor light emitting device according to clause D76, wherein the semiconductor light emitting element includes a vertical cavity surface emitting laser (VCSEL) element.

## DESCRIPTION OF THE REFERENCE NUMERALS

**1**, **1**B) semiconductor light emitting device
**2**, **2**B) electronic apparatus
**10**, **10**B) substrate
**11**) substrate front surface
**12**) substrate back surface
**13**) insulation portion
**13**L) back-half insulation portion
**13**U) front-half insulation portion
**20**, **20**B, **20**C) case
**21**) frame
**21***a*) first side wall
**21***b*) second side wall
**21***c*) third side wall
**21***d*) fourth side wall
**22**) cover
**22***a*) opening
**23**) accommodation space
**23**A) first accommodation space
**23**B) second accommodation space
**24**) partition wall
**25**) light-blocking wall
**30**, **30**B) common conductive portion
**30***a*) common contact front surface
**30***b*) common contact back surface
**30***c*) first common contact front surface portion
**30***d*) second common contact front surface portion
**30***e*) first common contact back surface portion
**30***f*) second common contact back surface portion
**31**) common front surface conductive layer
**32**) common back surface conductive layer
**33**) common conductive portion
**34***a* to **34***g*) projection
**35***a* to **35***e*) recess
**36**) flange
**40**) connection conductive portion
**40***a*) connection contact front surface
**40***b*) connection contact back surface
**41**) contact front surface conductive layer
**42**) contact back surface conductive layer
**43**) connection joint
**44***a* to**44***c*) projection

**50**, **50**B) element conductive portion
**50***a*) element contact front surface
**50***b*) element contact back surface
**51**) element front surface conductive layer
**52**) element back surface conductive layer
**53**) element joint
**54***a* to **54***f*) projection
**55**) recess
**56**) flange
**60**, **60**B) drive conductive portion
**60***a*) drive contact front surface
**60***b*) drive contact back surface
**60***c*) first drive contact front surface portion
**60***d*) second drive contact front surface portion
**60***e*) first drive contact back surface portion
**60***f*) second drive contact back surface portion
**60***r*) recess region
**61**) drive front surface conductive layer
**62**) drive back surface conductive layer
**63**) drive joint
**64***a* to **64***f*) projection
**65**) recess
**66**) flange
**70**, **70**B) control conductive portion
**70***a*) control contact front surface
**70***b*) control contact back surface
**71**) control front surface conductive layer
**72**) control back surface conductive layer
**73**) control joint
**74***a*, **74***b*) projection
**76**) flange
**80**) semiconductor light emitting element
**80***a*) element upper surface
**80***b*) element lower surface
**91**) element upper surface electrode
**92**) element lower surface electrode
**100**) electronic component
**100***a*) upper surface
**100***b*) lower surface
**101**) first drive electrode
**102**) control electrode
**103**) second drive electrode
**120**) capacitor
**130**) light diffusion plate
**131**) recess
**140**) encapsulation resin
**141**) coating agent
**160**) vent
**160**A) side wall recess
**160**B) substrate recess
**166**) first vent
**167**) second vent
**168**) third vent
**170**) encapsulation resin
**180**) frame
**181**) first receptacle
**182**) second receptacle
**183**) partition wall
**190**A) first encapsulation resin
**190**B) second encapsulation resin
**201**) transmissive portion
**202**) light-blocking portion
P**2**) conductive bonding material
S**1**) first opening region (inner opening region)
S**2**) second opening region (outer opening region)
W**1** to W**3**) wire



The invention claimed is:

**1**. A semiconductor light emitting device, comprising:

a substrate;

a common conductive portion formed on the substrate;

a semiconductor light emitting element mounted on the common conductive portion; and

an electronic component mounted on the common conductive portion and electrically connected to the semiconductor light emitting element by the common conductive portion, wherein

the electronic component is configured to drive the semiconductor light emitting element,

the semiconductor light emitting element has a lower surface on which an element lower surface electrode is formed,

the electronic component has an upper surface on which a first drive electrode and a control electrode are formed and a lower surface on which a second drive electrode is formed,

the element lower surface electrode and the second drive electrode are bonded to the common conductive portion,

the semiconductor light emitting element and the electronic component are arranged in a predetermined direction,

the common conductive portion includes a common contact front surface extending in thelain **1** has arrangement direction of the semiconductor light emitting element and the electronic component,

the semiconductor light emitting element and the electronic component are disposed on the common contact front surface,

the substrate includes a drive conductive portion and a control conductive portion,

the drive conductive portion includes a drive contact front surface electrically connected to the first drive electrode by a wire,

the control conductive portion includes a control contact front surface electrically connected to the control electrode by a wire, and

in a planar direction of the substrate, when the arrangement direction is referred to as a first direction, and a direction orthogonal to the first direction is referred to as a second direction, the drive conductive portion and the control conductive portion are separately disposed at opposite sides of the common contact front surface in the second direction.

**2**. The semiconductor light emitting device according to claim **1**, wherein the drive contact front surface and the control contact front surface are separately disposed at opposite sides of the electronic component in the second direction.

**3**. The semiconductor light emitting device according to claim **1**, wherein the common contact front surface is larger than the drive contact front surface and the control contact front surface.

**4**. The semiconductor light emitting device according to claim **2**, wherein

the semiconductor light emitting element includes an upper surface on which an element upper surface electrode is formed,

an element conductive portion is formed at one side of the common contact front surface in the second direction, the element conductive portion includes an element contact front surface electrically connected to the element upper surface electrode,

a connection conductive portion is formed at the other side of the common contact front surface in the second direction and is electrically connected to the common conductive portion, and

the connection conductive portion includes a connection contact front surface projecting in the second direction from one of opposite ends of the common contact front surface in the second direction that is located at a side opposite from the element conductive portion.

**5**. The semiconductor light emitting device according to claim **4**, wherein the element conductive portion and the drive conductive portion are disposed at the same side of the common contact front surface in the second direction.

**6**. The semiconductor light emitting device according to claim **5**, further comprising a capacitor disposed so as to span between the element contact front surface and the drive contact front surface.

**7**. The semiconductor light emitting device according to claim **4**,

wherein the connection conductive portion includes a connection contact back surface disposed on a back surface of the substrate at a position opposite from the connection contact front surface,

the element conductive portion includes an element contact back surface disposed on the back surface of the substrate at a position opposite from the element contact front surface,

the drive conductive portion includes a drive contact back surface disposed on the back surface of the substrate at a position opposite from the drive contact front surface, and

the control conductive portion includes a control contact back surface disposed on the back surface of the substrate at a position opposite from the control contact front surface.

**8**. The semiconductor light emitting device according to claim **7**, wherein the element contact back surface is larger than the drive contact back surface and the control contact back surface.

**9**. The semiconductor light emitting device according to claim **7**, wherein the common conductive portion includes a common contact back surface disposed on the back surface of the substrate at a position opposite from the common contact front surface.

**10**. The semiconductor light emitting device according to claim **9**, wherein the common contact back surface is separate from the connection contact back surface.

**11**. The semiconductor light emitting device according to claim **9**, wherein the common contact back surface is larger than the connection contact back surface, the element contact back surface, the drive contact back surface, and the control contact back surface.

**12**. The semiconductor light emitting device according to claim **7**, wherein

the substrate is formed from an insulative material,

each of the common contact front surface, the connection contact front surface, the element contact front surface, the drive contact front surface, and the control contact front surface is a surface of a conductive layer formed on a front surface of the substrate, and

each of the connection contact back surface, the element contact back surface, the drive contact back surface, and the control contact back surface is a surface of a conductive layer formed on a back surface of the substrate.

**13**. The semiconductor light emitting device according to claim **7**, wherein

the substrate is formed of a conductive material,

the common conductive portion, the connection conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion are portions of the substrate that are insulated and separated from each other by an insulation portion,

the common contact front surface, the connection contact front surface, the element contact front surface, the drive contact front surface, and the control contact front surface are formed of a front surface of the substrate, and

the connection contact back surface, the element contact back surface, the drive contact back surface, and the control contact back surface are formed of a back surface of the substrate.

**14**. The semiconductor light emitting device according to claim **13**, wherein

at least one of the common conductive portion, the connection conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion has a peripheral edge including a recess recessed in a direction orthogonal to the planar direction of the substrate, and

the insulation portion is disposed in the recess.

**15**. The semiconductor light emitting device according to claim **14**, wherein at least one of the common contact front surface, the connection contact front surface, the element contact front surface, the drive contact front surface, and the control contact front surface has a peripheral edge including the recess.

**16**. The semiconductor light emitting device according to claim **13**, wherein

at least one of the common conductive portion, the connection conductive portion, the element conductive portion, the drive conductive portion, and the control conductive portion has a peripheral edge including a flange,

the substrate includes a substrate back surface located opposite from a surface on which the semiconductor light emitting element is mounted, and

the insulation portion is disposed between the flange and the substrate back surface in a direction orthogonal to the planar direction of the substrate.

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